The Fermat tower in the metabelian approximation

Grothendieck, *La Longue Marche*, §51 and its refinements

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INCOMPLETE: the exact residual questions are retained in the paper and ledger.

We give proofs and reproducible calculations for the integral schemes, the Fermat modules and splitting, the cuspidal motive, the arithmetic Kummer images, and the covering comparisons. We preserve the arithmetic plus-Fermat model of the manuscript. Full profinite realization questions and the remaining cubic section calculation are explicitly unresolved; the finite and rational calculations below do not replace those questions.

Notation fixed on this page. C4 means cote 140-4. A.1 gives the expanded dictionary, including the changing early and final versions.

Our notation	Meaning and manuscript symbol	C4 scan; folio
Π_0, F, M	$F \xrightarrow{\sim} \Pi_0$, $x \mapsto \lambda_0$, $y \mapsto \lambda_1$; $M = \Pi_0^{\mathrm{ab}}$	34–38; 664–666
$\mathfrak{S}_{\mathrm{d}}, \mathcal{S}$	$\mathrm{SL}_2(\mathbb{Z})/\Pi_0'$ and its indicated finite schematic stage; his $\mathfrak{S}$ and underlined $\mathcal{G}^+$	34–38; 664–666
$\gamma = \mathbb{O}_{12}^*$	Units of his ring scheme O_{12}, with multiplier μ	51–58; 673–676v
$N, \Gamma, \mathcal{M}$	His corrected $N_{\rho,\sigma}$ and final $\Gamma_{\rho,\sigma}, M_{\rho,\sigma}$	62, 85–88; 678v, 690–691v
X_n, U_n	$X_n : x_0^n + x_1^n + x_\infty^n = 0$; U_n deletes its coordinate cusps	97–99; 696–697
Λ, c	$\widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}^2]]$; a specified generator of $\widehat{F}'/\widehat{F}''$	comparison with 95–99; 695–697
$G_{\mathbb{Q}}, \chi, \kappa_a$	Arithmetic $\mathbb{I}\Gamma_{\mathbb{Q}}$; cyclotomic multiplier; Kummer cocycle with chosen roots	156–170; 725v–732v

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Scope, choices, and verification

We use the supplied second-pass transcriptions and inspect the available scans at the mathematical decision points recorded in the readings ledger. A transcription is evidence for a reading, and a proved identity is evidence for the mathematical meaning; the latter does not settle unread handwriting. The manuscript passage for each numbered statement is also recorded in `LM7_FERMAT_TOWER_results.csv`. All named groups in an argument carry the discrete, profinite, pro-ℓ, or algebraic interpretation stated there. Prime-to-characteristic and fppf assertions are specified separately.

We label a proved calculation *Proved*, a false proposed assertion together with its proved replacement *Refuted*, and a reduction to an opened theorem *Known*. An *Open* label records an uncompleted step of this return, with its proved cases and attempted approaches. Such a label does not assert that the literature has no answer. The report has a row for each of the three items, three leads, lettered tasks, and follow-ups in the request. The bibliography records which original texts were opened; a bibliographic identification alone is labelled unverified.

The programs use integer arithmetic, exact rational linear algebra, finite permutation groups, and finite-field arithmetic. Their saved standard output and machine-readable witnesses accompany the source. Finite verification is used with the displayed general proofs, and a Lie-algebra computation is transferred to a group only where the requisite comparison is proved. The source and complete reproduction instructions are in `code/`.

A. Integral structure, level twelve, and the congruence actions

We use constant finite group schemes, including in residue characteristic dividing their order. Thus C_2 below means the disjoint union of two copies of $\operatorname{Spec}\mathbb{Z}$; the morphism $C_2 \to \mu_2$, $i \mapsto (-1)^i$, will always be written explicitly when it is used. All scheme exact sequences in this section are exact for the fppf topology. Our finite-stage notation distinguishes a group from its schematic envelope by specifying their comparison morphism.

A.1. Dictionary and the finite-stage comparison maps [Proved]

Fix the following matrices, over $\mathbb{Z}$:

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \tau = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \omega = -I.$$

Put

$$\lambda_0 = \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix}, \quad \lambda_1 = \begin{pmatrix} -1 & 0 \\ -2 & -1 \end{pmatrix}, \quad \Pi_0 = \langle \lambda_0, \lambda_1 \rangle, \quad M = \Pi_0^{\mathrm{ab}} = \mathbb{Z}e_0 \oplus \mathbb{Z}e_1.$$

The following dictionary fixes the successive versions in §51.

Our symbol	Definition and manuscript symbol	C4 scan; folio
$\mathfrak{S}_{\mathrm{d}}$	$\mathrm{SL}_2(\mathbb{Z})/\Pi_0'$, the determinant-one part of the discrete $\mathfrak{S}$; $\Pi_0' = [\Pi_0, \Pi_0]$	33–34; 664–664v
$\mathcal{S}$	Pushout of $1 \to M \to \mathfrak{S}_{\mathrm{d}} \to H \to 1$ along $M \to \underline{M} = \mathbb{G}_a^2$; his $\underline{\mathcal{G}}^+$	34–38; 664–666
H	$\mathrm{SL}_2(\mathbb{Z})/\Pi_0$, of order 12; the full determinant-sign quotient has order 24	33, 66; 664, 680v
$\mathbb{O}_N$	Additive pushout $\mathbb{Z}_{\mathrm{const}} \amalg_{N\mathbb{Z}} \underline{N\mathbb{Z}}$, with its ring law in A.5	51–54; 673–674v
γ	$\mathbb{O}_{12}^*$; $\mu : \gamma \to \mathbb{G}_m$ is evaluation; $\varepsilon_2, \varepsilon_3 : \gamma \to C_2$ are the two residue characters	55–58; 675–676v
M_0	$\{c_0e_0 + c_1e_1 : c_0 + c_1 \equiv 0 \pmod 3\}$, a lattice with its own vector group $\underline{M}_0$	60–62; 677v–678v
N	The corrected $\underline{N}_{\rho,\sigma}$, with the loop equation in $\underline{M}_0$	62; 678v
Z, G	$\underline{Z}_{\rho,\sigma}$ and $\underline{G}_{\rho,\sigma}$, namely $\mathbb{G}_a \rtimes \gamma$ and $\mathcal{S} \rtimes \gamma$	75–76; 685–685v
$\Gamma_{\mathrm{e}}, \mathcal{M}_{\mathrm{e}}$	The versions of $\Gamma_{\rho,\sigma}, \underline{M}_{\rho,\sigma}$ in (123)–(127)	70–72; 682v–683v
$N_\rho, N_\rho^?$	The smooth closure of the ρ-normalizer, and its final bisection (212); A.13 records the smaller literal version (171)	78–82, 94; 686v–688v, 694v
$N_\sigma, N_\sigma^\sharp$	σ-normalizer and its intersection with $G^\sharp = \underline{M} \rtimes \gamma$	83–84; 689–689v
$\Gamma, \mathcal{M}$	The final fibre products (183)–(184), retaining the central C_2	85–88; 690–691v
$\mathcal{M}(0)$, $\mathcal{M}(1/2)$, $\mathcal{M}(-\bar{j})$	The three graph subgroups in A.15	88–94; 691v–694v

Proof. The assertions about the discrete groups are proved in A.2. A section $h \mapsto s_h$ of $\mathfrak{S}_{\mathrm{d}} \to H$ gives the finite-stage scheme $\coprod_{h \in H} \mathbb{A}^2_{\mathbb{Z}}$, with multiplication

$$(m, h)(n, k) = (m + a_h n + c(h, k), hk), \qquad c(h, k) = s_h s_k s_{hk}^{-1} \in M.$$

Changing the section to $d_h s_h$ gives the explicit isomorphism $(m, h) \mapsto (m - d_h, h)$. Thus this construction has a specified, section-independent meaning. An integral cocycle is obtained from

$$\rho \longmapsto (R, 0), \quad \sigma \longmapsto (S, (1/3, -1/3)), \qquad R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Choose words s_h lifting the twelve elements of H and let t_h be their affine translations. Then $c(h, k) = t_h + a_h t_k - t_{hk} \in \mathbb{Z}^2$, since the indicated word lies in M. The affine representation kills ω; the retained component h records it. The pair consisting of the component and affine image is faithful: on its possible kernel in M, the affine image is translation by the lattice vector itself.

The envelope $E_{\mathfrak{S}_{\mathrm{d}}}$ of §50 is the inverse limit of the analogous stages obtained from nM. The transition from any of these stages to the stage above is induced by the inclusion $nM \hookrightarrow M$

and is a morphism of affine group schemes. It gives the comparison $E_{\mathfrak{S}_{\mathrm{d}}} \to \mathcal{S}$; the formulas below concern this specified finite stage. On profinite points the continuous map $\widehat{\mathfrak{S}_{\mathrm{d}}} \to \mathcal{S}(\widehat{\mathbb{Z}})$ uses one common H-component and the coordinate maps $\widehat{M} \to M \otimes \mathbb{Z}_p$ at every prime. This describes its image and the meaning of the diagonal points used later in the manuscript. Likewise $\widehat{\mathbb{Z}}^* \to \gamma(\widehat{\mathbb{Z}})$ sends μ to its single residue class r mod 12 and the unique $k \in \widehat{\mathbb{Z}}$ with $\mu = r + 12k$. The proof of A.5 shows directly that this is a continuous homomorphism. These comparison maps retain the integral and profinite data rather than identifying all points of the schemes with one profinite group. □

A.2. The subgroup, its free basis, and the action of the quotient [Proved]

The subgroup Π_0 is normal and free on the displayed λ_0, λ_1. Its indices in $\mathrm{SL}_2(\mathbb{Z})$ and $\mathrm{GL}_2(\mathbb{Z})$ are 12 and 24. Moreover

$$H \simeq \mathrm{SL}_2(\mathbb{Z}/4)/V_4 \simeq \mathrm{Dic}_3, \qquad V_4 = \langle \lambda_0 \bmod 4, \lambda_1 \bmod 4 \rangle.$$

Conjugation on M sends ρ, σ, τ respectively to $R, S, -I$; in particular $\tau' = \omega\sigma\tau$ acts as $-S$. For every divisor N of 24, the action of $\mathrm{GL}_2(\mathbb{Z})/\Pi_0$ on M/NM has the following orders.

N	1	2	3	4	6	8	12	24
$\lvert\,\mathrm{im}\,\rvert$	1	6	12	12	12	12	12	12
$\lvert\,\ker\,\rvert$	24	4	2	2	2	2	2	2

This proves the index and lattice-action portion of Lead 2 (§51.1–4; C4 scans 34–38, 64–67; fols. 664–666, 679v–681).

Proof. Matrix multiplication gives

$$\lambda_0 = \omega\sigma\rho\sigma\rho, \quad \lambda_1 = \omega\rho\sigma\rho\sigma,$$

$$\rho\lambda_0\rho^{-1} = \lambda_1, \quad \rho\lambda_1\rho^{-1} = (\lambda_1\lambda_0)^{-1}, \quad \sigma\lambda_0\sigma^{-1} = \lambda_1, \quad \sigma\lambda_1\sigma^{-1} = \lambda_0.$$

Conjugation by τ inverts each λ_i. These identities prove normality and the stated action after abelianization. We use the amalgam presentation $\mathrm{SL}_2(\mathbb{Z}) = C_4 *_{C_2} C_6$, with factors generated by σ, ρ and $\sigma^2 = \rho^3 = \omega$; it is recorded in [KLG13, §4, equation (1)]. Adding $\lambda_0 = 1$ forces $\sigma\rho\sigma^{-1} = \rho^{-1}$; $\lambda_1 = 1$ is its conjugate. The resulting group has the twelve normal forms $\rho^i\sigma^j$, $0 \le i < 6$, $0 \le j < 2$. Modulo four, V_4 has four elements, is normal, and the images of ρ, σ in $\mathrm{SL}_2(\mathbb{Z}/4)/V_4$ retain orders six and four. Since $|\mathrm{SL}_2(\mathbb{Z}/4)| = 48$, this realizes all twelve normal forms and proves the index claim and the stated quotient. Surjectivity of the reduction map follows also from elementary row operations over $\mathbb{Z}/4$.

The Bass–Serre tree of the amalgam has trivial Π_0-stabilizers: each factor, and therefore every conjugate of a factor, injects in H. Its quotient by Π_0 has $12/4 + 12/6 = 5$ vertices and $12/2 = 6$ edges. It is connected. Choosing a maximal tree shows directly that its fundamental group is free of rank $6 - 5 + 1 = 2$, and lifting paths identifies this group with Π_0. Since λ_0, λ_1 generate it, they are a free basis. For completeness, a surjective endomorphism of a finitely generated free group is injective: free groups are residually finite by completing the finite labelled path of a nonempty reduced word to permutations; on the finite set of subgroups of any fixed finite index, inverse image under a surjection is an injective permutation. A nontrivial element in its kernel would then lie in every finite-index subgroup, a contradiction. This applies to the map from a chosen free basis to the displayed generating pair.

The matrices R, S satisfy $R^3 = S^2 = (RS)^2 = I$ and give the six distinct matrices of S_3 modulo every $N \ge 2$. The additional $-I$ is already I modulo two. For $N \ge 3$, $-I$ is outside those six matrices: direct inspection of I, R, R^2, S, RS, R^2S proves this over $\mathbb{Z}/N$. Thus the

image has order twelve for each displayed $N \geq 3$. The central element ω is the order-two kernel integrally, and the table follows. The program `a_finite.py` constructs the twenty-four cosets modulo four and propagates the matrix action by a coupled breadth-first traversal; it also constructs the above quotient graph and checks its connectivity and Euler characteristic. This is a finite certificate for the presentation argument, not a finite test used in place of freeness. □

A.3. The precise source of level twelve and the group of order thirty-two [Proved]

Let Π_{00} be the inverse image in Π_0 of M_0, and put $\delta = \lambda_1\lambda_0^{-1}$. Then

$$\Pi_{00} = \langle\langle \delta \rangle\rangle_{\mathrm{SL}_2(\mathbb{Z})} = \Pi_0 \cap \mathrm{SL}_2(\mathbb{Z})', \qquad [\mathrm{SL}_2(\mathbb{Z}) : \Pi_{00}] = 36.$$

Its congruence level is exactly 12, and the reduction map gives

$$K_{32} := \Pi_{00}/\Gamma(12) = V_4 \times Q_8 \ \subset\ \mathrm{SL}_2(\mathbb{Z}/4) \times \mathrm{SL}_2(\mathbb{F}_3) = \mathrm{SL}_2(\mathbb{Z}/12).$$

Here Q_8 is the quaternion subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$. The subgroup K_{32} is normal in $\mathrm{GL}_2(\mathbb{Z}/12)$. This is the group of order 32 asked for at C4 scan 64, folio 679v; its exact congruence origin is the additional index-three lattice $M_0 \subset M$, together with the level-four realization of Π_0.

Proof. With $e_i = [\lambda_i]$ we have

$$M_0 = \mathbb{Z}(e_1 - e_0) + \mathbb{Z}(3e_1) = (1 - R)M + (1 - S)M.$$

The normal closure J of δ is contained in Π_{00}. In Π_0/J, the relation $\lambda_0 = \lambda_1$ and its ρ-conjugate give $\lambda_1 = (\lambda_1\lambda_0)^{-1}$, so the quotient is cyclic of order at most three. The map $M \to \mathbb{Z}/3$, $(c_0, c_1) \mapsto c_0 + c_1$, shows that its order is exactly three. Hence $J = \Pi_{00}$. Abelianizing the amalgam gives $\mathrm{SL}_2(\mathbb{Z})^{\mathrm{ab}} \simeq \mathbb{Z}/12$, with $[\rho] = 2$ and $[\sigma] = 3$ after choosing a generator. Each $[\lambda_i] = 4$. The restriction to Π_0 therefore has kernel precisely Π_{00}, proving the intersection formula. In particular the two occurrences of 12 in Lead 2 are related by the exact sequence $0 \to C_3 \to C_{12} \to H^{\mathrm{ab}} = C_4 \to 0$.

Chinese remainders give the displayed product decomposition. The matrix of δ is

$$\delta = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \qquad \delta^2 \equiv 5I \pmod{12}.$$

Modulo four its normal closure is V_4: conjugation cyclically permutes the three nonzero elements. Modulo three its normal closure is the eight-element group consisting of $\pm I$ and the six trace-zero determinant-one matrices; each of those six squares to $-I$. It is closed, as one checks on two anticommuting generators, and is the quaternion group. To see that the normal closure in the product is the whole product, take commutators of δ with $(g, 1)$: these generate $V_4 \times 1$. We can then isolate $(1, \delta \bmod 3)$, whose normal closure is $1 \times Q_8$. Consequently the image of Π_{00} is $V_4 \times Q_8$. Its full inverse image has index $48 \cdot 24/32 = 36$, equal to the already proved index of Π_{00}; the two subgroups coincide. This proves $\Gamma(12) \subset \Pi_{00}$.

For minimality, the upper transvection $T^m = \left(\begin{smallmatrix} 1 & m \\ 0 & 1 \end{smallmatrix}\right)$ belongs to Π_{00} only when $4 \mid m$, because the only such element of V_4 is I, and only when $3 \mid m$, because Q_8 has no nonidentity element of order three. Every $\Gamma(m)$ contains T^m, so $\Gamma(m) \subset \Pi_{00}$ implies $12 \mid m$. Finally V_4 is normalized by $\mathrm{GL}_2(\mathbb{Z}/4)$, and the trace description of Q_8 is invariant under every invertible conjugator. Their product is normal in the full level-twelve group. All thirty-two matrices, their orders, their conjugates, and the normal-closure assertions are exhaustively checked and printed by `a_finite.py`. □

A.4. All congruence arrows and the scope of Lead 2 [Proved]

The exact finite quotients obtained from A.3 have orders

group	order	quotient by K_{32}
$\mathrm{SL}_2(\mathbb{Z}/12)$	1152	36
$\mathrm{im}(\mathrm{GL}_2(\mathbb{Z}) \to \mathrm{GL}_2(\mathbb{Z}/12))$	2304	72
$\mathrm{GL}_2(\mathbb{Z}/12)$	4608	144

The middle image is $\det^{-1}\{1,-1\}$. Thus reduction induces isomorphisms

$$\mathrm{SL}_2(\mathbb{Z})/\Pi_{00} \simeq \mathrm{SL}_2(\mathbb{Z}/12)/K_{32}, \qquad \mathrm{GL}_2(\mathbb{Z})/\Pi_{00} \simeq \overset{-1}{\det}\{1,-1\}/K_{32}.$$

There is also a surjective homomorphism $\mathfrak{S}_{\mathrm{d}} \to \mathrm{SL}_2(\mathbb{Z}/12)/K_{32}$ with kernel M_0. Reduction to $\mathrm{SL}_2(\mathbb{Z}/12)$ itself does not factor through $\mathfrak{S}_{\mathrm{d}}$. These are the actual morphisms relating the action on M to the order-32 subgroup. The action images in A.2 have order at most 12; none is the order-32 group. They determine the lattice M_0 by $M_0 = (1-R)M + (1-S)M$, whose inverse image is used in A.3. This proves the strongest precise form of Lead 2 and the relation requested in item 1 (C4 scans 64–67; fols. 679v–681).

Proof. The determinant map on $\mathrm{GL}_2(\mathbb{Z}/12)$ is onto its four units, with kernel of order 1152. Integral invertible matrices have determinant ± 1, and the elementary generators of $\mathrm{SL}_2(\mathbb{Z}/12)$ lift to integral matrices. Adjoining τ therefore gives exactly the middle group. Taking inverse images of K_{32} proves each isomorphism and each kernel claim. For the last assertion, direct multiplication gives

$$[\lambda_0, \lambda_1] \equiv \begin{pmatrix} 1 & 4 \\ 4 & 5 \end{pmatrix} \not\equiv I \pmod{12}.$$

This element is killed in $\mathfrak{S}_{\mathrm{d}}$ but survives under reduction, proving the asserted failure of factorization by an explicit element. It dies after quotient by K_{32}, as the exact kernel calculation requires. This also verifies the qualification in the margin of scan 67. The scan 66 notation for the Sylow-two group is transcribed as D_4; the six order-four elements and unique involution prove that its abstract group is Q_8, independently of the notation intended by that handwriting. □

A.5. The ring scheme and its exact unit sequence [Proved]

For a positive integer N, the ring scheme $\mathbb{O}_N$ is $\coprod_{r \in \mathbb{Z}/N} \mathbb{A}^1_{\mathbb{Z}}$. A section is locally written $n + N[x]$, with the equivalence

$$(n, x) \sim (n + Nk, x - k) \quad (k \in \mathbb{Z}).$$

Addition is the additive pushout law, and multiplication is

$$(n + N[x])(m + N[y]) = nm + N[ny + mx + Nxy]. \tag{A.5.1}$$

Its unit scheme and unit exact sequence are

$$\mathbb{O}_N^* = \coprod_{r \in (\mathbb{Z}/N)^*} \operatorname{Spec} \mathbb{Z}[t, (r + Nt)^{-1}],$$

$$1 \longrightarrow D(1 + Nt) \longrightarrow \mathbb{O}_N^* \longrightarrow (\mathbb{Z}/N)^*_{\mathrm{const}} \longrightarrow 1. \tag{A.5.2}$$

The right arrow is smooth and surjective. The corresponding additive sequence is $0 \to \mathbb{G}_a \to \mathbb{O}_N \to (\mathbb{Z}/N)_{\mathrm{const}} \to 0$. These prove (64)–(66), (74), and (78)–(79), with the stated choice of constant finite schemes (C4 scans 51–55; fols. 673–675).

More generally, for a unital associative ring A and a two-sided ideal J that is finite free as an abelian group, the additive pushout $A[J]$ has a ring structure and exact sequences

$$0 \to \underline{J} \to A[J] \to (A/J)_{\mathrm{const}} \to 0, \quad 1 \to (1+\underline{J})^* \to A[J]^* \to ((A/J)^*)_{\mathrm{const}} \to 1. \tag{A.5.3}$$

The second surjection is smooth. This supplies the general version of (69)–(73), including a proof of its local surjectivity.

Proof. The pushout is the quotient of $\mathbb{Z}_{\mathrm{const}} \times \mathbb{G}_a$ by the constant subgroup $k \mapsto (Nk, -k)$. Choosing one integer representative of every residue identifies its components as claimed. Formula (A.5.1) is unchanged under either replacement $(n, x) \mapsto (n + Nk, x - k)$; direct expansion proves associativity and distributivity. Equivalently it is the multiplication inherited from the ring with square-zero-free ideal coordinate $N\mathbb{Z} \otimes R$, whose internal product is $N[x]\,N[y] = N[Nxy]$.

A unit must have unit residue and unit evaluation $n + Nx$. Conversely choose integers r, r' with $rr' = 1 + Nh$ and set

$$y = -\frac{h + r'x}{r + Nx}.$$

Then $(r + N[x])(r' + N[y]) = 1$, proving both sufficiency and the displayed localized coordinate ring. In residue class one the coordinate is $1 + N[t]$, with multiplication $t \star t' = t + t' + Ntt'$; this is the kernel in (A.5.2). Every geometric fibre of every component is nonempty: when the characteristic divides N, its representative is nonzero; otherwise at most one value of t is excluded. The arrow is an open subscheme of an affine-line bundle over the constant base, so it is smooth and surjective, and therefore fppf locally admits sections. This proves exactness without asserting surjectivity on points over every ring.

For the general construction, take $(A_{\mathrm{const}} \times \underline{J})/J_{\mathrm{const}}$, where $j \mapsto (j, -j)$, with multiplication $(a, x)(b, y) = (ab, ay + xb + xy)$. Each component is an affine space of rank $\operatorname{rank} J$. Fix a component $b \in (A/J)^*$ and a lift $a \in A$. On this component multiplication on J on the left and on the right is a pair of square matrices. The unit locus is the open set where both determinants are invertible. Indeed, a chosen lift a' of b^{-1} leaves both inverse errors in J; invertibility of left and right multiplication on J corrects the two errors, producing a left and a right inverse, which coincide.

To prove that this open set has a point in every geometric fibre, work over an algebraically closed field k. In the finite-dimensional algebra of pairs of endomorphisms of J_k, take the subalgebra generated by the left-right multiplication pairs from A and J_k. The image of J_k is an ideal, and the image of a is a unit modulo this ideal. In any finite-dimensional algebra, units lift modulo every ideal: reduce by the nilpotent Jacobson radical, decompose the semisimple quotient into matrix blocks, and choose arbitrary invertible entries in the blocks killed by the ideal; units then lift across the nilpotent radical by the finite geometric series. Lifting the adjustment back to J_k gives $a + j$ with both multiplication operators invertible. The preceding correction supplies its inverse. Thus every component-unit fibre is nonempty. It is open in affine space, so the map in (A.5.3) is smooth and surjective. The kernel is exactly the units of the residue-one component. This also proves the pair description (73): in the finite free unitization of J, a right inverse is a two-sided inverse, by the determinant criterion for multiplication on a finite free module. □

A.6. Three literal ring formulas and their exact replacements [Refuted]

The evaluation criterion in (67b), the multiplication printed in (68), and the full-kernel assertion (77) are false for the pushout (66). Their correct replacements are A.5 and

$$\ker(\mathbb{O}_N^* \xrightarrow{\mu} \mathbb{G}_m) = \coprod_{r \in (\mathbb{Z}/N)^*} \operatorname{Spec} \mathbb{Z}[t]/(r + Nt - 1). \tag{A.6.1}$$

The residue-one component alone is ${}_N\mathbb{G}_a$. After inverting N, evaluation and residue give an isomorphism of ring schemes $\mathbb{O}_N \simeq \mathbb{G}_a \times (\mathbb{Z}/N)_{\mathrm{const}}$, and of unit schemes $\mathbb{O}_N^* \simeq \mathbb{G}_m \times (\mathbb{Z}/N)_{\mathrm{const}}^*$. The putative global section of the unit-residue map using the four integers $\pm 1, \pm 5$ in (83) is also false over $\mathbb{Z}$. These are mathematical corrections to C4 scans 51–56, fols. 673–675v; they do not settle an unattached handwritten reading.

Proof. Over $\mathbb{F}_2$, the sections $0{+}2[0]$ and $0{+}2[1]$ are distinct points of the same affine component of $\mathbb{O}_2$, while their evaluations coincide. Thus evaluation does not detect equality. For (68), the product $12[1]\,12[1]$ evaluates to 144 over $\mathbb{Z}$; the printed formula evaluates to 12. Formula (A.5.1) inserts exactly the missing factor N.

Evaluation on the r-component is $r+Nt$, giving (A.6.1). For $N = 12$, the section $5{+}12[-1/3]$ over $\mathbb{Q}$ has evaluation one and residue five. It is a nonidentity element of the full kernel, although ${}_{12}\mathbb{G}_{a,\mathbb{Q}} = 0$. After N is inverted, $(\mu, r) \mapsto r + N[(\mu - r)/N]$ is the inverse of the displayed comparison, proving the stronger structural assertion. Finally a $\mathbb{Z}$-point of $\mathbb{O}_{12}^*$ evaluates to ± 1, and hence has residue 1 or 11. The other two residue components have no integral point. Thus there is no global section of the four-component residue map over $\mathbb{Z}$; the numbers ± 5 can be used as affine coordinate representatives only. All these counterexamples are checked by `a_schemes.py`. □

A.7. Integral coordinates on the multiplier scheme [Proved]

Write $a = \nu_2, b = \nu_3 \in \{0,1\}_{\mathrm{const}}$, so that $\varepsilon_2 = (-1)^a$ and $\varepsilon_3 = (-1)^b$. The four charts of γ have coordinates

$$r_0 = 1 - 6a + 4b, \qquad \mu = r_0 + 12k \in R^*.$$

The regular functions used in the manuscript are

$$\nu = -3a + 2b + 6k, \quad \nu' = -a + b + 3k, \quad \nu'' = -a + b + 2k, \tag{A.7.1}$$

and satisfy, integrally,

$$\mu = 1 + 2\nu = 1 - 2a + 4\nu' = 1 - 2b + 6\nu'', \qquad 2\nu' - a = 3\nu'' - b. \tag{A.7.2}$$

For the alternative representative $\mu_0 = (-5)^a 5^b$ used in (84)–(92), with $\mu = \mu_0 + 12\nu_0$, the required formulas instead are

$$\nu' = -a + b - 6ab + 3\nu_0, \qquad \nu'' = -a + b - 4ab + 2\nu_0. \tag{A.7.3}$$

The transcribed coefficients $-3ab, -2ab$ are false. The substitutions (A.7.3) correct the corresponding expansions in (138)–(141) as well. The map $\gamma \to \operatorname{Spec}\mathbb{Z}$ is smooth, flat, of relative dimension one, with four geometric components at every prime.

Proof. The representatives $1, -5, 5, -1$ are congruent to $(-1)^a$ mod 4 and $(-1)^b$ mod 3. Each equation in (A.7.2) follows by expansion. Conversely the simultaneous data in (63) give $k = \nu' - \nu''$, and substitution recovers (A.7.1) without division by two or three. Thus these are integral isomorphisms of schemes. Expanding $(-5)^a 5^b$ on the four constant components gives $1 - 6a + 4b - 24ab$, whence (A.7.3). The discrepancy is detected over $\mathbb{F}_7$ by $a = b = 1, \nu_0 = 0$: $\mu = -25 \equiv 3$ is a unit, whereas either of the transcribed formulas reconstructs $\mu = 1$. The exact polynomial checks are in `a_schemes.py`. Smoothness and the component claim follow from the four open affine-line charts of A.5. In characteristics two and three, each chart is the entire affine line since its evaluation is a constant nonzero element. These calculations concern C4 scans 51, 57–58, 74–75; fols. 673, 676–676v, 684v–685. □

A.8. The first requested exact sequence, over every base ring [Proved]

Set $\delta = e_1 - e_0$ and $e = 3e_1$, a basis (δ, e) of M_0. For $\alpha = a_0e_0 + a_1e_1$ and $\beta = b_0e_0 + b_1e_1$, put

$$F(\alpha, \beta) = (1 - R)\alpha - (1 - S)\beta = (-a_0 - a_1 + b_0 - b_1)\delta + a_1e. \tag{A.8.1}$$

Define N by $F(\alpha, \beta) = \nu''e$ *in* $\underline{M}_0$, with multiplier in γ. Then

$$(t, s, \underline{\mu}) \longmapsto \Big(\underline{\mu}, \alpha = te_0 + \nu''\delta, \beta = te_0 + s(e_0 + e_1)\Big) \tag{A.8.2}$$

is an isomorphism of schemes $\mathbb{G}_a^2 \times \gamma \simeq N$, and its group law is

$$(t, s, \underline{\mu})(t', s', \underline{\mu}') = (t + \mu t',\ s + \mu s',\ \underline{\mu\mu'}). \tag{A.8.3}$$

Consequently the sequence requested at the beginning of §51.4 is split exact over $\mathbb{Z}$:

$$1 \longrightarrow N^{0+} = \mathbb{G}_a^2 \longrightarrow N \longrightarrow \mathbb{O}_{12}^* \longrightarrow 1. \tag{A.8.4}$$

The middle group is smooth and flat of relative dimension three, including at two and three. This proves (104)–(105) after the author's correction in scans 60–62, fols. 677v–678v.

Proof. Multiplication by the displayed R, S gives (A.8.1). The two coordinates of its required equality say exactly $a_1 = \nu''$ and $a_0 + a_1 = b_0 - b_1$. Therefore the inverse to (A.8.2) is $t = a_0 + a_1$, $s = b_1$, with no division by an integer. In particular $F : M^2 \to M_0$ is a split surjection of free lattices.

We verify the group law and its automorphism meaning. For $\underline{\mu} \in \gamma(R)$, define an automorphism $u_{\underline{\mu}}$ of $\mathcal{S}_R$ by

$$u_{\underline{\mu}}(\rho) = \lambda_1^{\nu}\rho, \qquad u_{\underline{\mu}}(\sigma) = \omega^a\sigma, \qquad u_{\underline{\mu}}(m) = \mu m \quad (m \in \underline{M}). \tag{A.8.5}$$

Exponents on the lattice denote additive vector-group coordinates. These formulas define the morphism on every component of the finite-stage scheme. Indeed $(1 + R + R^2)e_1 = 0$ gives $(\lambda_1^{\nu}\rho)^3 = \omega$, and $(\omega^a\sigma)^2 = \omega$. Substituting in $\lambda_0 = \omega\sigma\rho\sigma\rho$ and $\lambda_1 = \omega\rho\sigma\rho\sigma$ gives multiplication by $1 + 2\nu = \mu$ on both lattice generators. The commutator subgroup of the lattice is therefore killed. These checks construct a compatible map on the twelve charts of $\mathcal{S}$, using the scalar map on their affine coordinates; they do not infer uniqueness merely from the values of two sections in a nonreduced fibre. The identity

$$\nu(\underline{\mu\mu'}) = \nu(\underline{\mu}) + \mu\nu(\underline{\mu}')$$

holds as an integral polynomial on every pair of components, by (A.7.1). The sign a adds modulo two. Thus $u_{\underline{\mu}}u_{\underline{\mu}'} = u_{\underline{\mu\mu'}}$; the formula for the inverse follows, so these are automorphisms.

The pair of elements

$$\Big(\mathrm{Int}(te_0)u_{\underline{\mu}}, \quad (-s(e_0 + e_1))u_{\underline{\mu}}\Big) \tag{A.8.6}$$

records the effective automorphism and the σ-normalizer conjugator of (A.8.2). To check this directly, $(1 - R)\delta = 3e_1$ and $\nu = 3\nu'' - b$ give the first generator formula after the prescribed τ^b correction; the second follows from $te_0 - \beta = -s(e_0 + e_1)$ and $(1 - S)(e_0 + e_1) = 0$. Each factor in (A.8.6) multiplies with scalar action μ on its one-dimensional translation coordinate. Their fibre product over γ therefore has exactly (A.8.3), which proves the transported group law on the manuscript's data. Its section is $t = s = 0$. Smoothness follows from (A.8.2) and A.7. □

A.9. The premature exact sequences and their flat replacement [Refuted]

The surjection onto $\mathbb{G}_m \times C_2^2$ in (54), and the surjection $M^2 \to M$ in (59), are false as written. The latter has cokernel $\mathbb{Z}/3$, and the correct integral lattice sequence is

$$0 \to P \to M^2 \xrightarrow{F} M \to \mathbb{Z}/3 \to 0, \quad P = \mathbb{Z}(e_0, e_0) \oplus \mathbb{Z}(0, e_0 + e_1). \tag{A.9.1}$$

The raw scheme obtained by imposing the loop equation only in $\underline{M}$ is

$$\widetilde{Z} \simeq \operatorname{Spec}_\gamma \mathcal{O}_\gamma[t, s, z]/(3z). \tag{A.9.2}$$

Its projection to γ is split surjective, with kernel $\mathbb{G}_a^2 \times {}_3\mathbb{G}_a$ as a scheme. The closed subscheme $z = 0$ is exactly N of A.8 and is the schematic closure of $\widetilde{Z}_{\mathbb{Z}[1/3]}$. The fine kernel sequence (56), after the sign in (55) is corrected to $1 - R$, is exact with raw kernel $\mathbb{G}_a^2 \times {}_3\mathbb{G}_a$ and quotient ${}_4\mathbb{G}_a$. These are the full corrections to (54)–(59), (94)–(103) (C4 scans 48–49, 59–62; fols. 671v–672, 677–678v).

Proof. The sum of the two coordinates of $F(\alpha, \beta)$ is $3a_1$. Its image is therefore M_0, and (A.8.1) proves surjectivity onto that lattice and the stated kernel. Tensoring this map with a ring does not preserve the monomorphism $\underline{M_0} \to \underline{M}$: in its basis (δ, e) the determinant is -3. Imposing the equation in $\underline{M}$ instead yields $3(a_1 - \nu'') = 0$; the other equation determines b_0 from a_0, a_1, b_1. Take $z = a_1 - \nu''$, $t = a_0 + a_1$, $s = b_1$ to obtain (A.9.2). Its ideal of functions vanishing after inverting three is (z), which proves the closure assertion. In particular the characteristic three raw fibre has dimension four, while the corrected fibre has dimension three. The kernel version has dimensions three and two. Finite point enumeration over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$ verifies these counts in `a_schemes.py`.

For (54), over the algebraic closure of $\mathbb{F}_2$, equation (63) forces $\mu = 1$ on every component, so any other nonzero element of that field has no lift, even fppf locally. Equivalently, the already specified map to the naive character base factors through γ; the correct exact target is the one in A.8. For (56), put $a = b = 0$ and $w = \nu'$. Its equations are $4w = 0$ and $F(\alpha, \beta) = 2we_1$. A section of the projection to ${}_4\mathbb{G}_a$ is $\alpha = 2we_0 - 2we_1$, $\beta = 0$: $F(\alpha, 0) = -6we_1 = 2we_1$ since $4w = 0$. The kernel is $F = 0$, already computed. The literal plus sign in $(1 + R)\alpha$ in (55) is inconsistent with (57), for example at $\alpha = \beta = e_0, w = 0$ over $\mathbb{Z}$. We have stated which replacement makes the sequence exact.

Scan 61 (100)–(101) contains another sign discrepancy visible in the attached image: the zero equation (96) requires $F(\alpha, \beta) - \nu'' e = 0$, or equivalently $(1 - R)(\alpha - \nu''\delta) - (1 - S)\beta = 0$. The plus sign in the shifted α gives $6e_1$ on $\alpha = \delta, \beta = 0, \nu'' = 1$ over $\mathbb{Z}$. Thus this replacement is forced by (94)–(96); it is not an unannounced choice between readings. □

A.10. The integral comparison with the fine scheme of LM1 [Proved]

The fine scheme of scan 41, with component bits a, b, coordinate $w = \nu'$, and loop equation $3a_1 = 2w - a + b$, is isomorphic over $\mathbb{Z}$ to the corrected N of A.8. The comparison and its inverse are

$$k = w - a_1, \quad \nu'' = a_1, \quad t = a_0 + a_1, \quad s = b_1, \qquad w = 3k - a + b, \quad a_1 = 2k - a + b, \tag{A.10.1}$$

with α, β recovered by (A.8.2). In particular it is an isomorphism on fibres at two and three and on points over every ring. The alternate LM1 coordinate using $\nu = 2w + a$ is carried to the present coordinate using $\nu = 2w - a$ by $w_{\text{here}} = w_{\text{LM1}} + a$. This settles the finite-stage comparison left in LM1 E.20 for this specific parameter scheme (C4 scans 40–41, 62, 71–72; fols. 667v–668, 678v, 683–683v).

Proof. The equation $2w-3a_1 = a-b$ implies $w = 3(w-a_1)-a+b$ and $a_1 = 2(w-a_1)-a+b$. Thus (A.10.1) is integral in both directions. Substituting gives $\mu = 1-2a+4w = 1-6a+4b+12k$, the same inverted element in both coordinate rings. The remaining loop coordinate equation is exactly $a_0 + a_1 = b_0 - b_1$, whose solutions are (A.8.2). This proves the claimed scheme isomorphism, including all nilpotent test rings. Both groups use the automorphism and conjugator multiplication (A.8.6), so the comparison also respects their group structures. The final coordinate translation follows by equating $2w_{\mathrm{here}} - a = 2w_{\mathrm{LM1}} + a$ as integral functions; substitution, not cancellation in a characteristic-two ring, proves the equality. □

A.11. Effective automorphisms and the early quotient sequences [Proved]

The effective group, its geometric extension, and the early quotient groups are

$$Z = \mathbb{G}_a \rtimes_\mu \gamma, \quad G = \mathcal{S} \rtimes_u \gamma, \quad \Gamma_{\mathrm{e}} = \mathbb{G}_a \rtimes_\mu \gamma,$$
$$\mathcal{M}_{\mathrm{e}} \simeq (\mathbb{G}_a \times \mathcal{S}) \rtimes \gamma, \qquad \mathcal{M}_{\mathrm{e}}^{\sharp} \simeq \mathbb{G}_a^3 \rtimes_\mu \gamma. \tag{A.11.1}$$

The γ-action on $\mathcal{S}$ is (A.8.5). In N, the subgroup L_0 is the t-axis and the quotient is the s-coordinate. We obtain the exact sequences

$$\begin{aligned} &1 \to L_0 \to N \to \Gamma_{\mathrm{e}} \to 1, && 1 \to \mathbb{G}_a \to \Gamma_{\mathrm{e}} \to \gamma \to 1, \\ &1 \to \mathcal{S} \to \mathcal{M}_{\mathrm{e}} \to \Gamma_{\mathrm{e}} \to 1, && 1 \to \underline{M} \to \mathcal{M}_{\mathrm{e}}^{\sharp} \to \Gamma_{\mathrm{e}} \to 1, \\ &1 \to L_0 \to Z \to \gamma \to 1, && 1 \to \mathcal{S} \to G \to \gamma \to 1. \end{aligned} \tag{A.11.2}$$

All are split and exact on points over every ring. The lattice kernel sequences (127)–(132) are $0 \to \mathbb{Z} \to P \to \mathbb{Z} \to 0$ and their vector-group base changes; the maps are $(t,s) \mapsto s$ and $t \mapsto (t,0)$. These prove (123)–(127), (131)–(132), (142), (145)–(150) (C4 scans 70–76; fols. 682v–685v).

Proof. Under (A.8.6), the first factor is $\mathrm{Int}(te_0)u_{\underline{\mu}}$ and the second has coordinate $-s$. The kernel of the effective projection is the s-axis; its image is the displayed Z. Inner conjugation by te_0 embeds $L_0 \simeq \mathbb{G}_a$ into Z, and (A.8.5) normalizes it by the scalar μ. Thus the quotients are represented by the displayed affine schemes, with the stated sections.

For the crossed quotient defining $\mathcal{M}_{\mathrm{e}}$, take $\mathcal{S} \rtimes N$ with the action of N through Z and quotient by $t \mapsto (-te_0, (t,0,1))$. This is a normal subgroup because its action on $\mathcal{S}$ equals inner conjugation by te_0. Every coset has exactly one representative with $t = 0$; its remaining coordinates are $(x, s, \underline{\mu})$ and multiply as in (A.11.1). The same calculation with $\underline{M}$ in place of $\mathcal{S}$ gives the sharp subgroup. All kernel and section assertions now follow by coordinate projection, so they are exact on arbitrary ring points and hence as fppf sheaves. The kernel lattice P was computed in A.9; its quotient by the diagonal e_0-line has the free generator $(0, e_0 + e_1)$. This proves every lattice and vector-group sequence asserted here.

The ambiguous exponent in scan 76 (151) is fixed by the verified formula $\lambda_1^{\nu}\rho$ in scan 93, and independently by $u(\lambda_i) = \mu\lambda_i$ in A.8. The candidate exponent $6\nu''$ fails this test, since the required relation is $\nu = 3\nu'' - b$. We retain this correction explicitly in the readings ledger. □

A.12. The two normalizers and the central extension of the multiplier scheme [Proved]

Let $d = \nu''$ and set, in $G = \mathcal{S} \rtimes \gamma$,

$$v_{\underline{\mu}} = \sigma^b \delta^{-d} u_{\underline{\mu}}.$$

For the raw normalizer of (152), before the closure in (159), the literal sequence (155) is

$$1 \to C_6 \times {}_3\mathbb{G}_a \to N_\rho^{\mathrm{raw}} \to \gamma \to 1, \qquad N_\rho^{\mathrm{raw}} \simeq {}_3\mathbb{G}_a \times N_\rho. \tag{A.12.0}$$

The smooth closure of the ρ-normalizer is represented by $\{\rho^k v_{\underline{\mu}} : k \in \mathbb{Z}/6\}$; hence it has the exact sequence

$$1 \to C_6 = L_\rho \to N_\rho \to \gamma \to 1. \tag{A.12.1}$$

The final subgroup specified in (212), scans 93–94, is

$$N_\rho^? = \{\omega^i v_{\underline{\mu}} : i \in \mathbb{Z}/2\}, \qquad 1 \to C_2 = \mu_\rho \to N_\rho^? \to \gamma \to 1. \tag{A.12.2}$$

Its multiplication on $C_2 \times \gamma$ is

$$(i, \underline{\mu})(i', \underline{\mu}') = (i + i' + c(\underline{\mu}, \underline{\mu}'), \underline{\mu\mu}'), \qquad c = (a+b)b' \pmod 2. \tag{A.12.3}$$

Thus (A.12.2) is the pullback along $(\varepsilon_2, \varepsilon_3) : \gamma \to C_2^2$ of the order-eight dihedral central extension $1 \to C_2 \to D_8 \to C_2^2 \to 1$. Moreover

$$N_\sigma^\sharp = \mathbb{G}_a \rtimes_\mu \gamma, \qquad N_\sigma = (C_4 \times \mathbb{G}_a) \rtimes \gamma, \tag{A.12.4}$$

where γ acts by ε_2 on C_4 and by μ on $\mathbb{G}_a$. These assertions prove (155), (159), and (173)–(181), and compute the extension asked for after (196) (C4 scans 78–84, 87, 93–94; fols. 686v–689v, 691, 694–694v).

Proof. The identities $(1-R)\delta = 3e_1$ and $\nu - 3d = -b$ give

$$\delta^{-d} u_{\underline{\mu}}(\rho) \delta^d = \lambda_1^{-b} \rho.$$

For $b = 1$, direct multiplication in the modular group gives $\sigma\lambda_1^{-1}\rho\sigma^{-1} = \rho^{-1}$. For $b = 0$ the displayed expression is ρ. Thus $v_{\underline{\mu}}$ has the required normalizing action $\rho \mapsto \rho^{\varepsilon_3}$. In the finite quotient H, the centralizer of ρ is its cyclic subgroup of order six. Within each of these six components, the remaining translation condition is $(1-R)m = 0$. Its kernel has equations equivalent to $3z = 0$, because $\det(1-R) = 3$. Explicitly $m = z\delta$, with $3z = 0$. The cyclic component C_6 fixes this vector. Conjugation by $v_{\underline{\mu}}$ acts on it as $\mu\varepsilon_3$, and $\mu\varepsilon_3 = 1 + 6\varepsilon_3 d$; hence it also fixes $z\delta$. This proves the central direct product (A.12.0) and the original sequence (155), including its extra fibre in characteristic three. The closure of its characteristic-not-three part is the zero translation. Translation by $v_{\underline{\mu}}$ therefore gives exactly the smooth normalizer stated above, including all its components, with kernel L_ρ.

The identity

$$d(\underline{\mu\mu}') = \varepsilon_3(\underline{\mu}')d(\underline{\mu}) + \mu d(\underline{\mu}') \tag{A.12.5}$$

is an integral polynomial identity on the sixteen pairs of charts; it follows by multiplying $\mu = \varepsilon_3 + 6d$ and using (A.7.1). In particular its proof does not cancel six in a test ring. Moving $\sigma^{b'}$ past δ^{-d} uses $S\delta = -\delta$, and $u_{\underline{\mu}}(\sigma) = \omega^a\sigma$. The two central factors are $\omega^{ab'}$ and $\omega^{bb'}$. Equation (A.12.5) then gives

$$v_{\underline{\mu}} v_{\underline{\mu}'} = \omega^{(a+b)b'} v_{\underline{\mu\mu}'}.$$

Consequently the multiplication in N_ρ is $(k, \underline{\mu})(k', \underline{\mu}') = (k + \varepsilon_3 k' + 3c, \underline{\mu\mu}')$. Restricting k to $0, 3$ proves (A.12.2)–(A.12.3). The bilinear function $((a,b),(a',b')) \mapsto (a+b)b'$ is a two-cocycle. The resulting eight-element group has five involutions and two elements of order four, and is generated by two involutions with noncommuting product of order four. This proves it is D_8; the full cocycle and element-order checks are in `a_finite.py`. Exactness of (A.12.2) holds even on ring points, because $v_{\underline{\mu}}$ is a global section as a scheme, although it is not a group section.

For σ, the centralizer in H is C_4, generated by σ. Its translation kernel is $\ker(1-S) = \mathbb{Z}(e_0 + e_1)$, also after every base change. These translations commute with σ, so the geometric centralizer is $C_4 \times \mathbb{G}_a$. Equation (A.8.5) supplies a group section over γ and its action on

both factors, which proves (A.12.4) and all the stated exact sequences. The sharp intersection retains just the translation component. Similarly $N_\rho \cap G^\sharp$ is the section over $\gamma' = \ker \varepsilon_3$, since the component $\rho^k\sigma^b$ is the identity precisely when $k = b = 0$. Put $\gamma'' = \ker \varepsilon_2$, $\gamma^0 = \gamma' \cap \gamma''$, and $\gamma''' = \{a = b\}$. The neutral partial section of (166)–(167) is $v(\gamma^0)$. The partial section of (168)–(169) is exactly

$$\underline{\mu} \longmapsto \omega^b v_{\underline{\mu}} \quad (\underline{\mu} \in \gamma'''),$$

because its cocycle vanishes on $a = b$, and at $\underline{\mu} = -1$ it is τ'. Conjugation by τ' sends $v_{\underline{\mu}}$ on γ' to $\omega^a v_{\underline{\mu}}$, as (A.12.3) shows. It preserves the partial section precisely on γ^0. This proves the prime and double-prime intersections and the normalization assertions of (160)–(170) with their exact domains. □

A.13. The literal definition (171) has only half the stated image [Refuted]

The group defined in (171) from $Z_\rho^{0\sharp}$, ω, and τ' maps onto

$$\gamma''' = \{\varepsilon_2 = \varepsilon_3\} \subset \gamma,$$

and is the inverse image of this subgroup in the final $N_\rho^?$ of (212). Thus (172) with target all of γ is false for the literal definition (171). Its exact replacement is $1 \to C_2 \to N^?_{\rho,\mathrm{literal}} \to \gamma''' \to 1$; the final fibres (212) yield the larger exact sequence (A.12.2). The surviving last line of (194), with N_ρ and no question mark, requires the replacement $1 \to C_3 \to N_\rho/\mu_\rho \to \gamma \to 1$; the isomorphism to γ holds for $N_\rho^?/\mu_\rho$. The final Γ in (184) also satisfies $\Gamma/C_2 \simeq \mathbb{G}_a \rtimes \gamma$; the unquotiented version of this isomorphism in the third row of (195) is false. These corrections concern C4 scans 82, 85–87, 94; fols. 688v, 690–691, 694v.

Proof. The identity component $Z_\rho^{0\sharp}$ maps isomorphically to $\gamma^0 = \{a = b = 0\}$ by A.12. The central ω has trivial character, and τ' has multiplier -1, hence $(a, b) = (1, 1)$. The literal generated subgroup therefore has image in $a = b$. Conversely multiplication by -1 carries the whole 00 chart to the whole 11 chart, so the image is exactly γ'''. At $\underline{\mu} = -1$, $d = 0$ and $\tau' = \omega v_{-1}$; hence both lifts over each of these charts occur. This proves the asserted inclusion and its exact image and kernel. Over $\mathbb{F}_2$, the γ-point with $a = 1, b = 0, k = 0$ has evaluation one and has no lift to this literal subgroup. This finite counterexample is recorded in `a_schemes.py`.

Quotienting (A.12.1) by its central order-two subgroup leaves kernel $C_6/C_2 = C_3$, which proves the correction to (194). The final fibre product defining Γ has kernel $C_2 \times \mathbb{G}_a$ over γ, by A.12. Its quotient by the indicated central C_2 is $N_\sigma^\sharp$, giving the claimed isomorphism. On $\mathbb{F}_2$ points the two groups have 32 and 16 elements, respectively; the explicit projection has kernel of order two. Thus the missing quotient in (195) is detected by an actual map and a finite counterexample. □

A.14. The final fibre products and every surviving extension sequence [Proved]

Using the final (212), define

$$\Gamma = N_\rho^? \times_\gamma N_\sigma^\sharp, \quad \mathcal{M} = \Gamma \times_\gamma G, \quad \mathcal{M}^\sharp = \Gamma \times_\gamma G^\sharp. \tag{A.14.1}$$

The coordinates on Γ are $(i, s, \underline{\mu})$ and its law is

$$(i, s, \underline{\mu})(i', s', \underline{\mu}') = (i + i' + (a + b)b',\ s + \mu s',\ \underline{\mu\mu}'). \tag{A.14.2}$$

There are central quotient morphisms, with the indicated kernels,

$$\Gamma \twoheadrightarrow \Gamma_{\mathrm{e}}, \qquad \mathcal{M} \twoheadrightarrow \mathcal{M}_{\mathrm{e}}, \qquad \mathcal{M}^\sharp \twoheadrightarrow \mathcal{M}^\sharp_{\mathrm{e}}, \qquad \ker = C_2 = \mu_\rho. \tag{A.14.3}$$

In addition to A.8, A.11, and A.12, the final exact sequences are

$$\begin{gathered} 1 \to \mathcal{S} \to \mathcal{M} \to \Gamma \to 1, \qquad 1 \to \underline{M} \to \mathcal{M}^\sharp \to \Gamma \to 1, \\ 1 \to C_2 \times \mathbb{G}_a \times \mathcal{S} \to \mathcal{M} \to \gamma \to 1, \\ 1 \to C_2 \times \mathbb{G}_a \times \underline{M} \to \mathcal{M}^\sharp \to \gamma \to 1, \qquad 1 \to C_2 \times \mathbb{G}_a \to \Gamma \to \gamma \to 1. \end{gathered} \tag{A.14.4}$$

For $X = \Gamma, \mathcal{M}, \mathcal{M}^\sharp$, the auxiliary group $S_0X = Z \times_\gamma X$ has split exact sequence $1 \to L_0 \to S_0X \to X \to 1$. These prove (182)–(193) and (196), with the corrected quotient versions of (194)–(195). The central extensions are precisely the pullbacks of the explicit D_8 extension in A.12.

Proof. Every map to γ in (A.14.1) is a homomorphism, and the multiplication of the fibre product is componentwise. Insert (A.12.3)–(A.12.4) to obtain (A.14.2). Forgetting i is a homomorphism, since its cocycle is central, and its kernel is the constant C_2. The resulting three groups have exactly the coordinates and actions in (A.11.1), proving the actual comparisons (A.14.3) between the early and final revisions. The projection $\mathcal{M} \to \Gamma$ has the group section $(a, b) \mapsto (a, b, u_{\underline{\mu}})$; its kernel is $\mathcal{S}$. Replacing $\mathcal{S}$ by $\underline{M}$ gives the sharp row. Kernels over γ are the products of the three displayed geometric kernels, proving (A.14.4). These projections are onto on ring points: use $v_{\underline{\mu}}$, $u_{\underline{\mu}}$, and zero translation as a scheme section. Finally $Z \to \gamma$ has the group section (A.8.5), so the auxiliary projections are split exact. This proves exactness and representability directly for the entire final diagram.

The central factor μ_ρ here is $(\omega, 1, 1)$ in (a, b, U)-coordinates. The central subgroup μ_G is $(1, 1, \omega)$ and has a separate inclusion. The quotient map in (A.14.3) removes the first of these, exactly as its kernel states; no central factors have been combined without a map. □

A.15. The three distinguished subgroups: all requested points and fibres [Proved]

The final subgroups of scans 88–94 are the graph subgroups

$$\mathcal{M}(0) = \{U = u_{\underline{\mu}}\}, \qquad \mathcal{M}(1/2) = \{U = b_\sigma\}, \qquad \mathcal{M}(-\bar{j}) = \{U = a_\rho\} \tag{A.15.1}$$

inside $\mathcal{M} = \{(a_\rho, b_\sigma, U)\}$. Projection on (a_ρ, b_σ) identifies each with Γ as a group scheme over $\mathbb{Z}$. They are smooth and flat of relative dimension two, with eight geometric components at every prime.

For any connected ring R, their common point set is exactly

$$\{(a, b, k, i, s) : a, b, i \in \{0, 1\},\ k, s \in R,\ 1 - 6a + 4b + 12k \in R^*\}\,. \tag{A.15.2}$$

For a general ring the bits are locally constant functions on its spectrum. The group law is (A.14.2), with multiplier product

$$\begin{aligned} a'' &= a + a' \pmod 2, \qquad b'' = b + b' \pmod 2, \\ k'' &= \frac{r_0 r_0' - r_0''}{12} + r_0 k' + r_0' k + 12kk'. \end{aligned} \tag{A.15.3}$$

The numerator is an integer divisible by twelve on every component. In particular the requested rings have the following complete parametrizations, with $s \in R$ and $i \in C_2$ always free.

R	Allowed multipliers and component bits; $k = (\mu - r_0)/12$
$\mathbb{Z}$	$\mu = 1 : (a, b, k) = (0, 0, 0)$; $\mu = -1 : (a, b, k) = (1, 1, 0)$.
$\mathbb{Z}[1/2]$	$\mu = \pm 2^m$ $(m \in \mathbb{Z})$; a arbitrary, $(-1)^b = \mu \bmod 3$.
$\mathbb{Z}[1/3]$	$\mu = \pm 3^m$ $(m \in \mathbb{Z})$; b arbitrary, $(-1)^a = \mu \bmod 4$.
$\mathbb{Z}[1/6]$	$\mu = \pm 2^m 3^n$ $(m, n \in \mathbb{Z})$; a, b arbitrary.
$\mathbb{F}_2$	a, b, k arbitrary; $\mu = 1$; $\lvert\mathcal{M}(p)(\mathbb{F}_2)\rvert = 32$.
$\mathbb{F}_3$	a, b, k arbitrary; $\mu = (-1)^b$; $\lvert\mathcal{M}(p)(\mathbb{F}_3)\rvert = 72$.

Here p denotes any of the three displayed points, not a prime. As abstract groups, the integral points are $C_2 \times (\mathbb{Z} \rtimes \{\pm 1\})$; the two finite groups are $D_8 \times C_2^2$ over $\mathbb{F}_2$ and $C_3 \times (C_3 \rtimes D_8)$ over $\mathbb{F}_3$, where D_8 acts on the second C_3 through ε_3.

Proof. The maps $\Gamma \to G$ given respectively by $(a_\rho, b_\sigma) \mapsto u_{\underline{\mu}}, b_\sigma, a_\rho$ are homomorphisms. Their graphs have the stated inverse projection, proving all three group-scheme isomorphisms, not just equality of dimensions or of point counts. Formula (A.15.2) follows from $\Gamma \simeq C_2 \times \mathbb{G}_a \times \gamma$ as a scheme. Equations (A.14.2) and multiplication of $r_0 + 12k$ give (A.15.3); its carry term is integral because residue classes multiply according to the two sign bits.

For $\mathbb{Z}$, invertibility forces $\mu = \pm 1$ and the congruences in (A.7.2) force exactly the displayed bits. The cocycle then vanishes since $a = b$. For $\mathbb{Z}[1/2]$, the unit group is $\{\pm 2^m\}$, and divisibility by twelve in that ring is equivalent to the displayed congruence modulo three; modulo four imposes no condition because two is inverted. The argument with the primes reversed proves the next row, and inverting six proves the fourth. Uniqueness of k in these torsion-free rings follows directly. In the two finite fields, twelve vanishes and all representatives r_0 are units, giving $4 \cdot 2 \cdot 2 \cdot 2 = 32$ and $4 \cdot 3 \cdot 2 \cdot 3 = 72$ points. These were also counted directly from the original lattice equations by `a_finite.py`.

There is an explicit ring isomorphism $\mathbb{O}_N(\mathbb{Z}/q) \to \mathbb{Z}/Nq$, $r + N[t] \mapsto r + Nt$. Therefore $\gamma(\mathbb{F}_2) = (\mathbb{Z}/24)^* \simeq C_2^3$ and $\gamma(\mathbb{F}_3) = (\mathbb{Z}/36)^* \simeq C_3 \times C_2^2$. The residue maps split as maps of these finite abelian groups, with kernels C_2 and C_3. Pulling back the D_8 extension yields the claimed finite-group descriptions, including its action on s. Finally each of the eight charts is an open subset of affine two-space, and over characteristics two and three it is the whole affine two-space. This proves smoothness, flatness, and the component count on the actual fibres.

Under (A.14.3), the first two graphs contain μ_ρ, while the third intersects it trivially. Indeed $U = a_\rho$ forces $U = \omega$ when $a_\rho = \omega$. Its image in $\mathcal{M}/\mu_\rho$ is a bisection over Γ_{e}, with kernel μ_G. This gives explicitly the change of descriptions in scans 91–94 rather than identifying the central quotients with all three unquotiented graphs. □

A.16. An actual comparison with the groups of §48 [Proved]

Let H_{48} be the group of pairs of quadratic-normalizer matrices

$$A = c\rho + dI, \quad B = \begin{pmatrix} u & t \\ -et & eu \end{pmatrix}, \quad Q_3 = c^2 + cd + d^2, \quad Q_2 = t^2 + u^2,$$

with Q_2, Q_3 units and $e^2 = (e')^2 = 1$, satisfying $\det B = eQ_2 = e'Q_3$. Its character map is

$$H_{48} \to \gamma_{48} = \mathbb{G}_m \times \mu_2^2, \quad (A, B) \mapsto (\det B, e, e').$$

There is a specified comparison of character schemes

$$\iota : \gamma \longrightarrow \gamma_{48}, \qquad \underline{\mu} \longmapsto (\mu, (-1)^a, (-1)^b), \tag{A.16.1}$$

which is an isomorphism over $\mathbb{Z}[1/6]$. Define the actual base change $H_{48}^\dagger = H_{48} \times_{\gamma_{48}} \gamma$. Projection followed by the zero-translation section is a morphism of affine group schemes

$$H_{48}^\dagger \longrightarrow \Gamma/\mu_\rho, \quad (A, B, \underline{\mu}) \longmapsto (0, \underline{\mu}). \tag{A.16.2}$$

Its kernel is $ST_\rho \times ST_\sigma$, the product of the two norm-one quadratic groups, and its image is exactly the displayed copy of γ. For the full groups the analogous map $\mathcal{M}_{48} \times_{\gamma_{48}} \gamma \to \mathcal{M}/\mu_\rho$ has kernel $ST_\rho \times ST_\sigma \times \mathrm{SL}_2$ and image the canonical γ section.

The §48 identifications $\mathcal{M}_{48}(0) \simeq H_{48}$ and $\mathcal{M}_{48}(1/2) \simeq H_{48}$ hold by their graph maps. The graph $U = A$ is defined there over $e' = 1$, and is identified with $H_{48}|_{e'=1}$. Its comparison is the

restriction of (A.16.2) over $b = 0$. Thus all three §51 groups are related to the earlier groups by these explicit base changes, character maps, and central quotients. The resulting schemes are not isomorphic: their generic relative dimensions are respectively two and three, and their bad fibres have the distinct descriptions proved below.

Proof. Solving $A\rho = \rho A$ and $B\sigma = e\sigma B$ gives the two displayed matrix forms, and their determinants give exactly the stated equations. The character map is fppf surjective: given its three coordinates, adjoining square roots of the required unit norms and taking scalar elements in the quadratic algebras supplies lifts. The extensions adjoining square roots are finite free and faithfully flat, including in characteristic two. Its kernel is precisely the two norm-one groups. Formula (A.16.1) is a homomorphism by the ring and sign formulas in A.5–A.7. Over $\mathbb{Z}[1/6]$, its inverse recovers the constant signs and sets $k = (\mu - r_0)/12$, proving the asserted isomorphism.

The base-changed character map retains its exact kernel, so its composition with the group section of Γ/μ_ρ has the kernel and image claimed in (A.16.2). For the full matrix triple, the additional determinant-prescribed U coordinate has kernel SL_2. The section $U = \mathrm{diag}(\det B, 1)$ gives the graph at zero, and $U = B$ gives the graph at one-half. For $U = A$, the determinant condition reads $e' = 1$, proving exactly the specified restriction. This derives the comparison from the equations used in LM1 B.2–B.5 instead of importing their conclusion.

Here is also a direct fibre computation. Eliminating the common determinant gives

$$H_{48} = \operatorname{Spec} \mathbb{Z}[c, d, t, u, e, e', Q_2^{-1}]/(Q_3 - ee'Q_2, e^2 - 1, (e')^2 - 1).$$

Relative to the sign scheme μ_2^2 its hypersurface is smooth: the gradient in c, d, t, u is $(2c + d, c + 2d, -2ee't, -2ee'u)$ and cannot vanish on $Q_2 \neq 0$. For characteristic different from two, use the last two entries; in characteristic two the first two become d, c and cannot both vanish because Q_3 is a unit. The geometric fibres are nonempty, so this is a smooth surjective map of relative dimension three. Thus the generic dimension is three. At three, $Q_3 = (c - d)^2$; setting $v = c - d \neq 0$ shows four smooth geometric components $\mathbb{G}_m^2 \times \mathbb{G}_a$. At two put $x = e - 1, y = e' - 1$. The sign algebra is $\mathbb{F}_2[x, y]/(x^2, y^2)$; after reduction, $Q_2 = (t + u)^2$ and the remaining equation is the norm-one equation $a^2 + ab + b^2 = 1$. Hence the reduced fibre is

$$\mathbb{G}_a \times \mathbb{G}_m \times R^1_{\mathbb{F}_4/\mathbb{F}_2}\mathbb{G}_m.$$

It is geometrically connected, and smoothness over the sign algebra gives generic length four before reduction. Restricting $e' = 1$ replaces that length by two. The full $\mathcal{M}_{48}$ is, as a scheme, the product of this scheme with SL_2, by $U = \mathrm{diag}(\det B, 1)h$. These fibre calculations and the explicit morphisms above prove the asserted comparison with A.15, whose characteristic-two fibre has eight reduced affine-plane components. They also prove flatness of the §48 models and their failure of smoothness exactly at two. No nonexistence of further mathematical comparisons is asserted. □

A.17. Linear congruence descent from the actual braid automorphism [Proved]

Use the standard tangential and braid normalization

$$B_3 = \langle \beta_1, \beta_2 : \beta_1\beta_2\beta_1 = \beta_2\beta_1\beta_2 \rangle, \quad x = \beta_1^2, \quad y = \beta_2^2.$$

For $(\mu, f) \in \widehat{\mathrm{GT}_0}$, where $f \in \widehat{\langle x, y \rangle}'$, let

$$\Phi(\beta_1) = \beta_1^\mu, \qquad \Phi(\beta_2) = f^{-1}\beta_2^\mu f. \tag{A.17.1}$$

For every $n \geq 2$, define the continuous surjection

$$q_n : \widehat{B}_3 \longrightarrow \mathrm{SL}_2(\mathbb{Z}/n), \quad \beta_1 \mapsto U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \beta_2 \mapsto W = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

There is a unique matrix

$$V_n(\Phi) = \begin{pmatrix} \mu & t_n(\Phi) \\ 0 & 1 \end{pmatrix} \quad \text{such that} \quad q_n\Phi = \mathrm{Int}(V_n(\Phi))q_n. \tag{A.17.2}$$

The maps $\Phi \mapsto V_n(\Phi)$ are continuous homomorphisms, compatible under reduction. In particular all linear congruence kernels are invariant. They give a specified action on the full $\mathrm{GL}_2(\mathbb{Z}/n)$ by the same conjugation formula. This supplies a direct proof of the descent needed in LM2 C.d1 and its requested independent check at $n = 12$.

Proof. The group-theoretic input is that (A.17.1) is an automorphism of $\widehat{B}_3$ for an actual $\widehat{\mathrm{GT}}_0$ pair. Lochak–Schneps prove the braid identity in Lemma 5 of [LS94]; Harbater–Schneps, §1.3, Proposition 4, proof of (i)⇒(iii), explicitly applies that lemma to the full profinite group, [HS00]. Both proofs were opened and checked. The input uses the two cycle relations and the automorphism requirement; no pentagon is used here. The following reduction from that theorem to (A.17.2) is elementary and also proves the requisite generation condition cannot be omitted.

Write $A = q_n\Phi(\beta_1) = U^\mu$ and $B = q_n\Phi(\beta_2) = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Conjugacy to W^μ gives $a + d = 2$, $ad - bc = 1$. The lower-left entry of $ABA = BAB$ gives

$$c(1 + \mu c) = 0. \tag{A.17.3}$$

The pair A, B generates all of $\mathrm{SL}_2(\mathbb{Z}/n)$, because q_n is onto and Φ is an automorphism. For each prime $p \mid n$, the element c mod p cannot vanish: otherwise both matrices would be upper triangular, and could not generate $\mathrm{SL}_2(\mathbb{F}_p)$. Thus c is a unit modulo n. Equation (A.17.3) yields $c = -\mu^{-1}$; the determinant and trace then give

$$d = 2 - a, \quad b = \mu(a-1)^2, \quad t_n = \mu(1-a).$$

Direct multiplication proves $V_nUV_n^{-1} = A$, $V_nWV_n^{-1} = B$. Agreement on the topological generators proves (A.17.2) on the full profinite group. For surjectivity of q_n, elementary row operations over each local ring $\mathbb{Z}/p^r$ reduce a determinant-one matrix to diagonal form, and diagonal determinant-one matrices are products of elementary matrices. Chinese remainders combine these operations. Powers of U, W give all the elementary matrices.

A matrix commuting with U, W is scalar, by comparing its four entries. A scalar relating two matrices with lower-right entry one is the identity. This proves uniqueness. The product $V_n(\Phi)V_n(\Psi)$ has the same normalization and implements the composite automorphism, which proves the homomorphism assertion. The formula for a in terms of $q_n(f)$ proves continuity; uniqueness proves compatibility in n. Taking the same inner conjugation on $\mathrm{GL}_2(\mathbb{Z}/n)$ is now a well-typed extension of the action on its determinant-one subgroup.

For comparison with the modular group, the central braid $z = (\beta_1\beta_2)^3$ maps to ω, and $\widehat{B}_3/\overline{\langle z^2 \rangle} \simeq \widehat{\mathrm{SL}_2(\mathbb{Z})}$. The cited proof also gives $\Phi(z) = z^\mu$, so this quotient is preserved. The subgroup $\widehat{\Pi}_0$ is the inverse image of V_4 modulo four, preserved by the matrices in A.18. Alternatively its generators are $\lambda_0 = \omega U^{-2}$, $\lambda_1 = \omega W^{-2}$; their images are respectively λ_0^μ and a conjugate of λ_1^μ by an element of $\widehat{\Pi}_0'$. To verify the last assertion, evaluate the commutator word f in $U^2 = \omega\lambda_0^{-1}$ and $W^2 = \omega\lambda_1^{-1}$: its central factors cancel. This constructs the comparison to the action on $M/12M$ used below.

The program `a_finite.py` independently enumerates every signed matrix-coboundary solution used in LM2 C.d1 at level twelve: there are twelve for each of the four multipliers and no failed intertwiner. The program `a_braid.py` additionally enumerates the eight possible values of f in the finite commutator image, for every multiplier. All thirty-two pairs satisfy the braid relation; eight pairs fail generation and are explicitly rejected. The remaining twenty-four pairs give the twelve distinct matrices in A.18. This records both the computation and the exact reason the discarded matrix solutions cannot be automorphisms of the profinite source. □

A.18. Exact images at level twelve, with the genuine normalization stated [Proved]

For the normalization (A.17.1)–(A.17.2), the images of $G_{\mathbb{Q}}$, $\widehat{\mathrm{GT}}$, and $\widehat{\mathrm{GT}}_0$ in $\mathrm{GL}_2(\mathbb{Z}/12)$ are all

$$\mathcal{V}_{12} = \left\{ \begin{pmatrix} \mu & t \\ 0 & 1 \end{pmatrix} : \mu \in \{1,5,7,11\}, \quad t \in \{0,4,8\} \right\}, \qquad |\mathcal{V}_{12}| = 12. \tag{A.18.1}$$

Their specified actions in $\mathrm{Aut}(\mathrm{GL}_2(\mathbb{Z}/12))$ therefore have image $\mathrm{Ad}(\mathcal{V}_{12})$ of order 12. Their images in $\mathrm{Out}(\mathrm{GL}_2(\mathbb{Z}/12))$ are trivial. The canonical outer images on $\mathrm{SL}_2(\mathbb{Z}/12)$ are all $(\mathbb{Z}/12)^*/((\mathbb{Z}/12)^*)^2 = C_2^2$, of order four. On $M/12M$ the genuine image of each group is the scalar subgroup $\{\mu I : \mu \in (\mathbb{Z}/12)^*\}$, also of order four.

For $G_{\mathbb{Q}}$, choose the tangential base point at $\lambda = 0$ with tangent $+1$ and the real interval path to the tangent at one; this is the Belyi normalization in (A.17.1). The kernel of its normalized level-twelve matrix representation is exactly

$$G_{\mathbb{Q}(\zeta_{12}, \sqrt[3]{2})}. \tag{A.18.2}$$

The extended group generated by the geometric modular group and any of the three arithmetic groups acts on the same linear congruence quotient with image $\mathrm{Ad}(\mathrm{GL}_2(\mathbb{Z}/12))$ of order 1152. These statements give the precise actions requested after LM2; the order 32 remains the normal subgroup K_{32} of A.3.

Proof. Modulo four, $q_4(x) = U^2$ and $q_4(y) = W^2$ commute, so $q_4(f) = I$. Hence $q_4\Phi(\beta_2) = W^\mu$, whose upper-left entry is one. The formula in A.17 gives $t \equiv 0 \pmod 4$. Thus the whole $\widehat{\mathrm{GT}}_0$ image lies in $\mathcal{V}_{12}$. This set is a group under matrix multiplication and has twelve elements. To prove equality for the arithmetic subgroup, we now compute a compatible finite geometric cover, including its base-point comparison.

Consider the Legendre curve $E_\lambda : y^2 = x(x-1)(x-\lambda)$ over $\mathbb{P}^1 - \{0,1,\infty\}$, and its finite etale cover of cyclic order-three subgroups. Its four points are the four x-coordinates of nonzero three-torsion modulo sign. With the oriented vanishing cycles above the intervals $[0,\lambda]$ and $[\lambda,1]$, a positive loop around either colliding pair of branch points is a full twist. On the double cover it acts by the square of the transvection in the corresponding vanishing cycle. Taking their intersection to be one gives the matrices U^2 and W^2. This constructs the geometric monodromy comparison from x, y in (A.17.1) to the projectivization of $E[3]$. Changing the simultaneous homology basis conjugates both sides; the four-point permutation representation and its coefficient field are unchanged. The central signs used to define λ_i disappear under this specified projectivization.

We also specify the arithmetic fibre functor. At the tangent $\lambda = 0, +1$, the fibre is obtained by adjoining formal roots $\lambda^{1/m}$ and letting $G_{\mathbb{Q}}$ act on their coefficients, fixing the symbols $\lambda^{1/m}$. The real interval path gives the usual second generator. In every abelian Kummer covering generated by $\lambda^{1/m}$ and $(1-\lambda)^{1/m}$, the leading coefficients at the two tangents are one. Thus the path cocycle has zero abelianization; the induced based Galois action has precisely $x \mapsto x^\chi$,

$y \mapsto f^{-1}y^{\chi}f$ with $f \in \widehat{F}_2'$. The arithmetic compatibility and the inclusions $G_{\mathbb{Q}} \subset \widehat{\mathrm{GT}} \subset \widehat{\mathrm{GT}}_0$ are the tangential actions described in [HS00, §0.3 and §1.2–1.3]. Consequently the arithmetic conjugation of this four-point Legendre monodromy is the projective reduction of V_{12}. There is no extra projective conjugator ambiguity here: a matrix centralizing the projective images of U^2, W^2 modulo three centralizes their actual matrices, since their trace two is nonzero; comparing entries then makes it scalar.

The three-division polynomial, obtained from the tangent-and-chord addition formula on the displayed cubic, is

$$\psi_3(x) = 3x^4 - 4(1+\lambda)x^3 + 6\lambda x^2 - \lambda^2. \tag{A.18.3}$$

At zero it is $x^3(3x-4)$. The root with constant term $4/3$ lies in $\mathbb{Q}[[\lambda]]$, by the formal implicit-function theorem. For the other roots put $\lambda = s^3$, $x = s^2 z$. Dividing the polynomial by s^6 gives

$$3s^2z^4 - 4(1+s^3)z^3 + 6sz^2 - 1. \tag{A.18.4}$$

Each root c of $c^3 = -1/4$ is simple at $s = 0$, and uniquely lifts to $z \in \mathbb{Q}(c)[[s]]$. Hence the exact field of all Puiseux coefficients is $L = \mathbb{Q}(\zeta_3, \sqrt[3]{2})$: its containment follows by the unique recursion for the coefficients, and the reverse containment follows from the three leading coefficients, because $(-2c)^3 = 2$. The polynomial $X^3 - 2$ is irreducible by Eisenstein at two, and its discriminant -108 is not a square. Therefore $[L : \mathbb{Q}] = 6$ and the arithmetic action on the four lines has image S_3, fixing the one rational branch.

Modulo four, V_{12} is $\mathrm{diag}(\chi, 1)$, so its remaining field is $\mathbb{Q}(i)$. The field L has unique quadratic subfield $\mathbb{Q}(\sqrt{-3})$, by the subgroup lattice of S_3, and therefore $L \cap \mathbb{Q}(i) = \mathbb{Q}$. The pair consisting of V mod 4 and [V mod 3] determines every $V \in \mathcal{V}_{12}$: a scalar with lower-right entry one is the identity, and Chinese remainders recover both entries. Thus the arithmetic kernel fixes exactly the degree-twelve field $L(i) = \mathbb{Q}(\zeta_{12}, \sqrt[3]{2})$, proving (A.18.1)–(A.18.2). The two containing GT groups have the same image by the already proved upper bound. The formal polynomial calculations (A.18.3)– (A.18.4), the leading field, and all twelve matrices are checked in `a_braid.py`.

The centralizer of $\mathrm{SL}_2(\mathbb{Z}/12)$ in $\mathrm{GL}_2(\mathbb{Z}/12)$ consists of its four scalar units. The normalization $V_{22} = 1$ makes their intersection with $\mathcal{V}_{12}$ trivial. This proves the genuine automorphism image count and the trivial outer image on the full GL group. A conjugation by V is inner on SL exactly when $V \in \mathrm{SL}_2(\mathbb{Z}/12) \cdot ((\mathbb{Z}/12)^* I)$; taking determinants gives its unit square class. All four units modulo twelve square to one, proving the outer SL count. The scalar action on $M/12M$ follows from the generator comparison at the end of A.17, and cyclotomic surjectivity realizes all four scalars. Finally the geometric image is SL and the determinants of $\mathcal{V}_{12}$ are all units, so $\langle \mathrm{SL}_2(\mathbb{Z}/12), \mathcal{V}_{12} \rangle = \mathrm{GL}_2(\mathbb{Z}/12)$. Quotienting its order 4608 by the scalar centralizer of order four gives 1152.

This specifies which choices give a genuine action and which quotients remove them. The congruence action is obtained from $\widehat{B}_3$ and its modular quotient; A.4 proves that the reduction to the unquotiented GL or SL group cannot instead be defined by first passing through $\mathfrak{S}_{\mathrm{d}}$. No equality between the normalized upper-right entry and the Kummer cocycle based at $1/2$ is used in this proof. □

A.19. The normalized cyclotomic arithmetic image in the minimum approximation [Proved]

In the tangential normalization of A.17–A.18, the arithmetic homomorphism into the final §51 minimum approximation has image in $\mathcal{M}(0)$: its effective coordinate is $U = u_{\underline{\mu}}$. Its restriction to $G_{\mathbb{Q}^{\mathrm{cyc}}}$ has image in the abelian group

$$\ker(\mathcal{M}(0)(\widehat{\mathbb{Z}})^{\Delta} \longrightarrow \widehat{\mathbb{Z}}^*) = C_2 \times \widehat{\mathbb{Z}}. \tag{A.19.1}$$

After the central quotient by μ_ρ, this kernel is $\widehat{\mathbb{Z}}$ under addition. In particular the closed commutator subgroup of $G_{\mathbb{Q}^{\mathrm{cyc}}}$ maps to the identity in this normalized minimum approximation. Here the superscript Δ requires the single global component choice described in A.1; it is essential to (A.19.1).

Proof. The congruence-normalized braid action has conjugator $f \in \widehat{F}_2'$. In the modular quotient its image is in $\widehat{\Pi}_0'$ by the evaluation calculation in A.17, so it vanishes in the minimum approximation. Thus both elementary generators U, W are sent to their μ-th powers. The automorphism (A.8.5) has exactly these images on diagonal profinite points: from $U = (\sigma\rho)^{-1}$, $W = (\rho\sigma)^{-1}$ and $\lambda_0 = \omega U^{-2}$, $\lambda_1 = \omega W^{-2}$, the possible central factor is $\omega^{\nu+a} = 1$, since $\nu = (\mu-1)/2 \equiv a \pmod 2$. This proves $U = u_{\underline{\mu}}$ for the effective coordinate, which is precisely the graph defining $\mathcal{M}(0)$. The retained normalizer conjugators give its other two coordinates as in (183), so its projection to those coordinates is the isomorphism of A.15.

For a diagonal cyclotomic point with $\mu = 1$, its residue is one and the unique quotient $(\mu-1)/12$ is zero in $\widehat{\mathbb{Z}}$: hence $a = b = k = 0$. Formula (A.14.2) on this fibre is $(i, s)(i', s') = (i + i', s + s')$. This proves (A.19.1), its central quotient, and the commutator assertion. It also explains why replacing diagonal profinite points by arbitrary $\gamma(\mathbb{F}_2)$-points would give the wrong kernel. □

B. The Fermat tower and its homology

We use the arithmetic model actually written in (220):

$$X_n^+ :\ x_0^n + x_1^n + x_\infty^n = 0, \qquad D_n = \{x_0 x_1 x_\infty = 0\}, \qquad U_n^+ = X_n^+ \setminus D_n.$$

The notation $X_n : x^n + y^n = z^n$, used in our geometric chain calculations, is related to this model explicitly in B.1. Throughout this part, $F = \langle x, y\rangle$, $c = [x, y] = xyx^{-1}y^{-1}$, and $F'' = [F', F']$. In a profinite group, a commutator subgroup means its closed version. Put

$$A_n = (\mathbb{Z}/n)^2, \quad R_n = \mathbb{Z}[A_n] = \mathbb{Z}[X^{\pm1}, Y^{\pm1}]/(X^n - 1, Y^n - 1), \quad K_n = \ker(F \longrightarrow A_n).$$

Capital letters X, Y in a group ring denote the images of x, y. We write $M = \mathbb{Z}^3/\mathbb{Z}(1, 1, 1)$, with basis the classes e_0, e_1 and $e_\infty = -e_0 - e_1$.

B.1. Arithmetic models and the precise problem [Proved]

The morphism

$$q_n : U_n^+ \longrightarrow U_1^+, \qquad [x_0 : x_1 : x_\infty] \longmapsto [x_0^n : x_1^n : x_\infty^n]$$

is defined over $\mathbb{Q}$, is equivariant for coordinate permutations, and has deck group μ_n^3/μ_n. The coordinate

$$z = -x_0/x_\infty, \qquad \psi(z) = [-z : z - 1 : 1]$$

identifies U_1^+ with $\mathbb{P}^1 - \{0, 1, \infty\}$. For $\alpha_n^n = -1$, the geometric comparison

$$\iota_n : X_n^+ \longrightarrow X_n, \qquad [x_0 : x_1 : x_\infty] \longmapsto [x_0 : x_1 : \alpha_n x_\infty]$$

is an isomorphism over $\mathbb{Q}(\alpha_n)$. At odd levels one can take $\alpha_n = -1$ over $\mathbb{Q}$; at level four the displayed comparison uses $\alpha_4 = \zeta_8$. In particular $X_4^+(\mathbb{R}) = \varnothing$, whereas $[0 : 1 : 1] \in X_4(\mathbb{Q})$.

The *position du problème* in §51.11 is to identify the affine representation

$$\widetilde{\mathcal{M}} \longrightarrow G_{\rho,\sigma}/\mu_G \longrightarrow \widehat{\mathbb{Z}}^* \times S_3$$

with the action on a fibre of the universal abelian Fermat covering, and separately to interpret its specialization at $z = 1/2$. Its translation group is $\widehat{M}(1)$. *Manuscript: §51.11, folios 695–697, scans 95–99, (218)–(230); §51.14, folio 723, scan 151, (315)–(317).*

Proof. The asserted equations follow on taking n-th powers. On the chart $x_\infty \neq 0$, division by α_n gives $(x_0/\alpha_n x_\infty)^n + (x_1/\alpha_n x_\infty)^n = 1$; the induced coordinate on the base is z. Compatible choices, for example $\alpha_n = \exp(\pi i/n)$ after fixing an embedding in $\mathbb{C}$, commute with the power maps. Thus all later geometric comparisons have an actual map behind them. The even-level real assertion follows because a sum of three real fourth powers vanishes only when all three coordinates vanish. Equations (228)–(230) explicitly identify the deck lattice and its cyclotomic action; (219) singles out the remaining rank-one affine specialization. These are the data just stated. □

B.2. The universal abelian covering [Proved]

Over $\mathbb{C}$, q_n corresponds to the subgroup K_n of $\pi_1(U_1) = F$. Its monodromy is $x \mapsto (1, 0)$, $y \mapsto (0, 1)$. Consequently the pro-étale tower $\varprojlim_n U_n^+$ is the universal abelian geometric covering, with deck module $\widehat{M}(1)$. Its deck and coordinate-permutation actions form $(\mu_n^3/\mu_n) \rtimes S_3$ at finite level. *Manuscript: §51.11, folios 696–697, scans 97–99, (220)–(230); §51.13, folio 715, scan 135.*

Proof. In the standard geometric chart, $u^n = z$ and $v^n = 1 - z$. Analytic continuation around 0 multiplies u by ζ_n and leaves v unchanged; continuation around 1 does the reverse. These two elements generate A_n, so the covering is connected and its subgroup is exactly K_n. Every finite quotient of $\widehat{\mathbb{Z}}^2$ factors through some A_n; this proves the universal assertion, rather than only the existence of a collection of abelian coverings. A change of compatible geometric roots identifies the deck group with $\widehat{\mathbb{Z}}^2$ by a change of basis; intrinsically it is $(\widehat{\mathbb{Z}}^3/\widehat{\mathbb{Z}}) \otimes \widehat{\mathbb{Z}}(1)$. Permutations act on the three coordinates, commute with power maps, and conjugate the diagonal multipliers by permuting them. This proves the stated semidirect product and the compatibility of the action with the inverse system. □

B.3. The withdrawn freeness and its exact replacement [Refuted]

For $n > 1$, the action of μ_n^3 on X_n^+ is not free: its diagonal μ_n fixes the whole curve. The quotient group scheme $A_n^{\mathrm{geom}} = \mu_n^3/\mu_n$ acts freely on U_n^+, and $U_n^+ \to U_1^+$ is an A_n^{geom}-torsor. At a coordinate cusp its stabilizer in A_n^{geom} is a copy of μ_n. Thus the quotient action on the compact curve is not free. These are assertions about actions on schemes; the corresponding integral homology modules are computed in B.5–B.7. *Manuscript: §51.11, folio* 696′*, scan 98, deletion before (224) and the inserted U_n in the paragraph following (225).*

Proof. A multiplier (a_0, a_1, a_∞) fixes a point with all coordinates invertible precisely when the three multipliers agree projectively. Its class in the quotient is then the identity. At a point $[0 : b : c]$, with $bc \neq 0$, the equality required is $a_1 = a_\infty$; a_0 is unrestricted. Modulo the diagonal this is exactly μ_n. This calculation also describes the stabilizer functors, and hence applies to group schemes, including their nonreduced fibres. On a torus chart the power map is given by adjoining independent n-th roots of two units. Its coordinate algebra is locally free of rank n^2, and the standard root-of-unit torsor identity proves faithful flatness and the torsor assertion. Over $\mathbb{Z}[1/n]$ it is finite étale. At residue characteristics dividing n it is still a torsor for the fppf topology; this does not assert étaleness or smoothness of those Fermat fibres. The finite checks in `code/b_freeness.py` enumerate the full and quotient stabilizers over finite fields for $1 \leq n \leq 8$. They give full-group stabilizer order n on the open part, quotient stabilizer order one there, and quotient stabilizer order n at a cusp. □

B.4. The discrete Alexander module [Proved]

Conjugation induces a free rank-one $R = \mathbb{Z}[X^{\pm 1}, Y^{\pm 1}]$-module structure

$$F'/F'' = R\,c.$$

With the convention for c fixed above, the injective Fox-chain map is

$$R\,c \longrightarrow R^2, \qquad fc \longmapsto ((1-Y)f, (X-1)f),$$

and its image is $\ker\big((X-1,\ Y-1) : R^2 \to R\big)$.

Proof. The graph with one vertex and oriented edges x, y has fundamental group F. Its universal abelian covering has vertex module R, edge module R^2, and the displayed boundary. Its first homology is the abelianization of its fundamental group F'. The chain of $xyx^{-1}y^{-1}$ is $(1-Y, X-1)$. For a cycle (a, b), reduction of $(X-1)a + (Y-1)b = 0$ modulo $X-1$ gives $(Y-1)\bar{b} = 0$ in the domain $\mathbb{Z}[Y^{\pm 1}]$. Thus $b = (X-1)f$. Substitution and cancellation of the nonzero divisor $X-1$ give $a = (1-Y)f$. The same cancellation proves uniqueness. This supplies both the generator and the proof of freeness. □

B.5. All-cusp-deleted integral homology [Proved]

There are canonical R_n-linear descriptions

$$H_1(U_n, \mathbb{Z}) = \ker\left(R_n^2 \xrightarrow{(X-1,\ Y-1)} R_n\right), \tag{B.1}$$

$$0 \longrightarrow R_n/\mathbb{Z}\mathcal{N}_n \xrightarrow{f \mapsto fc} H_1(U_n, \mathbb{Z}) \xrightarrow{q_{n*}} n\mathbb{Z}^2 \longrightarrow 0, \qquad \mathcal{N}_n = \sum_{a,b=0}^{n-1} X^a Y^b. \tag{B.2}$$

The deck action on the right-hand term is trivial. The group $H_1(U_n, \mathbb{Z})$ has rank $n^2 + 1$, and over $\mathbb{C}$ the trivial deck character occurs twice and every other character occurs once.

Proof. The finite covering of the two-circle graph has n^2 vertices and $2n^2$ edges, with boundary as in (B.1). A spanning tree proves that its cycle group is free of rank $2n^2 - n^2 + 1$. The inclusion $K_n \subset F$ induces $K_n^{\mathrm{ab}} \to F^{\mathrm{ab}}$. Its image is $n\mathbb{Z}^2$; its kernel is the image of F', because $F' \subset K_n$ and $K_n/F' = n\mathbb{Z}^2$. B.4 identifies that image with the orbit module generated by c. The kernel of $R_n \to R_n^2,\ f \mapsto ((1-Y)f, (X-1)f)$ consists of elements fixed by both translations. Equality of all coefficients says precisely that it is $\mathbb{Z}\mathcal{N}_n$. This proves (B.2), including its exactness over $\mathbb{Z}$. On a deck character (ζ_n^a, ζ_n^b), the boundary in (B.1) has rank one unless $a = b = 0$; in that case it has rank zero. This proves the character formula directly. □

B.6. The compact integral homology module [Proved]

Define

$$S_X = \sum_{i=0}^{n-1} X^i, \quad S_Y = \sum_{j=0}^{n-1} Y^j, \quad P_n = \sum_{0 \le j \le i < n} X^i Y^j, \quad I_n = (S_X, S_Y, P_n) \subset R_n.$$

The compact homology has the exact integral presentation

$$H_1(X_n, \mathbb{Z}) \simeq R_n/I_n, \qquad 1 \longmapsto c. \tag{B.3}$$

Equivalently, (B.1) is divided by the $3n$ cusp cycles

$$Y^j(S_X,0),\quad X^i(0,S_Y),\quad X^i(S_{XY},XS_{XY})\quad (0\le i,j<n),\qquad S_{XY}=\sum_{i=0}^{n-1}(XY)^i. \tag{B.4}$$

The cusp subgroup is primitive of rank $3n-1$. The rank of (B.3) is $(n-1)(n-2)$, and its complex characters are exactly those (a,b) with $a,b,a+b\neq 0 \pmod n$, each once.

Proof. Filling a puncture kills its oriented peripheral loop in first homology. Lifting the peripheral words $x^n,y^n,(xy)^n$ gives precisely (B.4). The homology sequence of the pair consisting of the compact surface and its punctured surface identifies the subgroup of peripheral classes with $\mathbb{Z}[D_n]/\mathbb{Z}\sum_{P\in D_n}P$. The quotient is the integral homology of an oriented compact surface, so is torsion-free. This proves primitivity and rank $3n-1$.

We give an integral derivation of the ring presentation. First fill the $2n$ cusps over $0,1$. The resulting affine curve $u^n+v^n=1$ retracts onto the inverse image of the segment $[0,1]$. To see this, take a closed regular neighbourhood of that segment in the z-plane. Outside it the finite map is unramified and its ends are collars. Inside it the local charts at the only branch points are $w\mapsto w^n$; their lifted discs and the intervening strips are a regular neighbourhood of the lifted segment. Shrinking these collars, discs and strips gives the stated retraction. That graph has vertices $A_n/\langle X\rangle$ and $A_n/\langle Y\rangle$, and one edge for each element of A_n; it is the complete bipartite graph on two sets of n vertices. Its cycles are the integral matrices whose row and column sums vanish. The map

$$R_n\longrightarrow\{\text{such matrices}\},\qquad f\longmapsto(1-X)(1-Y)f$$

is surjective: subtracting the last row and column expresses every such matrix as an integral sum of elementary rectangles. Its kernel is (S_X,S_Y), as one sees by successively taking differences of rows and columns. The elementary rectangle is the commutator cycle, up to the fixed choice of its orientation.

Now fill the n remaining cusps. In the original two-edge chain module the identities

$$(1-Y)P_n=S_X-YS_{XY},\qquad (X-1)P_n=S_Y-S_{XY}$$

give

$$(S_{XY},XS_{XY})=-Y^{-1}P_n\,c+Y^{-1}(S_X,0)+Y^{-1}(0,S_Y).$$

All infinity cusp cycles are translates of this one. Thus their classes in the bipartite graph kill exactly the ideal generated by P_n, proving (B.3). For a character with both $a,b\neq 0$, the last two displayed identities make P_n zero precisely when $a+b\neq 0$; when $a+b=0$, $(X-1)P_n=-n$ is nonzero. The character statement follows. Its cardinality is $(n-1)(n-2)$, agreeing with the topological rank. □

B.7. Finite integral computations and module freeness [Proved]

The exact calculations for $1\le n\le 8$ give

n	1	2	3	4	5	6	7	8
$g(X_n)$	0	0	1	3	6	10	15	21
$\operatorname{rk} H_1(U_n)$	2	5	10	17	26	37	50	65
$\operatorname{rk} H_1(X_n)$	0	0	2	6	12	20	30	42
$\operatorname{rk}\langle\text{cusps}\rangle$	2	5	8	11	14	17	20	23

For every $n>1$, $H_1(U_n,\mathbb{Z})$ is not a free R_n-module. For every $n\ge 3$, $H_1(X_n,\mathbb{Z})$ is cyclic but not free as an R_n-module. At $n=1,2$ the compact module is zero. The exact comparison maps, deck actions and S_3-actions are supplied as integer matrices in `code/b_fermat_modules.json`.

Proof. The program builds the lifted graph, chooses the tree obtained from the paths $x^i y^j$, and uses each remaining edge as a fundamental cycle. It checks the Fox boundary, computes the deck transformations and the two based substitutions in B.9, and expresses every cusp cycle in this integral basis. It checks that the nonzero Smith entries of the cusp matrix and of the compact annihilator presentation are all one. It also checks that the map to F^{ab} has Smith entries n, n. These are full integer presentations, rather than character counts. All output rows displayed here were generated and checked by that program.

A free R_n-module has integral rank divisible by n^2. The affine rank n^2+1 fails this test for $n > 1$. The positive compact rank $(n-1)(n-2)$ is strictly below n^2. These rank conclusions apply through the explicit presentations (B.1) and (B.3); they do not replace the scheme-action assertion of B.3. □

B.8. The full profinite rank-one theorem [Proved]

Let $\widehat{F}$ be the free profinite group on x, y, and put

$$\Lambda = \widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}^2]] = \varprojlim_{m,n} (\mathbb{Z}/m)[A_n].$$

There are continuous Λ-linear isomorphisms

$$\widehat{F}'/\widehat{F}'' \xrightarrow{\sim} \varprojlim_n H_1(U_n, \widehat{\mathbb{Z}}) \xrightarrow{\sim} \varprojlim_n H_1(X_n, \widehat{\mathbb{Z}}) \simeq \Lambda\, c. \tag{B.5}$$

The transition maps on homology are pushforwards of the covering maps. The same limits can be formed diagonally with coefficients $\mathbb{Z}/n$. This is the full profinite assertion, including the prime-to-ℓ factors of the completed group ring.

Proof. We first prove the group-to-tower comparison. Let $\widehat{K_n} = \ker(\widehat{F} \to A_n)$ and $H = \bigcap_n \widehat{K_n} = \widehat{F}'$. The map $H \to \varprojlim_n \widehat{K_n}^{\mathrm{ab}}$ is surjective: compatible abelianized cosets are nonempty compact subsets of $\widehat{K_n}$, nested along a cofinal divisibility chain; their intersection supplies a lift in H. Its kernel contains H'. To prove the reverse inclusion, take a continuous homomorphism $f : H \to C$ to a finite abelian group. Choose an open normal subgroup V of $\widehat{F}$ with $V \cap H \subset \ker f$. Since H is normal, $[V, H] \subset V \cap H$, and f extends to HV by $f(hv) = f(h)$. Compactness gives $\widehat{K_n} \subset HV$ for a sufficiently divisible n. It follows that every element in $\bigcap_n \widehat{K_n'}$ is annihilated by f. Finite abelian quotients separate H/H', proving the first isomorphism. The covering graph identifies $\widehat{K_n}^{\mathrm{ab}}$ with $H_1(U_n, \widehat{\mathbb{Z}})$: the finite-index topology on $K_n \subset F$ is its full profinite topology, because the intersection of the finitely many F-conjugates of a finite-index subgroup of K_n is finite-index normal in F. The extra quotient $n\widehat{\mathbb{Z}}^2$ in (B.2) has transition maps given by inclusion and has inverse limit $\bigcap_n n\widehat{\mathbb{Z}}^2 = 0$. Thus its rank-two contribution at every finite level does not survive in (B.5).

For the second isomorphism, work first modulo m. The kernel of $H_1(U_n, \mathbb{Z}/m) \to H_1(X_n, \mathbb{Z}/m)$ is the peripheral module of B.6. Under $U_{dn} \to U_n$, a small loop around a cusp maps to d times a small loop. Thus the image of that kernel is zero when $m \mid d$. The inverse system is pro-zero. All groups are finite, so taking the inverse limit preserves the short exact sequences. This gives the second isomorphism modulo m, and passage to the inverse limit over m gives the assertion.

Finally use (B.3) modulo m. It stays exact because its integral quotient is free. Under $R_{dn} \to R_n$,

$$S_{X,dn} \mapsto dS_{X,n}, \quad S_{Y,dn} \mapsto dS_{Y,n}, \quad P_{dn} \mapsto dP_n + \binom{d}{2}\mathcal{N}_n.$$

The last formula counts pairs $0 \le j \le i < dn$ with each prescribed pair of residues modulo n. A value of d divisible by $2m$ kills every image generator of I_{dn} modulo m. Hence $\varprojlim_n I_n(\mathbb{Z}/m) = 0$,

and the finite inverse-limit exact sequence identifies $\varprojlim_n H_1(X_n, \mathbb{Z}/m)$ with $\varprojlim_n (\mathbb{Z}/m)[A_n]$, by the compatible generator c. Passing to m proves freeness. This argument agrees with [Ots16, Theorem 6.1], whose full profinite statement was opened and checked. It does not use a power-series-domain argument. The diagonal $(m, n) = (r, r)$ is cofinal in the divisibility product, which proves the coefficient formulation in the statement. □

B.9. The exact permutation comparison [Proved]

On Λc the selected S_3-action is semilinear:

$$r(f(X,Y)c) = f(Y, X^{-1}Y^{-1})c, \qquad s(f(X,Y)c) = -f(Y,X)c, \tag{B.6}$$

where $r^3 = s^2 = (rs)^2 = 1$. It agrees with the coordinate-permutation action on the Fermat tower after a compatible deck translation. Changing that translation conjugates the action by the corresponding deck operator. The same comparison is represented integrally at every finite level by the matrices of B.7.

Proof. Conjugation by the modular generators acts on F by $\rho : x \mapsto y,\ y \mapsto x^{-1}y^{-1}$ and $\sigma : x \mapsto y,\ y \mapsto x$. Replace ρ by $x\rho$; its based substitution is

$$r : x \mapsto xyx^{-1}, \quad y \mapsto y^{-1}x^{-1}.$$

Direct free reduction gives $r(c) = c$ and $s(c) = c^{-1}$. Their actions on F^{ab} are

$$R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The defects in r^3 and $(rs)^2$ are inner automorphisms by elements of F'. Such automorphisms act trivially on F'/F'' and on K_n^{ab}, since $F' \subset K_n$. Formula (B.6) therefore gives a genuine S_3-action. At the level of deck extensions this is the complement constructed in B.11. The coordinate permutations provide another complement of the same A_n-extension. B.10 proves that there is a unique deck translation conjugating the two. Uniqueness makes these translations compatible as n varies. Passing to homology and then to (B.5) gives the claimed comparison, with its ambiguity stated. □

B.10. The two cohomology groups governing uniqueness [Proved]

For every abelian group I, with trivial S_3-action on I, put $M_I = I^3/I(1,1,1)$. Then

$$H^0(S_3, M_I) = H^1(S_3, M_I) = 0. \tag{B.7}$$

The same assertions hold for continuous cohomology with a profinite I. For any profinite group B acting on I and commuting with the permutation action,

$$H^0(B \times S_3, M_I) = H^1(B \times S_3, M_I) = 0.$$

Manuscript: §51.12–13, folios 702, 715–716, scans 109, 136–138, (295)–(297); §51.14, folio 721′, scan 148, (309).

Proof. Use coordinates in which r, s act by R, S above. An invariant (u, v) satisfies $u = v$, $u = -v$, and $u = 2v$. Thus $2v = 3v = 0$, so $u = v = 0$, over every I. A crossed homomorphism is specified by $f(r) = a = (a_0, a_1)$, $f(s) = b$. The relation $s^2 = 1$ gives $b = (t, -t)$. The relation $r^3 = 1$ imposes no condition, because $1 + R + R^2 = 0$. The relation $(rs)^2 = 1$ gives

$$a_0 = 2a_1 + 3t.$$

The element $m = (-a_1 - 2t, -a_1 - t)$ satisfies $(R-1)m = a$ and $(S-1)m = b$. Hence the cocycle is a coboundary, and the conjugating element is unique by the invariant calculation. Every formula uses only integral addition, so it also proves the continuous profinite assertion. For $B \times S_3$, first subtract the unique coboundary that kills the restriction to S_3. The cocycle identity for commuting $b \in B, s \in S_3$ then says that $f(b)$ is S_3-invariant. It is zero. This proves the last assertion without a spectral-sequence convergence assumption. □

B.11. An explicit integral splitting [Proved]

The extension $\mathfrak{S}^+/\mu_{\mathfrak{S}}$ of S_3 by M splits. In the affine realization

$$\rho = (R, 0), \qquad \sigma = (S, t), \qquad t = (1/3, -1/3),$$

a complement is generated by

$$\rho' = \lambda_0 \rho = (R, (1, 0)), \qquad \sigma' = \sigma.$$

It fixes $p = (2/3, 1/3)$. In particular its integral extension class, and the profinite class after completion, are zero. *Manuscript: §51.12, folios 700–702, scans 105–109; §51.13, folio* 715′*, scan 136.*

Proof. For affine multiplication $(A, u)(B, v) = (AB, u + Av)$, $1 + R + R^2 = 0$ gives $(\rho')^3 = 1$, $(1 + S)t = 0$ gives $(\sigma')^2 = 1$, and

$$RS = \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix}, \qquad (1, 0) + Rt = (4/3, 2/3), \qquad (1 + RS)(4/3, 2/3) = 0$$

gives $(\rho'\sigma')^2 = 1$. Their linear parts generate all of S_3, so this is a section. Its translations lie in the original affine cosets, since the only changed generator was multiplied by $\lambda_0 \in M$. Solving $(1 - R)p = (1, 0)$ and $(1 - S)p = t$ gives the stated fixed point.

For complete cocycle data choose $h = r^i s^j$ and put $t_h = R^i(jt)$. The original integral cocycle is

$$c(h, k) = t_h + A_h t_k - t_{hk}.$$

The section just found has translation $p - A_h p$. Thus the integral cochain $b_h = p - A_h p - t_h$ satisfies $c + \delta b = 0$. This proves vanishing as a cohomology class, not merely satisfaction of a few relations. All formulas commute with reduction modulo N and with profinite completion. □

B.12. Splitting and uniqueness for the full quotient [Proved]

The continuous extension

$$1 \longrightarrow \widehat{M} \longrightarrow G_{\rho,\sigma}/\mu_G \longrightarrow \widehat{\mathbb{Z}}^* \times S_3 \longrightarrow 1$$

has zero class in continuous H^2. Its set of continuous splittings is a torsor under $\widehat{M}$, by conjugation. For the canonical scalar lift u_μ of the manuscript, put $\nu = (\mu - 1)/2 \in \widehat{\mathbb{Z}}$. A section commuting with the complement of B.11 is

$$s(\mu) = (-\nu, -\nu)\, u_\mu.$$

*Manuscript: §51.12, folios 704–*704′*, scans 113–114, (264)–(266); §51.13, folios 715–*716′*, scans 135–138.*

Proof. The manuscript's scalar automorphism sends ρ to $\lambda_1^\nu \rho$ and fixes σ modulo the central involution, and it acts by multiplication by μ on M. In the affine realization it is conjugation by

$$(\mu I, w_\mu), \qquad w_\mu = (-\nu/3, \nu/3).$$

Indeed $(1-R)w_\mu = \nu(0,1)$ and $\mu t + (1-S)w_\mu = t$. The product law follows from $\nu(\mu\mu') = \nu(\mu) + \mu\nu(\mu')$. After multiplying by the integral translation $(-\nu, -\nu)$, its affine translation becomes $(1-\mu)p$. These are precisely the scalar homotheties about p; they commute with the finite complement fixing p, and multiply according to $\mu\mu'$. This gives a continuous section of the full product. The thirds in this verification are computed in $\widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$; the correcting cochain is integral at every prime, including 3. The half is integral because every profinite unit is odd at 2.

Two sections differ by a continuous 1-cocycle with coefficients $\widehat{M}$, acted on by (μ, h) as μA_h. B.10 makes that cocycle a coboundary and makes the conjugating element unique. Thus the asserted torsor and the vanishing of the actual H^2-class both hold. □

B.13. The finite 2- and 3-primary checks [Proved]

For $N \mid 24$, the actual extension class of B.12 in coefficients M/NM is zero, with trivial 2- and 3-primary parts. The independent normalized-cochain computation gives

$$\left|H^0(S_3, M/NM)\right| = \left|H^1(S_3, M/NM)\right| = 1, \qquad \left|H^2(S_3, M/NM)\right| = \gcd(N, 3).$$

In particular the zero answer for the actual class at $N = 3$ is a genuine cocycle computation in a nonzero obstruction group.

Proof. The integral cochain of B.11 is included in `code/b_splitting.json`. For the full product we use the reference affine lift

$$t_{\mu,h} = w_\mu + \mu t_h, \qquad b_{\mu,h} = (-\nu, -\nu) + \mu b_h.$$

The program verifies $c + \delta b = 0$ modulo N for every pair of elements of $(\mathbb{Z}/6N)^* \times S_3$. The modulus $6N$ retains the halves and thirds in these reference formulas. For $N = 1, 2, 3, 4, 6, 8, 12, 24$ the numbers of checked pairs are, respectively,

$$144,\ 576,\ 1296,\ 2304,\ 5184,\ 9216,\ 20736,\ 82944.$$

The normalized differentials $C^0 \to C^1 \to C^2 \to C^3$ are integer matrices. Their nonzero Smith diagonals are $1^2,\ 1^8,\ 1^{41}, 3$. For an integer matrix with nonzero Smith entries d_i, its kernel modulo N has cardinality $N^{b-r} \prod_i \gcd(N, d_i)$, where b is its column count and r its rank. Applying this formula to those three matrices gives the displayed cohomology orders. The code checks consecutive differentials compose to zero and verifies the full-product cocycle by exact rational arithmetic before reduction. □

B.14. Identification and rigidity of the arithmetic tower [Proved]

Let

$$E = \pi_1^{\mathrm{et}}([U_1^+/S_3], \bar{b}), \qquad E_{\mathrm{ab}} = E/\overline{[\pi_1(U_{1,\overline{\mathbb{Q}}}), \pi_1(U_{1,\overline{\mathbb{Q}}})]}.$$

Its exact extension

$$1 \to \widehat{M}(1) \to E_{\mathrm{ab}} \to \Gamma_{\mathbb{Q}} \times S_3 \to 1$$

splits, with one conjugacy class of splittings and a unique $\widehat{M}(1)$-conjugator between any two. The affine representation on a Fermat fibre has target $\widehat{M} \rtimes (\widehat{\mathbb{Z}}^* \times S_3)$. The representation

supplied by the quotient $G_{\rho,\sigma}/\mu_G$, with the same geometric translation map and cyclotomic character, agrees with it after a unique translation of the fibre. This proves the identification posed in B.1. *Manuscript: §51.13, folios 715–718′, scans 135–142, (294)–(298).*

Proof. The coordinate-permutation action is defined over $\mathbb{Q}$. Consequently the arithmetic quotient of the orbifold group by the geometric fundamental group is $\Gamma_{\mathbb{Q}} \times S_3$. The fibres of $U_n^+ \to U_1^+$ form compatible torsors under $M/nM(1)$. Choose one point in their inverse-limit fibre. Monodromy then acts by affine transformations with linear action $\chi(\gamma)A_h$, and its restriction to the geometric group is the abelianization map of B.2. Thus it factors through E_{ab}. The map from E_{ab} to the pullback of this affine group over $\Gamma_{\mathbb{Q}} \times S_3$ is an isomorphism: it is the identity on its kernel $\widehat{M}(1)$ and on its quotient, so injectivity and surjectivity follow by lifting a quotient element and correcting in the kernel. The pullback of a semidirect product is split.

B.10 with $B = \Gamma_{\mathbb{Q}}$, acting through χ, proves uniqueness of the conjugacy class and of its conjugator. The algebraically constructed representation and the Fermat representation have the same restriction to $\widehat{M}(1)$ and the same quotient action. Their translation difference is therefore a cocycle on $\Gamma_{\mathbb{Q}} \times S_3$. It is the unique coboundary given by B.10, which proves the claimed comparison. This identifies the resulting unbased torsor canonically; a based homomorphism still requires the fibre point chosen above. The group E in this argument is the arithmetic fundamental group, with quotient the actual Galois group. No equality of that group with a group of all loop-preserving outer automorphisms is used.

The source changes its tilde notation here. At scan 136 it describes the geometric-abelian quotient, while at scans 139–140, in (298), it uses $\widetilde{\mathcal{M}} = \mathcal{M}/\mu_{\mathcal{M}}$ for the orbifold group. Our notation retains the successive maps $\mathcal{M} \to E \to E_{\mathrm{ab}}$. Thus the argument uses the displayed kernels, rather than identifying the two quotients because they share a symbol in the manuscript. □

Scope of the two ranks. The map in B.14 first uses the quotient $\widehat{F} \to \widehat{F}^{\mathrm{ab}} = \widehat{M}(1)$. The Λ-module in B.8 is obtained by taking the homology *of the kernels* of its finite quotients. This is the exact passage from the universal abelian covering to the commutator module seen by the metabelian fundamental group. It is this map that connects the rank-two construction in §51 to the universal homology module.

C. The relevant 1-motives and the cuspidal extension

We keep the arithmetic plus model X_n^+ and its reduced divisor D_n of $3n$ coordinate cusps. All groups and motives in this part are over characteristic zero. The geometric comparison of B.1 transports the computations on $x^n + y^n = z^n$ to this model; its field and its effect on the arithmetic descent are retained. Write $J_n = \mathrm{Pic}^0(X_n^+)$ and $\mathcal{J}_n = \mathrm{Pic}^0(X_n^+, D_n)$, the degree-zero generalized Jacobian with reduced modulus D_n.

C.1. The actual Fermat 1-motive and its realizations [Known]

Put

$$L_n = \mathrm{Div}^0_{D_n}(X^+_{n,\overline{\mathbb{Q}}}), \qquad \mathcal{P}_n = [L_n \xrightarrow{D \mapsto \mathcal{O}(D)} J_n].$$

This is a Deligne 1-motive, with its natural descent over $\mathbb{Q}$. Its Cartier dual, using the principal polarization of J_n, is

$$\mathcal{A}_n = [0 \longrightarrow \mathcal{J}_n].$$

There is an exact sequence of semiabelian varieties

$$1 \longrightarrow T_{D_n} := (\mathrm{Res}_{D_n/\mathbb{Q}}\,\mathbb{G}_m)/\mathbb{G}_m \longrightarrow \mathcal{J}_n \longrightarrow J_n \longrightarrow 1. \tag{C.1}$$

The character lattice of T_{D_n} is L_n, and its cocharacter lattice is $\mathbb{Z}[D_n]/\mathbb{Z}\sum_{P\in D_n} P$. The Betti and Tate realizations of $\mathcal{A}_n$ are

$$T_{\mathbb{Z}}(\mathcal{A}_n) = H_1(U_n^+(\mathbb{C}),\mathbb{Z}), \qquad T_\ell(\mathcal{A}_n) = H_1^{\mathrm{et}}(U^+_{n,\overline{\mathbb{Q}}},\mathbb{Z}_\ell). \tag{C.2}$$

Its de Rham realization is the dual of $H^1_{\mathrm{dR}}(U_n^+/\mathbb{Q})$. Its weights are -2, with rank $3n-1$, and -1, with rank $(n-1)(n-2)$; its total rank is n^2+1. *Source: Deligne, [Del74, 10.1.3–10.1.8, 10.2.10–10.2.11, 10.3.4–10.3.8].*

Proof. We spell out the reduction to the cited construction. A point of the generalized Picard group is a line bundle on X_n^+, together with a trivialization along D_n, modulo scalar automorphisms of the line bundle. The fibre over a fixed degree-zero line bundle is thus the product of the nonzero fibres at the cusps modulo simultaneous scaling. This proves (C.1) and both of its lattice descriptions.

Deligne's definition 10.3.4 assigns to a smooth open curve $X - D$ the complex $[\ker(\mathbb{Z}[D]\to\mathbb{Z}) \to \mathrm{Pic}^0(X)]$. For $X = X_n^+$, this is precisely $\mathcal{P}_n$. Construction 10.3.6 identifies its m-torsion realization with $H^1(U_n^+,\mu_m)$: a representative is a pair $(\mathcal{L},D)$ with $\mathcal{L}^{\otimes m}\simeq\mathcal{O}(D)$, modulo pairs $(\mathcal{O}(E),mE)$. This is the same complex defining the finite realization of $[L_n\to J_n]$. Thus the comparison includes the cusp lattice and has no missing toric summand. Construction 10.2.11 dualizes this complex to the extension by $\mathrm{Hom}(L_n,\mathbb{G}_m)$ whose class evaluated at D is $\mathcal{O}(D)$; the line-bundle description above identifies this extension with $\mathcal{J}_n$. Dualizing the finite realizations with values in μ_m gives $H_1(U_n^+,\mathbb{Z}/m)$, and their inverse limits give (C.2).

The Betti construction 10.1.3 and the comparison 10.3.8 give the integral homology realization; 10.1.7–10.1.8 and 10.3.9 give the de Rham comparison. Concretely, its dual Hodge filtration has

$$F^1H^1_{\mathrm{dR}}(U_n^+) = H^0(X_n^+,\Omega^1_{X_n^+}(\log D_n)), \quad \dim F^1 = g_n + 3n - 1.$$

The torus in (C.1) contributes weight -2 and the abelian variety weight -1. Their ranks are $3n-1$ and $2g_n$, respectively. B.6 gives $2g_n=(n-1)(n-2)$. Every construction used here is functorial for the deck and permutation automorphisms, which proves compatibility of all the realizations with those actions. □

C.2. The rank-two identification in the lead [Refuted]

The assertion that the rank-two module $\widehat{M}(1)$ of §51, or its reduction modulo a nontrivial integer m, is the full Tate realization of the generalized Jacobian of X_n^+ with its cusps is false for $n>1$. Its exact replacement is

$$\widehat{M}(1) = T_\bullet(T), \qquad T = (\mathbb{G}_m^3)/\mathbb{G}_m = \mathcal{J}_1, \tag{C.3}$$

and the covering pushforward has the exact realization

$$0 \longrightarrow (R_n/\mathbb{Z}\mathcal{N}_n)\otimes\widehat{\mathbb{Z}} \longrightarrow T_\bullet(\mathcal{J}_n) \xrightarrow{q_{n*}} n\widehat{M}(1) \longrightarrow 0. \tag{C.4}$$

The Galois structure on the first term is the one transported by its inclusion into the geometric homology in (C.2). At level $n=2$, the ranks in (C.4) are $3,5,2$. *Manuscript: §51.11, folio 697, scan 99, (228); §51.13–14, folios 719–720, scans 143–146, (299)–(305).*

Proof. At $n = 1$ the compact curve is a projective line, so its ordinary Jacobian vanishes. Formula (C.1) then gives (C.3), with cocharacter lattice M. Abel–Jacobi sends a peripheral loop to the corresponding cocharacter; hence its Tate realization is the geometric abelianization used in B.2 and B.14. For arbitrary n, functoriality of the generalized Jacobian gives the norm map induced by q_n. On the Betti realization it is the map $K_n^{\mathrm{ab}} \to F^{\mathrm{ab}}$. B.5 computes its kernel and its image exactly. Tensoring (B.2) with the flat ring $\widehat{\mathbb{Z}}$ and using (C.2) proves (C.4). This is an explicit comparison morphism; at level two its matrix and its Smith entries $2, 2$ are supplied in B.7.

The geometric module in the middle has rank $n^2 + 1$; the proposed module has rank two. Modulo $m > 1$, their orders are m^{n^2+1} and m^2, respectively. Thus even an abstract isomorphism of the proposed finite realizations fails, with $n = m = 2$ giving the finite orders 32 and 4. The finite order comparison is checked in `code/b_freeness.py`. The norm comparison (C.4) gives the actual relationship that replaces that proposed identification. □

C.3. Complex multiplication with its actual ring map [Proved]

For $n \geq 3$, over a field containing μ_n, the deck action gives a homomorphism

$$R_n \longrightarrow \mathrm{End}(J_n)$$

whose kernel is I_n of B.6. The induced commutative semisimple algebra

$$\mathcal{E}_n = \mathbb{Q}[A_n]/(I_n \otimes \mathbb{Q}) \ \subset \ \mathrm{End}^0(J_n)$$

has dimension $(n-1)(n-2) = 2 \dim J_n$. Its factors are cyclotomic fields indexed by the Galois orbits of characters (a, b) with $a, b, a + b \neq 0 \pmod{n}$. This is the precise complex-multiplication algebra supplied by the Fermat automorphisms. The action on $\mathcal{J}_n$ restricts to the permutation action on the torus lattice in (C.1).

Proof. Automorphisms of the curve act on divisor classes and therefore on the Jacobian. An endomorphism of a complex abelian variety is zero as soon as it is zero on integral first homology: after lifting to the complex universal cover it is a complex linear map, and the integral lattice spans that cover over $\mathbb{R}$. B.6 identifies this homology with R_n/I_n, so the kernel is exactly I_n. The rational group algebra of a finite abelian group is a product of number fields, and its character decomposition selects exactly the characters in B.6. Each has multiplicity one. The dimension assertion follows by counting them. Complex conjugation pairs (a, b) with $(-a, -b)$; none of the selected characters is real: a real character would have a, b of order at most two and would force one of $a, b, a + b$ to vanish. Thus the selected cyclotomic factors are CM fields. This proves the full degree required for complex multiplication. The line-bundle-with-trivialization description of $\mathcal{J}_n$ shows directly that deck and coordinate automorphisms permute the cuspidal fibres, giving the torus action claimed. The notation $\mathbb{Z}[\mu_n]^2$ in Lead 3 is therefore replaced by this explicit group-ring homomorphism and its kernel, with no interpretation of that ambiguous expression left implicit. □

C.4. Exact order of cuspidal differences [Proved]

For every $n \geq 3$ and every two distinct geometric coordinate cusps P, Q of X_n^+, the class $[P - Q] \in J_n$ has exact order n. Consequently the subgroup generated by all cuspidal degree-zero divisor classes has exponent exactly n. At $n = 1, 2$ that subgroup is zero.

Proof. Let $\ell_Q = 0$ be the tangent line at a coordinate cusp Q. For example, at $Q = [0 : q : 1]$, with $q^n = -1$, its equation is $y - qz = 0$. Substitution into $x^n + y^n + z^n = 0$ gives $x^n = 0$; the

intersection divisor is exactly nQ. The same holds at every coordinate cusp. Therefore

$$\operatorname{div}(\ell_P/\ell_Q) = n(P-Q), \tag{C.5}$$

and the order divides n.

We prove exactness, including the even levels. At $Q=[0:q:1]$, the functions z/ℓ_Q and x/ℓ_Q have their only poles at Q, of orders n and $n-1$, respectively. Thus the pole semigroup $H(Q)$ contains $\langle n-1, n\rangle$. The latter numerical semigroup has exactly $(n-1)(n-2)/2$ positive gaps: the least element in residue class $r \pmod{n-1}$ is rn, for $0 \le r < n-1$, and counting the smaller positive members of those residue classes gives this number. Riemann–Roch gives exactly g_n gaps for $H(Q)$: for $a \ge 2g_n$, $\ell(aQ) = a+1-g_n$, and every successive dimension increase is zero or one. B.6 gives $g_n = (n-1)(n-2)/2$. Since the contained semigroup already has all g_n gaps, it equals $H(Q)$.

A proper positive divisor d of n satisfies $d \le n/2 < n-1$, hence is a gap. A principal divisor $d(P-Q)$ would supply a nonconstant function with its only pole at Q, of order d, a contradiction. The exact order is therefore n. The genus-zero cases have zero Jacobian. This gives a proof for every n, without a parity restriction. □

C.5. The extension class and its exact isogeny exponent [Proved]

The class of (C.1) is the homomorphism

$$L_n \longrightarrow J_n^\vee, \qquad D \longmapsto [D], \tag{C.6}$$

under the principal polarization and the extension–character correspondence $\operatorname{Ext}^1(J_n, T_{D_n}) \simeq \operatorname{Hom}(L_n, J_n^\vee)$. For every $n \ge 3$, this class has exact order n. Its pullback by $[n]: J_n \to J_n$ is split, and the resulting isogeny

$$T_{D_n} \times J_n \longrightarrow \mathcal{J}_n$$

has kernel isomorphic to $J_n[n]$ and degree n^{2g_n}. The splitting of that pullback descends to $\mathbb{Q}$. There is no single positive integer that kills these extension classes at every Fermat level.

Proof. Push (C.1) out by the torus character associated with $D = \sum a_P P$. The fibre of the resulting $\mathbb{G}_m$-extension over a line bundle $\mathcal{L}$ is the nonzero part of $\bigotimes_P \mathcal{L}(P)^{\otimes a_P}$. This is the restriction of the Poincare bundle corresponding to $[D] \in J_n^\vee$; it proves (C.6). Conversely, these character pushouts determine a torus extension, because a split torus is a product of copies of $\mathbb{G}_m$. Descent retains exactly the Galois-equivariant homomorphisms in (C.6); morphisms from an abelian variety to a torus vanish, so there is no ambiguity in descending a splitting. This is also the Cartier-duality description in [Del74, 10.2.11–10.2.12].

C.4 says exactly that (C.6) is killed by n, and that no proper divisor of n kills it. Pullback by the multiplication map $[m]$ multiplies the associated point of $J_n^\vee$ by m, as is immediate from the Poincare bundle. Thus $[n]^*$(C.1) is the trivial torus extension. Projection from that pullback to $\mathcal{J}_n$ has kernel $J_n[n]$, proving the degree and the isogeny statement. A splitting over $\overline{\mathbb{Q}}$ is unique because $\operatorname{Hom}(J_n, T_{D_n}) = 0$, so it is Galois-invariant and descends. Finally C.4 gives exponents n for unbounded n, proving the last assertion.

This statement describes the order in the algebraic extension group. Its realizations give a splitting of the weight filtration in the isogeny category, and an integral comparison whose finite defect is controlled by n. It does not replace that specified extension group by an unspecified Galois cohomology group. □

C.6. Rohrlich's theorem and the requested computations [Known]

For $n \geq 3$, the geometric cuspidal subgroup is

$$C_n \simeq \begin{cases} (\mathbb{Z}/n)^{3n-7}, & n \text{ odd}, \\ (\mathbb{Z}/n)^{3n-7} \times \mathbb{Z}/2, & n \text{ even}. \end{cases}$$

This is [Roh77, Theorem 1, p. 101], *Points at infinity on the Fermat curves*. We opened the original theorem and its definitions on p. 95. The odd case is also reproduced with an independent proof in [Vél79, Theorem 1, p. 28-02]. For the requested levels the exact orders of the extension class are

n	3	4	5	7
g_n	1	3	6	15
$\operatorname{rk} T_{D_n}$	8	11	14	20
$\operatorname{rk} T_\bullet(\mathcal{J}_n)$	10	17	26	50
ord[(C.1)]	3	4	5	7

At $n = 3, 5, 7$, the computed nontrivial Smith factors of the checked cuspidal presentation are respectively $3^2,\ 5^8,\ 7^{14}$.

Proof. Rohrlich defines $\mathcal{D}^\infty$ to be the degree-zero divisors supported on the $3n$ coordinate cusps of $x^n + y^n = z^n$, and $\mathcal{F}^\infty$ to be the principal divisors in this lattice. The geometric isomorphism ι_n of B.1 identifies their quotient with the image of $L_n \to J_n$. By (C.6), that is precisely the image of the homomorphism classifying the generalized-Jacobian extension. The exponent of the displayed group is n, also in the even case because $2 \mid n$. Therefore multiplication by n kills the extension class, and no smaller positive integer does. C.5 then constructs the splitting isogeny, over $\mathbb{Q}$ itself. This is the reduction from the original theorem to the asserted weight-extension statement.

For independent integral computations at the requested odd levels, choose a cusp P_*, and identify the degree-zero cusp lattice with the free lattice on $P - P_*$, for $P \neq P_*$. Velu's proof determines its principal-divisor sublattice modulo n, with six relation classes. In the standard ordering a_i, b_i, c_i, these can be written

$$\sum_i (a_i - P_*), \quad \sum_i (b_i - P_*), \quad \sum_i (c_i - P_*),$$
$$\sum_i i(a_i - b_i), \quad \sum_i i(a_i - c_i), \quad \sum_i i^2(a_i + b_i + c_i - 3P_*).$$

The same displayed presentation appears in [BM22, Theorem 29], which was also opened. Together with n times each basis vector it is the relation matrix used by `code/c_cuspidal_exponent.py`. Its exact integer Smith form gives the three listed odd-level groups, hence their exponents. Formula (C.6) identifies the image of the extension-class homomorphism with that same cuspidal group, so the exponent is again the order of the extension class.

For $n = 4$ the program does not apply an odd-level presentation. It checks instead the independent exact certificate of C.4: the tangent divisor is $4(P - Q)$, the pole semigroup $\langle 3, 4\rangle$ has gaps $1, 2, 5$, and the proper divisors $1, 2$ of 4 are gaps. Therefore the class has exact order four. The same certificate is checked for all four requested levels and, as an additional arithmetic verification, for every $3 \leq n \leq 32$. The proof of C.4 supplies its validity for all levels. □

C.7. The toric motive actually constructing the tower [Proved]

Let $T = \mathbb{G}_m^3/\mathbb{G}_m$, and let $\psi : U_1^+ \hookrightarrow T$ be the map of B.1. There is a Cartesian square over $\mathbb{Q}$

$$\begin{array}{ccc} U_n^+ & \longrightarrow & T \\ {\scriptstyle q_n}\downarrow & & \downarrow{\scriptstyle [n]} \\ U_1^+ & \xrightarrow{\psi} & T. \end{array} \tag{C.7}$$

For a point $e \in U_1^+(k)$, the 1-motive

$$\mathcal{K}_e = [\mathbb{Z} \xrightarrow{1\mapsto\psi(e)} T]$$

has finite realizations

$$0 \to T[n] \to T_{\mathbb{Z}/n}(\mathcal{K}_e) \to \mathbb{Z}/n \to 0, \tag{C.8}$$

whose fibre over 1 is $q_n^{-1}(e)$. The compatible fibres give exactly the Fermat torsor and its affine Galois action of B.14. Thus the rank-two module of §51 is the weight--2 part of this toric Kummer construction. *Manuscript: §51.13–14, folios 719–724, scans 143–153, (299)–(323), especially (317) and (323).*

Proof. A point $[a_0 : a_1 : a_\infty]$ of T lies over $[-e : e-1 : 1]$ under $[n]$ precisely when its n-th powers have that projective ratio. Their sum is zero, which is the equation of U_n^+. This proves the Cartesian square scheme-theoretically by its equations on a torus chart. Deligne's construction 10.1.5 represents $T_{\mathbb{Z}/n}([\mathbb{Z} \to T])$ by pairs (a, t), with $t^n = \psi(e)^a$, modulo $(a, t) \sim (a + nb, t\psi(e)^b)$. Projection to a mod n gives (C.8), and the fibre at $a = 1$ is the torsor of n-th roots of $\psi(e)$. By (C.7) it is the stated Fermat fibre. All formulas commute with automorphisms of k and with coordinate permutations, proving the claimed action and its compatibility with the representation in B.14.

This construction has weights $0, -2$, with quotient $\mathbb{Z}$ and toric Tate kernel of rank two. The generalized-Jacobian construction in C.1 is connected to it by the norm morphism (C.4); that map records the full additional kernel at level n. □

C.8. Lead 3 and the reading of the manuscript [Refuted]

Lead 3's proposed identification of §51.13–14 with a splitting of the generalized Jacobian of every Fermat curve is false. The source's explicit construction is (C.7) and the Kummer fibre (C.8). Its further elliptic-curve construction starts with an elliptic curve E/S and the three nonzero two-torsion points $D = E[2] \setminus \{0\}$, and has exact sequence

$$1 \to \mathbb{G}_m \otimes (p_*\mathbb{Z}_D/\mathbb{Z}_S) \to \mathcal{J}_{E,D} \to E \to 1. \tag{C.9}$$

The Tate kernel of (C.9) has rank two, as written in (303). It is identified with the torus of the three-point quotient $E/\{\pm 1\}$ punctured at the images of D. The weight-extension result for the Fermat Jacobians is C.5, with exact and unbounded exponents n; the rank-two group extension of B.12 has zero class and unique splitting conjugacy class. *Manuscript: §51.13, folios 719–720, scans 143–145, (299)–(304); §51.14, folios 720′–724, scans 146–153, (305)–(323).*

Proof. The divisor used in scan 144 is explicitly the degree-three subscheme of nonzero two-torsion points on the elliptic curve, and (302) has quotient that elliptic curve. Applying the line-bundle construction of C.1 to (E, D) proves (C.9). The three-point quotient map $E \to E/\{\pm 1\}$ identifies the two degree-three etale divisors. Hence it identifies their cocharacter lattices $p_*\mathbb{Z}_D/\mathbb{Z}_S$ and their associated tori. On the generalized Jacobians this torus isomorphism

is the pullback of a trivial line bundle with prescribed nonzero values at the three marked points; the three values are carried to the corresponding points of D, modulo common scaling. This is the explicit comparison explaining the repeated rank-two Tate module. The degree-three divisor is etale over bases on which 2 is invertible, precisely the qualification made at scan 152.

The other torus comparison written at scan 147 is also explicit:

$$[x_0 : x_1 : x_\infty] \longmapsto (x_1/x_\infty, x_\infty/x_0, x_0/x_1) \in \ker(\mathbb{G}_m^3 \xrightarrow{\text{product}} \mathbb{G}_m).$$

It is an isomorphism: its first two coordinates recover x_1/x_∞ and x_0/x_∞. For an even permutation it commutes with coordinate permutation; for an odd permutation its value is the inverse of the permuted value. This proves the manuscript's marginal qualification that it is not equivariant for the two untwisted full S_3-actions. The S_3-module used in B.10 is always the quotient lattice, so no product-one torus action is substituted in the vanishing proof.

At scans 151 and 153 the manuscript supplies ψ and the fibre of the universal multiplication tower at the marked section e. They are exactly the equations proved in C.7. For the full Fermat generalized Jacobian, the torus has rank $3n-1$ and the comparison with this rank-two torus is the noninjective norm morphism (C.4). Thus the proposed equality fails by that explicit morphism and its nonzero kernel already at $n = 2$. The exponents in C.5 further show that a fixed bounded torsion correction cannot describe all levels.

The supplied table of contents retains the unread initial letters in *[??]tivisation*; the body heading at folio 708, scan 121, reads *Retour sur la tour de Fermat*. The mathematics settles the interpretation just described. It does not settle the missing letters of the table-of-contents word. □

D. The arithmetic action, its Kummer coordinates, and the two CM factors

We use $G_k = \mathrm{Gal}(\overline{k}/k)$, $K = \mathbb{Q}^{\mathrm{cyc}} = \mathbb{Q}^{\mathrm{ab}}$, and the cyclotomic character $\chi : G_\mathbb{Q} \to \widehat{\mathbb{Z}}^\times$. Compatible roots of unity identify $T\mathbb{G}_m$ with $\widehat{\mathbb{Z}}(1)$. A coordinate in this Tate module depends on that choice. The group in scans156–170 is the arithmetic $G_\mathbb{Q}$; the larger loop-automorphism groups enter only in the explicitly designated comparisons below. The relevant scans157,166–170 have been inspected as images, independently of their transcription. A positive-weight calculation always refers to the kernel of the multiplier and to its lower-central filtration.

The homomorphism of scan157 and the rational-prime questions

D.1. The actual affine Kummer homomorphism [Proved]

Choose compatible roots $a^{1/n}$ of $a = -1/2$. The homomorphism of §51.15, (326), (329)–(335), scans156–160, is, after a choice of origin in its torsor,

$$\rho_a : G_\mathbb{Q} \longrightarrow \widehat{\mathbb{Z}}^\times \ltimes \widehat{\mathbb{Z}}(1), \qquad g \longmapsto (\chi(g), \kappa_a(g)), \qquad g(a^{1/n})/a^{1/n} = \zeta_n^{\kappa_a(g) \bmod n}.$$

Its image is the closed subgroup of index two

$$\{(u, v) : (-1)^v = \chi_{-8}(u)\}, \tag{D.1}$$

where χ_{-8} is the quadratic character of $\mathbb{Q}(\sqrt{-2})$, with values $+1$ at $1, 3 \pmod 8$ and -1 at $5, 7 \pmod 8$. Changing the root system translates the origin and changes κ_a by $(\chi - 1)b$. On G_K its image is exactly $2\widehat{\mathbb{Z}}(1)$.

Proof. The cocycle identity is obtained by applying g to $h(a^{1/n})$: $\kappa_a(gh) = \kappa_a(g) + \chi(g)\kappa_a(h)$. This is precisely the multiplication in the displayed affine group. In the homogeneous Fermat model of the manuscript, the point $1/2$ has coordinates $(-1/2, -1/2, 1)$; its fibre is the diagonal pair of root torsors of $-1/2$. Dividing the two coordinates removes the diagonal ambiguity and gives the rank-one torsor of (329)–(335). Thus this construction supplies the stated morphism, rather than merely a group of the same abstract form. The unread signs in (332), scan159, do not change this argument: the homogeneous equation and the specified point determine the two nth powers to be $-1/2$.

The square root $\sqrt{-2}/2$ lies in K, and hence the parity of $\kappa_a(g)$ is exactly the quadratic action on $\sqrt{-2}$. On G_K the roots of -1 are fixed, so $\kappa_a = -\kappa_2$. D.2 below proves that this has image $2\widehat{\mathbb{Z}}(1)$. Finally χ is onto. Every fibre over u is a coset of that translation image; the parity constraint therefore describes the entire fibre, proving (D.1). □

D.2. Exact joint images and independence for distinct rational primes [Proved]

For distinct positive rational primes $p_1, \dots, p_r$, the continuous homomorphism

$$(\kappa_{p_1}, \dots, \kappa_{p_r}) : G_K \longrightarrow \widehat{\mathbb{Z}}(1)^r$$

has image $2\widehat{\mathbb{Z}}(1)^r$, of index 2^r. Consequently every odd-primary projection is onto, the 2-primary image is $2\mathbb{Z}_2^r$, and the characters are linearly independent over $\widehat{\mathbb{Z}}$. On $G_{\mathbb{Q}}$ the full affine image consists exactly of the pairs $(u, (v_i))$ satisfying $(-1)^{v_i} = \chi_{\mathbb{Q}(\sqrt{p_i})}(u)$ for every i. These assertions answer the individual and simultaneous questions of scans168–169; integral surjectivity as written there is false.

Proof. Let $a = \prod p_i^{e_i} > 0$ and $n \geq 1$. We first compute the exact relation

$$a \in K^{\times n} \quad \Longleftrightarrow \quad n \mid 2e_i \quad \text{for every } i. \tag{D.2}$$

Choose the positive real root $\alpha = a^{1/n}$. Any other root differs by an element of K, so membership in K is independent of this choice. A finite subfield $\mathbb{Q}(\alpha)$ of the abelian extension $K/\mathbb{Q}$ is Galois and totally real. Every conjugate of α is a real root of $X^n - a$, of which there are at most two. Thus $\alpha^2 \in \mathbb{Q}$ and valuation at p_i gives $n \mid 2e_i$. Conversely this divisibility makes α a product of rational numbers and square roots of rational primes. Such square roots are cyclotomic: $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$, and the quadratic Gauss sum for an odd prime has square $(-1)^{(p-1)/2}p$. Multiplication by i removes the latter sign. This proves (D.2).

For completeness, the finite Kummer pairing used here needs no rank estimate. Let B be the subgroup generated by the classes of the p_i in $K^\times/K^{\times n}$ and let $L = K(p_i^{1/n} : 1 \leq i \leq r)$. The pairing $\mathrm{Gal}(L/K) \times B \to \mu_n$, $(g, a) \mapsto g(a^{1/n})/a^{1/n}$, is nondegenerate on both sides: a Galois element fixing every root is trivial, and a root fixed by every element belongs to K. Both groups have exponent dividing n; injections into their finite character groups therefore have equal orders and are isomorphisms. By (D.2) the annihilator in $(\mathbb{Z}/n)^r$ of the relation subgroup is exactly $2(\mathbb{Z}/n)^r$. These are all finite images. Compactness gives the inverse-limit image stated above. A linear relation vanishes on $2\widehat{\mathbb{Z}}^r$ only when all coefficients vanish, since multiplication by two on $\widehat{\mathbb{Z}}$ is injective. Parity and surjectivity of χ give the last assertion just as in D.1.

The program `d_kummer_soule.py` enumerates these annihilators for every $n \leq 24$ and records the indices for $r = 1, 2, 3, 4$. This checks, in particular, the finite obstruction to the proposed surjectivity. □

D.3. General openness and two corrections to the manuscript [Proved]

Let k be a field finitely generated over $\mathbb{Q}$ and $A \subset k^\times$ a finitely generated subgroup of torsion-free rank r. The Kummer homomorphism on $G_{k(\mu_\infty)}$ has open image in $\mathrm{Hom}(A, \widehat{\mathbb{Z}}(1)) \simeq \widehat{\mathbb{Z}}(1)^r$.

This proves the openness requested under (355), scans166–167, also for normal connected schemes essentially of finite type over $\mathbb{Q}$ and their groups of units. The stronger surjectivity proposed in scan167 is false even for the full unit group of a normal arithmetic scheme of dimension two. The criterion for cyclotomic surjectivity proposed in scan163 is also false.

Proof. Torsion in A contributes no homomorphism to the torsion-free Tate module, so choose a free complement. Write $H = \mathrm{Gal}(k(\mu_\infty)/k)$, an open subgroup of $\widehat{\mathbb{Z}}^\times$: the algebraic closure of $\mathbb{Q}$ inside k is a number field. Choose an integer m such that H contains all units congruent to one modulo m. There is $u \in H$ with $u - 1 = dv$, where $d = \mathrm{lcm}(2, m)$ and $v \in \widehat{\mathbb{Z}}^\times$: take $u_p = 1 + p^{v_p(m)}$ at $p \mid m$, $u_p = 2$ at odd $p \nmid m$, and $u_2 = 3$ when $2 \nmid m$.

Suppose $\alpha^n = a \in A$ with $\alpha \in k(\mu_\infty)$. Put $\tau(\alpha)/\alpha = \zeta_n^{c_\tau}$. Since H is abelian, comparison of its two cocycle products gives $(u-1)c_\tau = (\chi(\tau) - 1)c_\sigma$ for an element σ with character u. Setting $b = v^{-1}c_\sigma$ modulo n shows that $\alpha^d \zeta_n^{-b}$ is fixed by H. Thus

$$a^d \in k^{\times n}. \tag{D.3}$$

Choose a normal finitely generated $\mathbb{Z}$-algebra R with fraction field k containing A as units. Its unit group is finitely generated. One proof is to take a normal projective compactification over $\operatorname{Spec}\mathbb{Z}$: a unit has divisor supported on the finitely many codimension-one boundary components, while the kernel of this divisor map is the unit group of the ring of global functions of the compactification. That ring is finite over $\mathbb{Z}$; its units are finitely generated by the number-field unit theorem. Normalization is finite here, so this compactification argument applies to R.

If $b^n = a^d$ in k, both b and b^{-1} are integral over R, hence belong to R. In the free lattice $U = R^\times/\mu(R)$, the saturation of A modulo A is finite; let e annihilate it. Equation (D.3) then implies $de\, a \in nA$ in additive lattice notation. The kernel of $A/nA \to k(\mu_\infty)^\times/k(\mu_\infty)^{\times n}$ has order at most $(de)^r$. The perfect finite pairing in D.2 therefore bounds all finite indices of the Kummer image by this same number. A closed subgroup of $\widehat{\mathbb{Z}}^r$ with uniformly bounded finite indices has finite index: any collection of more distinct cosets is separated in one finite quotient. This proves openness.

For a normal connected scheme, the Kummer covers of units are finite etale and the generic-point map is onto their monodromy groups; the preceding field computation consequently gives the same image. For the counterexample take $S_1 = \operatorname{Spec}\mathbb{Z}[1/2][t]$ and $A = \mathcal{O}(S_1)^\times = \{\pm 2^j\}$. Its generic fibre has the Kummer image $2\widehat{\mathbb{Z}}$ of D.2. Moreover A is saturated in $\mathbb{Q}[t]^\times$ modulo its torsion, so the proposed saturated variant fails too. Finally $k = \mathbb{Q}(\sqrt{5})$ has only the roots of unity ± 1, but its cyclotomic image has index two because $k \subset \mathbb{Q}(\mu_5)$. □

D.4. Exact local inertia images [Proved]

Embed $G_{\mathbb{Q}_\ell}$ into $G_{\mathbb{Q}}$ and put $I_\ell^+ = I_\ell \cap \ker\chi$. For a positive rational prime p, the image of $\kappa_{p,\ell}$ on I_ℓ^+ is $\ell^{e_{p,\ell}}\mathbb{Z}_\ell(1)$, where

$$e_{p,\ell} = \begin{cases} 0 & \ell > 2, p = \ell, \\ v_\ell(p^{\ell-1} - 1) - 1 & \ell > 2, p \neq \ell, \\ 1 & \ell = p = 2, \\ v_2(p^2 - 1) - 2 & \ell = 2, p \text{ odd}. \end{cases} \tag{D.4}$$

Every image in scan170 is therefore open, and the formula determines exactly which are onto. The program `d_local_cm.py` records every pair $p, \ell \le 50$; for example the image for $(p, \ell) = (3, 11)$ has index 11, and for $(19, 7)$ index 49.

Proof. The fixed field of I_ℓ^+ is $L = \mathbb{Q}_\ell^{\mathrm{ur}}(\mu_{\ell^\infty})$, abelian over $\mathbb{Q}_\ell$. For odd ℓ, choose a cyclotomic multiplier with $u - 1 \in \mathbb{Z}_\ell^\times$. The cocycle argument in D.3, now with $d = 1$, proves that an ℓ^kth root of $a \in \mathbb{Q}_\ell^\times$ belongs to L precisely when $a \in \mathbb{Q}_\ell^{\times \ell^k}$. The decomposition $\mathbb{Q}_\ell^\times = \ell^{\mathbb{Z}} \times \mu_{\ell-1} \times (1 + \ell\mathbb{Z}_\ell)$ and the ℓ-adic logarithm give the first two cases.

For $\ell = 2$ use multiplier -1. The same argument proves that a having a 2^kth root in L implies $a^2 \in \mathbb{Q}_2^{\times 2^k}$, equivalently $a = \pm b^{2^{k-1}}$ for $b \in \mathbb{Q}_2^\times$. Conversely every quadratic extension of $\mathbb{Q}_2$ lies in L: the square classes are generated by $-1, 2, 5$, the first two have cyclotomic square roots and $\mathbb{Q}_2(\sqrt{5})$ is unramified. Thus $\pm b^{2^{k-1}}$ indeed has a 2^kth root in L. For $a = 2$ the largest such k is one. For odd p, write $p = \epsilon u$, $\epsilon \in \{\pm 1\}$, $u \in 1 + 4\mathbb{Z}_2$; logarithms give the largest k as $v_2(u-1) - 1 = v_2(p^2 - 1) - 2$. Finite Kummer duality converts these root heights into the displayed closed images. □

The universal Fermat cocycle and its actual comparison maps

D.5. Full profinite universal measure, and the precise correction to Lead 1 [Known]

Put $\Lambda = \widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}^2]]$. The compatible projective Fermat homology of Otsubo's Theorem6.1 is a free cyclic Λ-module. For its specified generator c, write $g(c) = F_g c$; then $F_{gh} = g(F_h)F_g$. On G_K this is a homomorphism into $\Lambda^\times$. The comparison with Anderson's relative-path measure B_g is the identity

$$(1 - s^r)(1 - t^m)F_g = (1 - s^{\chi(g)r})(1 - t^{\chi(g)m})B_g. \tag{D.5}$$

Here r, m are integers and s, t are the two deck generators. This is Otsubo's (7.2). In particular $F_g = B_g$ on G_K. Its ℓ-power specialization is related to Ihara's power series by the explicit derived-module basis comparison below.

The morphisms relating this invariant to D.1 are the exact quotient

$$1 \longrightarrow \widehat{\pi}'/\widehat{\pi}'' \longrightarrow \widehat{\pi}/\widehat{\pi}'' \longrightarrow \widehat{\pi}^{\mathrm{ab}} \longrightarrow 1 \tag{D.6}$$

and the root torsors obtained by evaluating the abelian covering at $1/2$. For rational primes the additional exact comparison is

$$p = \prod_{a=1}^{p-1}(1 - \zeta_p^a), \qquad \kappa_p\big|_{G_K} = \sum_{a=1}^{p-1} \kappa_{1-\zeta_p^a}\big|_{G_K}. \tag{D.7}$$

Lead1's proposed Kummer comparison is made exact by (D.5)–(D.7) and the coefficient extraction in D.23 below. In particular the rank-one affine coordinate is obtained from the full universal measure by a specified continuous map. This comparison retains the factor two in D.2 and the prime-to-ℓ deck components that a pure pro-ℓ series discards.

Proof. We use [Ots16, Theorem6.1, Corollary6.2, equation(7.2)], whose full text was opened. The non-zero-divisor assertion in Corollary6.2 permits cancellation in (D.5); when $\chi = 1$ the two measures coincide. The maps to ℓ-power deck quotients and $\mathbb{Z}_\ell$ coefficients are continuous quotient maps of completed group rings; hence this comparison is profinite before any pro-ℓ specialization. Part B constructs the derived-module comparison in (D.6); evaluation of its abelian quotient at the displayed point gives D.1 by the explicit root equations. Finally (D.7) is $\Phi_p(1) = p$ and functoriality of roots under multiplication. Applying valuations at the distinct primes to its rational norm products, followed by the exact pairing of D.2, proves independence through these cyclotomic elements. The possible quadratic relations, including the factor two, are retained; an assertion of unrestricted integral independence of all circular units would not give the required exact image.

Otsubo's Remark8.2 identifies the original results precisely as Ihara1986, Theorem7 for prime-power level, and Anderson1989, Theorem6 in general. Anderson's complete original has now been retrieved from the official GDZ scan and checked. Its §§1.3–1.4 use the full ring $\widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}(1)^2]]$; Proposition5.5.1, p.88, constructs the all-degree torsion comparison from relative interval homology and states $e_N(B_\sigma\,\sigma(v)) = \sigma(e_N(v))$. Theorem6, pp.118–124, then interpolates Gauss sums using a specified Frobenius lift and the hyperadelic gamma function; it is not by itself the torsion comparison.

The original comparison to Ihara is [And89, Proposition13.3.2 and §§13.4–13.6]. For odd ℓ it constructs a Galois-equivariant map $\theta : \pi'_\ell/\pi''_\ell \simeq \varprojlim_n T_\ell J_{\ell^n}$ and a basis-change unit $Q(u,v)$, with exact formula

$$F^{\mathrm{Ih}}_\sigma(u,v) = S_\sigma(u)S_\sigma(v)B_{\sigma,\ell}(u,v)\frac{Q((1+u)^{\chi_\ell(\sigma)}-1,(1+v)^{\chi_\ell(\sigma)}-1)}{Q(u,v)},$$
$$S_\sigma(T) = \frac{(1+T)^{\chi_\ell(\sigma)}-1}{T}.$$

The map θ sends a deck transformation to its translation on the compatible division points of the divisor $(\delta_n)-(\infty_n)$, where $\delta_n = (t_n, -t_n(1-t^{-1})_n)$ in §13.5. That section records the plus-model twist for $\ell = 2$; RemarkII in §13.3 proves $F^{\mathrm{Ih}}_\sigma = B_{\sigma,\ell}$ on the cyclotomic kernel for every ℓ. These maps specify both the basis coboundary and the arithmetic twist.

The original Ihara1986 and Coleman1989 proofs remain to be obtained; the separate coefficient formula used below was checked in Ihara–Kaneko–Yukinari's primary paper and now also in Anderson's original. The retrieved original [Iha00, Supplement] corrects Ihara1986, Theorem7 by replacing the Jacobi indices (a,b,c) with $(-a,-b,-c)$, and corrects the signs in Theorem10. It also instructs replacing Anderson1989, p.125 line11's group-ring identification by its inverse, and records the inverse deck convention on p.85. These corrections explain why a convention must be attached to the original theorem numbers; D.6 uses the explicitly checked Otsubo convention. □

D.6. Frobenius specialization with nonsplit primes and the plus model [Known]

Let $p \nmid n$, $f = \mathrm{ord}_n(p)$, $q = p^f$, and let $\theta : \mathbb{F}_q^\times \to \mu_n$ be fixed. For $a, b, a+b \neq 0$ modulo n, put

$$j_n^{a,b} = -\sum_{t\in\mathbb{F}_q} \theta(t)^a\theta(1-t)^b,$$

with nontrivial characters extended by zero. Otsubo's Theorem8.1 identifies this number with the character specialization of the product of F_{Frob_p} along the orbit $(a,b),(pa,pb),\dots,(p^{f-1}a,p^{f-1}b)$. For $x^n+y^n=z^n$ the sum of these $j_n^{a,b}$ is the trace of geometric Frobenius on H^1 over $\mathbb{F}_q$. For the manuscript's $x_0^n+x_1^n+x_\infty^n=0$ it is instead the sum of

$$j_{n,+}^{a,b} = \theta(-1)^{a+b}j_n^{a,b}. \tag{D.8}$$

Proof. The cited theorem [Ots16, Theorem8.1] is formulated for an arbitrary residue degree; its product, rather than a single value at p on a non-fixed character, is essential. The comparison with H^1 uses the ordinary arithmetic Frobenius action on the Tate module and its dual geometric action on cohomology; these have the same characteristic polynomial in this convention. The change of models is defined over $\mathbb{Q}(\epsilon_n)$ by $(x_0 : x_1 : x_\infty) \mapsto (x_0 : x_1 : \epsilon_n x_\infty)$, $\epsilon_n^n = -1$. Its Frobenius discrepancy is $\epsilon_n^{q-1} = (-1)^{(q-1)/n}$. Projectively this multiplies the first two coordinates by the same element. The (a,b) character is therefore multiplied by $\theta(-1)^{a+b}$, which proves (D.8). This retains the arithmetic descent of scan171, including even n. □

D.7. Complete finite-field computation in the requested range [Proved]

The files `d_jacobi.py`, `d_jacobi_field.cpp`, `d_jacobi.out` and `d_jacobi.json` compute every nondegenerate character value for every prime $p \le 50$ and $3 \le n \le 12$ with $p \nmid n$, over the exact field $\mathbb{F}_{p^{\mathrm{ord}_n(p)}}$. Every value satisfies $j\bar{j} = q$. Character sums and independent point counts agree for both models. Levels1 and2 have zero projective H^1, and the output separately retains the bad-reduction pairs $p \mid n$.

Proof. For non-semiprimitive pairs the code finds a monic polynomial for which multiplication by its residue class has order $q-1$. This certifies that all nonzero elements of the quotient ring form a cyclic group, hence that the quotient is the required finite field. It enumerates the cyclotomic-number array $C_{ij} = \#\{t \neq 0,1 : \theta(t) = \zeta_n^i, \theta(1-t) = \zeta_n^j\}$. Discrete Fourier summation gives each j as an integral polynomial modulo Φ_n. All norm identities are checked in this exact quotient ring. Independently, the standard curve has $n^2 C_{00} + 2n + n\,[\theta(-1) = 1]$ points, and the plus curve has $n^2 C_{ee} + 3n\,[e = 0]$, where $\theta(-1) = \zeta_n^e$.

For the larger semiprimitive fields, $p^h \equiv -1 \pmod{n}$, $q = p^{2h}$, the code uses and checks the exact identity $j = -p^h$. Indeed the characters are trivial on $\mathbb{F}_{p^h}^\times$. In the quadratic extension, summing the additive character over trace fibres gives the Gauss sum $G(\theta^a) = p^h \theta^a(\tau)$ for a nonzero trace-zero element τ; the quotient $G(\theta^a)G(\theta^b)/G(\theta^{a+b})$ is p^h. This proves the shortcut. The smaller semiprimitive cases are also enumerated directly. The largest direct field is $\mathbb{F}_{47^5}$; it was actually enumerated. For an explicit arithmetic-twist check, $(p,n) = (43,6)$ gives standard trace 32 and plus-model trace -64, with respective point counts 12 and 108. These computations verify the finite-field side of D.6 and its character conventions; they do not claim a numerical oracle evaluating an arbitrary abstract Galois element independently of the theorem. □

Circular Kummer characters, regularity, and the first irregular prime

D.8. The checked Ihara–Anderson–Coleman coefficient formula [Known]

For odd ℓ, let $U = \log(1+u)$, $V = \log(1+v)$, $W = -U - V$. On $G_{\mathbb{Q}(\mu_{\ell^\infty})}$ the universal series has the expansion

$$F_g(u,v) = \exp\left(\sum_{m \ge 3 \text{ odd}} \frac{\kappa_m(g)}{(1-\ell^{m-1})m!} (U^m + V^m + W^m) \right), \tag{D.9}$$

where κ_m is the compatible Kummer character of $\epsilon_n^{(m)} = \prod_{a \in (\mathbb{Z}/\ell^n)^\times} (\zeta_{\ell^n}^a - 1)^{\langle a^{m-1} \rangle_n}$. The integral factor $1 - \ell^{m-1}$ is a unit. Vandiver at ℓ is equivalent to surjectivity of all these κ_m for $m = 3, 5, \ldots, \ell - 2$.

Proof. This is the reduction of [IKY87, Theorems A_2 and B, equations(9)–(11)] to the root convention of D.1. Their character is defined by the same quotient of a chosen ℓ^nth root and its Galois translate, so no unnamed rescaling is involved. Their Theorem B identifies $\beta_m = (1-\ell^{m-1})^{-1}\kappa_m$, and Theorem A_2 gives the exponential with $\beta_m/m!$. The paragraph immediately following Theorem B proves the stated Vandiver equivalence. Both pages were inspected visually. Reducing modulo ℓ permits replacing $\zeta^a - 1$ by $1 - \zeta^a$: the resulting sign is an ℓth power. These are the exact characters computed next.

Anderson's original §11.6, Proposition11.6.1 and formula(13.7.3), pp.117–118 and130, give an independent normalization check. His β_m sums i^{m-1} times the Kummer cocycle of $1 - \zeta_{\ell^n}^{-i}$ over every $0 < i < \ell^n$. On the cyclotomic kernel this is the cocycle of $\zeta_{\ell^n}^i - 1$, since their

ratio is $\zeta_{\ell^n}^{-i}$ and all of its ℓ-power roots are fixed. Splitting the sum into $\ell \nmid i$ and $i = \ell j$ gives $\beta_m = \kappa_m + \ell^{m-1}\beta_m$. Thus his exponential coefficient is exactly the one in (D.9). □

D.9. An exact obstruction module, not an irregular-prime guess [Known]

Let $\mathcal{F}_\ell \subset \mathbb{Z}_\ell[[u,v]]^\times$ be the group defined by the five conditions of Ichimura–Kaneko1989, §1.2: congruence $F \equiv 1 \pmod{uvw}$, inversion $F\overline{F} = 1$, symmetry in u, v, w, symmetry of $F * F$ in four variables with product of 1+variables equal to one, and

$$\prod_{\zeta^\ell=1} F(\zeta(1+u)-1, \zeta(1+v)-1) = F((1+u)^\ell - 1, (1+v)^\ell - 1).$$

Their Theorem1 identifies equality $\operatorname{Im} F = \mathcal{F}_\ell$ with Vandiver at ℓ. Their Theorem2 computes the characteristic power series of the defect $(\mathcal{F}_\ell / \operatorname{Im} F)(-1)$ as that of $\operatorname{Hom}_{\mathbb{Z}_\ell}(A_\infty^+, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ over the cyclotomic Iwasawa algebra. Thus the relevant obstruction is the plus class-group module, not merely divisibility of a Bernoulli numerator. At every regular odd prime this defect is zero. D.10 proves the same equality at every odd prime at most50, including37.

Proof. We use [IK89, Theorems1–2]; the defining conditions and both theorem statements were checked on the original pages314–315. Equation(D.9) identifies their F and Kummer characters with ours. The plus part A_∞^+ is the direct limit of the ℓ-Sylow parts of the ideal class groups of $\mathbb{Q}(\mu_{\ell^n})$, with the authors' transition maps. The stated characteristic-series equality records exactly the size invariant proved there; it is not promoted to an isomorphism of modules. Regularity implies that the full class number at the first level is prime to ℓ, hence its plus part is, and so Vandiver holds. For the bounded range, the certificates below establish the equivalent surjectivities directly. □

D.10. Soulé characters modulo every prime at most50 [Proved]

For every odd prime $\ell \le 50$ and each odd $3 \le m \le \ell - 2$, the program `d_kummer_soule.py` exhibits an explicit nonzero value of $\kappa_m \bmod \ell$. Consequently all these individual characters are onto $\mathbb{Z}_\ell$, their distinct Teichmuller-weight components are independent, and Vandiver holds throughout this range. At $\ell = 37$ all seventeen characters are certified by the single prime $q = 5477$. The Bernoulli computation finds the irregular pair $(37, 32)$ and no other irregular pair below50; it causes no missing plus-part image here.

Proof. For a prime $q \equiv 1 \pmod{\ell^2}$, a primitive root $g \bmod q$, and $r = g^{(q-1)/\ell}$, the code evaluates $c_m(r) = \prod_{a=1}^{\ell-1}(1-r^a)^{a^{m-1} \bmod \ell}$ and finds the unique $k \bmod \ell$ such that $c_m(r)^{(q-1)/\ell} = r^k$. A nonzero k proves that the corresponding cyclotomic element is not an ℓth power. All choices, exponents and residues are retained in the JSON, so each witness is checked by modular exponentiation alone.

This is also a certificate on the infinite cyclotomic field, not only at its first layer. In $\mathbb{Q}(\zeta_\ell)^\times/\ell$ the class of c_m has Teichmuller character $b \mapsto b^{1-m}$. The only order-ℓ subextension of the cyclotomic $\mathbb{Z}_\ell$-extension has Kummer class ζ_ℓ, of character $b \mapsto b$. These characters differ in the indicated range. Thus the nonzero class cannot disappear in that extension. The character takes a unit value in $\mathbb{Z}_\ell$, proving surjectivity; projection by the distinct idempotents of $\mathbb{F}_\ell^\times$ proves joint independence of these weight components. D.8 now proves Vandiver without inferring it from the irregularity test.

The witness primes, grouped by ℓ, are

ℓ	5	7	11	13	17	19	23
q	101	197, 491	727	677, 2029	3469	10831	12697

ℓ	29	31	37	41	43	47
q	10093, 13457	15377	5477	16811	22189	8837.

For $\ell = 3$ the specified weight list is empty; for $\ell = 2$ there is likewise no weight in the requested range. No omitted character is hidden by these conventions. □

Metabelian leading terms and the lifting issue

D.11. Exact special-derivation images through weight10 [Proved]

Let L be the free Lie algebra over $\mathbb{Q}$ on x, y, with the three peripheral elements $x, y, -x-y$. Modulo inner derivations normalize a positive-degree special derivation by $D(x) = 0$, $D(y) = [f, y]$. For weight $n \geq 2$, it is special exactly when some $g \in L_n$ satisfies

$$[f, y] = [g, x + y]. \tag{D.10}$$

The metabelian quotient is $L/L'' = \mathbb{Q}x \oplus \mathbb{Q}y \oplus \mathbb{Q}[X, Y]c$, where $c = [x, y]$, $X = \operatorname{ad} x$, $Y = \operatorname{ad} y$; the last summand is abelian. The following exact rational ranks were computed, with source witnesses, by `d_metabelian.py`:

n	1	2	3	4	5	6	7	8	9	10
special outer source	0	0	1	0	3	0	6	4	13	12
its metabelian image	0	0	1	0	3	0	5	0	7	0
S_3-invariant source	0	0	1	0	1	0	2	1	4	2
its metabelian image	0	0	1	0	1	0	1	0	2	0

(D.11)

Thus all three weight-five directions survive. The extra invariant direction in weight seven dies. In weight nine two invariant directions survive. These are images of derivations of the free Lie algebra; special derivations defined only on L/L'' have dimension $n - 2$ for every $n \geq 3$, and are a larger calculation.

Proof. Subtracting the inner derivation that conjugates x gives the normalization. For $n \geq 2$, the centralizer of x has no component of degree n, so it is unique. The condition at the third peripheral element gives (D.10), with a harmless simultaneous sign determined by the convention for g. The free metabelian description follows by Jacobi: the commuting operators X, Y generate L'/L'' from c, and its monomials are independent, for instance in the usual upper-triangular matrix realization over $\mathbb{Q}[X, Y]$.

The program uses the Lyndon basis in each degree and expands its brackets in the free associative algebra. It forms the exact matrix of $(f, g) \mapsto [f, y] - [g, x + y]$ and computes its rational nullspace. The invariant subspace is separately defined by the checked LM2 equations

$$f(x, y) + f(y, x) = 0, \qquad [x, f(x, y)] - [x + y, f(-x - y, y)] = 0. \tag{D.12}$$

These follow by conjugating the normalized outer derivation by the transposition and the three-cycle of the three peripheral elements and then renormalizing $D(x) = 0$. Thus the matrix computes the actual infinitesimal centralizer, rather than the two linear cycle equations. Projection of each Lyndon bracket to $\mathbb{Q}[X, Y]c$ gives the image ranks. Every pivot-image vector retains an exact preimage in the JSON.

In the metabelian algebra (D.10) reads $Yp = (X + Y)q$, hence $(X + Y) \mid p$. This gives the stated $n - 2$ comparison dimension. At weight five the full image consists of $(X + Y)$ times all degree-two polynomials; the invariant image is spanned by $X^3 + 2X^2Y + 2XY^2 + Y^3$. At weight seven its invariant image is spanned by $X^5 + 3X^4Y + 5X^3Y^2 + 5X^2Y^3 + 3XY^4 + Y^5$. At weight nine one extra independent vector is $X^2Y^2(X + Y)^3$; the output gives the other vector and both preimages. The zero even-weight images, including weights eight and ten with nonzero sources, are computed by this projection, not assumed from parity. □

D.12. The reviewed pro-ℓ lifting assertion and its precise scope [Proved]

For a free pro-ℓ group on finitely many peripheral generators, every integral special derivation is the leading term of a special automorphism. For type $(0,3)$ and $\ell \geq 5$, every S_3-invariant integral leading term lifts to an element of the centralizer of the geometric S_3 in the pro-ℓ outer special-automorphism group. Thus the image ranks in (D.11) are ranks of actual pro-ℓ special-automorphism images, and of their centralizer for $\ell \geq 5$.

Proof. Write Γ_m for the closed lower-central series. Its associated graded is the free Lie algebra over $\mathbb{Z}_\ell$. Start with conjugators representing the given degree-n Lie elements. The product-of-peripheral-generators equation holds modulo Γ_{n+2} because its leading error is the special-derivation equation. At stage m the error lies in Γ_m/Γ_{m+1}; changing the conjugators by elements of Γ_{m-1} changes it by $\sum_i [a_i, x_i]$. This map is onto L_m: iterated Jacobi rewrites every Lie monomial of degree m as a sum $[x_i, b_i]$. In the two-generator normalization fixing x, the two available operators are $[\ , y]$ and $[\ , x+y]$; their sum still spans $[x, L_{m-1}] + [y, L_{m-1}] = L_m$. We can therefore cancel each error without altering the prescribed leading term.

The corrections converge in the free pro-ℓ group. Indeed every finite ℓ-group quotient is nilpotent, so Γ_m eventually lies in every open normal subgroup. The resulting endomorphism is the identity on the Frattini quotient and hence is onto by the pro-ℓ basis theorem; a finitely generated profinite group is Hopfian, so it is an automorphism. This independently proves the reviewed assertion.

For the invariant assertion, the group of IA automorphisms in this pro-ℓ setting is pro-ℓ, as are the successive kernels. A finite group of order prime to ℓ acts on it. For a quotient by a stable closed normal subgroup, fixed elements lift to fixed elements: for finite ℓ-groups this follows by induction on a central series and averaging the additive obstruction, and the profinite assertion follows by compactness. Apply this to the order-six group S_3 and successive filtration quotients. This proves the claimed lift. No such convergence argument applies to the full free profinite group: a finite nonnilpotent quotient need not kill any Γ_m. □

D.13. The remaining full profinite image comparison in D(e) [Open]

The maps from actual full profinite loop-preserving outer automorphisms, and from $\widehat{\mathrm{GT}}_0$, to their pro-ℓ leading terms land in the second and fourth rows of (D.11), respectively. D.12 proves the reverse inclusions for the pro-ℓ groups, not for those full profinite source groups. The full profinite realization of the extra weight-five metabelian directions, and of the extra invariant weight-nine direction, remains unproved here. Consequently D(e), as asked for the three full profinite groups, remains undecided in this return.

Established comparisons and attempted lifting. Taking the maximal pro-ℓ quotient is characteristic, so it defines the displayed continuous homomorphisms of outer automorphism groups. Taking a leading term is injective on each filtration quotient by its definition; the peripheral and centralizer equations give (D.10)–(D.12). D.12 realizes every admissible leading vector after the change of source to the pro-ℓ group. A proposed full profinite lift would have to solve the same conjugacy and generation equations in every finite quotient, including nonnilpotent quotients. The lower-central correction sequence controls none of the residual errors in such a quotient. We also checked the weaker route of solving just the metabelian Lie equations: its larger dimension $n-2$ proves that this route discards further lifting conditions already at weight four. These two failed transfers are exact gaps, not counterexamples to a full profinite lifting theorem.

For the Galois source, (D.9) gives at most one rational direction in each odd weight and zero in even weights in this metabelian image. D.10 supplies nonzero directions through weight

ten at $\ell = 37$; E.3 supplies the rational general-prime comparison. The weight-five pro-ℓ image therefore distinguishes the whole special group from its S_3 centralizer and from Galois. D.20–D.21 below add the full profinite beta/gamma compatibility and prove equality of the rational metabelian graded images of Galois and full $\widehat{\mathrm{GT}}$ through weight ten. The pentagon used there is an additional equation on $\widehat{\mathrm{GT}}_0$. The exact missing lift test for the latter is D.22; the pro-ℓ construction alone supplies only the subfamily of its tests with nilpotent quotients. D.24 gives an explicit test of the general automorphism-lifting route and its peripheral limitation. D.25 constructs nonarithmetical finite GT shadows with generation verified; it leaves their genuineness, and hence the indicated full profinite comparison, undecided. □

The integral signs, CM coordinates, and the genuine congruence action

D.14. What the first integral sign congruences say [Proved]

The norm equations give $\epsilon_2 = \chi_4(\mu)$ and $\epsilon_3 = \chi_{-3}(\mu)$, already at moduli4 and3. On the Galois image $\mu = \chi$. Thus these signs add no independent condition to the full universal datum (χ, F_g): the exact comparison is evaluation of χ modulo4 and3. The unadorned series restricted to G_K has $\chi = 1$, and consequently both signs there are one.

Proof. For a unit Gaussian norm $a^2 + b^2$, exactly one of a, b is odd; its value is one modulo4. For the Eisenstein norm $c^2 + cd + d^2$, reduction modulo3 is $(c - d)^2$, hence is one for a unit. The signed norm equalities in LM2 C.d2 therefore force the stated signs. Conversely the local norm images are $1 + 4\mathbb{Z}_2$ and $1 + 3\mathbb{Z}_3$: the norm on sufficiently deep principal units is the trace under logarithm, and inspection at moduli8 and9 supplies the first step. These finite images are also enumerated inside `d_local_cm.py`. Evaluating the already specified multiplier proves the asserted factorization; it does not equate an unmarked beta series with its missing multiplier. □

D.15. The two normalizer coordinates are the projective CM representations [Known]

Use the rigidified modular stack $\mathcal{M}_{1,1}/\{\pm 1\} \simeq [U_{0,3}/S_3]$ and the positive tangential point $\lambda = 0$ with tangent1 and the standard branch-cut paths on $0 < \lambda < 1$. Choose the same etale paths and homology basis in the modular congruence quotient and in LM2 C.d4. Then its coordinates $T = A^{-1}V$ and $W = \beta^{-1}V$, each modulo its single global sign, are respectively the projective adelic Tate representations of

$$E_3 : y^2 = x^3 + 1, \qquad E_4 : y^2 = x^3 - x. \tag{D.13}$$

Their determinants are χ; their actions on the CM orders are complex conjugation according to χ_{-3} and χ_4. Changing the paths conjugates the indicated representations by their transported bases. Quadratic twists of the displayed curves make exactly the same representations modulo the global sign.

Proof. The quotient of an elliptic curve by $[-1]$ is a four-pointed projective line, with the identity among the four branch points. Ordering the other three points gives the Legendre parameter and is a torsor under S_3. Conversely the double cover of this four-pointed line gives the elliptic curve; its two possible covering automorphisms are exactly ± 1. These two constructions are inverse on families and their isomorphisms after rigidification, proving the displayed stack equivalence over $\mathbb{Q}$.

The projective Tate local system consequently has geometric monodromy $\mathrm{SL}_2(\widehat{\mathbb{Z}})/\{\pm I\}$. At the elliptic points its inertia generators are the matrices R, S of LM2, satisfying $R^3 = S^2 = -I$.

Transport the Galois action at either elliptic point to the cusp by a fixed path. The resulting path cocycle is a conjugator A, respectively β, with $u_0(R) = AR^{\chi-3}A^{-1}$ and $u_0(S) = \epsilon_2\beta S\beta^{-1}$. These are precisely the conjugators in C.d4: their prescribed reductions modulo2 remove the finite inertia ambiguity. More explicitly, the centralizer of either torsion generator in the free profinite product $C_2 \amalg C_3$ is its cyclic factor, whose intersection with the level-two free subgroup is trivial. Hence two such normalized path conjugators coincide. Transport of the Tate action now gives exactly $A^{-1}V$ and $\beta^{-1}V$, proving the formulas.

At $\lambda = 1/2$, the change $X = 4x - 2$, $Y = 8y$ gives $Y^2 = X^3 - 4X$, the quadratic twist by2 of E_4. At $\lambda^2 - \lambda + 1 = 0$, shifting x by $(1 + \lambda)/3$ gives $y^2 = x^3 + (2\lambda - 1)/9$; this constant is the cube of $-(2\lambda - 1)/3$, so it is a quadratic twist of E_3 over $\mathbb{Q}(\sqrt{-3})$. The descent to $\mathbb{Q}$ is fixed by its action on the three nonzero two-torsion points: for $j = 0$ the splitting field $\mathbb{Q}(\sqrt{-3})$ forces the cubic constant to be a rational cube up to a quadratic twist, and for $j = 1728$ rational two-torsion forces the coefficient to be a negative square. Thus the descended projective representations are those in (D.13). This is a typed arithmetic comparison, not an inference from the shapes of the two centralizer rings. The CM reciprocity used next is [LR22, Corollary3.4], or equivalently [Mil07, Theorem3.10], both opened. □

D.16. Exact joint CM image: one additional quadratic relation [Proved]

Let $K_3 = \mathbb{Q}(\sqrt{-3})$, $K_4 = \mathbb{Q}(i)$ and $L = K_3K_4$. In the product of the two normalizer groups of D.15 with common determinant and the modulo-two conditions of LM2 C.d4, the joint Galois image has index two. Its exact additional condition is equality of the two restrictions to

$$M = \mathbb{Q}(i, \sqrt[4]{3}); \qquad K_3^{\mathrm{ab}} \cap K_4^{\mathrm{ab}} = \mathbb{Q}^{\mathrm{ab}}M. \tag{D.14}$$

Thus the two-tori lead has an open image, but the Galois image does not fill the proposed full pair of equal norms. The missing coset is a single C_2; the existence of an element of $\widehat{\mathrm{GT}}_0$ realizing that coset is a separate unresolved realization question.

Proof. Both quadratic rings have class number one, as follows already from their Euclidean norms. CM reciprocity identifies the Weber-function torsion fields with the ray class fields; taking all levels in [LR22, Corollary3.4] gives K_j^{ab}. The field defined by the Tate representation modulo ± 1 over K_j contains these Weber values (x^3 for E_3, x^2 for E_4), and is contained in K_j^{ab} since the Tate action over K_j is linear over its commutative CM order. It is therefore exactly K_j^{ab}. An element outside G_{K_j} cannot act as $\pm I$, because it conjugates a nonreal CM endomorphism to its distinct complex conjugate. Hence the field of the projective representation over $\mathbb{Q}$ is also exactly K_j^{ab}.

More precisely the image fills the specified individual factor. Reciprocity identifies $\mathrm{Gal}(K_j^{\mathrm{ab}}/K_j)$ with $\widehat{\mathcal{O}}_{K_j}^{\times}/\mathcal{O}_{K_j}^{\times}$. Reduction modulo2 maps $\mu_6/\{\pm 1\}$ isomorphically to $(\mathcal{O}_{K_3}/2)^{\times}$, and $\mu_4/\{\pm 1\}$ isomorphically to $(\mathcal{O}_{K_4}/2)^{\times}$. Thus removing that finite root of unity gives an isomorphism with the units congruent to one modulo2, modulo the single global sign. The curves in (D.13) have their full two-torsion over the stated fields, so their projective representations give exactly this isomorphism. The determinant is restriction to the cyclotomic subfield by the Weil pairing. Outside G_L, the two conjugation signs are already forced by that determinant and are all realized by $G_{\mathbb{Q}}$.

It remains to compute the intersection; this step determines the joint image, not merely its openness. Put $E = K_3^{\mathrm{ab}} \cap K_4^{\mathrm{ab}}$. It contains $\mathbb{Q}^{\mathrm{ab}}$. In $G = \mathrm{Gal}(E/\mathbb{Q})$ the two distinct index-two subgroups fixing K_3, K_4 are abelian. Their intersection is central, and together they generate G. Choosing representatives x, y of their two nontrivial cosets, the closed commutator group is generated by the central element $[x, y]$. Since x^2 and y^2 are central, this element has order at most two. Therefore $[E : \mathbb{Q}^{\mathrm{ab}}] \leq 2$.

The splitting field M of X^4-3 has dihedral group of order eight: its automorphisms are (a,b), $0\le a<4$, $0\le b<2$, acting by $\sqrt[4]{3}\mapsto i^a\sqrt[4]{3}$, $i\mapsto(-1)^b i$, with product $(a,b)(c,d)=(a+(-1)^b c, b+d)$. The subgroups fixing K_4 and K_3 are respectively $b=0$ and $a+b=0 \pmod 2$, both abelian. Thus $M\subset E$, and M is nonabelian over $\mathbb{Q}$. This realizes the bound and proves (D.14). Galois theory now identifies the joint image with the fibre product over this exact intersection. The common-determinant target is the fibre product over $\mathbb{Q}^{\mathrm{ab}}$, so its index is two. The program checks the dihedral group and this finite index explicitly. □

D.17. The local indices at two and three [Proved]

For $\mathcal{O}_L=\mathbb{Z}[i,z]$, $z^2-z+1=0$, the paired local norm map to the equal-norm units of $\mathcal{O}_{K_3}\otimes\mathbb{Z}_p$ and $\mathcal{O}_{K_4}\otimes\mathbb{Z}_p$ has index two at each $p=2,3$, and is onto at every other prime. Its two local obstruction classes are identified by the global root-of-unity pairs, leaving the single global relation of D.16. Exact finite enumeration gives

p^k	2	4	8	16	32	3	9	27
equal-norm pairs	6	48	384	3072	24576	24	648	17496
norm image	6	24	192	1536	12288	12	324	8748.

(D.15)

Proof. At either bad prime one quadratic subfield is unramified and the other ramified. Given a compatible pair (a,b), first lift its coordinate in the unramified subfield through the ramified quadratic norm. It is a norm because the quadratic Hilbert-symbol obstruction equals the Hilbert symbol of its rational norm, and that rational norm is already a norm from the ramified quadratic subfield. Its valuation is zero, so a lift is a unit. The remaining discrepancy has norm one in the ramified subfield. Hilbert90 writes it as $c/\bar{c}$. It can be corrected by a unit from the biquadratic field exactly when the valuation of c in that ramified subfield is even: norms from its unramified quadratic extension have even valuation, and multiplying c by a rational scalar changes its valuation by an even integer. This gives precisely one C_2 obstruction. The unit norm in an unramified extension is onto, so no further obstruction remains.

The pairs $(-1,1)$ and $(1,i)$ represent the nonzero class at each prime. The identity $N_{L/K_3}(z+i\bar{z})=-1$, $N_{L/K_4}(z+i\bar{z})=-i$ shows that their sum is zero. At other primes the integral norm torus has connected smooth kernel; norm surjectivity on residue fields and on successive principal-unit quotients proves surjectivity. The code enumerates the rings directly: for $a+bz+i(c+dz)$ its two norms are $(a+bz)^2+(c+dz)^2$ and $(a+ci)^2+(a+ci)(b+di)+(b+di)^2$. It tests invertibility by the rational norm and records the entire image size. The signs above are also tested at every displayed level. The global roots of unity contribute the same nonzero class at both bad primes, hence remove only the diagonal C_2. This agrees with, and does not replace, the field-intersection proof of D.16. □

D.18. The genuine cusp congruence action retains the full rank-one Kummer coordinate [Known]

In the positive tangential normalization $\lambda=0$, tangent1, take $q^{1/n}$ with positive real leading coefficient $256^{-1/n}$ and $\zeta_n=\exp(2\pi i/n)$. Use the resulting Tate basis, matched to the Legendre cycles over $[0,\lambda]$ and $[\lambda,1]$, with positive local monodromy $\left(\begin{smallmatrix}1&2\\0&1\end{smallmatrix}\right)$. The genuine normalized congruence representation is

$$V_g=\begin{pmatrix}\chi(g) & -8\kappa_2(g)\\ 0 & 1\end{pmatrix}. \tag{D.16}$$

Compatible roots may be chosen so that $\kappa_{-1/2}=(\chi-1)/2-\kappa_2$. Consequently the continuous

inverse comparison on this image is

$$(\chi, V_{12}) \longmapsto \left(\chi, \frac{\chi-1}{2} + \frac{V_{12}}{8}\right). \tag{D.17}$$

Multiplication by eight is a homeomorphism from $\widehat{\mathbb{Z}}$ onto $8\widehat{\mathbb{Z}}$, so no part of this Kummer coordinate is lost in the genuine normalized congruence action. In the basis of the manuscript with local monodromy $\left(\begin{smallmatrix} 1 & -2 \\ 0 & 1 \end{smallmatrix}\right)$, conjugation by $\mathrm{diag}(-1,1)$ reverses the off-diagonal sign. On *outer* congruence actions this translation is forgotten: the comparison retains only χ modulo squares.

Proof. The Legendre invariant is $j(\lambda) = 256(1-\lambda+\lambda^2)^3/(\lambda^2(1-\lambda)^2)$. Inverting $j(q) = q^{-1} + 744 + \cdots$ gives $q = \lambda^2/256\,(1 + \lambda\mathbb{Q}[[\lambda]])$. Tate uniformization gives the exact torsion sequence with basis $\zeta_n, q^{1/n}$ and matrix $\left(\begin{smallmatrix} \chi & \kappa_q \\ 0 & 1 \end{smallmatrix}\right)$; see [Jos, §11, Theorem11.17 and the preceding uniformization]. For the formal field used here, the comparison follows by base change of the universal Tate equation and its torsion representatives over $\mathbb{Z}((q))$: the convergent uniformization identities are identities of Laurent series with integral coefficients, and remain identities after $q = q(\lambda)$. Thus the torsion representatives are $\zeta_n^a q^{b/n}$ over $\overline{\mathbb{Q}((\lambda))}$. The tangential section fixes compatible $\lambda^{1/n}$ and the unique roots with constant coefficient one of the last power-series unit. Its only remaining Kummer action on $q^{1/n}$ is therefore that on $256^{-1/n}$, namely $-8\kappa_2$. The Legendre curve is a quadratic twist of the split Tate curve at this cusp; that twist multiplies the entire matrix by a single sign and disappears in the projective congruence local system. Normalizing its lower-right entry to one yields (D.16). To fix the residual shear, let $K(m) = \int_0^{\pi/2}(1 - m\sin^2\theta)^{-1/2}d\theta$. The branch-cut periods have ratio $\tau = iK(1-\lambda)/K(\lambda)$, purely imaginary and positive for $0 < \lambda < 1$. Hence $q = \exp(2\pi i\tau)$ and its chosen roots are positive real. A shear would replace $q^{1/n}$ by $\zeta_n^b q^{1/n}$ and change the off-diagonal entry by $b(1-\chi)$; positivity for every n forces $b = 0$. This proves that the Tate basis is the specified global Legendre branch-cut basis, rather than merely a basis with the same local monodromy.

Choose $(-1)^{1/n} = \zeta_{2n}$. Its Kummer cocycle is $(\chi-1)/2$, which proves (D.17) by multiplication of root systems. The stated change of basis explicitly handles the orientation convention. Finally conjugation by V_g on the special linear congruence quotient has outer class determined by its determinant modulo squares, as established in Part A; its unipotent upper-triangular factor is inner. Thus the often proposed claim that the rank-one translation is visible only metabelianly is false for genuine normalized actions, while the precise outer quotient does discard it. □

D.19. The residual GT and full profinite realization questions [Open]

D.1–D.4 settle item3, including general openness, the quadratic failure of surjectivity, joint independence, and the local inertia images. D.15–D.18 settle the stated Galois comparison of the two CM coordinates and the rank-one congruence coordinate. The unresolved parts of D(e) and D-after-LM2(1) concern the full profinite GT beta/gamma presentation and actual realization outside Galois: whether the extra leading metabelian directions of D.11 lift from $\widehat{\mathrm{GT}}_0$ or the full profinite loop-preserving group, and whether $\widehat{\mathrm{GT}}_0$ supplies the missing index-two CM coset of D.16. These assertions remain on the coverage ledger; none follows from solving only matrix or Lie equations.

Concrete residual instances. The first extra metabelian special vector occurs at weight five, where the free special source has dimension three and its invariant part dimension one; exact preimages are in `d_metabelian.json`. The first extra invariant metabelian vector occurs at weight nine. D.12 supplies their pro-ℓ realizations and identifies the nonnilpotent-quotient

obstruction to transferring that construction. For the CM problem, D.17 supplies an explicit finite matrix coset, represented locally by $(-1,1)$ at one bad prime and the identity at the other, whose two obstruction classes disagree. D.16 proves that this coset has no Galois lift. Its possible lift to an actual full profinite $\widehat{\mathrm{GT}}_0$ element must solve the conjugator and generation equations before congruence reduction; the matrix reconstruction of LM2 C.d4 does not do so. The source search reached the original Ihara Part II through the RIMS1224 PostScript file, and the exact modern profinite lift criterion through Dolgushev–Guynee. Part I's RIMS1177 catalogue entry has no download; the PDF, PostScript and DVI locations returned404. D.20–D.22 record the additional results actually obtained. The unresolved steps are the complete lift tests for the two indicated metabelian vectors and for the missing CM coset. □

The original profinite beta/gamma relations and their graded consequence

D.20. Ihara's exact cyclic-cover compatibility and gamma multiplication [Known]

Write a full profinite GT element as (λ, f) acting by $x \mapsto x^\lambda$, $y \mapsto f^{-1}y^\lambda f$. For $n \geq 1$ let $H_n = \ker(\widehat{F}_2 \to \mathbb{Z}/n)$, where $x \mapsto 1$, $y \mapsto 0$, and define the continuous epimorphism

$$\phi_n : H_n \longrightarrow \widehat{F}_2, \qquad x^n \mapsto x, \quad y \mapsto y, \quad x^{-j}yx^j \mapsto 1 \quad (0 < j < n).$$

For every $g \in G_{\mathbb{Q}}$ one has

$$\phi_n(f_g) = y^{-\kappa_n(g)} f_g, \qquad \phi_n \circ g|_{H_n} = g \circ \phi_n. \tag{D.18}$$

Here the roots of the positive integer n use the positive real branches of D.2. These are [Iha00, Theorem1 and Proposition1].

The full profinite beta function of $\sigma \in \widehat{\mathrm{GT}}$ has Ihara's factorization

$$B_\sigma(s,t) = \frac{\Gamma_\sigma(s)\Gamma_\sigma(t)}{\Gamma_\sigma(s+t)}, \qquad s,t \in \mathbb{Q}/\mathbb{Z}. \tag{D.19}$$

The coefficient ring and completed group-ring interpretation are those of the hyperadelic gamma function; equality precedes every prime-power specialization. For Galois elements Anderson's multiplication formula, in Ihara's corrected convention, is

$$\frac{\prod_{nc=0} \Gamma_g(s+c)/\Gamma_g(c)}{\Gamma_g(ns)} = 1 \otimes \exp(-2\pi i\, ns\kappa_n(g)). \tag{D.20}$$

Ihara Part II, Introduction and §2, states (D.19) and (D.20), attributes (D.20) to Anderson's Corollary8.6.3, and gives the following exact relations among its intermediate subgroups: $G_{\mathbb{Q}} \subset \widehat{\mathrm{GT}}_K \subset \widehat{\mathrm{GT}}_{A'}$ and $G_{\mathbb{Q}} \subset \widehat{\mathrm{GT}}_A \subset \widehat{\mathrm{GT}}_{A'}$. Here K means all the commuting squares in (D.18), A means (D.20), and A' means the two distribution identities in the proof below.

Proof. The original RIMS1224 version was downloaded, decompressed, converted from PostScript, and read. Its proof of Theorem1 identifies H_n with the fundamental group of $\mathbb{P}^1 - \{0, \mu_n, \infty\}$, and ϕ_n with the map induced by filling the punctures different from1, after the change $t \mapsto t^n$. Coefficientwise Galois action on Puiseux series commutes with this change, proving the second equality of (D.18). Applying the two sides to y shows that $\phi_n(f_g)f_g^{-1}$ centralizes $y^{\chi(g)}$, hence is a power of y. Its exponent is the logarithmic Kummer quasi-cocycle $\Psi_n^{(0)}(g)$, characterized in the original Theorem1 by $(n^{-r})^{g-1} = \exp(2\pi i r \Psi_n^{(0)}(g))$ for $r \in \mathbb{Q}$. Thus $\Psi_n^{(0)} = -\kappa_n$, proving the first equality with its sign.

For completeness the identities defining A' are

$$\Psi^{(0)}_{mn} = \Psi^{(0)}_m + \Psi^{(0)}_n, \qquad \Psi^{(in)}_{mn} = \Psi^{(i)}_m \quad (i \neq 0 \bmod m).$$

The functions $\Psi^{(i)}_n$ are defined by the action of f on $(1-\zeta_n^{-i}t^{1/n})^r$; Proposition2 proves these identities directly from (D.18). Proposition3 proves that they are exactly the logarithmic derivative of (D.20), using Part I's Theorem2, whose formula is reproduced there. The original Introduction explicitly says that the proof of (D.19) uses the five-cycle relation. These statements specify which equation is used and which group it constrains.

Anderson's original Corollary8.6.3, p.103, was also visually checked. With $\mathcal{W} = \prod_p W(\overline{\mathbb{F}}_p)$ from his §7.1, it states, in $(\overline{\mathbb{Q}} \otimes \mathcal{W})^\times$,

$$\prod_{nb=a} \Gamma_g(b) = ((n^{\langle -a \rangle})^{g-1} \otimes 1)\Gamma_g(a) \prod_{nc=0} \Gamma_g(c).$$

The roots of n are positive real. Put $a = ns$ and divide by the two displayed products; their Kummer factor is $\exp(-2\pi i\, ns\kappa_n(g)) \otimes 1$. Ihara's coefficient convention places $\mathcal{W}$ first; the tensor flip $z \otimes w \mapsto w \otimes z$ gives exactly (D.20), including its sign.

There is a precise obstruction to reversing the logarithmic derivative in the hyperadelic ring. For complementary nonzero idempotents $e, e' \in \widehat{\mathbb{Z}}$, $ee' = 0$, the nonconstant element $ex^{e'}$ has derivative zero. It is nonconstant already after projection to a coefficient prime supported on e and a finite deck quotient supported on e'. Thus the kernel of the derivation contains more than constants. Ihara's final Remark explicitly records that $\widehat{\mathrm{GT}}_K \subset \widehat{\mathrm{GT}}_A$ was not proved there. We use only the two inclusions actually proved. □

D.21. Galois and full GT have the same rational metabelian graded image through weight ten [Proved]

For every prime ℓ, the rational graded images of the actual groups $G_{\mathbb{Q}}$ and $\widehat{\mathrm{GT}}$ in the metabelian outer action, in weights1 through10 and after removing the cyclotomic multiplier, both have dimensions

$$0, 0, 1, 0, 1, 0, 1, 0, 1, 0. \tag{D.21}$$

In weight $n \geq 3$ odd the common line, expressed in the $p(X, Y)[x, y]$ coordinate of D.11, is

$$\mathbb{Q}_\ell \, \frac{(X+Y)^n - X^n - Y^n}{XY}. \tag{D.22}$$

The inclusion of this line in the $\widehat{\mathrm{GT}}_0$ upper-bound space is the explicit polynomial inclusion in D.11. The latter upper bound has an additional line in weight nine.

Proof. We give the graded comparison explicitly. Map the free profinite group to its pro-ℓ quotient, take logarithms in each rational nilpotent quotient, and then take the first nonzero homogeneous term of a filtered GT element. Expanding the two-cycle, three-cycle and five-cycle equations gives their infinitesimal equations for this term. The infinitesimal pure-braid generators are the initial classes of the logarithms of the pure-braid generators. Their relations are $[t_{ij}, t_{kl}] = 0$ for four distinct indices and $[t_{ij}, t_{ik} + t_{jk}] = 0$; the successive split fibrations forgetting a strand identify their free Lie kernels, so these relations give the associated graded pure-braid Lie algebra used in the expansion. Thus the displayed map sends the leading term into $\mathfrak{grt}_1(\mathbb{Q}_\ell)$.

Furusho's [Fur11, Corollary0.3, Theorem0.5, Proposition5.1 and Lemma5.2] gives the gamma constraint on its metabelian image. Here is the comparison with our coordinate, including the

factor that matters. For a Lie element $h = h_x x + h_y y$, put $M(h) = (h_y y)^{\mathrm{ab}}$. Direct expansion gives

$$M([x,y]) = XY, \quad M([x,h]) = XM(h), \quad M([y,h]) = YM(h) \quad (h \in L').$$

It follows that $M(p(X,Y)[x,y]) = XYp(X,Y)$ and that M kills L''. Furusho's gamma constraint is $M(h) = \gamma(X) + \gamma(Y) - \gamma(X+Y)$. In homogeneous degree n it is a scalar multiple of $X^n + Y^n - (X+Y)^n$. Since multiplication by XY is injective, this bounds the image by the line (D.22). Equivalently the actual module action has leading multiplier $-XYp$, because $D([x,y]) = [x,[p[x,y],y]] = -XYp[x,y]$. This is the leading term of the full profinite beta multiplier in (D.19), rather than an unlabelled identification of the two series.

The program `d_gamma.py` checks $M(h) = XY\bar{h}$ on all224 Lyndon basis words of degrees2 through10, and solves for each gamma polynomial in the exact invariant image basis of `d_metabelian.json`. In weight nine its coordinates in that basis are $(15/2, 9/2)$. The first basis vector $X^2Y^2(X+Y)^3$ is independent of the gamma line, verified by an exact rank-two calculation. Thus the comparison with the extra vector is computed in the same coordinate system as D.11.

The special equation forces $X+Y$ to divide p, as proved in D.11. At $Y = -X$ the numerator of (D.22) is $-(1+(-1)^n)X^n$, so every positive even-degree scalar must vanish in characteristic zero. Weight one vanishes directly. For each odd weight3,5,7,9, E.3 supplies a nonzero Galois class for every ℓ, using [HM03, Theorem8.4, §8.4, Lemma8.5 and Remark8.6]; at $\ell = 37$ D.10 also supplies direct finite-field certificates. Since $G_{\mathbb{Q}} \subset \widehat{\mathrm{GT}}$, these lower bounds equal the upper bounds just proved. This proves (D.21) for actual profinite source groups with rational graded target. It does not replace either source by its pro-unipotent automorphism group. □

D.22. The exact finite-quotient criterion for the remaining GT0 lifts [Known]

Let $N \triangleleft B_3$ have finite index and lie in PB_3, and let $\mathrm{GT}(N)$ be the finite set of pairs satisfying the two hexagon equations, the commutator condition on f, the unit condition on λ, and surjectivity onto B_3/N. Write $R_{K,N}$ for reduction from a finer such quotient. Then the image of the actual full profinite group is exactly

$$\mathrm{Im}\big(\widehat{\mathrm{GT}}_0 \longrightarrow \mathrm{GT}(N)\big) = \bigcap_{K \triangleleft B_3,\ [B_3:K]<\infty,\ K \subset N} R_{K,N}\big(\mathrm{GT}(K)\big). \tag{D.23}$$

This is [DG24, Theorem5.2, Corollary5.4 and Proposition5.5]. It includes all finite quotients of the braid group, including the nonnilpotent ones. Consequently the finite Lie and CM solutions in D.11 and D.17 specify candidates and concrete finite tests, but their realization in $\widehat{\mathrm{GT}}_0$ requires the intersection in (D.23).

Proof. One inclusion is reduction of a genuine element. For the other, the fibres over a fixed member of the right side form an inverse system of nonempty finite sets indexed by all finer quotients. Any finite collection is dominated by their intersection, so the system has the finite intersection property. Compactness gives a compatible family. Theorem5.2 identifies these compatible pairs with the profinite group of hexagon-preserving invertible automorphisms, and Proposition5.5 identifies that group with Harbater–Schneps' $\widehat{\mathrm{GT}}_0$. Surjectivity in each finite quotient supplies the invertibility condition; it cannot be omitted. This proves the equality, including the exact hypothesis that the pro-ℓ lift of D.12 alone does not supply. □

D.23. The exact coefficient map from the profinite universal measure to every rational Kummer character [Proved]

Use the positive interval paths and the Ihara convention of D.20. Write $c = [x,y]$, $\Lambda = \widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}^2]]$, and let s,t act on $\widehat{F}_2'/\widehat{F}_2'' = \Lambda c$ by conjugation by x,y. For $g \in G_{\mathbb{Q}}$ put $f_g \bmod \widehat{F}_2'' = u_g c$ and

$A_\lambda(s) = (s^\lambda - 1)/(s-1)$, where $\lambda = \chi(g)$. Then the universal measures of D.5 satisfy the exact identities

$$F_g = A_\lambda(s)A_\lambda(t)(1 + (s-1)(t-1)u_g), \qquad B_g = 1 + (s-1)(t-1)u_g. \tag{D.24}$$

For a positive integer n, reduce u_g by $t = 1$, $s^n = 1$, and write $u_{g,n} = \sum_{j=0}^{n-1} a_j(g)s^j \in \widehat{\mathbb{Z}}[\mathbb{Z}/n]$. The coefficient map is

$$\kappa_n(g) = a_0(g) - a_{n-1}(g). \tag{D.25}$$

Consequently the homomorphism of scan157 is recovered from (χ, F) by

$$g \longmapsto \left(\chi(g), \frac{\chi(g)-1}{2} - (a_0(g) - a_1(g))\right), \tag{D.26}$$

where the two coefficients in (D.26) use $n = 2$. For $p \neq \ell$, the extraction map in (D.25) on the $\mathbb{Z}_\ell$ coefficient component does not factor through the quotient with only pro-ℓ deck groups; the proof gives an explicit element of its kernel on which extraction is1. For every odd ℓ, this failure already occurs on actual Galois elements: a pro-ℓ tame-inertia subgroup at2 has trivial full cyclotomic character and trivial normalized pro-ℓ action on $\pi_1(\mathbb{P}^1 - \{0, 1, \infty\})$, but its $\kappa_{2,\ell}$ image is all of $\mathbb{Z}_\ell$.

Proof. All divisions in (D.24) are well-defined continuous operations on their stated images. The product $(s-1)(t-1)$ is a non-zero-divisor by Otsubo's Corollary6.2, hence its individual factors are also non-zero-divisors. The finite geometric sums defining A_m for integers m extend by continuity to $\lambda \in \widehat{\mathbb{Z}}$. For a unit λ, $A_\lambda(s)A_{\lambda^{-1}}(s^\lambda) = 1$, by multiplication by $s-1$ and cancellation, so A_λ is a unit. The inverse on a closed divisible image is continuous because the completed group ring is compact and Hausdorff.

In additive notation for the derived module, $f_g^{-1}y^\lambda f_g = (t^\lambda - 1)u_g c \cdot y^\lambda$. The commutator identities give

$$[x^\lambda, y^\lambda] = A_\lambda(s)A_\lambda(t)c, \qquad [x^\lambda, (t^\lambda - 1)u_g c] = (s^\lambda - 1)(t^\lambda - 1)u_g c.$$

This proves the first identity of (D.24); Otsubo's comparison (D.5) proves the second. To fix the path convention, the lift of $[x, y]$ along the positive interval has homology chain $(s-1)(1-t)\delta = -(1-s)(1-t)\delta$. Thus it is the negative of Otsubo's Pochhammer generator. The factor is the constant -1, so their Galois multipliers agree for every χ, not only on the cyclotomic kernel. The leading derivation convention in D.11 uses $D(y) = [f, y]$, so its f is the negative of the leading term of this Ihara conjugator; this accounts for the sign $-XYp$ in D.21.

Consider the Schreier generators $e_j = x^{-j}yx^j$ of H_n, besides x^n. In H_n^{ab}, $c = e_{n-1} - e_0$. Conjugation by x^a sends this to $e_{-a-1} - e_{-a}$, with indices modulo n; conjugation by y acts trivially on this abelianization. Therefore the coefficient of y in $\phi_n(u_g c)$ is $a_{n-1} - a_0$. By Ihara's exact equality (D.18), that coefficient is $-\kappa_n(g)$, proving (D.25). Formula (D.26) now follows from the chosen roots $(-1/2)^{1/m} = \zeta_{2m}/2^{1/m}$ and $\kappa_{-1} = (\chi - 1)/2$. The formulas give a continuous map from the actual metabelian action to the affine image of D.1, with no assertion that arbitrary coefficients satisfy Galois relations.

Here is the promised prime-to-ℓ kernel element. In $\mathbb{Z}_\ell[[\widehat{\mathbb{Z}}^2]]$, take the normalized Haar idempotent e_p of the $\mathbb{Z}_p$ factor of the first deck group. Its images at finite p^r-levels are $p^{-r}\sum_{j=0}^{p^r-1} s^j$; these are compatible since p is a unit in $\mathbb{Z}_\ell$. Projection to the pro-ℓ deck quotient sends e_p to1. At $t = 1, s^p = 1$ its image is $p^{-1}\sum_{j=0}^{p-1} s^j$. Hence $1 - e_p$ lies in the kernel of the pro-ℓ deck projection but its coefficient extraction is1. This proves the exact failure of that factorization on the ambient module.

For the asserted arithmetic witness, take an odd ℓ and a pro-ℓ Sylow subgroup J of inertia at2. Its tame quotient is $\mathbb{Z}_\ell(1)$. Every root of unity of odd order lies in the maximal unramified extension of $\mathbb{Q}_2$, and the action on the 2-power roots has pro-2 image. Therefore $\chi(J) = 1$. The extensions obtained by adjoining $2^{1/\ell^r}$ over the maximal unramified field are totally tamely ramified of degree ℓ^r; thus $\kappa_{2,\ell}(J) = \mathbb{Z}_\ell$. The pro-$\ell$ outer action of the thrice-punctured line is unramified at2, as in E.3 and [HM03, §8.4]. It follows that the normalized Ihara action of any $g \in J$ is inner. An inner automorphism fixing x has conjugator in the centralizer $\langle x \rangle$. Comparing its action on y with $f_g^{-1} y f_g$, where $f_g \in F'_\ell$, shows that f_g differs from a power of x by a power of y. Abelianization forces both powers to vanish, so this normalized action is the identity. Hence its pure pro-ℓ universal measure is1, while the full profinite coefficient extraction ranges over $\mathbb{Z}_\ell$. This proves the asserted failure on actual Galois elements, even when the whole cyclotomic character is retained.

The program `d_kummer_measure.py` independently computes abelian Fox derivatives and Schreier coefficients on160 explicit free words at all24 levels $1 \le n \le 24$, giving3840 exact checks of (D.25)'s coefficient identity. It also checks210 commutator-action identities for positive and negative integer exponents, which extend continuously as above. Its output does not substitute computation for the Galois theorem (D.18); that theorem supplies the arithmetic meaning of the checked coefficients. □

An additional full profinite lifting test

D.24. What unrestricted profinite lifting supplies, with an exact peripheral failure [Known]

Put $P = \widehat{F}_2/\widehat{F}_2''$. The canonical homomorphism

$$\mathrm{Aut}(\widehat{F}_2) \longrightarrow \mathrm{Aut}(P)$$

is surjective. Its kernel contains an automorphism which does not preserve the conjugacy class of the peripheral subgroup generator x, although it acts identically on the entire metabelian quotient. Thus the unrestricted lifting theorem supplies lifts in the displayed source; it supplies no peripheral conclusion for the chosen lift. The existence of peripheral lifts for the additional directions in D.13 remains a separate unsolved step of that calculation.

Proof. We use the profinite Gaschütz lemma in Bleher–Chinburg–Lubotzky, Lemma 2.3 and Corollary 2.4 [BCL22]. Applied to a quotient of a two-generated free profinite group, it lifts any generating pair of the quotient to a generating pair of the free profinite group. For an automorphism of the quotient, lift the images of x, y in this way. The resulting endomorphism of $\widehat{F}_2$ is surjective and hence invertible, by Hopficity of finitely generated profinite groups. This proves the asserted surjectivity and explains the imported theorem in our precise setting.

Here is a lift on which the peripheral condition fails. Let

$$q : \widehat{F}_2 \longrightarrow A_5, \qquad q(x) = (12345), \quad q(y) = (123),$$

and let $t = (12) \in S_5$. The map

$$(p, q) : \widehat{F}_2 \longrightarrow P \times A_5$$

is onto. Both projections are onto, and Goursat's argument identifies any proper subdirect product with a nontrivial common quotient of P and A_5. The latter group is simple and perfect, whereas every quotient of P has trivial second derived subgroup. There is no such common quotient, proving the assertion for this actual map.

The continuous automorphism $\mathrm{id}_P \times \mathrm{Ad}(t)$ of $P \times A_5$ lifts by the same lemma to an automorphism α of $\widehat{F}_2$. It lies in the kernel of the displayed map on automorphism groups. The two elements $q(x)$ and $tq(x)t^{-1}$ belong to the two different conjugacy classes of five-cycles in A_5. Consequently $\alpha(x)$ cannot be conjugate to x in $\widehat{F}_2$. A loop-preserving lift here would have multiplier one, because its action on $P^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$ is the identity. It would therefore have to send x to a conjugate of x, which this lift does not do.

The program `d_lifting.py` verifies the order sixty quotient, generation by the specified pair, simplicity by the normal closures of all five conjugacy classes, perfectness, and the two disjoint classes of twelve five-cycles. It also checks all 3600 multiplication identities for $\mathrm{Ad}(t)$. This example diagnoses the missing condition in the attempted unrestricted-lifting route. It makes no claim that a different lift of a given metabelian automorphism cannot preserve all three loops. □

The CM field and explicit nonarithmetical finite shadows

D.25. A finite CM entanglement field and nonarithmetical GT0 shadows [Proved]

Put $r_0 = (3/4)^{1/4}$ and $M' = \mathbb{Q}(i, r_0)$. There are exact equalities

$$\mathbb{Q}(E_3[4]) = M' = \mathbb{Q}(x(E_4[3])), \qquad M'\mathbb{Q}^{\mathrm{ab}} = M\mathbb{Q}^{\mathrm{ab}}, \tag{D.25.1}$$

where $E_3 : y^2 = x^3 + 1$, $E_4 : y^2 = x^3 - x$, and M is D.16's field. The projective $E_3[12]$ representation recovers the full $E_3[4]$ representation: choose its unique global sign fixing the rational point $(0, 1)$ of order three. Thus the extra relation of D.16 is already detected by T mod 12 and W mod 3, with this sign rule.

Here is a finite refinement of the matrix problem, including generation and the two hexagons. Write

$$B = \langle s, r : s^2 = r^3 = 1 \rangle, \quad Q = \mathrm{SL}_2(\mathbb{Z}/12)/\{\pm I\}, \quad N = \ker(B \to Q), \quad G_j = B/[N, N]N^j.$$

Use $s \mapsto \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $r \mapsto \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ projectively. Then $|Q| = 576$, N is free of rank97, and $|G_j| = 576j^{97}$ for $j = 2, 3, 6$. Among the48 pairs of matrix conjugators with multiplier one and $V_{12} = I$,24 satisfy the CM relation and24 violate it. Exact computation in G_2 and G_3 gives

condition	CM-compatible	extra CM coset
matrix conjugators	24	24
cusp relation in G_2	12	12
cusp relation in G_3	24	24
automorphisms of G_6	12	12.

(D.25.2)

Each surviving row has an explicit certificate whose induced matrix on N^{ab}/p has rank97 for $p = 2, 3$. The twelve extra pairs yield twelve distinct nonarithmetical finite $\widehat{\mathrm{GT}}_0$ shadows for the pullback of $[N, N]N^6$ to B_3. One representative has

$$A_{12} = I, \qquad \beta_{12} = \begin{pmatrix} 9 & 8 \\ 4 & 9 \end{pmatrix}, \qquad V_{12} = I. \tag{D.25.3}$$

Here "nonarithmetical" means that no element of $G_{\mathbb{Q}}$ has that finite shadow. Whether any of these shadows is genuine, that is, comes from an actual full profinite $\widehat{\mathrm{GT}}_0$ element, is not established.

Proof. For a nonzero two-torsion point of E_3 with x-coordinate $e \in \{-1, -\zeta_3, -\zeta_3^2\}$, the coordinates of its halves satisfy $x = e(1 \pm \sqrt{3})$. Their y-coordinates have squares $-9 \pm 6\sqrt{3}$. For the plus sign choose y with

$$y = \sqrt{3}(\sqrt{3} - 1)r_0.$$

The other square root is obtained using the product $3\sqrt{-3}$, up to sign. These formulas prove $\mathbb{Q}(E_3[4]) = \mathbb{Q}(i, r_0)$ in both directions. For E_4 the three-division polynomial is $3x^4 - 6x^2 - 1$. Choose two roots a, b with $a^2 = 1 + 2/\sqrt{3}$, $b^2 = 1 - 2/\sqrt{3}$ and $ab = i/\sqrt{3}$. Then $w = a + ib$ satisfies $w^4 = 4/3$; conversely

$$a = (w + 2/w)/2, \qquad b = (w - 2/w)/(2i).$$

The splitting field is therefore the same M'. Since $r_0 = 3^{1/4}/\sqrt{2}$, adjoining the abelian field $\mathbb{Q}(\sqrt{2})$ proves the second equality in (D.25.1).

The single global sign in this comparison is essential. For $P \in E_3[4]$ and $Q_0 = (0, 1) \in E_3[3]$, the addition law gives

$$y(P) = \frac{x(P)^2}{4}\big(x(P - Q_0) - x(P + Q_0)\big).$$

The points on the right belong to $E_3[12]$, proving directly that its projective coordinate field contains the full four-torsion field. Equivalently every Galois matrix on $E_3[3]$ fixes Q_0, so exactly one of the two simultaneous signs on $E_3[12]$ has that property. The separate projective $E_3[4]$ field is only $\mathbb{Q}(\zeta_{12})$; the displayed addition formula is the precise map supplying the missing sign.

We describe the finite calculation so that its certificates can be verified without enumerating $576 \cdot 6^{97}$ elements. Let $F = \ker(B \to S_3)$ and $H = q(F) \subset Q$; then $|H| = 96$. The graph $N\backslash\mathcal{T}$ of the Bass–Serre tree of $C_2 * C_3$ has edges Q and vertices $Q/\langle s\rangle \amalg Q/\langle r\rangle$. There are576 edges and480 vertices. A spanning tree gives97 integral fundamental cycles, a basis of N^{ab}. Write e_q for the oriented edge from its s-vertex to its r-vertex. The path cocycle into $\mathbb{Z}[Q]$ is

$$P(s) = 0, \quad P(r) = e_1 - e_r, \quad P(gh) = P(g) + q(g)P(h). \tag{D.25.4}$$

Its restriction identifies N^{ab} with the cycle group. Reduction modulo j realizes G_j faithfully as the image of B in $(\mathbb{Z}/j)[Q] \rtimes Q$.

For $A \in C_H(r)$ and $\beta \in C_H(s)$ choose the recorded paths P_A, P_β. There are respectively six and eight such choices. A lift of the cusp-preserving relation

$$\beta s \beta^{-1} A r A^{-1} = sr$$

exists modulo p exactly when

$$(1-s)P_\beta + s\big((1-r)P_A + (A-1)P(r)\big) \in (1-s)C_p + s(1-r)C_p, \quad C_p = H_1(N\backslash\mathcal{T}, \mathbb{F}_p). \tag{D.25.5}$$

This is an explicit linear system. The right side has rank96 at $p = 2$ and97 at $p = 3$. Its solution counts are those in (D.25.2). For every surviving pair the code gives corrections $a, b \in C_p$ and checks generation as follows. The induced equivariant map on edges sends

$$e_q \longmapsto q\big(P_A + a - P_\beta - b + e_A\big).$$

Its restriction to the97 fundamental cycles has rank97 in each characteristic. The homomorphism already induces the identity on Q, so it is an automorphism of G_p. Chinese remainders combine the two recorded corrections into single conjugators in G_6, giving an automorphism there. The code also multiplies the resulting group elements in the semidirect model and checks

both hexagon identities directly. All spanning-tree choices, matrix pairs, cycle coefficients and ranks are reproducible from `d_cm_shadow.py`; its JSON retains the certificates.

The CM test in these coordinates is particularly small. Use the unique representative of an element of H whose modulo-four image lies in the four-group generated by U^2, W^2. For every one of the six A choices, $A_4 = I$. The rational order-three line is $\ker(R - I)$ modulo3, spanned by $(1, -1)$. If A_3^{-1} acts on this line by -1, the global-sign normalization changes $T_4 = I$ to $-I$, and the restriction to M' sends r_0 to $-r_0$; otherwise it fixes r_0. On the other factor β_3^{-1} is in $\{\pm I, \pm S\}$; its class is nontrivial exactly for $\pm S$, again sending r_0 to $-r_0$. Comparing these two bits gives the24/24 split. In (D.25.3) the bits are zero and one, respectively.

The conjugator bits are invariants of the finite action, not arbitrary choices in a matrix centralizer. The same linear computation of (D.25.4) gives

$$q\big(C_{G_2}(s) \cap \operatorname{im} F\big) = 1, \qquad q\big(C_{G_3}(r) \cap \operatorname{im} F\big) = 1. \tag{D.25.6}$$

For example the first test asks whether $(1 - s)P_\beta$ lies in $(1 - s)C_2$; only $q(\beta) = 1$ survives. The second uses $(1 - r)P_A + (A - 1)P(r)$ and $(1 - r)C_3$, again with only the identity surviving. Thus G_6 uniquely determines the reductions of both normalized conjugators in Q.

Finally we give the exact passage to finite GT shadows. Put $u = (sr)^{-1}$, $v = sus$, $x = u^2$, $y = v^2$; these are the images of the standard braid generators and their squares under $B_3 \to B_3/Z(B_3) = B$. Let α be one of the finite automorphisms just constructed. It fixes u and satisfies $\alpha(s) = \beta s \beta^{-1} = sf$, where $f = s\beta s\beta^{-1}$. Choose a lift of β to the discrete free group $F = \langle x, y\rangle$ and write its abelianization as $a_0 x + b_0 y$. Since s swaps x, y, $f^{\mathrm{ab}} = (b_0 - a_0)(x - y)$. Set $k = b_0 - a_0$, $\beta' = x^k\beta$, and $f' = s\beta' s(\beta')^{-1}$. Then $f' \in [F, F]$. The gauge $\alpha' = \mathrm{Int}(x^k)\alpha$ still fixes u and satisfies

$$\alpha'(s) = sf', \qquad \alpha'(r) = (f')^{-1}r, \qquad \alpha'(v) = (f')^{-1}vf'. \tag{D.25.7}$$

The last two equalities, with $r = uv$ and $s = uvu$, are exactly the two $m = 0$ hexagons (2.1)–(2.2) of [BDHP26, Definitions2.1 and2.3]. The pullback $\widetilde{N}$ of $[N, N]N^6$ to B_3 is normal, finite index, and contained in PB_3, since $G_6 \to S_3$ is the standard permutation quotient. Here $\widetilde{N}_{\mathrm{ord}} = 36$: the cusp sr has orders 24,36,72 in G_2, G_3, G_6, respectively, its square has order36 in G_6, and the braid center is killed. Thus $m = 0$ gives the required unit multiplier one, f' has the required commutator image, and α' being an automorphism gives surjectivity. This proves that $(0, f' \bmod \widetilde{N})$ is a finite GT shadow.

The gauge multiplies A, β, V on the left by the same x^k, so it fixes $T = A^{-1}V$ and $W = \beta^{-1}V$. Equation(D.25.6) makes their reductions recoverable from the finite shadow. The twelve different extra pairs have different such coordinates, hence give distinct shadows. No Galois element can give one of them: its two restrictions to the single field M' in (D.25.1) must agree. This proves nonarithmeticality without assuming that an arbitrary finite shadow lifts to $\widehat{\mathrm{GT}}_0$. For these specific shadows the remaining genuine-lift test is precisely the intersection over all finer finite quotients in D.22. □

Isolated shadows and a complete tower of homology refinements

D.26. The CM witnesses survive every abelian-kernel refinement [Proved]

Keep B, Q, N, F and G_j from D.25, and let $\widetilde{N}_j \lhd B_3$ be the inverse image of $[N, N]N^j$. Every $\widetilde{N}_j$ is an isolated object of the gentle GT-shadow groupoid: every shadow with that target induces an automorphism of G_j.

Put $C = N^{\mathrm{ab}} \simeq \mathbb{Z}^{97}$, and let s, r act on C by conjugation. For $t = sr$ define $c = [t^{12}] \in C$. The integral maps

$$L : C \oplus C \longrightarrow C, \quad L(b, a) = (1 - s)b + s(1 - r)a, \qquad \widetilde{L}(b, a, h) = L(b, a) - hc \tag{D.26.1}$$

have Smith invariant factors $(1^{96}, 2)$ and (1^{97}), respectively. In particular $\operatorname{coker} L = C_2$ and $\widetilde{L}$ is onto over $\mathbb{Z}$.

There is an exact sequence and a specified quotient map

$$1 \longrightarrow \widehat{\mathbb{Z}}^{97} \longrightarrow \mathcal{G} := \varprojlim_{j} G_j \longrightarrow Q \longrightarrow 1, \qquad \widehat{B} \twoheadrightarrow \mathcal{G} = \widehat{B}/\overline{[\widehat{N}, \widehat{N}]}. \tag{D.26.2}$$

Each of D.25's twenty-four certified automorphisms of G_6 lifts to a continuous automorphism of $\mathcal{G}$ inducing the identity on Q and sending t to t^λ, for some $\lambda \in \widehat{\mathbb{Z}}^\times$ with $\lambda \equiv 1 \pmod{72}$. A single cusp gauge supplies a compatible family of GT shadows at all $\widetilde{N}_j$. For each of the twelve extra CM pairs, its member at level six can be chosen to be one of D.25's nonarithmetical shadows, with $m = (\lambda - 1)/2 \equiv 0 \pmod{36}$. Thus all twelve extra CM pairs survive the entire family of abelian-kernel refinements of N.

This family does not exhaust the finite quotients in D.22: there is an explicit finite quotient of B below G_6 whose kernel does not contain $[N, N]$. The existence of a genuine full profinite GT_0 lift of these shadows remains undecided.

Proof. First we verify isolation directly from finite shadow equations. Let (m, f) be a shadow at $\widetilde{N}_j$, put $\lambda = 2m + 1$, and choose a representative $f \in [F, F]$. The hexagons give a surjection $\phi : B_3 \to G_j$ and send the braid center to its λth power, which is trivial in G_j. Hence ϕ factors through B. The cusp t has order $12j$ in G_j: its image in Q has order twelve, and the cycle c is primitive in C. The last assertion also follows from the integral cycle coordinates in the calculation below, one of which is one and all the others are zero. Thus $(\widetilde{N}_j)_{\mathrm{ord}} = 6j$ and the shadow's multiplier is a unit modulo twelve.

Evaluate the same word f in the linear matrices U^2, W^2 of A.17 modulo twelve. Its reduction modulo four is I, since U^2, W^2 generate an abelian four-group modulo four. Consequently the linear matrices $A = U^\lambda$ and $D = f^{-1} W^\lambda f$ satisfy the braid identity with positive sign. Indeed the projective identity first gives $ADA = \pm DAD$, and modulo four the choice $\lambda \equiv \pm 1$ gives the positive identity for U^λ, W^λ; an invertible matrix cannot equal its negative modulo four. The pair A, D generates all of $\mathrm{SL}_2(\mathbb{Z}/12)$: its generated subgroup surjects onto Q, and a proper such subgroup would be a complement to $\{\pm I\}$. That complement cannot exist, because both lifts $\pm S$ of the order-two element $s \in Q$ have square $-I$.

Write $D = \left(\begin{smallmatrix} a & b \\ c_0 & d \end{smallmatrix}\right)$. The trace, determinant, braid identity and generation now give

$$a + d = 2, \quad ad - bc_0 = 1, \quad c_0(1 + \lambda c_0) = 0, \quad c_0 \in (\mathbb{Z}/12)^\times.$$

Thus $c_0 = -\lambda^{-1}$ and the matrix $V = \left(\begin{smallmatrix} \lambda & \lambda(1-a) \\ 0 & 1 \end{smallmatrix}\right)$ conjugates U, W to A, D, respectively. This repeats the elementary matrix argument of A.17 using only the finite shadow hypotheses. It proves $q\phi = \mathrm{Int}(V)q$. In particular $\phi(N) \subset C/jC$. Since this subgroup is abelian of exponent j, ϕ kills $[N, N]N^j$. It therefore induces a surjective endomorphism of the finite group G_j, hence an automorphism. Its kernel is precisely $\widetilde{N}_j$. This is isolation in [DG24, Definition3.13].

For the integral assertion use the same oriented graph and spanning tree as in D.25. The ninety-seven fundamental cycles form a matrix with the identity matrix in the non-tree-edge coordinates, so projection to those coordinates is an exact integral inverse on C. The columns of L are $(1 - s)C$ and $s(1 - r)C$ in this basis. The program `d_cm_refine.py` computes their integral Smith form as $(1^{96}, 2)$. Appending the column $-P(t^{12})$ gives Smith form (1^{97}); an independent Hermite-normal-form calculation of this augmented matrix returns I_{97}. The program retains the complete integer coordinates and reproduces the spanning tree. Thus these are integer-image computations, not deductions from ranks modulo two and three.

We construct the lift, retaining the multiplier's full residue. Fix one of the twenty-four certified matrix pairs A_0, β_0 and choose the fixed integral paths P_A, P_β of D.25. Let $E \in C$ be

its integral cusp error, the left side of (D.25.5). With cycle corrections (b, a) the exact cusp equation for multiplier $\lambda = 1 + 12h$ is

$$E + L(b, a) - hc = 0. \tag{D.26.3}$$

To see the exponent, $P(t^{1+12h}) = P(t) + hc$, since $q(t)c = c$. This proves (D.26.3) in every finite quotient, and by continuity for $h \in \mathbb{Z}_p$.

At $p = 2, 3$ begin with D.25's correction modulo p and $h = 0 \pmod p$. Suppose coefficients have been constructed modulo p^k; their error in (D.26.3) is divisible by p^k. Surjectivity of $\widetilde{L}$ supplies a correction $(\delta b, \delta a, \delta h)$ modulo p equal to the negative error divided by p^k. Add p^k times that correction. This constructs compatible solutions over $\mathbb{Z}_p$. Their induced matrix on $C \otimes \mathbb{Z}_p$ reduces to D.25's invertible matrix modulo p, hence is invertible. The associated map on the quotient Q is the identity, so it is an automorphism of the whole extension. The initial values $h = 0 \pmod 2$ and $h = 0 \pmod 3$ are preserved. Accordingly the combined multiplier satisfies $\lambda \equiv 1 \pmod{72}$, not just modulo twelve. As a check of the power and sign conventions, the program carries every one of the twenty-four solutions through exponents $2, 4, 8, 16$ and $3, 9, 27, 81$, and directly verifies both general-m hexagons in their faithful group models: there are384 successful identity checks.

For $p > 3$ take the identity automorphism on $\mathcal{G}_p = \varprojlim_k G_{p^k}$, with multiplier one, but choose conjugators with the required images A_0, β_0 in Q. Such choices exist. For example, an arbitrary lift a_0 of $A_0 \in C_Q(r)$ has $a_0 r a_0^{-1} r^{-1} = z \in C \otimes \mathbb{Z}_p$ and $(1 + r + r^2)z = 0$. Multiplication by three is invertible, and the identity $(1 - r)(2 + r) = 3 - (1 + r + r^2)$ shows that $z = (1 - r)((2 + r)z/3)$. Adjusting a_0 by the negative of this cycle gives a lift centralizing r. The same argument for s uses $(1 - s)z = 2z$ when $(1 + s)z = 0$ and division by two. These are compatible p-adic choices, and both belong to the preimage of $H = q(F)$.

The extensions $\mathcal{G}_p$ have the same quotient Q, and their fibre product over Q is precisely $\mathcal{G}$. This follows either from the path model or from Chinese remainders on C/jC. The constructed automorphisms and conjugators therefore assemble to $\alpha \in \mathrm{Aut}(\mathcal{G})$ and $A, \beta \in \overline{F}$, with $\alpha(s) = \beta s \beta^{-1}$, $\alpha(r) = ArA^{-1}$ and $\alpha(u) = u^\lambda$. Here $\overline{F}$ denotes the image of $\widehat{F}$ in $\mathcal{G}$. Its abelianization is $\overline{F}^{\mathrm{ab}} = \widehat{\mathbb{Z}}x \oplus \widehat{\mathbb{Z}}y$, because the kernel $\overline{[\widehat{N}, \widehat{N}]}$ is contained in $\widehat{F}'$.

Write $\beta^{\mathrm{ab}} = ax + by$, set $k = b - a$, and conjugate α by x^k. With $\beta' = x^k\beta$ and $f' = s\beta' s(\beta')^{-1}$, the same abelianization calculation as in D.25 gives $f' \in \overline{F}'$. The resulting automorphism satisfies

$$\alpha'(u) = u^\lambda, \qquad \alpha'(v) = (f')^{-1} v^\lambda f', \qquad \alpha'(s) = sf' = (f')^{-1} s. \tag{D.26.4}$$

Since $r = uv$, $s = uvu = vuv$, and $m = (\lambda - 1)/2$, its two hexagons are exactly

$$u^\lambda (f')^{-1} v^\lambda f' = (f')^{-1} r x^{-m}, \qquad (f')^{-1} v^\lambda f' u^\lambda = vuy^{-m} f'. \tag{D.26.5}$$

Reduction to every G_j gives a GT shadow: the center is killed, the unit condition holds, f' has commutator image, and the induced map is an automorphism. The single global pair makes these reductions compatible.

For precision concerning D.25's finite gauge, choose its discrete representative of β to agree with the constructed profinite β both in G_6 and in $\overline{F}^{\mathrm{ab}}/36$. Such a representative exists because F is dense in $\overline{F}$ and this is a finite joint quotient. Then its integer $b - a$ agrees with k modulo36, the order of x in G_6. Thus its D.25 gauge is exactly the reduction of (D.26.4), with $m = 0 \pmod{36}$. The gauge preserves $T = A^{-1}V$ and $W = \beta^{-1}V$. Consequently all twelve extra CM pairs remain nonarithmetical at level six in the compatible tower. This specifies the finite gauges that extend; it does not make an assertion about an independently chosen gauge.

Finally we locate the remaining finite tests by an actual quotient. Choose a free basis of the rank97 group N and map its first two generators to $(12), (23) \in S_3$ and the others to the

identity. Let J be the normal core in B of the intersection of this map's kernel with $[N,N]N^6$. Then $J \triangleleft B$ has finite index, $J \subset [N,N]N^6$, but $[N,N] \not\subset J$, since the commutator of the first two basis elements has nontrivial image in S_3. The quotient map $B \to B/J$ therefore cannot factor through any G_j: every such factorization would kill $[N,N]$. This also proves the kernel in (D.26.2) is nontrivial and that the homology tower is not cofinal in the system of D.22.

The source search supplies no additional lifting assertion for these quotients. In particular isolation gives reduction *homomorphisms*, as stated in [DG24, Remark3.16]; the all-finer-quotient criterion is its Theorem5.2 and Corollary5.4. The abelian sufficient criterion in [DLL24, TheoremB.2] requires the pure quotient to be abelian. Here its image already surjects onto $H \simeq C_2^2 \times \mathrm{SL}_2(\mathbb{F}_3)$: the isomorphism sends the unique lift with modulo-four image in V_4 to its modulo-four and modulo-three components. The latter group's second derived subgroup has order two. Similarly the dihedral quotients of [BDHP26, Remark3.2] embed their F images into products of three dihedral groups and are metabelian. No such pure quotient can surject onto this H, since surjections preserve the derived-series images. These explicit maps explain why those sufficient families do not supply the remaining lift. □

E. The next approximations and exactness on coverings

We write $F = \langle x, y \rangle$ for the discrete group, $F^{(i)}$ for its derived series, and $\gamma_i F$ for its lower central series, with $F^{(0)} = \gamma_1 F = F$. All commutators in this section are $[a,b] = aba^{-1}b^{-1}$. The puncture used to define depth is the puncture labelled 1; it is additional data. The manuscript anchors are §51.16, folios 733v–736v, scans 172–178, especially (371)–(379). The arithmetic model in those passages is $X_n : x_0^n + x_1^n + x_\infty^n = 0$, with U_n obtained by deleting the coordinate divisors. We retain that model over $\mathbb{Q}$.

Derived quotients and the depth filtration

E.1. The third derived quotient and its exact relation module [Proved]

Put $Q = F/F''$ and $R = \mathbb{Z}[Q]$. There is an exact sequence of discrete groups and an exact sequence of left R-modules

$$1 \longrightarrow F''/F''' \longrightarrow F/F''' \longrightarrow Q \longrightarrow 1, \qquad 0 \longrightarrow F''/F''' \longrightarrow R^2 \xrightarrow{d} R \xrightarrow{\epsilon} \mathbb{Z} \longrightarrow 0,$$

where $d(a,b) = a(\bar{x}-1) + b(\bar{y}-1)$, with right factors in this formula and the regular left action on each term. This is an explicit module description over the noncommutative ring R.

Proof. The cover of the two-loop graph corresponding to the normal subgroup F'' has vertices Q and oriented edges $(q,x), (q,y)$ from q to $q\bar{x}, q\bar{y}$. Its cellular differential is the displayed map. The graph is connected, so its reduced zeroth homology vanishes. Its fundamental group is F'', and its first homology is $F''/[F'',F''] = F''/F'''$. Lifting paths identifies conjugation by F with the left deck action of Q. These identifications give every arrow, its kernel and its image. The group sequence is the quotient sequence for the nested normal subgroups $F''' \subset F''$.

For the algebraic constructions below we use the rational relative unipotent completion over a specified finite quotient, constructed in E.14. For an unbased finite-dimensional representation problem one can also use the affine proalgebraic group with coordinate Hopf algebra the matrix coefficients of all finite-dimensional rational representations of F/F'''. Evaluation is the canonical group map into its rational points, and factoring matrix coefficients proves its universal property. These constructions specify an envelope and its comparison maps; they do not require a finite-index abelian subgroup of F/F'''. □

E.2. The depth-two Lie algebra and its comparison with the derived quotient [Proved]

Let $L = \mathbb{L}_{\mathbb{Q}}(x, y)$, graded by total degree, and let $D^r L$ be the span of Lie monomials with at least r occurrences of y. Put $v_i = \mathrm{ad}_x^i(y)$ and $V = \bigoplus_{i \geq 0} \mathbb{Q} v_i$. There is an isomorphism of graded Lie algebras

$$L/D^3 L \simeq \mathbb{Q}x \ltimes (V \oplus \Lambda^2 V).$$

Here $[v_i, v_j] = v_i \wedge v_j$, the bracket of V with $\Lambda^2 V$ vanishes, and $[x, v_i] = v_{i+1}$; the action of x on $\Lambda^2 V$ is the induced derivation. In weight $n \geq 2$ the depth-two dimension is $\lfloor (n-1)/2 \rfloor$. The identity on x, y induces the surjection $L/L''' \twoheadrightarrow L/D^3 L$ with kernel $D^3 L/L'''$.

Proof. We give the elimination argument. Jacobi moves every occurrence of x in a bracket in the ideal generated by y onto a bracket entry. Consequently that ideal is generated as a Lie algebra by the v_i. In the tensor algebra, the leading words of the v_i are $x^i y$. The words obtained by concatenating these blocks have unique factorizations at the occurrences of y. Thus the homomorphism from the tensor algebra on the symbols v_i into $\mathbb{Q}\langle x, y \rangle$ is injective, by its triangular leading words. Its primitive Lie subalgebra is the free Lie algebra on the v_i. Keeping at most two such symbols yields $V \oplus \Lambda^2 V$. The derivation formula follows from Jacobi. A basis in weight n is $v_i \wedge v_j$ with $i < j$ and $i + j = n - 2$, which has the stated cardinality.

Every element of L' has depth at least one, hence $L'' \subset D^2 L$ and $L''' \subset D^4 L \subset D^3 L$. This proves the displayed quotient map and its kernel. The element $[y, [y, [y, x]]]$ has weight four and belongs to $D^3 L$; its tensor expansion is nonzero, whereas L''' has no weight-four part. Thus the kernel is nonzero. Also $[[x, y], [x, [x, y]]]$ survives in $\Lambda^2 V$, so the identity on x, y does not descend from L/L'' to $L/D^3 L$: its defining kernel L'' contains this explicitly nonzero image. Completion for the total-degree filtration gives the same maps of pronilpotent Lie algebras, and exponentiation gives maps of rational prounipotent groups. No integral exponential is used. □

E.3. The rational Galois calculation in depth two through weight fourteen [Known]

Normalize the depth-one generators by $\bar{\sigma}_{2m+1} = \mathrm{ad}_x^{2m}(y)$. For the positive-weight Lie algebra of the outer pro-ℓ Galois image, tensored with $\mathbb{Q}_\ell$, the depth-one dimensions are one in weights $3, 5, 7, 9, 11, 13$ and zero in the other weights at most fourteen. Its depth-two dimensions in weights $1, \ldots, 14$ are

$$0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 2.$$

The first relation among the depth-one brackets is

$$\{\bar{\sigma}_3, \bar{\sigma}_9\} - 3\{\bar{\sigma}_5, \bar{\sigma}_7\} = 0 \quad \text{in depth two.}$$

This is a statement after tensoring with $\mathbb{Q}_\ell$ and taking the specified filtrations; it asserts no integral saturation index.

Proof. We use Hain–Matsumoto, Theorem 8.4, its application in §8.4, Lemma 8.5 and Remark 8.6 of the opened version [HM03], and Brown's depth-two bracket formula and exact sequence (7.8), Example 7.2 [Bro21]. Here U_1 has $H^1_{\mathrm{et}}(U_{1,\overline{\mathbb{Q}}}, \mathbb{Q}_\ell(1)) = \mathbb{Q}_\ell^2$ with trivial Galois action: the two Kummer covers are those of t and $1 - t$. Its outer pro-ℓ action is unramified away from ℓ. The indicated theorem therefore identifies the weight-graded Lie algebra of its Zariski closure with the rationalized graded Galois image. The weight-one generator maps to zero by Lemma 8.5; the odd generators in weights at least three have nonzero depth-one coefficient, as recorded in Remark 8.6. This supplies actual Galois elements after rationalization, rather than only solutions of a linear equation.

The polynomial form of the depth-two bracket is

$$B_{a,b}(X,Y) = X^a\big(Y^b - (Y-X)^b\big) + (Y-X)^a\big(X^b - Y^b\big) + Y^a\big((Y-X)^b - X^b\big), \qquad a,b \text{ positive and even.}$$

One can derive this formula directly by substituting $D_f(x) = 0$, $D_f(y) = [y,f]$ into $[f,g] + D_f(g) - D_g(f)$ in the tensor algebra, and reading the two y-blocks from the right. The program `e_depth.py` does both expansions independently over the integers and compares every coefficient. The only pairs in weights $8, 10, 12, 14$ are, respectively, $(3,5)$, $(3,7)$, $(3,9)$, $(5,7)$, and $(3,11)$, $(5,9)$. Their exact ranks are $1, 1, 1, 2$. The other weights contain no pair.

For completeness of this span in depth two, every bracket word of length at least three in the Galois generators has depth at least three. A generator itself has a nonzero depth-one image in its weight; different generator weights cannot cancel in a weight-graded piece. Thus the generation theorem reduces the calculation exactly to the displayed pairs. The polynomial

$$X^2Y^2(X-Y)^3(X+Y)^3 = X^8Y^2 - 3X^6Y^4 + 3X^4Y^6 - X^2Y^8$$

satisfies the antisymmetry and the three-term period equation. Substitution in $B_{a,b}$ gives the displayed relation, also checked word by word in the free tensor algebra. Brown's exact sequence identifies its line with the weight-twelve cusp-form period line. This verifies the requested modular relation without making a freeness assertion for an integral Galois Lie algebra. □

E.4. What the metabelian representation forgets about these brackets [Proved]

Under the canonical Lie homomorphism $L \to L/L''$, every Ihara bracket of two elements of L' maps to zero. The depth-two image of $\{\bar\sigma_3, \bar\sigma_5\}$ is nonzero. Thus the specified comparison maps exhibit information already added in weight eight; the first dependence between these new bracket directions occurs in weight twelve as in E.3.

Proof. Write the derived module of L/L'' as $\mathbb{Q}[X,Y]c$, where $c = [x,y]$ and X, Y act by $\mathrm{ad}_x, \mathrm{ad}_y$. For $f = pc$ one has $D_f(c) = [x,[y,pc]] = XYpc$. The derivation D_f vanishes on the operators X, Y when applied to the derived module: its correction there is a bracket of two derived elements. Hence $D_f(qc) = XYpq\,c = D_g(pc)$ for $g = qc$, while $[f,g] = 0$. The Ihara bracket therefore vanishes. In E.3 the polynomial $B_{2,4} = 2X^5Y - 5X^4Y^2 + 5X^2Y^4 - 2XY^5$ is explicitly nonzero. Both claims concern the images of the same derivations through the two quotient maps just constructed. □

Localizers and finite coverings

E.5. Two-out-of-three and relative Theorem A [Proved]

For each of abelianization, the metabelian quotient, a fixed nilpotent quotient $G/\gamma_{c+1}G$ with $c \geq 1$, pro-p completion, and rational prounipotent completion, the class $W_{\mathfrak{a}}$ in the question satisfies two-out-of-three, retract closure, the terminal-object axiom and relative Theorem A for all small categories.

Proof. We first specify the tests determining each class. They are the classifying groupoids BH for, respectively, all abelian groups, all metabelian groups, all groups of nilpotence class at most c, all finite p-groups, and all groups $U(\mathbb{Q})$ of rational points of finite-dimensional unipotent rational algebraic groups. We include all discrete sets as tests to detect components. For a category C, functors $C \to BH$ invert every arrow and therefore factor uniquely through

its fundamental groupoid. For a connected pointed category the homotopy fiber of evaluation $\mathrm{Fun}(C, BH) \to BH$ is equivalent to the discrete groupoid $\mathrm{Hom}(\pi_1 C, H)$. The universal properties of the five approximations, followed by Yoneda, prove that a component-bijective functor is in $W_{\mathfrak{a}}$ precisely when it induces an equivalence of these functor groupoids for every indicated test. Conjugations cause no loss here: the evaluation fiber restores the pointed homomorphisms. Two-out-of-three and retract closure follow from equivalences of groupoids. A category with a terminal object has contractible nerve and passes every test.

Here is the relative argument. For $C \to S$, the Grothendieck construction of $s \mapsto C/s$ has objects $(s, c, u : f(c) \to s)$. Its forgetful functor to C has a section $c \mapsto (f(c), c, 1)$ and a natural transformation from section followed by forgetful to the identity, furnished by u. It therefore has the same fundamental groupoid as C. Functors from this construction to a groupoid T consist exactly of functors $C/s \to T$ with invertible coherent transition transformations along the arrows of S. This is the pseudolimit of the groupoids $\mathrm{Fun}(C/s, T)$. A pointwise equivalence of such diagrams induces an equivalence of pseudolimits: choose objectwise lifts, transport the transitions through full faithfulness, and use faithfulness to verify their coherence; the same argument applies to morphisms. A map $C \to D$ over S which passes every test on every comma category consequently passes the test on $C \to D$. This is the stated relative Theorem A. □

E.6. The exact comparison maps for quotient approximations [Proved]

Let $q : F \twoheadrightarrow G$ be any finite quotient, $N = \ker q$, and $V(H)$ one of H', H'', or $\gamma_{c+1}H$. There is a natural short exact sequence

$$1 \longrightarrow \frac{N \cap V(F)}{V(N)} \longrightarrow \frac{N}{V(N)} \longrightarrow \ker\left(\frac{F}{V(F)} \longrightarrow \frac{G}{V(G)}\right) \longrightarrow 1.$$

The comparison requested with target G itself is the specialization $V(G) = 1$ of this exact sequence. The map $F/V(F) \to G$ with composite q exists exactly when $V(G) = 1$.

Proof. Surjectivity of q gives $q(V(F)) = V(G)$, by the definitions of the verbal subgroups. The inverse image of $V(G)$ is $NV(F)$, so the kernel in the display is $NV(F)/V(F)$. The map from N onto it has kernel $N \cap V(F)$ and kills $V(N)$. This proves exactness and identifies every map. A factorization of q through $F/V(F)$ kills $q(V(F)) = V(G)$; the converse is the same calculation. The formulas thus also give an explicit replacement target for every quotient for which the arrow to G written in the question does not descend. □

E.7. All nontrivial finite quotients have a nonzero comparison kernel [Proved]

For every finite quotient $F \twoheadrightarrow G$ of order $d > 1$, each of the three comparison kernels in E.6 is nonzero. For abelianization it is a free abelian group of rank $d - 1$; the image is a rank-two lattice of index $|G^{\mathrm{ab}}|$ in F^{ab}. For the nilpotent quotient of class c its rational Hirsch rank is

$$\sum_{j=1}^{c} \bigl(w_j(d+1) - w_j(2)\bigr), \qquad w_j(r) = \frac{1}{j} \sum_{e \mid j} \mu(e) r^{j/e}.$$

Proof. The connected Cayley graph of G has d vertices and $2d$ edges. A maximal tree gives $d+1$ free generators of N, hence $N^{\mathrm{ab}} = \mathbb{Z}^{d+1}$. The abelian image and its index follow from E.6. A subgroup of a free abelian group is free, and the rank of the kernel is $(d+1) - 2 = d - 1$.

For the nilpotent case the domain is the free nilpotent group on $d + 1$ generators and the image is a finite-index subgroup of the free nilpotent group on two generators. Both are

torsion-free. Their lower central ranks are the Witt numbers: the tensor algebra identity $1/(1-rt) = \prod_{j\geq 1}(1-t^j)^{-w_j(r)}$, obtained by ordering a homogeneous Lie basis in the universal enveloping algebra, gives the displayed formula by logarithms and Möbius inversion. Hirsch rank is additive in a short exact sequence of torsion-free nilpotent groups. The first summand is already $d-1$.

For the metabelian case choose $a \in N \cap F'$ with nonzero image A in N^{ab}; such an element exists by the abelian calculation. Choose $t \in N$ whose abelian class T is linearly independent from A over $\mathbb{Q}$. In the free metabelian quotient of N the module N'/N'' injects into $\mathbb{Z}[N^{\mathrm{ab}}]^{d+1}$ by the covering-graph differential. It is therefore torsion-free over this integral domain. The image of $[t,a]$ is nonzero: its degree-two term is $T \wedge A$. The element

$$w = [a, tat^{-1}] \in N \cap F''$$

equals $[a,[t,a]]$ and has image $(A-1)[t,a]$ in that module for our convention $[u,v] = uvu^{-1}v^{-1}$. This image is nonzero. Thus $w \notin N''$, giving a specific construction of a nontrivial element of $(N \cap F'')/N''$ for every such quotient. For $d = 1$, $N = F$ and every comparison is the identity. □

E.8. The pro-p comparison and the precise exactness criterion [Proved]

Write $G^{(p)}$ for the maximal p-group quotient of the finite group G. There is a continuous surjection

$$\widehat{N}_p \longrightarrow \ker(\widehat{F}_p \longrightarrow G^{(p)}).$$

Its source is free pro-p of rank $|G|+1$ and its target is free pro-p of rank $|G^{(p)}|+1$. It is an isomorphism exactly for p-groups G. For p-groups the target is the open subgroup corresponding to the given cover. More explicitly, put $r = |G|+1$ and $s = |G^{(p)}|+1$. There are free pro-p bases $a_1,\dots,a_s,b_1,\dots,b_{r-s}$ in the source and $\bar{a}_1,\dots,\bar{a}_s$ in the target for which the comparison sends $a_i \mapsto \bar{a}_i$ and $b_j \mapsto 1$. Its kernel is exactly the closed normal subgroup generated by those $r-s$ elements b_j. This description is noncanonical; the original continuous comparison is canonical.

Proof. The image is the closure of N. The quotient by that closure is the maximal pro-p quotient of the finite group F/N, namely $G^{(p)}$. This proves surjectivity. For a normal subgroup of p-power index the induced pro-p topology is its full topology: the core in the ambient group of a normal subgroup of finite p-power index is the intersection of finitely many conjugates, and its quotient is an extension of finite p-groups. Completing therefore identifies $\widehat{N}_p$ with the open subgroup when G is a p-group. The Schreier rank calculation applied to finite p-quotients, or the covering tree, gives the two ranks. An isomorphism would identify their Frattini quotients, whose dimensions are these ranks, and hence forces $|G| = |G^{(p)}|$. This equality says precisely that G is a p-group. All arrows used here are continuous homomorphisms of compact pro-p groups.

For the kernel description, lift a target basis to elements $a_1,\dots,a_s$. Their classes in the source Frattini quotient are independent. Extend them to a basis of that vector space and choose lifts b_j. The pro-p basis theorem and Hopficity show that the resulting r elements form a free basis. The subgroup generated by the a_i maps isomorphically onto the target: the induced surjection of free groups of rank s is an isomorphism by Hopficity. For each b_j, choose the unique word $w_j(a_1,\dots,a_s)$ with its target image and replace b_j by $b_j w_j^{-1}$. The Frattini classes remain a basis, so these are again free generators. The quotient by their closed normal closure is now the free group on the a_i, proving the stated kernel and image, not merely their ranks. □

E.9. Ordinary rational completion of a finite-index subgroup [Proved]

For $|G| = d$, the canonical morphism of rational prounipotent groups $N^{\mathrm{un}} \to F^{\mathrm{un}}$ is surjective. Its map on abelianized Lie algebras is a surjection $\mathbb{Q}^{d+1} \to \mathbb{Q}^2$ with kernel of dimension $d-1$. It is an isomorphism exactly for $d = 1$. In suitable noncanonical free pronilpotent coordinates, its Lie algebra map sends X, Y to the two target generators and $z_1, \dots, z_{d-1}$ to zero. Its kernel is the closed ideal generated by the z_j; by free-Lie elimination it is the completed free Lie algebra on $\mathbb{Q}\langle X, Y\rangle \otimes \langle z_1, \dots, z_{d-1}\rangle$, where a word acts by iterated adjoints. Exponentiation is the exact kernel group. Every morphism from a connected unipotent rational algebraic group to the constant finite group G is trivial; thus the specified map to G in a kernel formula descends only for the trivial quotient. The preceding surjection is the comparison obtained from the trivial completion of G.

Proof. A homomorphism from F to a unipotent group has the same Zariski closure on N as on F: the quotient of the closures has finite order, while a unipotent group in characteristic zero is connected and admits no nontrivial finite quotient. Equivalently, every $f \in F$ has a positive power in N, and taking roots in the one-parameter unipotent subgroup containing its image recovers that image. The resulting map of completions is surjective. Both Lie algebras are completed free Lie algebras on their abelianizations, of ranks $d+1$ and two. The abelian comparison is the rationalization of E.7. A morphism from a connected scheme to a constant finite scheme is constant, which proves the last assertion and types the replacement map.

For the coordinate assertion, lift the target free Lie generators and complete their independent linear terms to a basis of the source abelianization. The homomorphism of completed free Lie algebras determined by these lifts is invertible: its map in each graded degree is invertible, and degreewise inversion converges. Its first two generators give a section of the target free Lie algebra. Subtract from every remaining generator the section of its image. This triangular change is again invertible and puts that image equal to zero. The kernel is consequently the closed ideal generated by the remaining z_j. Jacobi writes this ideal as the Lie algebra generated by the iterated X, Y adjoints of the z_j. In the free associative algebra their leading words have unique cuts at the letters z_j, proving freeness exactly as in E.2. This also computes the whole kernel for every finite quotient, including those for which the proposed map to the original G has no descent. □

E.10. Which finite regular coverings preserve the localizers [Proved]

For a regular covering with monodromy $q : F \twoheadrightarrow G$, membership in $\mathrm{Fib}_{W_{\mathfrak{a}}}$ holds only for $G = 1$ for abelianization, the metabelian quotient, every fixed positive nilpotent class, and rational prounipotent completion. For pro-p completion it holds exactly for finite p-groups G. These are exactly the cases of isomorphism in E.7–E.9, with the properly descended target of E.6 or E.8.

Proof. We first give uniform counterexamples over BF, so the universal base-change requirement is tested, not merely the given fiber. For a nontrivial finite G, choose a prime $r \nmid |G|$ and let V be the augmentation-zero summand of the regular $\mathbb{F}_r[G]$-module. Averaging shows $V_G = 0$, and $V \neq 0$. Let $A = V \rtimes F$ for the action through q, with retraction $A \to F$. The inclusion $F \to A$ is an abelian and every nilpotent equivalence: $[F, V] = V$ puts V in every term $\gamma_j A$, $j \geq 2$. It is also a rational unipotent equivalence, since the finite normal subgroup V maps trivially to every rational unipotent group. Pullback along the G-cover replaces the inclusion by $N \to V \times N$. Its abelian and nilpotent quotients retain V, so they fail those tests. For the rational test use instead the integral augmentation ideal $V = \ker(\mathbb{Z}[G] \to \mathbb{Z})$. Every homomorphism $A = V \rtimes F \to U(\mathbb{Q})$, with U a rational unipotent group, kills V. To prove this, take the Zariski closure of the image of V. It is an abelian unipotent group. Conjugation by

each image of an element of F acts unipotently on its Lie algebra and has finite order, since the action on V factors through G. A finite-order unipotent linear operator in characteristic zero is the identity. Hence the image factors through V_G, which is finite because its rationalization is zero. A unipotent group in characteristic zero has no nontrivial finite subgroup, so that image is trivial. The universal property therefore identifies the rational prounipotent completions of F and A. After pullback, $N \to V \times N$ retains the nonzero vector quotient $V \otimes \mathbb{Q}$, of dimension $d-1$.

For the metabelian test form a finite quotient $H = F/(N'N^r)$, with r any prime. The inclusion $N \cap F' \hookrightarrow N$ has nonzero primitive abelian image of rank $d-1$, by E.7. Its reduction modulo r remains nonzero. Thus H' is nontrivial and the image of N' is trivial. Choose a prime $s \nmid |H|$ and put $V = (1 - e_{H'})\mathbb{F}_s[H]$, where $e_{H'} = |H'|^{-1} \sum_{h \in H'} h$. This is a nonzero H-module with $[H', V] = V$. For $A = V \rtimes F$ one has $A'' = V \rtimes F''$, so $F \to A$ induces an isomorphism of metabelian quotients. On the pulled-back group $V \rtimes N$, N' acts trivially on V; its second derived subgroup is $\{0\} \rtimes N''$. Consequently the metabelian quotient retains V, and $N \to V \rtimes N$ fails the test. These are explicit finitely presented group constructions. Their standard finite presentation complexes and mapping cylinders give finite simplicial-category models with the same fundamental-group calculation.

For a finite p-group cover, E.8 applies to every group occurring after any base change, not just F. An isomorphism of pro-p completions identifies the finite orbit sets and the appropriate open stabilizer subgroups, whose completions are those open subgroups. Hence pullback preserves W_p.

For a regular cover with G not a p-group, let P be a Sylow p-subgroup. The image of $F \to G \times (F/F'F^p)$ is the fiber product over $G/G'G^p$: this follows by taking the two projections and their kernels, or directly from $q(F'F^p) = G'G^p$. The map $P \to G/G'G^p$ is surjective, because its cokernel has order both a power of p and prime to p. Choose words u, v mapping into P whose images in $F/F'F^p$ are x, y. The endomorphism $x \mapsto u, y \mapsto v$ induces a pro-p automorphism: it is onto the Frattini quotient, hence onto the finitely generated free pro-p group, and that group is Hopfian. Its pulled-back regular G-set has image contained in the proper subgroup P, and is therefore disconnected. This violates the component condition in W_p. Finally, a trivial covering is a product with a discrete set and preserves all five classes. □

E.11. The explicit three-letter obstruction in every nilpotent approximation [Proved]

The endomorphism $\alpha(x) = x$, $\alpha(y) = y[x, y]$ induces the identity on every graded lower central quotient and an automorphism on every nilpotent quotient, every pro-p completion and the rational prounipotent completion of F. Pullback of the action $x \mapsto (12), y \mapsto (13)$ changes its orbit set from one orbit to $\{1, 2\}, \{3\}$. It consequently refutes intrinsic-fibration membership for this cover for each of these approximations, and for every functor of their inverse system which sends isomorphisms to isomorphisms.

Proof. Substituting a generator times a commutator in an iterated commutator changes it only in the next lower central degree. Thus the graded action is the identity. Induction in the central extensions gives an automorphism on each nilpotent quotient. The Frattini argument of E.10 proves the pro-p statement; the completed free rational Lie algebra gives the unipotent statement by degreewise inversion. The inverses commute with the transition maps, so the map of inverse systems is an isomorphism. For $a = (12), b = (13)$, multiplication gives $b[a, b] = a$. Both pulled-back generators therefore act by (12). The program `e_obstructions.py` verifies these permutations and the orbit sets. The ordinary metabelian quotient retains the map to S_3, since $S_3'' = 1$; the map of E.11 is not substituted for a metabelian equivalence. □

E.12. Exhaustive small-quotient and Schreier verification [Proved]

For all ordered generating pairs in every group model of order at most twenty-four, the program `e_smallgroups.py` constructs an integral cycle basis of N^{ab}, the comparison to F^{ab}, and the deck character; it computes the free-metabelian relation-module generic rank and the first four lower central ranks. Every generated kernel has rank $|G|+1$, the comparison lattice has index $|G^{\mathrm{ab}}|$, and the rational deck character is $1+\mathrm{reg}_G$. All checks pass.

Proof. The input is the complete order-at-most-twenty-four list in the opened primary GroupNames database [Gro]. It contains seventy-four groups. The accompanying input file records the permutation generators and exact source URLs. The program closes each set of generators, checks its order, constructs its multiplication table, and enumerates every ordered pair. It tests generated subgroup size without a presentation heuristic. For every generating pair it constructs the $2|G|$ edges and a spanning tree, and checks that every fundamental cycle has zero boundary. The non-tree edge coefficients form an integral basis, so this proves the integral calculation. The gcd of all two-by-two minors of the pushforward matrix checks the Smith index. Deck traces are calculated for every group element. Independent tests exhaust the images of a generating pair in every other two-generated model of the same order to verify pairwise nonisomorphism of the models used.

The lower central ranks are calculated from the proved Witt formula in E.7. For the free group N on $r=|G|+1$ generators, the graph over N^{ab} gives $N'/N''=\ker(R_N^r\to R_N)$, $R_N=\mathbb{Z}[N^{\mathrm{ab}}]$; localization to its fraction field has dimension $r-1=|G|$. Thus the printed free-metabelian invariant has an exact module presentation behind it. The output is accompanied by `e_all_surjections.csv`, with a row for every ordered epimorphism, and integral cycle witnesses for every group. The formulas E.6–E.10 prove the assertions for all finite quotients, so the bounded computation also has a general proof. □

E.13. Specialization to the Fermat coverings [Proved]

For $N_n=\ker(F\to(\mathbb{Z}/n)^2)$ the abelian comparison is

$$0\longrightarrow\mathbb{Z}^{n^2-1}\longrightarrow N_n^{\mathrm{ab}}\longrightarrow n\mathbb{Z}^2\longrightarrow 0.$$

The metabelian and class-c comparisons are the exact sequences of E.6 with kernels F''/N_n'' and $\gamma_{c+1}F/\gamma_{c+1}N_n$, respectively; both are nonzero for $n>1$. Pro-p completion is exact exactly when n is a power of p. With $n=p^am$, $(m,p)=1$, its replacement target is the open subgroup of $\widehat{F}_p$ of index p^{2a}, and the source has rank n^2+1. The rational unipotent comparison is the surjection of E.9, with abelian kernel dimension n^2-1.

Proof. The abelian covering map sends a closed edge cycle to its two exponent sums. Each sum is divisible by n, and the loops x^n,y^n realize the two generators of $n\mathbb{Z}^2$. The source rank is n^2+1 by the graph calculation. Since N_n contains F', it contains both F'' and every $\gamma_{c+1}F$ for $c\geq 1$, giving exactly the two kernels in the statement. E.7 proves their nonvanishing. The maximal p-quotient of $(\mathbb{Z}/n)^2$ is $(\mathbb{Z}/p^a)^2$, so E.8 gives the asserted ranks and the exactness criterion. E.9 gives the last assertion. These are the maps induced by the actual covering inclusion, and coincide with the Fox matrices computed in section B. □

E.14. Relative completion retains the entire covering kernel [Proved]

The rational relative unipotent completion $\mathcal{R}_q$ of $F\twoheadrightarrow G$, for finite G, fits into

$$1\longrightarrow N^{\mathrm{un}}\longrightarrow\mathcal{R}_q\longrightarrow G_{\mathbb{Q}}\longrightarrow 1.$$

It splits over $\mathbb{Q}$, with splittings conjugate by $N^{\mathrm{un}}(\mathbb{Q})$. Its quotient with abelian kernel has vector kernel $H_1(N,\mathbb{Q})$, of dimension $|G|+1$ and character $1+\mathrm{reg}_G$. For $q_2 : F \to (\mathbb{Z}/2)^2$ its four character multiplicities are $2,1,1,1$.

Proof. Choose a transversal of G in F. Its conjugation maps on N extend uniquely to N^{un}, and its products give a factor set with values in $N \to N^{\mathrm{un}}$. The associated multiplication on the disjoint union of $|G|$ copies of N^{un} is associative by the factor-set identities. This constructs $\mathcal{R}_q$ and the displayed exact sequence as rational proalgebraic group schemes. Every map from F into an extension of G by a unipotent group extends uniquely on N and then on the chosen components. It respects multiplication by the same identities. This proves the relative universal property.

At each successive central vector quotient of the unipotent kernel, a section lifts because the cohomology in positive degree of the finite group G with rational coefficients vanishes. To see the vanishing, average a cochain contraction over G and divide by $|G|$ in the standard bar resolution. The same averaging in degree one conjugates two sections. Induction gives compatible sections and conjugators through the central quotients, and their inverse limits give the assertions for N^{un}. This specifies the conjugacy ambiguity; uniqueness of the conjugating element is not claimed. Finally, the chain modules of the finite Cayley graph are $\mathbb{Q}[G]^2$ and $\mathbb{Q}[G]$, with zeroth homology $\mathbb{Q}$. Semisimplicity of $\mathbb{Q}[G]$ gives $H_1(N,\mathbb{Q}) \simeq \mathbb{Q} \oplus \mathbb{Q}[G]$. For $G=(\mathbb{Z}/2)^2$, expanding its regular character gives the four stated multiplicities. This action has a -1 eigenvalue; it survives in the relative completion, whereas the map of E.9 takes coinvariants and loses these characters. □

E.15. The genuine action on a relative completion [Proved]

Let $\mathrm{Aut}(F,q)$ consist of the automorphisms of F carrying N to itself. Its homomorphism to $\mathrm{Aut}_{\mathbb{Q}}(\mathcal{R}_q)$ is injective, and its image is exactly the subgroup preserving the embedded discrete subgroup $F \subset \mathcal{R}_q(\mathbb{Q})$ setwise. The automorphism $\mathrm{Int}([x,y])$, at $q=q_2$, acts nontrivially on the relative completion, although its action on F^{ab} and the image of its conjugator there are both the identity.

Proof. The universal property in E.14 extends an automorphism and its inverse, with their induced automorphism of G, so the extension is an automorphism. The map $N \to N^{\mathrm{un}}$ is injective. Write a nontrivial reduced word in maximal nonzero-power blocks $z_{i_1}^{a_1}\cdots z_{i_k}^{a_k}$, with adjacent i_j different. Under $z_i \mapsto 1+Z_i$, the coefficient of $Z_{i_1}\cdots Z_{i_k}$ with these k alternating blocks is $a_1\cdots a_k \neq 0$: omitting a block cannot yield that sequence of k blocks, and each retained block must contribute its linear term. A finite truncation of the power-series ring detects the word in its unipotent group $1+(Z_i)$. Since components detect F/N, the map $F \to \mathcal{R}_q(\mathbb{Q})$ is injective as well. This proves faithfulness. An algebraic automorphism preserving this subgroup restricts to an automorphism of F preserving its identity component N; the universal property identifies its extension with the given automorphism. This proves the image description.

At q_2 the commutator $w=[x,y]$ has a nonzero cycle class in $H_1(N_2,\mathbb{Q})$, as can be read from the unit-square cycle in the four-vertex graph. Choose another cycle with independent class. Their bracket is nonzero in the degree-two free Lie algebra of N_2^{un}, so conjugation by w is nontrivial. Passing from a genuine action to its outer class kills this particular inner action. Thus this example also specifies the actual quotient map responsible for the ambiguity, rather than identifying an outer action with a genuine one. □

Sections and arithmetic obstructions

E.16. All sections of the rank-two toric quotient [Proved]

Let $T = \mathbb{G}_m^{\{0,1,\infty\}}/\mathbb{G}_m$ over $\mathbb{Q}$, with its rational identity. Its arithmetic fundamental-group extension has geometric kernel $\widehat{\mathbb{Z}}(1)^2$. Its continuous sections modulo conjugacy by this kernel form the abelian group

$$\left(\varprojlim_m \mathbb{Q}^*/\mathbb{Q}^{*m}\right)^2.$$

One factor has the explicit description

$$\{\pm 1\} \times \left\{(a_p)_p \in \prod_{p \text{ prime}} \widehat{\mathbb{Z}} : \forall m,\ a_p \in m\widehat{\mathbb{Z}} \text{ for all but finitely many } p\right\}.$$

This computes the cohomology in (379) for its specified rank-two quotient at every level n.

Proof. The section at the identity gives the semidirect product $\widehat{\mathbb{Z}}(1)^2 \rtimes G_{\mathbb{Q}}$. A section has form $g \mapsto (c(g), g)$; multiplication says $c(gh) = c(g) + gc(h)$. Conjugation changes c by a coboundary. At finite level the Kummer sequence and Hilbert 90 identify $H^1(G_{\mathbb{Q}}, \mu_m)$ with $\mathbb{Q}^*/\mathbb{Q}^{*m}$. To verify the reduction, a multiplicative cocycle of a finite Galois extension is a coboundary by choosing b for which $\sum_\tau c(\tau)\tau(b) \neq 0$ and translating the sum; linear independence of field embeddings supplies such a b. Every algebraic cocycle is defined over a finite extension. Taking limits gives the formula. The possible $\varprojlim^1 H^0(\mu_m)$ vanishes because these groups are finite and their descending images stabilize.

Unique prime factorization gives $\mathbb{Q}^* = \{\pm 1\} \times \bigoplus_p p^{\mathbb{Z}}$. An element modulo m has finitely many nonzero valuation residues. Compatible residues are exactly the families in the second display. Even m retain the sign and odd m kill it; their compatible limit is $\{\pm 1\}$.

The map $U_n \to T$ is inclusion on the open with all coordinates nonzero. On geometric groups it is the covering pushforward of E.13 divided by n, hence surjective onto $\widehat{\mathbb{Z}}(1)^2$. The map to the base torus in coordinates is $(u, v) \mapsto (-u^n, -v^n)$, with $t = -u^n$, $1 - t = -v^n$. It induces multiplication by n on geometric kernels and retains the arithmetic translation $(-1, -1)$. Thus the quotient extension is the arithmetic fundamental group of T even when U_n has no rational point. This supplies the comparison underlying (373)–(379). □

E.17. Pairwise separation of cusp sections by the minimum quotient [Refuted]

For every odd n, the rational cusps $P_0 = [0 : 1 : -1]$ and $P_\infty = [1 : -1 : 0]$ admit rational tangent choices whose sections become conjugate in the rank-two quotient of (373). Both toric Kummer classes are $(1, -1)$ in E.16. For the three punctures of U_1 the complete families of toric images are $(w, 1)$, $(1, -w)$ and $(w^{-1}, -w^{-1})$, with $w \in \mathbb{Q}^*$, in the conventional coordinates $(t, 1 - t)$ and the tangent coordinates specified below.

Proof. Use $u = x_0/x_\infty, v = x_1/x_\infty$. Near P_0 take $s = u$; the branch is $v = -(1 + s^n)^{1/n}$ with constant coefficient -1. The unit tangent has leading torus coefficients $(1, -1)$. Near P_∞ take $s = x_\infty/x_0$; then $(u, v) = (s^{-1}, -s^{-1}(1 + s^n)^{1/n})$, giving the same leading coefficients. Adjoining compatible roots of these units computes the tangential cocycle: coefficientwise Galois action fixes the formal parameter and acts on roots of the leading coefficients. Both classes therefore coincide. This also derives the applicable case of Wickelgren's tangential formula, Lemma 4 [Wic12].

For U_1 in coordinate t, at 0 with tangent w the units $t, 1 - t$ give $(w, 1)$; at 1 with coordinate $t - 1$ they give $(1, -w)$; at infinity with coordinate $1/t$ they give $(w^{-1}, -w^{-1})$. These exhaust

rational tangential choices. E.16 determines equality of their images. The unit tangent at 0 and tangent -1 at 1, for example, both give $(1,1)$. The exhibited coincidences are recorded in `e_obstructions.py`. They concern precisely this quotient, with the quotient map specified. □

E.18. The real obstruction for the manuscript quartic [Proved]

The geometrically abelianized arithmetic fundamental group of $U_4/\mathbb{Q}$ has no continuous section. The section-class sets are also empty for its geometrically metabelian, full profinite and geometric pro-2 quotients. The obstruction already lies in a group of order two in $H^2(C_2, H_1(U_4(\mathbb{C}), \mathbb{Z}_2))$.

Proof. The plus quartic has no real points, by positivity of fourth powers. Complex conjugation acts freely on $U_4(\mathbb{C})$. Its quotient is a connected nonorientable punctured surface of Euler characteristic -8, since the double cover has Euler characteristic -16. It retracts to a graph, so its fundamental group J is free of rank nine. The geometric subgroup N of index two is free of rank seventeen. Finite complex topological covers with conjugation descent data identify the real arithmetic extension with the profinite completion of $1 \to N \to J \to C_2 \to 1$: descend the complex cover with its antilinear involution; in the other direction analytify a finite etale cover. This is the equivariant finite-cover comparison, and finite-index completion preserves this extension.

Free-generator changes put $J \to C_2$ in the form $a \mapsto \bar{1}$, $b_i \mapsto \bar{0}$ for $1 \le i \le 8$. A Schreier basis of N is

$$a^2, \quad b_i, \quad ab_ia^{-1} \quad (1 \le i \le 8).$$

Conjugation by a fixes the first abelian basis vector and exchanges each remaining pair. The factor set at $(\bar{1}, \bar{1})$ is a^2. Replacing the lift by na changes its square by $(1 + \operatorname{Ad} a)[n]$. Its first coordinate is $1 + 2n_0$, which cannot vanish in $\mathbb{Z}_2$. Each exchanged pair has invariant diagonal equal to the image of the norm. Hence the indicated cohomology group is $\mathbb{Z}/2$, and the extension is its nonzero class. The finite program tests the factor-set equation for every modulus dividing twenty-four and for additional odd powers; precisely the even moduli are obstructed.

A rational section restricts to a real decomposition group, and each quotient in the statement projects to this 2-adic abelian quotient. All are therefore obstructed. Its image under $N^{\mathrm{ab}} \to 4\widehat{\mathbb{Z}}(1)^2$ is zero in H^2: conjugation acts as -1 on that torsion-free rank-two module, whose invariants vanish. This explicit pushout explains the splitting of (373) together with its failure to lift. The odd-adic and rational pushouts of the first-coordinate obstruction also vanish because two is invertible there; this cohomology calculation makes no assertion of a global section there. □

E.19. The cubic cusps are separated by their own 3-adic abelianization [Proved]

The three rational cusps $[0:1:-1]$, $[1:0:-1]$, $[1:-1:0]$ determine pairwise nonconjugate sections of the geometric pro-3 abelianization of $\pi_1(U_3/\mathbb{Q})$, for every rational tangent choice. They remain distinct in the geometric abelian, metabelian and full profinite quotients.

Proof. Filling the cusps maps the affine fundamental group to the Tate module of the elliptic curve X_3, based at one cusp. A cusp section maps to the point section there, independently of its tangent. Each difference of two distinct cusps has exact order three: the ratio of their tangent lines has divisor three times their difference, and a genus-one curve has no function with a single simple pole. Section C gives the same exact-order calculation.

The cubic has good reduction at two. Its equation over $\mathbb{F}_2$ has exactly the three projective points with two nonzero coordinates, as the program enumerates. The kernel of good reduction over $\mathbb{Q}_2$ is pro-2, from its formal-parameter congruence filtration. Odd torsion therefore injects into this group of three points. A rational point of exact order three cannot be divisible by three: a preimage would have order nine. Its Kummer class modulo three is nonzero. The difference of the two point sections is this class in $H^1(G_{\mathbb{Q}}, X_3[3])$. Projection from affine 3-adic homology detects it and proves the assertion. This map also identifies information lost in the quotient of E.17. □

E.20. A new obstruction to lifting a toric Kummer section [Proved]

For U_1 based at the tangent from 0 towards 1, the abelian section with Kummer coordinates $(2, 3)$ does not lift to the class-two quotient with geometric group

$$\langle x, y, z \mid x^2 = y^2 = z^2 = 1, [x, y] = z,\ [x, z] = [y, z] = 1\rangle.$$

Its obstruction is the nonzero class $\{2\} \cup \{3\}$ in $H^2(G_{\mathbb{Q}}, \mu_2)$.

Proof. The eight normal forms $x^a y^b z^c$ have central multiplication term ba'. The code verifies this group law and presentation. The tangential Galois action is trivial on it: $x \mapsto x^\chi = x$ and $y \mapsto f^{-1} y^\chi f = y$, since χ is odd and the commutator f is central here. Pulling the central factor set back along the two Kummer characters gives their cup product (the order of factors does not change its class modulo two).

Over $\mathbb{Q}_3$, the quadratic extension generated by $\sqrt{2}$ is unramified because 2 is a nonsquare modulo three. Norms have even valuation, so 3 is not a norm. The crossed-product description of the cup product identifies its vanishing with the norm equation $3 = u^2 - 2v^2$ over $\mathbb{Q}_3$: represent $\sqrt{2}$ by its multiplication matrix, and the matrix equation for the second quaternion generator is exactly this norm equation. Thus the local and the global class are nonzero. A lifted section would give a cochain whose coboundary is this factor set, a contradiction. □

E.21. The unmarked depth assignment fails to descend [Refuted]

The quotient of $\mathbb{L}(x, y)$ by D_y^3, with its canonical projection, is not functorial for all automorphisms of the unmarked free group. It is functorial for maps preserving the ideal of the marked puncture and its lower central filtration. Thus E.5 tests the five specified ordinary group approximations; the depth construction additionally uses this marking and these morphisms.

Proof. The swap of x, y sends $[y, [y, [y, x]]] \in D_y^3$ to $[x, [x, [x, y]]]$, a nonzero depth-one element of E.2. It therefore fails to preserve the kernel of the canonical projection, which proves the non-descent. A map preserving the marked ideal preserves its lower central terms and does induce the quotient map. No LM8 return specifying an additional unmarked functor is present in the supplied archive. □

E.22. The scheme coordinates do not carry the requested relative action [Refuted]

Put $q_2 : F \to (\mathbb{Z}/2)^2$ and extend the relative completion of E.14 to $\mathbb{Q}_\ell$. There is no homomorphism from the normalized diagonal arithmetic image in $\mathcal{M}(0)(\widehat{\mathbb{Z}})$ to $\mathrm{Out}(\mathcal{R}_{q_2,\mathbb{Q}_\ell})$ whose composite with $G_{\mathbb{Q}}$ is its actual outer action. In particular no such genuine action descends from those coordinates. The actual action is defined on the continuous geometric fundamental group, and its relative universal property gives its action on $\mathcal{R}_{q_2,\mathbb{Q}_\ell}$. E.15 gives the faithful discrete version and its exact image.

Proof. First we type the comparison with the arithmetic action. The continuous unipotent completion of the free profinite group $\widehat{N}_2$ over $\mathbb{Q}_\ell$ is the free prounipotent group on its five Schreier generators. Indeed each finite tuple in a unipotent $\mathbb{Q}_\ell$-group generates a relatively compact subgroup, as follows from upper-triangular coordinates and their polynomial group law. Its homomorphism from the discrete free group consequently extends continuously to the free profinite group. The two universal properties identify this completion with $N_2^{\mathrm{un}} \otimes \mathbb{Q}_\ell$. The transversal construction in E.14 then gives the continuous relative completion and its functorial arithmetic action. The normalized Galois action is the identity on $(\mathbb{Z}/2)^2$, because its abelian multiplier is $\chi \equiv 1 \pmod 2$.

The homomorphism $F \to G \times F^{\mathrm{un}}$, $g \mapsto (q_2(g), g)$, and the relative universal property give the canonical surjection

$$\mathcal{R}_{q_2,\mathbb{Q}_\ell} \longrightarrow F^{\mathrm{un}}_{\mathbb{Q}_\ell}.$$

It is the maximal unipotent quotient, so its kernel is characteristic. Every genuine automorphism descends through this map and inner automorphisms descend to inner automorphisms. This gives the needed map of outer automorphism groups. A nontrivial outer action on its target is therefore a nontrivial outer action on its source.

A.19 proves that the normalized arithmetic image is contained in $\mathcal{M}(0) \simeq \Gamma$. On $G_{\mathbb{Q}^{\mathrm{cyc}}}$ its multiplier-scheme coordinate is the identity, and the exact law is that of $C_2 \times \widehat{\mathbb{Z}}$, an abelian group. Every commutator of two elements of $G_{\mathbb{Q}^{\mathrm{cyc}}}$ thus maps to the identity in these scheme coordinates.

We now exhibit a commutator whose actual outer unipotent action is nontrivial. E.3 gives nonzero actual graded Galois classes in weights three and five, with depth-one coefficients proportional to $\bar{\sigma}_3, \bar{\sigma}_5$. Choose representatives g_3, g_5 of nonzero such classes and choose $\tau \in G_{\mathbb{Q}}$ with $\chi_\ell(\tau) = 1 + \ell$. The cyclotomic action on weight w is multiplication by χ_ℓ^w. Hence $h_w = [\tau, g_w]$, for $w = 3, 5$, has leading class $((1+\ell)^w - 1)$ times that of g_w, which is nonzero over $\mathbb{Q}_\ell$. Both h_w lie in $G_{\mathbb{Q}^{\mathrm{ab}}} = G_{\mathbb{Q}^{\mathrm{cyc}}}$, since they are commutators. Their commutator has nonzero weight-eight, depth-two outer class: its polynomial is a nonzero scalar multiple of $B_{2,4}$ computed in E.3. This assertion uses actual graded Galois classes, not arbitrary special derivations.

Consequently $[h_3, h_5]$ is killed by $\mathcal{M}(0)$ and acts nontrivially on $\mathrm{Out}(F^{\mathrm{un}}_{\mathbb{Q}_\ell})$, and thus on the relative completion. A homomorphism agreeing with both actions would send the identity to this nonidentity element, which proves the claimed non-descent. For a separate genuine geometric witness, $w = [x, y] \in \Pi_0'$ has trivial final coordinates $(a, b, U) = (1, 1, \bar{w})$ but its inner action on the relative completion is nontrivial by E.15. The arithmetic witness above also survives passage to outer automorphisms. □

E.23. The remaining cubic section calculation [Open]

The complete set of continuous section classes for the geometric metabelian and marked depth-two arithmetic quotients of $U_3/\mathbb{Q}$ is not determined in this return. The remaining calculation is the image and every fiber of their section maps to E.16. The first unresolved concrete nonabelian case here is the class-two pro-3 quotient of the cubic's rank-ten geometric fundamental group, with its integral arithmetic action.

Work and obstruction. E.18 determines the empty set for the plus quartic and all quotients retaining its 2-adic geometric abelianization; E.25 extends this to every even plus-Fermat level. E.24 computes the entire cubic abelian section set and every fiber over the minimum torus. E.19 separates the three rational cubic cusps inside that set, while E.17 proves their collision under the stated toric map.

The relation-module presentation E.1, the rank-two factor-set calculation, local cup products, the exact generalized-Jacobian extension, and a new three-punctured-line map on the cubic

were successive attempts to compute the nonabelian fibers. They give the following actual refinements. E.28 exhibits a whole nonempty abelian toric fiber with no class-two lifts. E.29 classifies every rational toric pair for the inherited cubic class-two quotient. E.32 completes that calculation for every completed pro-3 Kummer pair, including the exact convergence condition and all three classes in each nonempty fiber; it also exhibits an actual obstruction invisible at every finite level. E.31 computes a genuine intrinsic quotient with its arithmetic origin term: in one toric fiber the zero abelian class lifts, whereas an explicit other class and its entire congruence coset have no intrinsic class-two lift. Thus the two minimum coordinates do not determine intrinsic lifting, as witnessed by this exact same-fiber comparison. E.34 computes the entire intrinsic abelian family unramified outside three and proves that every one of its classes lifts to the full geometric pro-3 group. E.35 evaluates every section of the intrinsic finite Heisenberg quotient by a cubic norm equation and gives all of its nonempty fibers in explicit Kummer coordinates. E.36 proves that every abelian class lifts through the cuspidally central quotient retaining all ten abelian coordinates; each fiber is the specified completed group of rational units. Its quotient map from the full intrinsic class-two group has a central kernel of rank 44.

These calculations use the quotient homomorphisms specified in E.25. They do not supply every obstruction and every fiber for the full rank-ten intrinsic class-two group in the statement. After E.36 the next calculation is the actual obstruction in that rank-44 kernel, including its integral lattice and continuous inverse-limit conditions. Wickelgren's opened formulas apply directly to the ambient rank-two group; the additional intrinsic quotient in E.31 requires its separately computed arithmetic action. E.22 settles the proposed descent from the manuscript scheme by an actual Galois counterexample. The next unresolved step is to evaluate the full intrinsic central extension on every class of the explicitly computed set E.24, retaining the origin terms and the geometric directions killed by E.25's quotient map. A name for that nonabelian cohomology is not its evaluation.

The complete cubic abelian calculation and the precise depth maps

In this subsection $E = X_3^+$ is the cubic $x^3 + y^3 + z^3 = 0$, $D = E \setminus U_3$, and $G = J_{E,D}$ is the generalized Jacobian for the reduced divisor D. We use the origin $O = [1 : -1 : 0]$, and set $P = [0 : -1 : 1]$, $Q = [-1 : 0 : 1]$. For an abelian group A we write $A^\wedge = \varprojlim_m A/mA$, with m ordered by divisibility. This is the completion for multiplication by m; the quotients need not be finite. All inverse limits of cohomology groups below carry their inverse-limit prodiscrete topology. For a countable set $\mathcal{I}$ define

$$\mathcal{R}_{\mathcal{I}}(\widehat{\mathbb{Z}}) = \left\{ (a_j) \in \prod_{j \in \mathcal{I}} \widehat{\mathbb{Z}} : \text{for every } m,\ a_j \in m\widehat{\mathbb{Z}} \text{ for all but finitely many } j \right\}.$$

Its topology is the one coming from $\varprojlim_m \bigoplus_{\mathcal{I}} \mathbb{Z}/m$. In particular this space is not asserted to be compact.

E.24. All cubic abelian sections and every fiber over the toric minimum [Proved]

Put $K = \mathbb{Q}(\zeta)$, $\zeta^2 + \zeta + 1 = 0$, and $\lambda = 1 - \zeta$. There is a retraction of algebraic groups over $\mathbb{Q}$

$$r : G \longrightarrow \mathbb{G}_m^2, \qquad G \simeq \mathbb{G}_m^2 \times G', \qquad 1 \longrightarrow (\mathrm{Res}_{K/\mathbb{Q}}\, \mathbb{G}_m)^3 \longrightarrow G' \longrightarrow E \longrightarrow 0.$$

The map r is the map induced by the two units $x/z, y/z$ on U_3. The group $E(\mathbb{Q})$ is cyclic of order three, and

$$G'(\mathbb{Q}) \simeq \mu_6^3 \oplus \bigoplus_{j \in \mathcal{I}} \mathbb{Z}.$$

Here one may take $\mathcal{I}$ to consist of three copies of the nonzero prime ideals of $\mathbb{Z}[\zeta]$, with the first copy of (λ) replaced by one new generator. The displayed abstract isomorphism includes explicit choices of prime generators.

Choose the unit rational tangent at O, with local parameter z/x, as the origin of the set of sections of the geometrically abelian arithmetic fundamental group. Up to geometric conjugation, that entire set is then the group

$$H^1(G_{\mathbb{Q}}, T_{\bullet}G) \simeq (\mathbb{Q}^{\times,\wedge})^2 \times \mu_6^3 \times \mathcal{R}_{\mathcal{I}}(\widehat{\mathbb{Z}}) \times \widehat{\mathbb{Z}}. \tag{E24.1}$$

The last factor and its splitting are noncanonical. The actual map to the toric minimum of E.16 is the projection onto the first two factors followed by translation by the Kummer class of $(1,-1)$. Consequently it is surjective, and every fiber is a torsor under

$$\mu_6^3 \times \mathcal{R}_{\mathcal{I}}(\widehat{\mathbb{Z}}) \times \widehat{\mathbb{Z}}. \tag{E24.2}$$

For the pro-3 geometric abelianization the same formulas hold with $\mathbb{Q}^{\times,\wedge}$ replaced by $\varprojlim_a \mathbb{Q}^{\times}/\mathbb{Q}^{\times 3^a}$, μ_6^3 replaced by μ_3^3, and each $\widehat{\mathbb{Z}}$ replaced by $\mathbb{Z}_3$. The translation by -1 is zero in this pro-3 version.

Proof. The rational elliptic-curve calculation. The maps

$$X = -\frac{3z}{x+y}, \qquad Y = \frac{4x-5y}{x+y}, \qquad [x:y:z] = [Y+5 : 4-Y : -3X]$$

identify the smooth cubic with $Y^2+Y = X^3-7$, with O at infinity. Indeed the exact identity is

$$Y^2+Y-X^3+7 = \frac{27(x^3+y^3+z^3)}{(x+y)^3}.$$

The inverse formula proves the birational assertion, which extends to an isomorphism of the smooth projective curves.

For the rank calculation we use the explicit identification over $\mathbb{Q}$ of the Fermat cubic with $X_0(27)$ in Li–Long–Tu, §2.4 and Remark 4.2 [LLT18]. The corresponding normalized differential has newform $f(\tau) = \eta(3\tau)^2\eta(9\tau)^2$. The Mellin formula therefore gives

$$L(E,1) = 2\pi \int_0^{\infty} \eta(3it)^2\eta(9it)^2\, dt > 0.$$

The product formula for η makes the integrand strictly positive for $t>0$, and cuspidality gives convergence at both ends. The automorphism induced by multiplication of one Fermat coordinate by ζ gives complex multiplication by $\mathbb{Z}[\zeta]$; this is the maximal order of the norm-Euclidean field K. Coates–Wiles, Theorem 1 on p. 223 [CW77], now gives rank zero. The smooth cubic has exactly three points over $\mathbb{F}_2$, so the prime-to-2 rational torsion injects into a group of order three. An order-two point on the displayed Weierstrass model would have $Y=-1/2$ and $X^3=27/4$, which has no rational solution. Thus there is no 2-primary torsion. The distinct points O,P,Q already give three rational points, so $E(\mathbb{Q}) = \langle P\rangle \simeq \mathbb{Z}/3$.

The retraction and the exact rational-point presentation. The cusp scheme is the disjoint union of three rational points and three copies of $\operatorname{Spec} K$. Taking the frame at O as one identifies the torus of G with

$$T_D = \mathbb{G}_m^2 \times (\operatorname{Res}_{K/\mathbb{Q}} \mathbb{G}_m)^3.$$

We write its coordinates as (a,b,A,B,C), at P,Q and the three quadratic cusp orbits, respectively. The convention is that a frame is a nonzero vector in the fiber of the line bundle.

Here is an explicit construction of the characters used in r. For a unit h on U_3, write $\operatorname{div}(h) = \sum_{d\in D} m_d d$. For a framed line bundle $(L, (e_d))$, choose a rational section s nonvanishing along D, and set

$$\chi_h(L, (e_d)) = h(\operatorname{div} s) \prod_{d\in D} (s(d)/e_d)^{-m_d}. \tag{E24.3}$$

Products over nonrational points mean the corresponding norms. Replacing s by fs leaves this expression unchanged, since Weil reciprocity gives $h(\operatorname{div} f) = f(\operatorname{div} h)$. Tensor products multiply (E24.3). The construction in families, or the divisor presentation of the generalized Jacobian, consequently gives a morphism of algebraic groups. On the torus it is $\prod e_d^{m_d}$, and on a divisor $u - u_0$ in U_3 it is $h(u)/h(u_0)$. Thus it is precisely the character induced by h.

For $h = x/z, y/z$ their divisors are the respective coordinate-cusp orbits minus the orbit at infinity. Hence

$$r_T(a, b, A, B, C) = \big(aN(A)/N(C),\ bN(B)/N(C)\big). \tag{E24.4}$$

The map $i(a, b) = (a, b, 1, 1, 1)$ is an algebraic section of r. The isomorphism with the direct product is explicitly $g \mapsto (r(g),\, g\, i(r(g))^{-1})$. The kernel torus is parametrized by

$$(A, B, C) \longmapsto \big(N(C)/N(A),\ N(C)/N(B),\ A, B, C\big). \tag{E24.5}$$

Hilbert 90 gives $H^1(\mathbb{Q}, T_D) = 0$ and $H^1(\mathbb{Q}, (\operatorname{Res}_{K/\mathbb{Q}} \mathbb{G}_m)^3) = 0$.

To calculate the extension on rational points, let $z_0 \in G(\mathbb{Q})$ be the framed line bundle $\mathcal{O}(P - O)$ with the local frames specified next. Put $f = (y + z)/(x + y)$, so $\operatorname{div}(f) = 3P - 3O$. At P use $s = x/z$ and the frame s^{-1} for $\mathcal{O}(P - O)$; at O use $s = z/x$ and the frame s. At the other cusps use the canonical section 1. Multiplication by f trivializes its third tensor power. The two leading values of f are

$$\lim_P f/s^3 = 1/3, \qquad \lim_O fs^3 = 3, \qquad f(Q) = -1.$$

For example $y/z = -(1 + s^3)^{1/3}$ at P, which proves the first equality; the same expansion at O proves the second. At the nonrational cusps use the representatives $[0 : -\zeta : 1]$, $[-\zeta : 0 : 1]$, $[1 : -\zeta : 0]$. Dividing every frame value by the value 3 at O gives

$$3z_0 = \left(3^{-2}, -3^{-1}, \lambda^{-1}, \lambda^{-2}, \zeta^2\lambda^{-3}\right) \in \mathbb{Q}^{\times 2} \times K^{\times 3}. \tag{E24.6}$$

This calculation also follows directly from $3 = -\zeta^2\lambda^2$. Equation (E24.4), with $N(\lambda) = 3$, sends (E24.6) to $(1, -1)$. The rational cube roots of 1 and -1 are unique, so $r(z_0) = (1, -1)$. For $z' = z_0 - i(1, -1)$ the presentation becomes

$$G'(\mathbb{Q}) = (K^{\times 3} \oplus \mathbb{Z}z')/\left\langle 3z' - (\lambda^{-1}, \lambda^{-2}, \zeta^2\lambda^{-3})\right\rangle. \tag{E24.7}$$

There are no additional relations: the quotient onto $E(\mathbb{Q}) = \mathbb{Z}/3$ is surjective by Hilbert 90, the class of z' generates it, and (E24.7) specifies its third multiple in the kernel.

The norm-Euclidean ring $\mathbb{Z}[\zeta]$ has units μ_6, so prime factorization gives $K^\times = \mu_6 \oplus \bigoplus_{\mathfrak{p}} \mathbb{Z}$ after choosing one generator for each prime ideal. In (E24.7) the coefficient of the first λ generator is -1. Eliminate that generator and retain z' instead. This operation changes no torsion relation and proves the asserted description of $G'(\mathbb{Q})$. As an independent finite-support check, `e_cubic_abelian.py` obtains the Smith diagonal $1, 6, 6, 6$ for the relation involving the three μ_6 generators, the three λ generators and z'.

The full cohomology calculation. Write $B = (\operatorname{Res}_{K/\mathbb{Q}} \mathbb{G}_m)^3$. The extension of E by B in G' is killed by three: the extension of E by T_D is classified by the cusp-difference map, all

of whose values are killed by three by C.5, and (E24.5) and the direct product exhibit the component defining G'. It follows that the connecting map

$$\delta : H^1(\mathbb{Q}, E) \longrightarrow H^2(\mathbb{Q}, B)$$

is killed by three. Since $H^1(\mathbb{Q}, B) = 0$, $H^1(\mathbb{Q}, G')$ injects into $H^1(\mathbb{Q}, E)$. Every element of the maximal divisible subgroup of the latter is three times an element of that subgroup; hence its image under δ is zero. Consequently the injection identifies maximal divisible subgroups:

$$\operatorname{Div} H^1(\mathbb{Q}, G') = \operatorname{Div} H^1(\mathbb{Q}, E) \simeq \mathbb{Q}/\mathbb{Z}. \tag{E24.8}$$

The last equality is exactly Çiperiani–Stix, Theorem 18(2), printed p. 18 [ÇS13], applied to the analytic-rank-zero curve already established above. Their additional restrictions on primes concern the separate assertion about the Tate–Shafarevich intersection. They impose no restriction on (E24.8).

The Kummer sequences and inverse limit give

$$0 \longrightarrow G'(\mathbb{Q})^\wedge \longrightarrow H^1(G_\mathbb{Q}, T_\bullet G') \longrightarrow \operatorname{Hom}(\mathbb{Q}/\mathbb{Z}, H^1(\mathbb{Q}, G')) \longrightarrow 0. \tag{E24.9}$$

For precision, take the countable cofinal subsystem $m = a!$. The transitions $G'(\mathbb{Q})/m \to G'(\mathbb{Q})/d$, $d \mid m$, are surjective, so their first derived inverse limit vanishes. The finite groups $H^0(\mathbb{Q}, G'[m])$ satisfy the Mittag–Leffler condition, which identifies continuous H^1 of $T_\bullet G'$ with the inverse limit of the finite-coefficient H^1 groups. No finiteness of $G'(\mathbb{Q})/m$ or $H^1(\mathbb{Q}, G'[m])$ is used. The last term in (E24.9) is the limit of $H^1(\mathbb{Q}, G')[m]$ under multiplication transitions. Every homomorphic image of $\mathbb{Q}/\mathbb{Z}$ is divisible, so (E24.8) identifies it with $\operatorname{End}(\mathbb{Q}/\mathbb{Z}) = \widehat{\mathbb{Z}}$.

Choose a lift of 1 in (E24.9). Multiplication by $\widehat{\mathbb{Z}}$ then gives a continuous splitting: modulo m its value depends only on the scalar modulo m. The description (E24.7) gives

$$G'(\mathbb{Q})^\wedge = \mu_6^3 \times \mathcal{R}_\mathcal{I}(\widehat{\mathbb{Z}}). \tag{E24.10}$$

Indeed the image at each m has finite support in $\mathcal{I}$, and compatibility is exactly the restricted-sequence condition in the definition. This proves both the algebraic and topological assertions in (E24.9). The identical argument with the subsystem $m = 3^a$ gives the pro-3 version, whose divisible quotient is $\operatorname{Hom}(\mathbb{Q}_3/\mathbb{Z}_3, \mathbb{Q}_3/\mathbb{Z}_3) = \mathbb{Z}_3$.

Finally, $T_\bullet G = H_1(U_{3,\overline{\mathbb{Q}}}, \widehat{\mathbb{Z}})$ by the generalized-Jacobian realization in C.1. A rational tangential section splits the corresponding arithmetic extension; its conjugacy classes of splittings are exactly continuous one-cocycles modulo coboundaries. Thus (E24.9), (E24.10), Hilbert 90 and the algebraic retraction give (E24.1). At O, with $s = z/x$, the leading terms of $(x/z, y/z)$ are $(s^{-1}, -s^{-1})$. This makes the toric origin the Kummer class of $(1, -1)$ and proves the claimed affine map. The covering pushforward q_{3*} is multiplication by three after r_* on geometric homology; the map used here is the actual coordinate-unit map r_*, as in E.16. This fixes the comparison and the translation, and proves the fiber assertion. □

E.25. Intrinsic depth, inherited depth, and the even-level obstruction [Proved]

Let $N_n = \pi_1(U_{n,\overline{\mathbb{Q}}})$, embedded in the geometric profinite group $\widehat{F}$ of U_1 by the covering. In this statement write $F = \widehat{F}$ in the profinite formulas. Use y for inertia at the base puncture 1. For an intrinsic depth filtration choose a Galois-stable set of cusps above 1, and let I_n be the closed normal subgroup generated by their inertia. Put $D^r_{\mathrm{intr}} = \gamma_r I_n$. One allowed choice for $n = 3$ is the rational cusp Q alone; for $n = 4$ take the entire four-point coordinate-cusp orbit. There are actual quotient maps

$$N_n/D^3_{\mathrm{intr}} \longrightarrow \operatorname{im}(N_n \longrightarrow F/D^3_y), \qquad \ker = \frac{N_n \cap D^3_y}{D^3_{\mathrm{intr}}}, \tag{E25.1}$$

where $D_y^3 = \gamma_3\langle\langle y\rangle\rangle$. The same formulas apply to the corresponding Lie ideals and their completed rational Lie algebras.

Over each field $\mathbb{Q}_\ell$ the inherited prounipotent image is all of $F^{\mathrm{un}}_{\mathbb{Q}_\ell}/D_y^3$. The corresponding inherited arithmetic extension is the base arithmetic extension and has a rational tangential section, for every n. On abelianized Lie algebras the map from the intrinsic quotient has a kernel of dimension n^2-1.

For every even n, the inherited pro-2 depth quotient, the inherited pro-2 class-two quotient, and every intrinsic quotient retaining the curve's own pro-2 abelianization have empty section sets over $\mathbb{Q}$. The obstruction is the explicit central extension with class $\kappa_2(-1)\cup\kappa_2(-1)$ described below.

Proof. Inertia above 1 maps to a conjugate of a power of y. Consequently I_n maps into $\langle\langle y\rangle\rangle$, and lower central series give (E25.1), including its exact kernel. The chosen sets of cusps are Galois stable, so their normal closures and lower central terms are preserved by the full arithmetic fundamental group. These are therefore arithmetic quotients, not just quotients of unmarked geometric groups.

By E.9 the map $N^{\mathrm{un}}_{n,\mathbb{Q}_\ell} \to F^{\mathrm{un}}_{\mathbb{Q}_\ell}$ is surjective: positive powers of x, y lie in N_n, and unique roots in characteristic-zero unipotent groups recover x, y. Its inherited quotient is thus the entire base quotient. The map on arithmetic extensions induces this surjection on geometric kernels and the identity on $G_{\mathbb{Q}}$. Quotienting by the geometric kernel of that surjection identifies the inherited arithmetic extension with the base one, which is split by the unit tangent at 0 towards 1. Equivalently its fibers are the full fibers of $(F^{\mathrm{un}}_{\mathbb{Q}_\ell}/D_y^3)\rtimes G_{\mathbb{Q}}$. Since both depth ideals lie in the respective derived Lie algebras, their abelianizations are the original ones. The kernel dimension is therefore exactly the n^2-1 of B.5 and E.13.

The integral obstruction uses the quotient

$$F \longrightarrow D_8 = \langle x, y, c \mid x^2 = y^2 = c^2 = 1,\ c \text{ central},\ [x,y] = c\rangle. \tag{E25.2}$$

With the standard rational tangential section, the Galois action on this quotient is trivial: in the normalized action of E.20 the cyclotomic exponent is odd, while the conjugating element lies in F' and hence has central image in D_8. The arithmetic base quotient is $D_8 \times G_{\mathbb{Q}}$.

In the plus model, with $t = -x^n/z^n$, the two covering roots are

$$(x/z)^n = -t, \qquad (y/z)^n = -(1-t). \tag{E25.3}$$

At the unit tangent $t = 0$, their constant root factors are nth roots of -1. For even n, their reductions to quadratic root factors both give $\epsilon = \kappa_2(-1) : G_{\mathbb{Q}} \to \mathbb{Z}/2$. The sign in the stabilizer condition has no effect modulo two. The geometric image of N_n in D_8 is exactly $\langle c\rangle$: its two exponent sums are even, and $[x,y] \in N_n$ maps to c. The arithmetic image is consequently the explicit pullback

$$\mathcal{P} = \{(d, g) \in D_8 \times G_{\mathbb{Q}} : d^{\mathrm{ab}} = (\epsilon(g), \epsilon(g))\}. \tag{E25.4}$$

For each g there is a lift in $\pi_1(U_n)$, and multiplication by the geometric element c gives both lifts in (E25.4); this proves equality of the image, not just its containment. Using $(xy)^{\epsilon(g)}$ as a set-theoretic lift gives the factor set $c^{\epsilon(g)\epsilon(h)}$. Thus (E25.4) represents $\kappa_2(-1)\cup\kappa_2(-1)$.

Complex conjugation has $\epsilon = 1$. Both possible lifts, xy and xyc, square to $c \neq 1$. The extension cannot have a section even after restriction to $G_{\mathbb{R}}$, hence cannot have one over $\mathbb{Q}$. The group D_8 kills $\gamma_3 F$ and hence D_y^3. Its geometric image from N_n is abelian, so it also kills N_n'. These two explicit factorizations prove all three nonexistence assertions in the statement. They recover E.18 at level four and prove the assertion for all even levels. Pushing to the

rank-two toric lattice kills the central element c, exactly as in E.18; extending scalars to $\mathbb{Q}_\ell$ gives the inherited comparison already proved. Thus neither operation is silently identified with the integral intrinsic quotient.

For clarity the intrinsic marked-depth quotient also has a precise comparison with the third derived quotient. Their common quotient is $N_n/(N_n''' D^3_{\mathrm{intr}})$. For the markings just specified neither kernel contains the other. Indeed filling the marked cusps gives free groups of ranks 9 and 13, respectively. On six distinct remaining generators use the word $w = [[[a,b],[a,c]],[[d,e],[d,f]]]$. Its degree-eight Magnus term has the monomial *abacdedf* with coefficient one, and is nonzero. This third-derived word survives the filling map, so N_n''' is not contained in D^3_{intr}. In the other direction choose a marked cusp generator z_0 and an unmarked free generator a. The element $[z_0,[z_0,[z_0,a]]]$ lies in $\gamma_3 I_n$. Its degree-four Magnus term is nonzero, whereas every element of N_n''' has zero Magnus terms below degree eight. Thus it does not lie in N_n'''. The corresponding maps from N_n, and the common quotient, state the relation without imposing the base rank-two containment from E.2 on a different free group. □

The first obstruction beyond the splitting of §51.12

E.26. The class-two splitting obstruction is the generator modulo three [Proved]

Let

$$P = \langle \rho, \sigma \mid \rho^3 = \sigma^2 = 1 \rangle = \mathrm{PSL}_2(\mathbb{Z}), \qquad q : P \longrightarrow S_3,$$

where $q(\rho)$ is a three-cycle and $q(\sigma)$ a transposition. The kernel is the free group

$$F = \langle x, y \rangle, \qquad x = (\sigma\rho)^2, \quad y = (\rho\sigma)^2.$$

The extension of S_3 by F/F' splits by B.11. Its class-two refinement

$$1 \longrightarrow F/\gamma_3 F \longrightarrow P/\gamma_3 F \longrightarrow S_3 \longrightarrow 1 \tag{E.26a}$$

does not split. Neither does its profinite completion, its relative pro-3 completion, or any of the discrete or profinite extensions $P/F^{(j)} \to S_3$ for $j \geq 2$.

Write $c = [x,y]$ and $r' = x\rho$. The inverse image in (E.26a) of the chosen abelian complement is

$$1 \longrightarrow \mathbb{Z}(\mathrm{sign}) \xrightarrow{1 \mapsto c} \langle r', \sigma \mid \sigma^2 = 1,\ \sigma r' \sigma = (r')^{-1} \rangle \longrightarrow S_3 \longrightarrow 1, \qquad (r')^3 = c. \tag{E.26b}$$

Its cohomology class is 1 in $H^2(S_3, \mathbb{Z}(\mathrm{sign})) \simeq \mathbb{Z}/3$. The finite quotient

$$P/(\gamma_3 F\, F^3) \longrightarrow S_3 \tag{E.26c}$$

has order 162, Heisenberg kernel of order 27, and no section. The inverse image of the chosen abelian complement is the dihedral group of order 18.

For the class-two relative pro-p extension, the set of continuous sections modulo conjugation by its kernel is empty at $p = 3$, has two elements at $p = 2$, and has one element at every $p \geq 5$. Over $\mathbb{Q}$ or $\mathbb{Q}_\ell$, the rational class-two relative extension does split, with one conjugacy class of sections. An explicit section in every one of these cases where it exists is

$$\bar\rho \longmapsto c^{-1/3} x\rho, \qquad \bar\sigma \longmapsto \sigma. \tag{E.26d}$$

Here c^a denotes the actual central coordinate a of the class-two kernel; it is defined over $\mathbb{Z}_p$ for $p \neq 3$ and over the indicated fields.

Manuscript: §51.12, folios 700–704 and 715–716, scans 105–114 and 135–138; in particular the abelian splitting on scan 136. This answers the splitting part of E(c) with its integral, profinite, and rational comparison maps.

Proof. The free-kernel identification and the matrices of its conjugation action were proved in A.2 and B.9. In the present central quotient of the modular group the exact free-group substitutions are

$$\rho x \rho^{-1} = y, \qquad \rho y \rho^{-1} = x^{-1}y^{-1}, \qquad \sigma x \sigma = y, \quad \sigma y \sigma = x. \tag{E.26e}$$

They also follow immediately by substituting the displayed words for x, y and using $\rho^3 = \sigma^2 = 1$. Consequently, already in P,

$$(x\rho)^3 = xyx^{-1}y^{-1} = c, \qquad ((x\rho)\sigma)^2 = 1. \tag{E.26f}$$

The second identity follows alternatively by reducing $(\sigma\rho\sigma\rho^2\sigma)^2$.

We specify the integral class-two group rather than replacing it by a vector space. Every element of $F/\gamma_3 F$ has a unique normal form $x^a y^b c^k$, $a, b, k \in \mathbb{Z}$, with law

$$(a, b, k)(a', b', k') = (a + a', b + b', k + k' - a'b). \tag{E.26g}$$

Existence is collection using the central commutator c. Uniqueness follows from the injective matrix model

$$x^a y^b c^k \longmapsto \begin{pmatrix} 1 & a & ab + k \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus c has infinite order. Conjugation by r' fixes c and conjugation by σ sends c to c^{-1}. The inverse image of the complement generated by $\overline{x\rho}, \bar{\sigma}$ in P/F' has elements $c^k(r')^i\sigma^j$, with $0 \le i < 3$, $0 \le j < 2$. Since $c = (r')^3$, these are exactly the normal forms of the infinite dihedral group in (E.26b). There are no additional relations: the image in S_3 first determines i, j, and $c^k = 1$ then forces $k = 0$.

For clarity we compute the cohomology class. Write elements of S_3 as $(i, j) = \bar{\rho}^i\bar{\sigma}^j$, with $0 \le i < 3$ and $0 \le j < 2$. The section $(i, j) \mapsto (r')^i\sigma^j$ gives the normalized integral two-cocycle

$$f((i, j), (i', j')) = \left\lfloor \frac{i + (-1)^j i'}{3} \right\rfloor, \qquad (i, j) \cdot a = (-1)^j a. \tag{E.26h}$$

Its restriction to the cyclic subgroup has carry 1, namely $(r')^3 = c$. We can compute the whole cohomology group directly. In an arbitrary extension by $A(\mathrm{sign})$, for $A = \mathbb{Z}$ or $\mathbb{Z}_3$, choose lifts r, s. The elements s^2 and $(rs)^2$ lie in A and are fixed by conjugation by s and rs, respectively. Those conjugations act as -1 on A; because A has no 2-torsion, both elements vanish. The only remaining parameter is $r^3 = a \in A$. Replacing r by br, with additive notation on A, changes a to $a + 3b$. Conversely every $a \in A$ defines an extension: take $(A \oplus \mathbb{Z}r)/(3r - a)$ and adjoin the involution acting by negation. This proves $H^2(S_3, A(\mathrm{sign})) = A/3A$ and identifies (E.26b) with 1. Equivalently, the one-cochain $(i, j) \mapsto i$ has coboundary $3f$.

To exclude sections of the whole group, we must also exclude other abelian complements. B.10 proves, over every coefficient group used here,

$$H^0(S_3, M) = H^1(S_3, M) = 0, \qquad M = F^{\mathrm{ab}}.$$

Thus conjugation by a unique element of M carries any abelian section to the B.11 section. The surjection $F/\gamma_3 F \to M$ lifts this conjugating element. After this conjugation any proposed section of (E.26a) must send $\bar{\rho}$ to $c^a r'$ and $\bar{\sigma}$ to $c^b\sigma$. Its order-three relation is

$$(c^a r')^3 = c^{1+3a} = 1. \tag{E.26i}$$

There is no solution $a \in \mathbb{Z}$ or $a \in \mathbb{Z}_3$. The other two defining relations hold for all a, b, because σ and $r'\sigma$ invert c and have order two. This proves the integral and relative pro-3 assertions.

Here are the exact completion and quotient comparisons. Since F is normal in P and $\gamma_3 F$, F^3, and $F^{(j)}$ are characteristic in F, every displayed quotient is a group extension over the same S_3. The relative pro-p class-two kernel is obtained by completing the normal form (E.26g) to coordinates in $\mathbb{Z}_p$. Conjugation (E.26e) extends continuously: its formulas use addition, multiplication, and $\binom{a}{2}$, the continuous integer-valued polynomial on $\mathbb{Z}_p$. The finite cosets of F give the corresponding continuous extension over S_3. The quotient (E.26c) receives (E.26a) by reducing its three kernel coordinates modulo three. It therefore also receives the profinite completion of (E.26a), continuously. A section of that completion would give a section of (E.26c).

To construct (E.26c) completely, over $\mathbb{F}_3$ the law (E.26g) is the Heisenberg group of order 27 and exponent three. It is exactly $F/(\gamma_3 F F^3)$: collection bounds its order by 27, and the matrix model realizes all 27 normal forms. The quotient has six such cosets, hence order 162. With states $x^a y^b c^k \rho^i \sigma^j$, right multiplication by σ toggles j. Right multiplication by ρ increments i when $j = 0$, and when $j = 1$ it multiplies the kernel coordinate by $\rho^i x \rho^{-i}$ and replaces i by $i + 2$ modulo three. The three kernel factors are $x, y, x^{-1}y^{-1}$. These rules construct its regular permutation representation without a presentation-order guess. The program `code/e_splitting.py` checks closure of the 162 normal-form permutations under both generators, all 4 251 528 associativity triples, and all $27^2 = 729$ pairs of possible section-generator images. There are no sections. Its abelian quotient of order 54 has exactly nine sections; all 81 pairs lifting these have their $\bar{\rho}$ image of order nine. The chosen complement has preimage $\langle r', \sigma \rangle = D_{18}$, with $(r')^3 = c$. The code also checks all 216 two-cocycle identities and all 243 normalized one-cochains modulo three: none has coboundary f.

For $j \geq 2$, the commutator identities $[\gamma_a F, \gamma_b F] \subset \gamma_{a+b} F$ give $F^{(j)} \subset \gamma_{2^j} F \subset \gamma_3 F$. Consequently the identity on generators gives the exact quotient map

$$P/F^{(j)} \twoheadrightarrow P/\gamma_3 F, \quad \ker = \gamma_3 F/F^{(j)}. \tag{E.26j}$$

A section of the source extension would compose to a section of (E.26a). For the profinite version, use the continuous map to the finite group (E.26c); its kernel contains the closed jth derived subgroup. The same composition gives the contradiction. This establishes the transfer to the requested third-derived quotient by an actual surjection.

Finally, the normalized lift calculation (E.26i) works with central coordinates in $\mathbb{Z}_p$, $\mathbb{Q}$, or $\mathbb{Q}_\ell$. It has the unique solution $a = -1/3$ exactly in the coefficient cases specified in the statement. The parameter b remains arbitrary. Any kernel conjugation preserving the normalized abelian complement has zero image in M, by $H^0(S_3, M) = 0$, and is therefore c^u. It changes b to $b + 2u$. Thus the conjugacy classes are $\mathbb{Z}_p/2\mathbb{Z}_p$ for $p \neq 3$, and a singleton over the indicated characteristic-zero fields. Formula (E.26d) gives their specified representative. In the cochain description, $-i/3$ cancels (E.26h), so the comparison to rational coefficients explicitly kills this integral obstruction. The rational group here is the class-two relative unipotent quotient over S_3: its six cosets are copies of the Heisenberg group (E.26g) over the field. To verify the universal comparison, a homomorphism from F to a class-two unipotent group, with images X, Y, extends uniquely by $x^a y^b c^k \mapsto X^a Y^b [X, Y]^k$. Rational powers are given by the finite exponential and logarithm polynomials. The finite-coset relations then give the unique extension of the original homomorphism from P over S_3. The program also checks the normalized central equations $1 + 3a = 0$ and the conjugacy relation $b \sim b + 2u$ modulo p^e for $p = 2, 3, 5, 7$ and $1 \leq e \leq 4$.

We also record precisely how the marked-depth construction in E.2 compares. There is a surjective Lie homomorphism

$$L/D^3 L \longrightarrow L/\gamma_3 L, \qquad \ker = \gamma_3 L/D^3 L,$$

because a monomial with at least three occurrences of y has total degree at least three.

The specific full permutation action does not descend to that marked quotient: the swap $x \leftrightarrow y$ sends $[y,[y,[y,x]]] \in D^3L$ to $[x,[x,[x,y]]]$, a nonzero depth-one element modulo D^3L. Thus (E.26j) is an S_3-extension comparison for the derived quotients, whereas the displayed marked-depth map has the stated Lie source and target. The rational section (E.26d) and the integral obstruction (E.26i) retain these exact domains and coefficient rings. □

Boundary types and the scope of the covering criterion

E.27. The four boundary types and their actual comparison maps [Proved]

Over an algebraically closed field of characteristic zero the geometric fundamental groups of types $(0,0),(0,1),(0,2),(1,0)$ are, respectively,

$$1, \qquad 1, \qquad \widehat{\mathbb{Z}}(1), \qquad TE \simeq \widehat{\mathbb{Z}}^2,$$

where a chosen origin gives the elliptic curve E in the last case. The last two groups are their own centers and their own abelian and metabelian quotients. The assertions about nontrivial comparison kernels in E.7–E.9 have the stated rank-two free-group source. For the discrete group $F_1 = \mathbb{Z}$ and its subgroup $d\mathbb{Z}$, the abelian, metabelian and fixed nilpotent comparisons to their correctly descended kernels are isomorphisms for every d. The rational completion map is multiplication by d on $\mathbb{Q}$, hence is also an isomorphism. For $d = p^a m$ with $(m,p) = 1$, the pro-p comparison is the isomorphism

$$\mathbb{Z}_p \longrightarrow p^a\mathbb{Z}_p, \qquad u \longmapsto du.$$

The same statements hold for a finite-index lattice in $\mathbb{Z}^2$, with its inclusion matrix in place of d.

For the corresponding nontrivial finite covers of $B\mathbb{Z}$ or $B\mathbb{Z}^2$, intrinsic-fibration membership for the five localizers of E.5 still holds only for the p-group covers in the pro-p case. Thus the equivalence between the two criteria in E.10 is proved for its specified source F_2; the displayed lattice comparisons compute what happens at these boundary sources.

Proof. For the projective line a connected degree-d finite etale cover has $2g - 2 = -2d$ by Riemann–Hurwitz, forcing $d = 1$. For the affine line, extend a finite etale cover to smooth projective curves. Ramification can occur only over infinity, and its total contribution is at most $d - 1$. Hence $2g - 2 \le -d - 1$, again forcing $d = 1$. For the twice-punctured line, ramification is allowed at two points. The same formula forces $g = 0$ and total ramification over each of them. Choosing coordinates at their unique preimages makes the covering $z \mapsto z^d$. The cofinal inverse system of these maps has deck groups μ_d, which proves the asserted Tate-module description and its action.

A finite etale cover of a genus-one curve has genus one by the same formula. Choose a point over the fixed origin. The resulting pointed map of elliptic curves is an isogeny: its map to the target Jacobian is the pointed curve map under the degree-zero divisor identification. Its kernel is finite; its dual isogeny shows that a multiplication map dominates it. The multiplication coverings are therefore cofinal, and their inverse deck system is TE. In characteristic zero $E[n] \simeq (\mathbb{Z}/n)^2$, giving the displayed geometric group. For the cubic over $\mathbb{Q}$ the specified origin is a rational coordinate cusp; E.24 computes its required arithmetic abelian cohomology. We impose no rational-point conclusion on a genus-one torsor without such an origin.

For the rank-one discrete group, every verbal commutator subgroup is zero, so the covering inclusion itself is exactly the displayed kernel map. Rationalization gives multiplication by d. In $\mathbb{Z}_p$, multiplication by m is a unit and multiplication by p^a maps bijectively onto the closed

subgroup $p^a\mathbb{Z}_p$. For a rank-two lattice, integral Smith normal form gives the same calculation in two coordinates. These arguments compute both the image and zero kernel, rather than applying the free-group rank formula outside its hypotheses. The program `e_boundary.py` checks the rank-one formulas for $d \le 24$ and the inverse unit coordinates modulo p^k, for $p = 2, 3, 5, 7$ and $k \le 4$.

Intrinsic-fibration membership is preserved by pullback, directly from its universal base-change definition. Pull a nontrivial lattice cover back by $BF_2 \to B\mathbb{Z}$ or $BF_2 \to B\mathbb{Z}^2$, using a surjection onto the lattice. Its monodromy is still the given finite quotient, so E.10 excludes the pulled-back cover in all the claimed cases. The general p-group sufficiency in E.10 proves the other cases. This supplies the comparison between the two criteria at every boundary source under discussion.

Real descent also retains its arithmetic data. The plus conic $X_2 : x_0^2+x_1^2+x_\infty^2 = 0$ has no real point, but its geometric projective fundamental group is trivial, so its arithmetic fundamental group maps isomorphically to $G_{\mathbb{Q}}$ and has its unique section. Deleting the coordinate divisors gives the nontrivial real obstruction treated by E.25. The exact open-to-proper quotient kills that obstruction. The elliptic and multiplicative centers above, the retained central C_2 in A.14, and the real orientation-reversing extension in E.18 are consequently part of the calculations rather than omitted exceptional cases. □

A cubic Kummer obstruction at the prime 109

E.28. A nonempty cubic abelian fiber with no class-two lifts [Proved]

In the pro-3 version of E.24, consider the toric section class

$$(\kappa(6), \kappa(109)) \in H^1(\mathbb{Q}, \mathbb{Z}_3(1)^2). \tag{E28.1}$$

Its fiber of geometrically abelian sections is nonempty and is a torsor under $\mu_3^3 \times \mathcal{R}_{\mathcal{I}}(\mathbb{Z}_3) \times \mathbb{Z}_3$. No point of this fiber lifts to a section of the curve's own class-two pro-3 quotient. No such lift exists for the intrinsic metabelian quotient, the intrinsic marked-depth quotient of E.25, or the inherited class-two pro-3 quotient either.

Proof. Let F_3 be the free pro-3 group on the base loops x, y, and put $\mathcal{H} = F_3/\gamma_3 F_3$. In its class-two Lie coordinates over $\mathbb{Z}_3$ write $X = \log x$, $Y = \log y$, $C = [X, Y]$. The Baker–Campbell–Hausdorff formula has only the denominator two, which is a unit in $\mathbb{Z}_3$. The image H of N_3 consists exactly of

$$\exp(3aX + 3bY + cC), \qquad a, b, c \in \mathbb{Z}_3. \tag{E28.2}$$

The exponent-sum condition proves containment, and the elements $x^3, y^3, [x, y] \in N_3$ generate the full lattice in (E28.2). In coordinates (a, b, c) its multiplication is

$$(a, b, c)(a', b', c') = \left(a + a', b + b', c + c' + \frac{9}{2}(ab' - ba')\right). \tag{E28.3}$$

In particular the commutator is $(0, 0, 9(ab' - ba'))$. The coordinates a, b are the two coordinate-unit homology coordinates of U_3; the covering pushforward to base homology is $(3a, 3b)$.

The normalized Galois action sends x to x^χ and y to $f_g^{-1} y^\chi f_g$, with $f_g \in F_3'$. In $\mathcal{H}$ this conjugator is central, so the action on (E28.2) is exactly

$$(a, b, c) \longmapsto (\chi(g)a, \chi(g)b, \chi(g)^2 c). \tag{E28.4}$$

The unit tangent at $t = 0$ lifts to the plus cubic by $w = -x/z$, $t = w^3$ and $y/z = -(1 - w^3)^{1/3}$. Its toric leading coefficients are $(-1, -1)$. Thus the base rational tangential section lifts to the

cubic and the inherited extension is $H \rtimes G_{\mathbb{Q}}$. The pro-3 Kummer classes of both these signs, and of the sign at the origin used in E.24, are zero. Changing between these two tangential origins therefore fixes the toric coordinates of every abelian class, so it leaves (E28.1) and its entire fiber unchanged.

Reduce all three coordinates in (E28.3) modulo 27. This gives a Galois-stable finite quotient H_{27} of order 27^3, of exponent 27, whose commutator of lifts of $(1,0)$ and $(0,1)$ is $(0,0,9) \neq 1$. At the local field $\mathbb{Q}_{109}$ the action (E28.4) on H_{27} is trivial: the 3-adic cyclotomic character is unramified there, with Frobenius $109 \equiv 1 \pmod{27}$. The integer 6 has exact order 108 in $\mathbb{F}_{109}^{\times}$. Consequently its Kummer class modulo 27 is unramified with Frobenius value a generator of μ_{27}, while the Kummer class of 109 has tame-inertia value a generator. Choose the common generator of μ_{27} so that the Frobenius value for 6 is 1; a generator of tame inertia can then be chosen with the value for 109 also equal to 1. We may choose Frobenius fixing the selected roots of the uniformizer. The two projected values are thus $(1,0)$ and $(0,1)$.

A section would therefore give a homomorphism $G_{\mathbb{Q}_{109}} \to H_{27}$ with those two projections. Wild inertia is pro-109 and maps trivially to this finite 3-group. In the tame quotient, choose Frobenius σ and tame inertia τ using roots of the uniformizer. Frobenius acts on their roots of unity by the 109th power, so $\sigma\tau\sigma^{-1} = \tau^{109}$. Since H_{27} has exponent 27, this relation forces the two images to commute. Equation (E28.3) gives their commutator as $(0,0,9)$, independently of their central coordinates, a contradiction. The program `e_cubic_class2.py` checks the order of 6 modulo 109, the exponent and the full multiplication formula, and all 27^2 choices of the two central lifts.

Finally the map $N_3 \to H_{27}$ kills $\gamma_3 N_3$, N_3'' and $\gamma_3 I_3$. These are the actual factorizations through the quotients in the statement. The explicit toric projection is (a,b), so every hypothetical lift of any point of the E.24 fiber would give the forbidden homomorphism. E.24 proves the asserted nonemptiness and size of the abelian fiber independently of this obstruction. There is no reduction map from the rational group $H_{\mathbb{Q}_3}$ to H_{27} extending this reduction on its integral lattice: every element of $H_{\mathbb{Q}_3}$ is a 27th power, whereas every 27th power in H_{27} is one. Thus any such homomorphism is trivial. The finite obstruction is used only for the indicated integral pro-3 quotients. The result does not classify the remaining cubic class-two sections whose toric pairs pass this local test. □

Every rational toric pair in the inherited cubic class-two quotient

E.29. The exact tame-residue criterion and the three classes in each nonempty fibre [Proved]

Let F be the free pro-3 geometric fundamental group of $\mathbb{P}^1 - \{0,1,\infty\}$, based at the rational tangent $t=0$, $dt=1$, and let $N_3 = \ker(F \to (\mathbb{Z}/3)^2)$. The image of N_3 in $F/\overline{\gamma_3 F}$ is the pro-3 group H with coordinates $(u,v,w) \in \mathbb{Z}_3^3$, law

$$(u,v,w)(u',v',w') = \left(u+u', v+v', w+w' + \frac{9}{2}(uv' - vu')\right), \tag{E.29a}$$

and arithmetic action

$$g(u,v,w) = (\chi_3(g)u, \chi_3(g)v, \chi_3(g)^2 w).$$

It is the inherited class-two quotient of the plus cubic; its arithmetic extension, with the lifted tangent, is $H \rtimes G_{\mathbb{Q}}$.

For $a, b \in \mathbb{Q}^*$ let $\mathscr{F}(a,b)$ be the fibre over (κ_a, κ_b) of the continuous section-class map

$$H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, H) \longrightarrow H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(1)^2).$$

For a rational prime q, put

$$\theta_q(a,b) = (-1)^{v_q(a)v_q(b)}\overline{a^{v_q(b)}/b^{v_q(a)}} \ \in \mathbb{F}_q^*. \tag{E.29b}$$

The expression under the bar is a q-adic unit. Let $o_3(\theta)$ denote the largest power of three dividing its multiplicative order. Then the complete answer on rational toric pairs is

$$|\mathscr{F}(a,b)| = \begin{cases} 3, & o_3(\theta_q(a,b)) \le 9 \text{ for every prime } q, \\ 0, & o_3(\theta_q(a,b)) > 9 \text{ for at least one prime } q. \end{cases} \tag{E.29c}$$

Only primes dividing a numerator or denominator of a or b need to be tested. For each nonempty fibre, choose one continuous cocycle $z(g) \in H$. Its three distinct classes are represented by

$$z_j(g) = \left(0, 0, j\frac{\chi_3(g)^2 - 1}{3}\right) z(g), \qquad j = 0, 1, 2. \tag{E.29d}$$

The choice of z is the stated ambiguity in this parametrization. The obstruction itself is exactly $9\kappa_a \cup \kappa_b \in H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(2))$.

For example, $(6, 109)$ and $(2, 163)$ have empty fibres, whereas $(6^3, 109)$ and $(2^9, 163)$ each have three classes. The source of this classification is the displayed inherited quotient. The intrinsic rank-ten class-two quotient maps onto it by

$$N_3/\overline{\gamma_3 N_3} \twoheadrightarrow H, \qquad \ker = (N_3 \cap \overline{\gamma_3 F})/\overline{\gamma_3 N_3}. \tag{E.29e}$$

Thus its section classes map to the fibres computed here.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379); the Kummer coordinates of the minimum quotient are retained, and their next inherited integral obstruction is computed. This statement treats every rational toric pair.

Proof. We first fix the comparison and its factors of three. Put $X = \log x$, $Y = \log y$, and $C = [X, Y]$ in the integral class-two pro-3 group; the denominator 2 in its Baker–Campbell–Hausdorff law is a unit in $\mathbb{Z}_3$. The image of N_3 has the lattice

$$A = 3X, \qquad B = 3Y, \qquad C = [X, Y], \qquad [A, B] = 9C.$$

Indeed its abelian coordinates are divisible by three, and $x^3, y^3, [x, y] \in N_3$ generate all these coordinates. This proves (E.29a) and the exact surjection (E.29e). For the normalized arithmetic action, $x \mapsto x^{\chi_3(g)}$ and $y \mapsto f_g^{-1} y^{\chi_3(g)} f_g$, where $f_g \in F'$. The element f_g is central after quotienting by $\overline{\gamma_3 F}$, giving the displayed diagonal action.

On the plus cubic put $u = x_0/x_\infty$, $v = x_1/x_\infty$. The covering is $t = -u^3$, $1 - t = -v^3$. At its rational cusp above $t = 0$, the parameter $s = -u$ gives $t = s^3$ and $v = -(1 - s^3)^{1/3}$ with constant coefficient -1. The tangential Galois section at $s = 0$, $ds = 1$ maps to the chosen base section: the corresponding embedding of Puiseux fields sends $t^{1/m}$ to $s^{3/m}$ and commutes with coefficientwise $G_{\mathbb{Q}}$-action. This specifies the comparison at the ramified cusp. Both leading toric units are -1, whose pro-3 Kummer classes are zero, since -1 has compatible rational 3^nth roots. Thus a toric cocycle (κ_a, κ_b) has inherited coordinates $\kappa_a A + \kappa_b B$ exactly. The comparison to base toric coordinates is multiplication by three, coming from $t = -u^3$, $1 - t = -v^3$.

Write $V = \mathbb{Z}_3(1)^2$, $Z = \mathbb{Z}_3(2)$, so that $1 \to Z \to H \to V \to 1$ is a central extension of pro-3 groups with $G_{\mathbb{Q}}$-action. For a continuous V-cocycle $v_g = (\alpha_g, \beta_g)$, use its continuous set-theoretic lift $(\alpha_g, \beta_g, 0)$. Its obstruction two-cocycle is

$$\omega(g, h) = \frac{9}{2}\left(\alpha_g \chi_3(g) \beta_h - \beta_g \chi_3(g) \alpha_h\right). \tag{E.29f}$$

The symmetric sum of the two cup cocycles is the negative coboundary of $g \mapsto \alpha_g \beta_g$. Hence the class of (E.29f) is $9[\alpha \cup \beta]$. Vanishing of this class means precisely that a continuous central one-cochain cancels (E.29f), which is the cocycle condition for a lift to H.

We now prove the criterion in continuous cohomology itself. The pairing

$$(a, b) \longmapsto \kappa_a \cup \kappa_b \quad \text{in } H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, Z)$$

is bilinear and satisfies the integral Steinberg relation. For the latter, the rational point $t = a$ of $\mathbb{P}^1 - \{0, 1, \infty\}$, for $a \neq 0, 1$, supplies an actual arithmetic section. Its class-two pro-3 quotient has toric cocycle (κ_a, κ_{1-a}) and commutator coefficient one. The calculation (E.29f) with 9 replaced by 1 therefore gives $\kappa_a \cup \kappa_{1-a} = 0$ in continuous integral cohomology. Bilinearity now defines a homomorphism

$$\beta : K_2^M(\mathbb{Q}) \longrightarrow H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, Z), \qquad \{a, b\} \longmapsto \kappa_a \cup \kappa_b. \tag{E.29g}$$

This constructs (E.29g) before taking any inverse limit of degree-two cohomology groups.

We use Matsumoto's presentation and the localization calculation in the opened source [Wei13, III.6.1, III.6.5–III.6.5.1, III.5.2.2]. For $\mathbb{Q}$ these give the exact sequence

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow K_2^M(\mathbb{Q}) \xrightarrow{(\partial_q)_q} \bigoplus_q \mathbb{F}_q^* \longrightarrow 0. \tag{E.29h}$$

The kernel is generated by $\{-1, -1\}$. In the convention of that source, $\partial_q\{a, b\} = \theta_q(a, b)^{-1}$; coordinatewise inversion is the explicit automorphism of the right-hand side relating the two conventions. Every element of the middle group is torsion: its finitely supported image has finite order, and its remaining kernel is killed by two. In particular (E.29h) identifies its 3-primary subgroup with $\bigoplus_q (\mathbb{F}_q^*)\{3\}$. The target of (E.29g) is a $\mathbb{Z}_3$-module, so β kills all primary components at primes different from three.

We give the local detection argument to prove injectivity on the remaining 3-primary subgroup. Let $q \neq 3$. For $T_n = \mu_{3^n}^{\otimes 2}$, wild inertia has no positive cohomology with these coefficients, and the 3-part of tame inertia is $\mathbb{Z}_3(1)$. The quotient Galois group of the finite residue field and this tame 3-part have 3-cohomological dimension one. Their two-term procyclic resolutions, followed by Hochschild–Serre, give

$$H^2(\mathbb{Q}_q, T_n) \simeq H^1(\mathbb{F}_q, \mu_{3^n}) \simeq (\mathbb{Z}/3^n)/(q - 1). \tag{E.29i}$$

In particular the degree-one finite-coefficient local groups are finite, so their first derived inverse limit vanishes. Passing to the inverse limit of (E.29i) therefore gives an actual continuous-cohomology isomorphism

$$\mathrm{res}_q^{\mathrm{tame}} : H^2_{\mathrm{cts}}(\mathbb{Q}_q, Z) \xrightarrow{\sim} H^1_{\mathrm{cts}}(\mathbb{F}_q, \mathbb{Z}_3(1)) \simeq \mathbb{Z}_3/(q - 1). \tag{E.29j}$$

Here is the cup-product normalization of this map. Choose arithmetic Frobenius ϕ and a tame generator τ with $\phi\tau\phi^{-1} = \tau^q$; these are the operators on compatible roots of a uniformizer. The conjugation formula is also given in the opened source [BC09, (8.2.2)]. For a unit u, its Kummer class is unramified. Write $a = \kappa_u(\phi)$ and choose ϕ fixing the chosen uniformizer roots, so $\kappa_q(\phi) = 0$ and $\kappa_q(\tau) = 1$. The extension two-cocycle $\kappa_u \cup \kappa_q$ has the relation defect qa between $\phi\tau\phi^{-1}$ and τ^q. Changing the central lift of τ by b changes that defect by $q(q-1)b$. Thus (E.29j), which divides the defect by q, sends this cup product to the unit's residue-field Kummer class. This specifies the map and its sign, and verifies the edge-map calculation directly. Two unit Kummer classes have zero cup product because they factor through the

procyclic residue Galois group. Also $\kappa_q \cup \kappa_q = 0$, since graded commutativity kills it by two and two is a unit in $\mathbb{Z}_3$. Writing $a = q^\alpha u$, $b = q^\beta v$ now yields

$$\operatorname{res}_q^{\mathrm{tame}}(\kappa_a \cup \kappa_b) = \kappa_{\overline{u}^\beta/\overline{v}^\alpha} = \kappa_{\theta_q(a,b)}. \tag{E.29k}$$

The sign in (E.29b) has zero pro-3 Kummer class. Finite-field Kummer theory identifies the image of an element of $\mathbb{F}_q^*$ in (E.29j) with its 3-primary component, preserving its order.

Every nonzero 3-primary element of $K_2^M(\mathbb{Q})$ has a nonzero tame component at some $q \neq 3$, by (E.29h), since $\mathbb{F}_3^*$ has order two. Equation (E.29k) detects that component under β. Consequently β is injective on this subgroup. It follows that

$$9\kappa_a \cup \kappa_b = 0 \quad \Longleftrightarrow \quad 9\{a,b\}_{\{3\}} = 0 \quad \Longleftrightarrow \quad o_3(\theta_q(a,b)) \leq 9 \quad \text{for every } q. \tag{E.29l}$$

This proves both necessity and sufficiency in continuous cohomology, including sufficiency for the existence of a continuous central cochain. No global inverse-limit injectivity assertion was used.

It remains to compute each nonempty fibre. We first compute

$$H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(2)) \simeq \mathbb{Z}/3. \tag{E.29m}$$

A continuous cocycle with values in $\mathbb{Q}_3(2)$ restricts on inertia at $q \neq 3$ to a homomorphism. Wild inertia contributes zero. If a is its value on a tame 3-generator, Frobenius equivariance gives $q^2 a = qa$, so $(q-1)a = 0$, hence $a = 0$. It therefore vanishes on the closed normal subgroup generated by these inertia groups and factors through $G_{\mathbb{Q},\{3\}}$. Theorem 7.2 and Theorem B.1 of the opened Hain–Matsumoto paper [HM03] give

$$\dim_{\mathbb{Q}_3} H^1_{\mathrm{cts}}(G_{\mathbb{Q},\{3\}}, \mathbb{Q}_3(2)) = r_2(\mathbb{Q}) = 0.$$

This proves the vanishing over the full $G_{\mathbb{Q}}$. The exact coefficient sequence $0 \to \mathbb{Z}_3(2) \to \mathbb{Q}_3(2) \to \mathbb{Q}_3/\mathbb{Z}_3(2) \to 0$ thus identifies (E.29m) with $H^0(G_{\mathbb{Q}}, \mathbb{Q}_3/\mathbb{Z}_3(2))$. The image of χ_3 is $\mathbb{Z}_3^*$; the ideal generated by $u^2 - 1$ for all units u is $3\mathbb{Z}_3$. The invariants are consequently $(\frac{1}{3}\mathbb{Z}_3)/\mathbb{Z}_3$, and the boundary of $j/3$ is exactly the central cocycle in (E.29d).

To see that this calculation gives the whole fibre, choose one lifted cocycle. The ratio of two lifts with the same toric cocycle is a central one-cocycle, and every such cocycle gives another lift. A kernel conjugation which preserves the exact toric cocycle has projection in $H^0(G_{\mathbb{Q}}, V) = 0$: complex conjugation acts by -1 on the torsion-free group V. Such a conjugation is therefore central and changes the ratio only by a central coboundary. When the toric cocycles initially differ by a coboundary, the surjection $H \to V$ supplies the conjugation which makes them equal. Thus the fibre is a torsor under the explicitly computed group (E.29m), with exactly the representatives (E.29d).

The program `code/e_rational_pairs.py` computes (E.29b) by exact rational arithmetic. Among all 40 000 ordered positive integer pairs up to 200, it finds 39 378 nonempty and 622 empty fibres. The histogram of the largest 3-primary residue order is

order	1	3	9	27	81
number of pairs	19036	15198	5144	364	258.

It also checks all 101 124 ordered pairs of the 318 nonzero signed reduced rationals of numerator and denominator height at most 16, and all 317 applicable Steinberg pairs in that set. It checks (E.29d) as a cocycle modulo 3^n for $1 \leq n \leq 4$, and the three twist-two invariants and absence of twist-one invariants at those finite levels. All output and every obstructed integer pair are retained in the accompanying JSON file. □

Arithmetic completion over the proconstant base

E.30. Arithmetic completion over the proconstant base kills the geometric group [Proved]

Put $U = \mathbb{P}^1_{\mathbb{Q}} - \{0, 1, \infty\}$ and $\mathcal{E} = \pi_1([U/S_3])$, with the positive rational tangential base point lifted to U. The quotient by its geometric free subgroup $\widehat{F}_2$ is $S_3 \times G_{\mathbb{Q}}$. For $m \geq 1$, let $K_m = \mathbb{Q}(\mu_m)$, $C_m = \mathrm{Gal}(K_m/\mathbb{Q})$ and $B_m = S_3 \times C_m$. There are two precise versions of the following construction: abstract relative completion over $k = \mathbb{Q}$, and continuous relative completion over $k = \mathbb{Q}_\ell$. Use the corresponding version throughout each formula, and let $\mathcal{A}_{m,k}$ be the relative completion of $G_{\mathbb{Q}} \to C_m$. The relative completion of $\mathcal{E} \to B_m$ is exactly

$$\mathcal{R}_{m,k} = S_3 \times \mathcal{A}_{m,k}, \qquad 1 \longrightarrow (G_{K_m})^{\mathrm{un}}_k \longrightarrow \mathcal{A}_{m,k} \longrightarrow C_m \longrightarrow 1. \tag{E.30.1}$$

For $m \mid n$ the transition maps are induced by the identity on $G_{\mathbb{Q}}$; on the displayed kernels they are induced by $G_{K_n} \hookrightarrow G_{K_m}$. Consequently the completion over the proconstant group scheme $B = S_3 \times \widehat{\mathbb{Z}}^\times_k$ is the specified inverse limit

$$\mathcal{R}_{B,k} = \varprojlim_m \mathcal{R}_{m,k}, \qquad 1 \longrightarrow \varprojlim_m (G_{K_m})^{\mathrm{un}}_k \longrightarrow \mathcal{R}_{B,k} \longrightarrow B \longrightarrow 1. \tag{E.30.2}$$

The natural homomorphism $\mathcal{E} \to \mathcal{R}_{B,k}(k)$ kills the whole geometric subgroup $\widehat{F}_2$. In particular its restriction to every Fermat subgroup is trivial; it retains none of the geometric homology computed in E.14.

There is a different, explicit completion retaining that geometry. Let $\mathcal{A}_\ell$ be the negatively weighted completion of $G_{\mathbb{Q}} \xrightarrow{\chi_\ell} \mathbb{G}_m(\mathbb{Q}_\ell)$, with central cocharacter $t \mapsto t^{-2}$. For the arithmetic group $\pi_1(U) = \widehat{F}_2 \rtimes G_{\mathbb{Q}}$ with the stated tangential section, its negatively weighted completion is

$$(\widehat{F}_2)^{\mathrm{un}}_{\mathbb{Q}_\ell} \rtimes \mathcal{A}_\ell. \tag{E.30.3}$$

The geometric factor is the free prounipotent group on two generators. No algebraic morphism from the proconstant B to $\mathbb{G}_m$ induces χ_ℓ on its cyclotomic factor.

Proof. The morphism $[U/S_3] \to [\mathrm{Spec}\,\mathbb{Q}/S_3]$ gives the stated quotient $S_3 \times G_{\mathbb{Q}}$, since the permutation action is defined over $\mathbb{Q}$. Its geometric kernel is $\widehat{F}_2$. Thus the kernel N_m of $\mathcal{E} \to B_m$ has the tangentially split sequence

$$1 \longrightarrow \widehat{F}_2 \longrightarrow N_m \longrightarrow G_{K_m} \longrightarrow 1.$$

On $\widehat{F}_2^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$ the arithmetic action is scalar multiplication by the full cyclotomic character.

First work continuously over $\mathbb{Q}_\ell$. Let $r : N_m \to U_0(\mathbb{Q}_\ell)$ be any continuous homomorphism to a unipotent algebraic group. Let A be the Zariski closure of $r(\widehat{F}_2)$. It is a connected unipotent normal subgroup of the Zariski closure of $r(N_m)$; unipotent groups in characteristic zero have no nontrivial finite algebraic quotients. The map to A^{ab} factors through $\mathbb{Z}^2_\ell$ and spans its Lie algebra over $\mathbb{Q}_\ell$. Choose $g \in G_{K_m}$ with $\chi_\ell(g) \neq 1$, possible because the cyclotomic image of this finite-index subgroup is infinite. Conjugation by $r(g)$ on $\mathrm{Lie}(A^{\mathrm{ab}})$ is both multiplication by $\chi_\ell(g)$ and a unipotent linear operator. The latter assertion follows, for example, by putting U_0 in upper triangular unipotent matrices and considering its conjugation action on its Lie algebra and invariant subquotients. It follows that $A^{\mathrm{ab}} = 1$. A nonzero nilpotent Lie algebra has nonzero abelianization, so $A = 1$. Every such r therefore kills $\widehat{F}_2$.

For the abstract rational version, the required abelianization is still exactly $\widehat{\mathbb{Z}}^2$: the abstract commutator subgroup of a finitely generated profinite group is closed by [NS07, Theorem1.4]. Choose $u \in \widehat{\mathbb{Z}}^\times$ with $u \equiv 1 \pmod m$ and $u - 1 = dv$, where $d \neq 0$ is an integer and

$v \in \widehat{\mathbb{Z}}^\times$, as in D.3. Explicitly use $u_p = 2$ for odd $p \nmid m$, $u_p = 1 + p^{v_p(m)}$ for odd $p \mid m$, and $u_2 = 1 + 2^{\max(1, v_2(m))}$. Cyclotomic surjectivity supplies $g \in G_{K_m}$ with $\chi(g) = u$. For an abstract homomorphism $r : N_m \to U_0(\mathbb{Q})$, there is an equivariant surjection

$$\widehat{\mathbb{Z}}^2 \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathrm{Lie}(A^{\mathrm{ab}}).$$

On the source $u - 1 = dv$ is invertible. Therefore $\mathrm{Ad}(r(g)) - 1$ on the target is surjective. It is also nilpotent, since $r(g)$ is unipotent. The target is zero, and the preceding nilpotent Lie algebra argument again gives $A = 1$. This proves the rational assertion without assuming continuity of a homomorphism to $U_0(\mathbb{Q})$.

For a group mapping onto a finite group C, its relative completion is the extension of C by the unipotent completion of the kernel. One can construct it directly by choosing coset representatives: multiplication on the finite disjoint union of copies of the completed kernel is given by the induced conjugation automorphisms and the original factor-set cocycle. This is the construction and universal-property proof used in E.14; it applies to the abstract and continuous categories just considered. The result above identifies N_m^{un} with $(G_{K_m})^{\mathrm{un}}$. Hence every representation defining $\mathcal{R}_{m,k}$ kills $\widehat{F}_2$ and factors uniquely through $S_3 \times G_{\mathbb{Q}}$. Completing this direct product gives $S_3 \times \mathcal{A}_{m,k}$: the finite S_3 image commutes with the arithmetic image, and these two homomorphisms have exactly the product universal property. This proves (E.30.1) and the specified transition maps.

An algebraic homomorphism from a proconstant profinite group scheme to a finite-type affine group scheme factors through a finite quotient. Indeed its finitely many coordinate generators belong to one finite stage of the coordinate-ring direct limit; the multiplication identities also hold at a sufficiently large stage. Every finite-dimensional algebraic representation of B therefore factors through some B_m. Defining relative completion by representations whose semisimplifications are algebraic representations of B gives precisely the inverse system (E.30.2): every such finite-dimensional representation occurs at one of its stages. This also proves its universal property. The transition maps on the unipotent kernels are faithfully flat: a finite-index subgroup has Zariski-dense image in the connected unipotent closure of the larger group's image. Inverse limits therefore give the exact proalgebraic sequence displayed in (E.30.2). The geometric image was already killed at every stage.

For (E.30.3), the genuine action of $G_{\mathbb{Q}}$ on the geometric prounipotent group has scalar χ_ℓ on its abelianization. Its kernel on that abelianization is prounipotent, with negative weights for $t \mapsto t^{-2}$: the degree-d part of its IA Lie algebra maps a degree-one generator to a degree-d word and has weight $-2(d-1) < 0$ for $d \geq 2$. Thus the universal property of weighted completion extends this action uniquely to $\mathcal{A}_\ell$; the weighted conventions and construction are those of [HM03, §§2–4]. The semidirect product in (E.30.3) receives the actual arithmetic group. Given any lift of that group into a negatively weighted extension of $\mathbb{G}_m$, its restriction to $\widehat{F}_2$ extends uniquely by continuous unipotent completion, and its restriction to the tangential $G_{\mathbb{Q}}$ extends uniquely by weighted completion. Their equivariance holds on the dense original images, hence algebraically. Thus these two extensions define a unique homomorphism from the semidirect product, proving its weighted universal property. Its geometric kernel is visibly the free prounipotent group, proving its injection as well. Finally an algebraic map $B \to \mathbb{G}_m$ has finite image by the finite-stage argument, whereas $\chi_\ell(G_{\mathbb{Q}})$ is infinite. This proves the claimed obstruction to that specific base-change comparison. We do not identify an abstract rational completion after scalar extension with a continuous ℓ-adic completion; the two constructions above specify their source categories independently. □

An intrinsic cubic obstruction inside the zero toric fiber

E.31. Two abelian classes in one toric fiber have different liftability [Proved]

Use the pro-3 version of E.24, with its unit rational tangent at $O = [1 : -1 : 0]$. Put $K = \mathbb{Q}(\zeta)$, $\lambda = 1 - \zeta$ and $\pi = 3 + \zeta$, so $N_{K/\mathbb{Q}}(\pi) = 7$. In the frame coordinates of (E24.4) take the rational torus point

$$t = (7, 7, 1, 1, \pi) \in T_D(\mathbb{Q}) \subset G(\mathbb{Q}). \tag{E31.1}$$

Its Kummer class $\alpha \in H^1(G_{\mathbb{Q}}, T_3G)$ has zero image in the toric minimum. Let $\mathcal{K}$ denote the kernel of that toric map on section classes. Every class in the nonempty coset $\alpha + 3\mathcal{K}$ fails to lift to the cubic's own class-two pro-3 arithmetic quotient, its metabelian quotient, and its marked-depth quotient of E.25. The zero class in the same toric fiber lifts to a full tangential section.

The obstruction is visible in an explicit quotient of the intrinsic based Galois action over K. Let H_3 be the group of order 27 with coordinates $(a, b, c) \in \mathbb{F}_3^3$ and multiplication

$$(a, b, c)(a', b', c') = \left(a + a', b + b', c + c' + \tfrac{1}{2}(ab' - ba')\right). \tag{E31.2}$$

There is a surjection from the cubic geometric group to H_3 whose Galois action, with the tangent at O as origin, is

$$g : (a, b, c) \longmapsto (a, b, c + \kappa_3(\zeta)(g)b), \qquad g \in G_K. \tag{E31.3}$$

Thus (E31.3) gives the arithmetic cross term explicitly, as well as a finite obstruction to the indicated section classes.

Proof. Equation (E24.4) sends (E31.1) to $(7/N(\pi), 7/N(\pi)) = (1, 1)$. Kummer functoriality therefore puts α in $\mathcal{K}$. This is a genuine geometrically abelian section class by E.24. It is nonzero modulo three: the valuation at the prime (π) in the third $K^\times$ factor of (E24.7) is zero on its displayed λ relation, and takes the value one on t. It extends to an integral homomorphism on that rational-point presentation, so t is not three times a rational point of G'.

Over K use the actual morphism

$$h = \frac{x + y}{x + \zeta y} : U_3 \longrightarrow \mathbb{P}^1 \setminus \{0, 1, \infty\}, \qquad 1 - h = \frac{(\zeta - 1)y}{x + \zeta y}. \tag{E31.4}$$

The equations $h = 0$ and $h = \infty$ force $z = 0$ on the cubic; $h = 1$ forces $y = 0$. All these points have been removed from U_3, so the displayed target and morphism are exact. Write $C_i = [1 : -\zeta^i : 0]$ and $B_i = [-\zeta^i : 0 : 1]$. The numerator and denominator tangent lines give

$$\operatorname{div}(h) = 3O - 3C_2, \qquad \operatorname{div}(1 - h) = B_0 + B_1 + B_2 - 3C_2. \tag{E31.5}$$

The frame at C_2 is $\bar{\pi}$, not π. Formula (E24.3) consequently gives the exact character values

$$\chi_h(t) = \bar{\pi}^{-3}, \qquad \chi_{1-h}(t) = 7\bar{\pi}^{-3}. \tag{E31.6}$$

The based arithmetic map is obtained by applying (E31.4) to the tangential fiber functor at O and choosing a geometric path to the standard unit tangent at zero in the target. Its affine effect can be computed without a path convention. At O put $s = z/x$; then $y/x = -(1 + s^3)^{1/3}$, and hence

$$h = -\frac{s^3}{3(1 - \zeta)} + O(s^6) = \zeta\lambda^{-3}s^3 + O(s^6), \qquad 1 - h = 1 + O(s^3). \tag{E31.7}$$

The identity in the second expression is $\lambda^2 = -3\zeta$. The map of Puiseux fiber functors under (E31.7) has the constant Kummer factors $(\zeta\lambda^{-3}, 1)$. Thus the image of the tangential section at O has, modulo three, abelian coordinates $(\kappa_3(\zeta), 0)$ relative to the standard target origin. Combining (E31.6) and (E31.7), the abelian section with class α has target coordinates

$$(\kappa_3(\zeta\lambda^{-3}\bar{\pi}^{-3}), \kappa_3(7\bar{\pi}^{-3})) = (\kappa_3(\zeta), \kappa_3(7)). \tag{E31.8}$$

This retains the nonzero affine origin term explicitly.

Take the quotient of the target free pro-3 group in which the two generators and their central commutator have order three; it is (E31.2). Its standard based Galois action over K is trivial. Indeed $\mu_3 \subset K$, and the normalized action of E.20 has cyclotomic exponent one modulo three and a conjugator with central image. Let $\phi_h : N_3 \to H_3$ be the geometric map induced by (E31.4). It is surjective. The two geometric Kummer characters $h, 1-h$ are independent modulo cubes: the valuation at a B_i forces the coefficient of $1-h$ to be zero modulo three; h is not a cube, since such a cube root would have divisor $O - C_2$, a degree-one map from an elliptic curve to $\mathbb{P}^1$. Their independent abelian images generate $H_3/[H_3, H_3]$, and their commutator generates its central subgroup, proving surjectivity.

Let $q_g \in H_3$ be the first component of the image of the tangential section at O in $H_3 \times G_K$. Equivariance of the arithmetic map gives $\phi_h(gng^{-1}) = q_g\phi_h(n)q_g^{-1}$. Its abelian coordinate is $(\kappa_3(\zeta)(g), 0)$ by (E31.7); its central coordinate does not affect conjugation. Conjugation in (E31.2) is therefore exactly (E31.3). The character $\kappa_3(\zeta)$ is nonzero: a cube root of ζ would be a unit in $\mathbb{Z}[\zeta]$, whose unit group μ_6 has only ± 1 as cubes. This proves the asserted nonzero arithmetic cross term.

Now pass to the prime (π) over 7. Its residue field is $\mathbb{F}_7$ and $\zeta \equiv -3 \equiv 4 \pmod{\pi}$, so $K_{(\pi)} = \mathbb{Q}_7$. The character $\kappa_3(\zeta)$ is unramified and nonzero: choose $\zeta_9^3 = \zeta$ in the unramified cubic extension. Arithmetic Frobenius sends ζ_9 to ζ_9^7, whose ratio to ζ_9 is ζ^2. The Kummer character of the uniformizer 7 has tame-inertia value ζ after a choice of generator, and Frobenius can be chosen to fix its selected roots. Consequently (E31.8) sends Frobenius and tame inertia to $(2, 0)$ and $(0, 1)$ in H_3^{ab}. Every pair of lifts has commutator $(0, 0, 2) \neq 1$.

An arithmetic section lifting α would yield a homomorphism $G_{\mathbb{Q}_7} \to H_3$, because the standard target action is trivial. Wild inertia is pro-7 and maps trivially to this finite 3-group. The tame relation is $\sigma\tau\sigma^{-1} = \tau^7$. Every element of H_3 has exponent three, so this relation forces its two images to commute, a contradiction. The exact field identities, the local leading constants, all 27 group elements and all nine choices of the two central lifts are checked by `e_intrinsic_obstruction.py`.

The map ϕ_h kills $\gamma_3 N_3$, N_3'' and $\gamma_3 I_3$. Thus it factors through each intrinsic quotient named in the statement. These factorizations also apply to the arithmetic maps after restriction to G_K, which suffices to rule out a section over $\mathbb{Q}$. There is also a single map over $\mathbb{Q}$ displaying both conjugate comparisons: the adjoint of (E31.4) is

$$U_3 \longrightarrow \mathrm{Res}_{K/\mathbb{Q}}(\mathbb{P}^1 \setminus \{0, 1, \infty\}). \tag{E31.9}$$

After base change to $\overline{\mathbb{Q}}$ it is $(h, \bar{h})$, giving the finite geometric image inside $H_3 \times H_3$. The arithmetic map to this image is defined over $\mathbb{Q}$ and its geometric kernel contains $\gamma_3 N_3$. Its restriction has the first projection used above. This supplies an actual global finite quotient comparison as well.

Adding any element of $3\mathcal{K}$ to α leaves (E31.8) unchanged, since the target abelian coefficient group has exponent three. The same obstruction applies to the whole coset $\alpha + 3\mathcal{K}$. The zero class is the original rational tangential section and lifts to the full arithmetic fundamental group. Both lie over the zero pro-3 toric class, as asserted. In particular this proof distinguishes liftability inside one specified fiber, using a quotient that detects an intrinsic direction beyond the inherited toric comparison of E.28. □

All completed toric pairs in the inherited cubic class-two quotient

E.32. The complete convergence criterion, including the invisible inverse-limit obstruction [Proved]

Retain the arithmetic group $H \rtimes G_{\mathbb{Q}}$ of E.29. For a set I write

$$\mathcal{R}_I = \{(z_i) \in \textstyle\prod_{i\in I} \mathbb{Z}_3 : \text{for every } n,\ z_i \in 3^n\mathbb{Z}_3 \text{ outside a finite subset of } I\}.$$

Let $\mathcal{P}$ be the rational primes and $\mathcal{P}^* = \mathcal{P} - \{3\}$. The Kummer isomorphism is

$$H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(1)) = \mathcal{R}_{\mathcal{P}}, \qquad (a_p)_p \longmapsto \sum_p a_p \kappa_p. \tag{E.32a}$$

Choose a compatible system of roots for each κ_p; the sum in (E.32a) converges uniformly as a cocycle.

Here is an explicit criterion for every pair $a = (a_p), b = (b_p) \in \mathcal{R}_{\mathcal{P}}$. For $q \neq 3$ put $e_q = v_3(q-1)$ and choose a primitive 3^{e_q}th root $\eta_q \in \mathbb{F}_q^*$ when $e_q > 0$. For $p \neq q$ let $r_{q,p} \in \{0, \dots, 3^{e_q} - 1\}$ be specified by

$$\overline{p}^{(q-1)/3^{e_q}} = \eta_q^{r_{q,p}}.$$

When $e_q = 0$, put $r_{q,p} = 0$. Define convergent sums and a central correction by

$$A_q = \sum_{p\neq q} a_p r_{q,p}, \qquad B_q = \sum_{p\neq q} b_p r_{q,p}, \qquad c_q = -\frac{9(A_q b_q - B_q a_q)}{q-1} \in \mathbb{Q}_3. \tag{E.32b}$$

The fibre of $H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, H)$ over the pair in (E.32a) has exactly three classes precisely when

$$c_q \in \mathbb{Z}_3 \quad (q \neq 3), \qquad (c_q)_{q\neq 3} \in \mathcal{R}_{\mathcal{P}^*}; \tag{E.32c}$$

in all other cases that fibre is empty. The criterion is independent of the roots and generators used above. For a nonempty fibre its three representatives are given by (E.29d), starting with any one lift.

The continuous degree-two group, and the morphism used to compute this obstruction, are

$$\begin{aligned} H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(2)) &\xrightarrow[\sim]{\mathrm{loc}_{\mathrm{cts}}} \mathcal{R}_{\mathcal{P}^*}/d\mathcal{R}_{\mathcal{P}^*}, \\ 9\Big(\sum_p a_p\kappa_p\Big) \cup \Big(\sum_p b_p\kappa_p\Big) &\longmapsto [\,9(A_q b_q - B_q a_q)\,]_q. \end{aligned} \tag{E.32d}$$

Here $d(z)_q = (q-1)z_q$. These are isomorphisms of $\mathbb{Z}_3$-modules; the denominator is the actual image of d, with no closure added. The map from this quotient to the product of local groups is coordinatewise reduction modulo $q-1$. Its exact kernel is

$$\frac{\{z \in \mathcal{R}_{\mathcal{P}^*} : z_q \in (q-1)\mathbb{Z}_3 \text{ for all } q\}}{d\mathcal{R}_{\mathcal{P}^*}}. \tag{E.32e}$$

This kernel contains an obstruction of an actual completed toric pair. Choose recursively increasing primes q_j, $j \geq 3$, with

$$q_j \equiv 1 \pmod{3^j}, \qquad 3^{(q_j-1)/3} \not\equiv 1 \pmod{q_j},$$

and write $r_j = r_{q_j,3} \in \mathbb{Z}_3^*$. Take $a_3 = 1$, $a_p = 0$ for $p \neq 3$, and

$$b_{q_j} = \frac{q_j - 1}{9 r_j}, \qquad b_p = 0 \text{ at the other primes.} \tag{E.32f}$$

Then $b \in \mathcal{R}_{\mathcal{P}}$ and every local obstruction and every finite-coefficient global obstruction vanishes. There is nevertheless no continuous section with this completed toric pair: its corrections are $c_{q_j} = -1$ for all j. Every finite-support partial pair obtained from (E.32f) does have three continuous section classes.

The classification concerns the inherited quotient H. The map from the cubic's intrinsic quotient is exactly (E.29e); the additional intrinsic obstruction E.31 remains present. *Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379), with its completed Kummer parameters retained.*

Proof. We first calculate the continuous cohomology group and its localization morphism, retaining the derived inverse limit. Put

$$U = \operatorname{Spec} \mathbb{Z}[1/3], \qquad G_S = G_{\mathbb{Q},\{3,\infty\}}, \qquad T_n = \mathbb{Z}/3^n(2), \qquad T = \mathbb{Z}_3(2).$$

Milne's comparison for locally constant sheaves [Mil06, II.2.9] identifies $R\Gamma(U, T_n)$ with $R\Gamma(G_S, T_n)$. We need the following actual vanishing:

$$H^2(G_S, T_n) = H^3(G_S, T_n) = 0, \quad H^1(G_S, T_n) \text{ finite}, \qquad n \geq 1. \tag{E.32g}$$

For $n = 1$, $T_1 = \mathbb{F}_3$ with trivial action, since $\chi_3^2 \equiv 1 \pmod 3$. The Kronecker–Weber theorem [Mil20, I.4.16] gives $H^1(G_S, \mathbb{F}_3) = \mathbb{F}_3$. More explicitly, an abelian extension unramified away from three is contained in a cyclotomic field, and its quotient kills the inertia factors at all other primes. Its cubic quotient therefore factors through $\operatorname{Gal}(\mathbb{Q}(\zeta_{3^m})/\mathbb{Q})$, which has a unique quotient of order three. The nontrivial quotient already occurs in $\mathbb{Q}(\zeta_9)^+$. Both $H^0(G_S, \mathbb{F}_3)$ and $H^1(G_S, \mathbb{F}_3)$ have order three. The global Euler characteristic [Mil06, I.5.1] equals one here: its real factor is $|H^0(G_{\mathbb{R}}, \mathbb{F}_3)|/3 = 1$. Consequently $H^2(G_S, \mathbb{F}_3) = 0$. Filter T_n by its powers of three. Each successive quotient is the same trivial $\mathbb{F}_3$, so the long exact cohomology sequence proves $H^2(G_S, T_n) = 0$ for all n. Milne I.4.10(c) identifies every degree at least three with real-place cohomology, which is zero for a module of odd order. Finiteness is I.4.15. This proves (E.32g).

For a finite set $J \subset \mathcal{P}^*$, apply the localization triangle to $U_J = U - \{q : q \in J\}$, using [Mil06, II.0.1(f), II.1.7(b)]. Here the local purity identification can also be checked directly: over the strict Henselization at q, a locally constant T_n is constant; the fraction field has tame 3-inertia $\mathbb{Z}_3(1)$, while its wild inertia is pro-q. Its cohomology is T_n in degree zero, $T_n(-1)$ in degree one, and zero above degree one. The local support complex is therefore $T_n(-1)[-2]$, with its residue-field action. Thus there is the compatible distinguished triangle

$$R\Gamma(U, T_n) \longrightarrow R\Gamma(U_J, T_n) \longrightarrow \bigoplus_{q \in J} R\Gamma(\mathbb{F}_q, \mathbb{Z}/3^n(1))[-1] \longrightarrow R\Gamma(U, T_n)[1]. \tag{E.32h}$$

First take the filtered direct limit over finite J. Filtered direct limits of abelian groups are exact. On the generic-field term this gives $R\Gamma(G_{\mathbb{Q}}, T_n)$: every continuous cochain with finite coefficients factors through a finite quotient of $G_{\mathbb{Q}}$, whose number field is ramified at only finitely many primes. In particular the third term becomes a direct sum at each fixed coefficient level n; no infinite product is substituted.

Next take $R\varprojlim_n$ of these triangles. The comparison with continuous cochains is Jannsen's theorem [Jan88, (2.2), together with (1.6)]. For the first term (E.32g), the finiteness of the degree-one groups kills their first derived inverse limit. It follows that

$$H^2(R\varprojlim_n R\Gamma(U, T_n)) = H^3(R\varprojlim_n R\Gamma(U, T_n)) = 0. \tag{E.32i}$$

For the support term, the procyclic residue Galois group has the compatible two-term cochain model

$$R\Gamma(\mathbb{F}_q, \mathbb{Z}/3^n(1)) \simeq [\,\mathbb{Z}/3^n \xrightarrow{q-1} \mathbb{Z}/3^n\,] \quad \text{in degrees } 0, 1. \tag{E.32j}$$

Choose arithmetic Frobenius for this model. After taking the direct sum over $q \neq 3$, the transition in each of its two terms is the surjective reduction $\bigoplus_q \mathbb{Z}/3^{n+1} \to \bigoplus_q \mathbb{Z}/3^n$. The inverse systems of terms are therefore acyclic for $\varprojlim$. Their derived inverse limit is the literal complex

$$[\,\mathcal{R}_{\mathcal{P}^*} \xrightarrow{d} \mathcal{R}_{\mathcal{P}^*}\,]. \tag{E.32k}$$

Indeed a compatible sequence of finitely supported residues is exactly a family of 3-adic integers tending to zero. The degree-two segment of the resulting triangle, using (E.32i), proves the first isomorphism in (E.32d). This calculation takes the inverse limit of complexes. It does not replace it by the inverse limit of their cohomology.

We next compute the image of the actual obstruction under this morphism. The Kummer sequence and $\mathbb{Q}^* = \{\pm 1\} \times \bigoplus_p p^{\mathbb{Z}}$ give (E.32a); the sign factor has trivial pro-3 completion. Multiplying a fixed cocycle κ_p by a_p and summing is legitimate on continuous cochains: for each n, all but finitely many summands vanish modulo 3^n, uniformly on $G_{\mathbb{Q}}$. Write these sums as α, β. Their obstruction is the continuous cocycle

$$\omega(g,h) = \frac{9}{2}\{\alpha(g)\chi_3(g)\beta(h) - \beta(g)\chi_3(g)\alpha(h)\}, \tag{E.32l}$$

representing $9\alpha \cup \beta$ as in E.29.

Fix $q \neq 3$ and a decomposition group. Choose tame inertia τ_q and arithmetic Frobenius ϕ_q such that

$$\phi_q \tau_q \phi_q^{-1} = \tau_q^q, \qquad \kappa_q(\tau_q) = 1, \qquad \kappa_q(\phi_q) = 0. \tag{E.32m}$$

These choices use compatible roots of a uniformizer, as in E.29. Put $\alpha(\tau_q) = a_q$, $\beta(\tau_q) = b_q$ and $\alpha(\phi_q) = A'_q$, $\beta(\phi_q) = B'_q$. On the full inertia group the two cocycles are proportional to the same tame character; wild inertia maps to zero. Consequently (E.32l) is identically zero on two inertia arguments, giving its canonical central-zero inertia splitting. In the pullback of $H \rtimes G_{\mathbb{Q}}$, let the central coordinates of the two lifts be w at τ_q and z at ϕ_q. Direct multiplication in (E.29a) gives the defect between the two sides of (E.32m) as

$$q\{(q-1)w + 9(A'_q b_q - B'_q a_q)\}. \tag{E.32n}$$

It is independent of z. In particular the central-zero inertia splitting represents the residue cochain with value $9(A'_q b_q - B'_q a_q)$ at Frobenius. The division by q in (E.32n) is the identification $\mathrm{Hom}(\mathbb{Z}_3(1), T) = T(-1)$: the conjugation defect at τ_q^q is pulled back to its value at τ_q. Changing the inertia splitting by w adds precisely $(q-1)w$, the differential in (E.32j). This is the chain-level localization morphism in (E.32h): it records the failure of an inertia splitting to be invariant under residue-field Frobenius.

At each finite coefficient level these inertia splittings agree with an unramified splitting outside finitely many primes, since α, β factor through a finite quotient modulo 3^n. They therefore form a compatible representative in the direct-sum complexes used in (E.32h)–(E.32k). The resulting family tends to zero, since it contains a factor a_q or b_q in each summand. This establishes the localization representative itself, including its class in the quotient (E.32k).

For a unit p at q, finite-field Kummer theory gives

$$\kappa_p(\phi_q) \equiv r_{q,p} \pmod{q-1}.$$

Choose the place over q to send the selected global 3^{e_q}th root of unity to η_q. The formula follows by applying Frobenius to a 3^{e_q}th root of p: its quotient is $\overline{p}^{(q-1)/3^{e_q}}$. Write $\kappa_p(\phi_q) - r_{q,p} = (q-1)t_{q,p}$ with $t_{q,p} \in \mathbb{Z}_3$. The sums of $a_p t_{q,p}$ and $b_p t_{q,p}$ converge and are uniformly bounded in $\mathbb{Z}_3$. Hence replacing A'_q, B'_q by A_q, B_q changes the representative in (E.32d) by d of a family tending to zero: the change divided by $q-1$ has a factor a_q or b_q. This proves the second

formula in (E.32d). The same argument proves independence of root choices. Changing a residue generator scales the corresponding coordinate by a 3-adic unit and adds a term of this form, so it also preserves (E.32c).

For completeness, changing global cocycles by $\alpha \mapsto \alpha + du$, $\beta \mapsto \beta + dv$, with $du(g) = (\chi_3(g) - 1)u$, changes the residue representative by

$$(q-1)\,9(ub_q - va_q).$$

The family in the second factor lies in $\mathcal{R}_{\mathcal{P}^*}$. The corresponding central inertia correction changes by $-9(ub_q - va_q)$, exactly as the change $z_g \mapsto h^{-1} z_g g(h)$ with $h = (u, v, 0)$ gives. Also replacing ϕ_q by $\tau_q^r \phi_q$ leaves $A'_q b_q - B'_q a_q$ unchanged. These formulas specify both the coboundary sign and the actual coefficient comparisons.

The obstruction vanishes precisely when its representative lies in $d\mathcal{R}_{\mathcal{P}^*}$. Since $q - 1 \neq 0$ in $\mathbb{Z}_3$, its unique possible central correction is (E.32b). This proves (E.32c) and the empty-fibre assertion. The central group has $H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, T) = \mathbb{Z}/3$ by E.29, and $H^0(G_{\mathbb{Q}}, \mathbb{Z}_3(1)^2) = 0$. The exact nonabelian central-extension sequence therefore makes each nonempty fibre a torsor under this group. Its three explicit twists are (E.29d).

We finally verify the inverse-limit example. The field $\mathbb{Q}(\sqrt[3]{3})$ cannot be contained in a cyclotomic field: a subfield of an abelian Galois extension is Galois, whereas this irreducible real cubic does not contain the two nonreal roots of $X^3 - 3$. Thus $\mathbb{Q}(\zeta_{3^j}, \sqrt[3]{3})/\mathbb{Q}(\zeta_{3^j})$ has degree three. A nonidentity element of this subgroup has a conjugacy class restricting trivially to the cyclotomic field. Chebotarev [Mil20, V.3.23] provides infinitely many rational primes with that class, giving the two requirements on q_j. Choose each q_j to be the least such prime larger than the previous choice. Noncubicity says r_j is a 3-adic unit, so $v_3(b_{q_j}) = e_{q_j} - 2 \geq j - 2$. This proves that (E.32f) is an actual completed Kummer pair. Its representative in (E.32d) is

$$z_{q_j} = q_j - 1, \qquad z_q = 0 \text{ otherwise.} \tag{E.32o}$$

It is a restricted family, but its only preimage under coordinatewise multiplication by $q - 1$ is the family with value one at every q_j, which is not restricted. Thus its class is nonzero in (E.32d), despite lying in the kernel (E.32e). At every finite level its class is zero in the cohomology of the direct-sum complex: only finitely many entries of (E.32o) survive modulo 3^n, and they are obtained by applying d to the finitely supported family with value one at those entries. The finite-level global obstruction is zero by (E.32g)–(E.32h). At $q \neq 3$ the local obstruction is zero by coordinatewise divisibility. At three, finite local duality [Mil06, I.2.3] gives $H^2(\mathbb{Q}_3, T_n) = H^0(\mathbb{Q}_3, \mathbb{Z}/3^n(-1))^* = 0$: the local cyclotomic character contains -1, whose action has no fixed vector on this odd-order module. Finite local degree-one groups then give $H^2_{\mathrm{cts}}(\mathbb{Q}_3, T) = 0$ as well. Real cohomology with these coefficients is zero. Every finite-support partial pair in (E.32f) has finitely many corrections equal to -1, so it satisfies (E.32c). All assertions follow. □

The continuous arithmetic kernel and its trace maps

E.33. The continuous arithmetic kernels are free, with explicit generators as a Galois module [Proved]

Fix a prime ℓ and put $K_m = \mathbb{Q}(\mu_m)$, $d_m = [K_m : \mathbb{Q}]$ and $r_m = r_2(K_m) + 1$. The continuous unipotent completion occurring in E.30 is

$$(G_{K_m})^{\mathrm{un}}_{\mathbb{Q}_\ell} \simeq \exp \widehat{\mathbb{L}}(V_m), \qquad V_m = \frac{K_m \otimes_{\mathbb{Q}} \mathbb{Q}_\ell}{\log_\ell(\mathcal{O}^\times_{K_m} \otimes_{\mathbb{Z}} \mathbb{Q}_\ell)}. \tag{E.33.1}$$

Here $\widehat{\mathbb{L}}$ is completion of the free Lie algebra by bracket length. The vector space V_m is the canonical abelianization of this Lie algebra; choosing lifts of a basis gives the displayed noncanonical free presentation. It has dimension r_m. For $m \geq 3$, complex conjugation c gives the explicit $\mathrm{Gal}(K_m/\mathbb{Q})$-equivariant identification

$$V_m \xrightarrow{\sim} \mathbb{Q}_\ell \oplus (K_m \otimes \mathbb{Q}_\ell)^-, \qquad [z] \longmapsto \left(\frac{\mathrm{Tr}_{K_m/\mathbb{Q}}(z)}{d_m}, \frac{z - cz}{2} \right). \tag{E.33.2}$$

For $m = 1, 2$ it is $V_m = \mathbb{Q}_\ell$. For $m \mid n$, the actual homomorphism of completed groups is induced by $G_{K_n} \hookrightarrow G_{K_m}$, as in E.30, and its map on abelianizations is

$$[z] \longmapsto [\mathrm{Tr}_{K_n/K_m} z]. \tag{E.33.3}$$

In (E.33.2) this is $(a, w) \mapsto ([K_n : K_m]a, \mathrm{Tr}_{K_n/K_m} w)$. Thus every transition is faithfully flat. These groups and these specified homomorphisms compute the continuous arithmetic kernel in (E.30.2); no identification of the abstract rational completion with this continuous completion is used. More explicitly, along the cofinal conductors $m_j = 4j!$, free generators can be chosen compatibly so that each transition keeps the first r_{m_j} generators and sends every additional generator to 1. Thus that kernel is an inverse limit of free prounipotent groups of the explicit ranks r_{m_j} with these deletion maps.

Proof. Write $K = K_m$, and let S consist of the places above ℓ and the infinite places. Every continuous homomorphism $\rho : G_K \to U(\mathbb{Q}_\ell)$, for U unipotent algebraic, is unramified outside S. Indeed its compact image is pro-ℓ: use a central series of U with additive quotients and the fact that a compact additive subgroup of a finite-dimensional $\mathbb{Q}_\ell$-space is pro-ℓ. At a finite place $v \nmid \ell$, wild inertia therefore dies, and tame inertia maps through its $\mathbb{Z}_\ell(1)$ quotient. If N is the logarithm of the image of a tame generator and F_v is arithmetic Frobenius, the exact relation $F_v \tau F_v^{-1} = \tau^{q_v}$ gives

$$\mathrm{Ad}(\rho(F_v))N = q_v N.$$

The left-hand operator is unipotent and $q_v \neq 1$ in $\mathbb{Q}_\ell$. Hence $N = 0$. The normal closed subgroup generated by these inertia groups is the kernel of $G_K \to G_{K,S}$, which proves the claimed factorization, with its universal property.

The class-field-theoretic exact sequence of local principal units, global units and the maximal abelian pro-ℓ extension unramified outside S gives

$$H_1(G_{K,S}, \mathbb{Q}_\ell) = \mathrm{coker} \left(\mathcal{O}_K^\times \otimes \mathbb{Q}_\ell \xrightarrow{\log_\ell} \bigoplus_{v \mid \ell} K_v \right). \tag{E.33.4}$$

The finite ideal class group disappears after tensoring. The exact sequence, including its proof from reciprocity, is [Sha, Theorem1.5.4 and Theorem1.5.7]. The diagonal logarithm is injective here: $K_m/\mathbb{Q}$ is abelian, so the proved abelian-field case of Leopoldt applies [Sha, Theorems1.5.17 and1.5.21]. To specify the reduction, the unit representation of the maximal real subfield is its augmentation representation. A rational normal generator gives conjugate units whose logarithms have just the norm relation over $\mathbb{Q}$; Brumer's linear-independence theorem retains independence over algebraic coefficients. Diagonalizing the abelian group matrix by characters makes every nontrivial character eigenvalue nonzero. Thus the logarithm has the full Dirichlet rank. This is precisely the proof of the cited Theorem1.5.21, and applies to every ℓ, including 2. Consequently (E.33.4) has dimension $[K : \mathbb{Q}] - (r_1(K) + r_2(K) - 1) = r_2(K) + 1$.

We also need degree two, rather than infer freeness from this dimension. Apply the finite-coefficient Euler characteristic [Mil06, TheoremI.5.1] to $\mathbb{Z}/\ell^a$ with trivial action. Its formula is exactly

$$\frac{|H^0(G_{K,S}, \mathbb{Z}/\ell^a)||H^2(G_{K,S}, \mathbb{Z}/\ell^a)|}{|H^1(G_{K,S}, \mathbb{Z}/\ell^a)|} = \ell^{-ar_2(K)}. \tag{E.33.5}$$

All these finite-coefficient groups are finite, by [Mil06, CorollaryI.4.15]. The class-field computation says that H^1 has order $\ell^{a(r_2+1)}$ times a bounded factor, because its abelian pro-ℓ source is finitely generated with finite torsion. Since $|H^0| = \ell^a$, (E.33.5) bounds the orders of H^2 independently of a. Inverse limits are exact on these finite systems; the resulting $H^2(G_{K,S}, \mathbb{Z}_\ell)$ is torsion, and tensoring with $\mathbb{Q}_\ell$ gives

$$H^2_{\mathrm{cts}}(G_{K,S}, \mathbb{Q}_\ell) = 0. \tag{E.33.6}$$

Continuous cochains with $\mathbb{Q}_\ell$ coefficients are the union of the cochain complexes with values in $\ell^{-b}\mathbb{Z}_\ell$, by compactness of their domain; this justifies the last tensoring step. The real places at $\ell = 2$ cause no exception: (E.33.5) is exactly Milne's degree-zero-through-two formula, whose footnote explicitly treats this case.

For completeness, the universal unipotent completion has the same H^1 and injects on H^2 into continuous group cohomology. The first assertion follows from homomorphisms to the additive group. For the second, an algebraic continuous Lie-cohomology class factors through a finite-dimensional quotient. Represent it by a central extension by $\mathbb{G}_a$; exponential coordinates give a continuous set-theoretic section of the projection. Its pullback along the dense continuous group homomorphism is a continuous group extension. A zero pullback class supplies a continuous lift; the universal property of unipotent completion extends that lift algebraically. Its composite with the projection is the identity on the dense original group, hence the identity, so it splits the original extension. This proves injectivity, with no comparison theorem left implicit. Equation (E.33.6) therefore makes the second cohomology of the completed Lie algebra zero.

Choose lifts of a basis of its abelianization V_m. They give a surjection from $\widehat{\mathbb{L}}(V_m)$, with a closed kernel R contained in its commutator ideal. The free-Lie five-term sequence identifies the continuous dual of $R/\overline{[\widehat{\mathbb{L}}(V_m), R]}$ with degree-two cohomology. A nonzero R has a nonzero term of least bracket degree, which cannot lie in $\overline{[\widehat{\mathbb{L}}(V_m), R]}$. Thus $R = 0$, proving (E.33.1).

For the more explicit module in (E.33.2), a unit in a CM field has unit/conjugate-unit a root of unity. Its logarithm is therefore fixed by c and has trace zero. The injectivity already proved and the equal dimensions $d_m/2 - 1$ show that the logarithmic image is exactly the trace-zero subspace of $(K_m \otimes \mathbb{Q}_\ell)^+$. The kernel and image of the displayed formula are now as asserted. The scalar $1/2$ is legitimate also over $\mathbb{Q}_2$.

Finally reciprocity sends the inclusion $G_{K_n} \to G_{K_m}$ to the local norm maps. Logarithms send norms to traces and send norms of global units to the logarithms of global units. This proves (E.33.3), including its compatibility with every tower. Trace is onto on the displayed quotient: on an embedded element of K_m it is multiplication by the nonzero degree. A morphism of prounipotent groups surjective on this finite-dimensional abelianization is surjective, successively on the lower central quotients. The transition maps are consequently faithfully flat. To obtain the final presentation, choose generators at m_1. At the next stage lift their logarithms through the surjection. Choose a basis of the remaining kernel on abelianizations and lift it into the Lie kernel: an arbitrary lift can be corrected by a commutator, because the map on commutator ideals is onto. Together these elements form a basis modulo commutators, and hence a free set of topological Lie generators by the freeness already proved. Iterating gives exactly the claimed deletion maps.

The accompanying exact cyclotomic-field program verifies the conjugation projections, all divisor trace maps and their tower compositions through conductor 24, and prints the corresponding free-Lie ranks through degree 8. This checks the explicit maps, not merely their dimensions. □

A complete intrinsic family with full geometric pro-3 lifts

E.34. Every cubic abelian class unramified outside three lifts fully [Proved]

Let N be the free pro-3 geometric fundamental group of the plus Fermat cubic with its nine cusps removed, of rank ten. Use the rational tangential section above the base tangent $t = 0$, $dt = 1$, specified by $s = -x/z$, $t = s^3$, $y/z = -(1 - s^3)^{1/3}$. Write $V = N^{\mathrm{ab}} = T_3 J_{E,D}$ and $G_S = G_{\mathbb{Q},\{3,\infty\}}$. The genuine action on N factors through G_S. Here a section is called unramified outside three when its geometric cocycle relative to this tangent factors through G_S.

The entire group of such geometrically abelian section classes is

$$H^1_{\mathrm{cts}}(G_S, V) \simeq \mathbb{Z}_3^6 \oplus (\mathbb{Z}/3)^3. \tag{E34.1}$$

Every one of these classes lifts to a section of $N \rtimes G_{\mathbb{Q}}$ whose geometric cocycle also factors through G_S. In particular all of them lift to the intrinsic metabelian, intrinsic class-two and marked-depth quotients. The actual coordinate-unit map sends (E34.1) onto

$$\{(a\kappa_3, b\kappa_3) : a, b \in \mathbb{Z}_3\} \simeq \mathbb{Z}_3^2 \tag{E34.2}$$

in the toric minimum. Its kernel is $\mathbb{Z}_3^4 \oplus (\mathbb{Z}/3)^3$, and every abelian class in each of these fibers has a full geometric pro-3 lift. These are assertions about the specified unramified family, not about all the ramified classes of E.24.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379); this is an intrinsic rank-ten geometric pro-3 family compared through the actual coordinate-unit map.

Proof. The actual action and its relative pro-3 quotient. The projective Fermat cubic over $\mathbb{Z}[1/3]$ is smooth: its three partial derivatives cannot vanish simultaneously. Each of its three coordinate-cusp schemes is finite etale of degree three over that ring, and they are pairwise disjoint. The specified cusp and parameter are integral away from three. For the based comparison, the displayed Puiseux formulas send $t^{1/m}$ to $s^{3/m}$ and commute with coefficientwise Galois action. The constant cubic roots are -1 and have compatible rational 3-power roots. Thus N is the actual index-nine subgroup of the base free pro-3 group F and its action is the restriction of the standard based action on F, with no additional pro-3 Kummer twist.

The based unramifiedness used here is stated explicitly in Hain–Matsumoto, §8.5, printed p. 28 of the opened version [HM03]: take $X_N = \mathbb{P}^1 - \{0, \mu_N, \infty\}$, $N = 1$ and $\ell = 3$ there. Smooth base change gives the *genuine* action on the fundamental group with tangent $\overrightarrow{01}$ over G_S. Restricting to the above subgroup proves the asserted unramifiedness for N.

Put $K = \mathbb{Q}(\zeta)$ and let P be the maximal pro-3 quotient of $G_{K,\{3\}}$. Its defining kernel is stable under G_S, so the relative quotient

$$\Gamma = G_S / \ker(G_{K,\{3\}} \longrightarrow P) \quad \text{fits into} \quad 1 \longrightarrow P \longrightarrow \Gamma \longrightarrow C_2 \longrightarrow 1. \tag{E34.3}$$

Complex conjugation splits this sequence. Over K the action on $F/\Phi(F) \simeq \mathbb{F}_3^2$ is trivial, because $\mu_3 \subset K$. The kernel of $\mathrm{Aut}(F) \to \mathrm{GL}(F/\Phi(F))$ is pro-3. One way to see this is to use the characteristic lower exponent-3 series: successive automorphism kernels embed in homomorphism groups between its finite $\mathbb{F}_3$-graded pieces and are 3-groups; the finite Burnside basis theorem starts the induction. The inverse limit is the asserted pro-3 kernel. The action of $G_{K,\{3\}}$ on F, and hence on N, therefore factors through P. The full action factors through (E34.3).

The same factorization holds for every V-cocycle on G_S. Indeed the map $g \mapsto (\alpha(g), \rho(g))$ takes $G_{K,\{3\}}$ into the pro-3 group $V \rtimes \rho(G_{K,\{3\}})$. By the defining universal property of P it kills the kernel in (E34.3). This proves that both the action and all the cocycles under consideration factor through the same actual quotient.

The free Galois kernel and vanishing of obstructions. The norm-Euclidean ring $\mathbb{Z}[\zeta]$ is a PID, its units are μ_6, and it has the single prime (λ), $\lambda = 1 - \zeta$, above three. Kummer theory gives

$$H^1(G_{K,\{3\}}, \mathbb{F}_3) = \langle \zeta, \lambda \rangle \simeq \mathbb{F}_3^2. \tag{E34.4}$$

For completeness, an element defining a cubic extension unramified away from three has valuation divisible by three at every other prime. Unique factorization then writes its class as $\zeta^a \lambda^b$ modulo cubes. Conversely these classes define extensions unramified away from three. Their independence follows first from the λ valuation and then from the unit group.

The global Euler characteristic of Milne, I.5.1 [Mil06], for this imaginary quadratic field and the trivial module $\mathbb{F}_3$ is $1/3$. The groups in degrees zero and one have orders three and nine, respectively, by (E34.4). Therefore

$$H^2(G_{K,\{3\}}, \mathbb{F}_3) = 0. \tag{E34.5}$$

Inflation embeds $H^2(P, \mathbb{F}_3)$ into this group. To verify that point, let J be the kernel defining P. An invariant nonzero character $J \to \mathbb{F}_3$ would produce a quotient of $G_{K,\{3\}}$ which is an extension of P by a central group of order three. That quotient is pro-3, contradicting maximality of P. The five-term sequence therefore proves the claimed injection.

These calculations prove that P is free pro-3 of rank two. Here is a direct explanation of the criterion being used. The dimension in (E34.4) gives a minimal surjection from a free pro-3 group on two generators to P. The vanishing of $H^2(P, \mathbb{F}_3)$ gives vanishing in degree two for every finite 3-primary P-module: each such module has a filtration with trivial $\mathbb{F}_3$ factors, since its action image is a finite 3-group. Every finite embedding problem with 3-group kernel is then soluble. Indeed use a nonzero characteristic elementary abelian subgroup of the center of its kernel, solve the quotient problem by induction, and split the remaining abelian extension using this H^2 vanishing. For a characteristic open subgroup U of the minimal free group F_0 and its surjection $f : F_0 \to P$, the precise finite-kernel problem is the pullback of $F_0/U \to P/f(U)$ along $P \to P/f(U)$. Its solution gives the required homomorphism $P \to F_0/U$. Apply this to all those finite quotients. There are finitely many homomorphisms to each finite target, because P has two generators; compactness gives a compatible section of the minimal surjection. Its resulting idempotent endomorphism of the free group is the identity on the Frattini quotient, hence is an automorphism by the pro-3 basis theorem. An idempotent automorphism is the identity, so the minimal surjection is an isomorphism.

In particular P has 3-cohomological dimension one. The Hochschild–Serre sequence for (E34.3), with averaging over C_2 (two is invertible on every 3-primary module), gives

$$H^2(\Gamma, M) = 0 \quad \text{for every finite 3-primary } \Gamma\text{-module } M. \tag{E34.6}$$

The same conclusion holds over G_S for every finite module whose restriction to $G_{K,\{3\}}$ has 3-group action image: filter the restriction by trivial $\mathbb{F}_3$ factors, apply (E34.5), and again take C_2 invariants.

Now take any cocycle $\alpha : \Gamma \to V$. Choose a decreasing characteristic series of N with finite 3-group quotients N_j and intersection one. At each level, pull the surjection $N_j \rtimes \Gamma \to N_j^{\mathrm{ab}} \rtimes \Gamma$ back along the reduction of α. Its kernel is a finite 3-group. The same central-subgroup induction using (E34.6) supplies a section. There are finitely many choices at every level: Γ has the two generators of P and complex conjugation, and the geometric target is finite. The compatible-section sets form a nonempty inverse limit. Explicitly, every finite collection of

levels is supplied by a section at their largest common refinement; compactness of the product of the finite sets gives a coherent choice. The limit is a continuous N-cocycle lifting α. Inflating along $G_{\mathbb{Q}} \to G_S \to \Gamma$ gives the asserted section of $N \rtimes G_{\mathbb{Q}}$. This constructs lifts in the full geometric pro-3 group, not merely compatible names for degree-two obstruction classes.

Counting the entire unramified abelian family. The genuine action on $V/3^nV$ has 3-group image over $G_{K,\{3\}}$, by the action factorization already proved. The last assertion following (E34.6) gives $H^2(G_S, V/3^nV) = 0$ for every n. These finite-coefficient degree-one groups are finite (Milne I.4.15); hence the inverse-limit sequence also gives $H^2_{\mathrm{cts}}(G_S, V) = 0$. Consequently

$$H^1_{\mathrm{cts}}(G_S, V)/3^n \simeq H^1(G_S, V/3^nV). \tag{E34.7}$$

Complex conjugation has four plus eigenvalues and six minus eigenvalues on V. To compute them, it fixes the three rational cusps and interchanges three pairs among the other six cusps. Its permutation action on degree-zero cusp divisors therefore has multiplicities five and three. The toric Tate twist reverses their signs, giving multiplicities three and five on the rank-eight toric part. On T_3E, its order is two and its determinant is -1 by the Weil pairing, so its two eigenvalues are $1, -1$. Adding the exact toric and elliptic pieces gives the claimed multiplicities. Since two is a unit in $\mathbb{Z}_3$, the same ranks hold for the integral eigensublattices.

By E.24 the rational 3-primary torsion of $G = J_{E,D}$ is $(\mathbb{Z}/3)^3$. Thus $H^0(G_S, V/3^nV) = G(\mathbb{Q})[3^n]$ has order 27 for every $n \geq 1$. Milne's Euler formula I.5.1, now over $\mathbb{Q}$, has real factor $3^{4n}/3^{10n} = 3^{-6n}$. Together with the preceding H^2 vanishing it yields

$$|H^1(G_S, V/3^nV)| = 3^{6n+3}. \tag{E34.8}$$

The continuous degree-one group is a finitely generated $\mathbb{Z}_3$-module: its cocycles factor through the three-generated group Γ and form a closed submodule of V^3, modulo the closed coboundary image. Equations (E34.7)–(E34.8) give rank six and torsion of order 27 and exponent three. Equivalently that torsion is $H^0(G_{\mathbb{Q}}, V \otimes \mathbb{Q}_3/V) = G(\mathbb{Q})[3^\infty]$. This proves (E34.1).

Finally the algebraic retraction of E.24 gives a split map of these Galois modules to $\mathbb{Z}_3(1)^2$. Kummer theory over $\mathbb{Z}[1/3]$ gives $H^1(G_S, \mathbb{Z}_3(1)) = \mathbb{Z}_3\kappa_3$. Its rank-two target is torsion-free, and the splitting therefore proves (E34.2) and the stated rank-four kernel with its three torsion factors. The change from the origin at O in E.24 to the present rational cusp is also unramified outside three. The framed cusp-difference relation (E24.6) involves only the units λ, powers of 3 and signs, so this framed point extends over $\mathbb{Z}[1/3]$ and its Kummer cocycle is unramified there. Equivalently both integral tangential fiber functors give the same unramified comparison. Both origins have zero toric pro-3 class. Thus changing origins preserves the unramified family and the displayed linear toric map. The matrix and order calculations are recorded in `e_intrinsic_positive.py`; the Galois vanishing and lifting arguments above establish the infinite assertions. □

All sections of the intrinsic Heisenberg quotient

E.35. A norm equation and every conjugacy class in the finite intrinsic quotient [Proved]

Let $K = \mathbb{Q}(\zeta)$ and retain the quotient H_3 with the based G_K-action (E31.3). Write $\mathcal{A} = K^\times/K^{\times 3}$ multiplicatively. For an abelian section pair $(a, b) \in \mathcal{A}^2$ put $A = \zeta a$, choose nonzero representatives $A, b \in K$, and form the etale cubic algebra $L_A = K[t]/(t^3 - A)$. The fiber of section conjugacy classes of $H_3 \rtimes G_K$ over (a, b) is nonempty exactly when the explicit equation

$$X^3 + AY^3 + A^2Z^3 - 3AXYZ = b \quad \text{has a solution } X, Y, Z \in K. \tag{E35.1}$$

Every nonempty fiber is a torsor under the explicitly specified group

$$\mathcal{A}/\langle A,b\rangle = K^{\times}/(K^{\times 3}\langle A,b\rangle). \tag{E35.2}$$

The subgroup in the denominator has rank zero, one or two over $\mathbb{F}_3$, according to the span of the two indicated cube classes. There are concrete coordinates for this group. Choose one generator $\pi_{\mathfrak{p}}$ for each nonzero prime ideal of $\mathbb{Z}[\zeta]$, and put $J = \{\zeta\} \amalg \{\mathfrak{p}\}$. Unique factorization gives the explicit isomorphism

$$\mathcal{A} \longrightarrow \bigoplus_{j\in J} \mathbb{F}_3, \qquad [\zeta^{e} \prod_{\mathfrak{p}} \pi_{\mathfrak{p}}^{e_{\mathfrak{p}}}] \longmapsto (e,(e_{\mathfrak{p}})) \pmod 3. \tag{E35.2a}$$

The sign in the factorization is a cube and is omitted. Thus (E35.2) is obtained by Gaussian elimination of the two finite-support vectors $[A],[b]$: choose zero, one or two pivot coordinates according to their rank, subtract the corresponding row-reduced basis vectors, and retain all remaining coordinates. This is an explicit parameterization by the direct sum of $\mathbb{F}_3$ over the coordinates in J outside those pivots. After choosing the lift n_0 constructed from a norm solution, all classes are represented by $n_d(g) = (0,0,\kappa_3(d)(g))n_0(g)$ with $d \in K^{\times}/(K^{\times 3}\langle A,b\rangle)$.

All dependent cases are included. If A is a cube, (E35.1) is always soluble and the fiber group is $\mathcal{A}/\langle b\rangle$. If A is not a cube and b is in its span, (E35.1) is always soluble and the fiber group is $\mathcal{A}/\langle A\rangle$. If both classes are trivial the group is $\mathcal{A}$. For two independent classes, a solution of (E35.1) constructs an actual Heisenberg Galois extension as in the proof.

The quotient is the one obtained from the actual map (E31.4) and the based arithmetic comparison of E.31. For example the pair $(a,b) = (1,7)$ has empty fiber, while $(a,b) = (2\zeta^2,3)$ has nonempty fiber, with norm solution $1 + \sqrt[3]{2}$. No assertion is made that every pair here is the image of a class in the full intrinsic rank-ten abelian section group.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379); this computes a specified finite intrinsic quotient reached by the additional morphism (E31.4).

Proof. Choose the path in E.31 within the formal punctured disc at the target zero. Equation (E31.7) then gives the image of the source tangential section as

$$q_g = (\kappa_3(\zeta)(g),0,0) \in H_3. \tag{E35.3}$$

To see the zero central coordinate directly, write $h = \zeta\lambda^{-3}s^3u(s)$ with $u(0) = 1$. Its chosen cube root is $(\zeta_9/\lambda)s\,u(s)^{1/3}$. The last factor has coefficients in K, so the only Galois change is the inertia power supplied by ζ_9. The formal puncture has only that one geometric generator and gives no commutator term. Thus $q : G_K \to H_3$ is the homomorphism in (E35.3), and the intrinsic action is $\mathrm{Ad}(q_g)$.

There is an explicit isomorphism of arithmetic groups

$$H_3 \rtimes_{\mathrm{Ad}\,q} G_K \longrightarrow H_3 \times G_K, \qquad (n,g) \longmapsto (nq_g,g). \tag{E35.4}$$

Its inverse sends (h,g) to (hq_g^{-1},g). It is the identity on the geometric group. Consequently it identifies section classes with continuous homomorphisms $G_K \to H_3$ modulo conjugation by H_3, and sends the intrinsic abelian pair (a,b) to the ordinary Kummer pair $(A,b) = (\zeta a,b)$.

Write these latter Kummer characters as $u,v : G_K \to \mathbb{F}_3$. A homomorphism with those abelian coordinates has the form (u_g,v_g,w_g), where its central coordinate solves

$$dw(g,h) = -\tfrac{1}{2}(u_gv_h - v_gu_h). \tag{E35.5}$$

The two-cocycle on the right, with sign reversed, differs from u_gv_h by $d(u_gv_g/2)$; hence its obstruction is exactly $\kappa_3(A) \cup \kappa_3(b)$. The Kummer exact sequence and Hilbert 90 embed this group of classes into $\mathrm{Br}(K)[3]$.

We spell out the resulting norm condition. Suppose first that A is not a cube, let $\alpha^3 = A$ and let $\sigma(\alpha) = \zeta\alpha$ generate $\mathrm{Gal}(L_A/K)$. The cyclic algebra is

$$\mathcal{C} = L_A \oplus L_A U \oplus L_A U^2, \qquad U\ell = \sigma(\ell)U, \quad U^3 = b. \tag{E35.6}$$

Its factor set is $b^{\lfloor (i+j)/3 \rfloor}$ on σ^i, σ^j. Choose $\beta^3 = b$ and represent the values of u by $0, 1, 2$. The multiplicative coboundary of $g \mapsto \beta^{u_g}$ is

$$b^{\lfloor (u_g + u_h)/3 \rfloor} \zeta^{v_g u_h}. \tag{E35.7}$$

The identity for $d(u_g v_g)$ in (E35.5) shows that the class of this factor set is precisely $\kappa_3(A) \cup \kappa_3(b)$. The crossed-product comparison with the Brauer group is Milne, *Class Field Theory*, IV.3.14 [Mil20]; its construction uses exactly the multiplication and factor set just displayed.

The algebra (E35.6) is split exactly when b is a norm. This equivalence also has a direct proof here. A norm solution $b = N_{L_A/K}(w)$ represents $\mathcal{C}$ on the three-dimensional K-space L_A, with ℓ acting by multiplication and U acting as $w\sigma$. The relations hold and the resulting homomorphism is an isomorphism with $\mathrm{End}_K(L_A)$, by central simplicity and the equal dimension nine. Conversely, a splitting realizes $\mathcal{C}$ on a three-dimensional K-space which, by its embedded degree-three field L_A, is one-dimensional over L_A. The operator U is necessarily $w\sigma$ in a choice of L_A-basis. Its cube is $N(w)$, so $b = N(w)$. The norm of $X + Y\alpha + Z\alpha^2$, by its multiplication matrix on this same basis, is $X^3 + AY^3 + A^2Z^3 - 3AXYZ$. This proves (E35.1).

For independent cube classes A, b the norm solution also yields an explicit finite Galois realization. Write $w = X + Y\alpha + Z\alpha^2$ with $N(w) = b$ and put

$$\beta^3 = b, \qquad v = w^2\sigma(w), \qquad \gamma^3 = v. \tag{E35.8}$$

The field $M = L_A(\beta, \gamma)$ has the automorphisms

	α	β	γ
$\widetilde{\sigma}$	$\zeta\alpha$	β	$\beta\gamma/w$
τ	α	$\zeta\beta$	γ.

(E35.9)

Indeed $\sigma(v)/v = b/w^3$, so these formulas preserve the defining equations, and $\widetilde{\sigma}^3(\gamma) = b\gamma/N(w) = \gamma$. Both displayed automorphisms have order three. The root γ does not already lie in $L_A(\beta)$. Otherwise the fact that its cube is in L_A would write $\gamma = c\beta^j$ for $c \in L_A$. The equation $\sigma(v)/v = b/w^3$ would then imply that b is a cube in L_A, contradicting the independence of A, b. Thus $[M : K] = 27$, the formulas make M/K Galois, and their commutator acts on γ as multiplication by ζ^{-1}. Taking that automorphism as the central generator identifies its Galois group with (E31.2), with the two stated Kummer characters. Composing the Galois restriction map with this identification and then $(\mathrm{E35.4})^{-1}$ gives an actual arithmetic section.

The dependent cases require no degree-27 field. If A is a cube, replace its representative by one. Then $L_A \simeq K^3$ and the norm is the product of its three coordinates, so it is surjective. In (E35.1) the choice $X = (b+2)/3$, $Y = Z = (b-1)/3$ gives norm b. The homomorphism $g \mapsto (0, \kappa_3(b)(g), 0)$ provides a lift. If A is nontrivial and $b = A^j c^3$ for $j = 0, 1, 2$, the element $w = c\alpha^j$ has norm b. The homomorphism $g \mapsto (\kappa_3(A)(g), j\kappa_3(A)(g), 0)$ has the required coordinates; its alternating term is zero. This includes $b = 1$ and all other dependent cases. The construction is independent of the selected representatives modulo cubes.

Finally choose one lift h_g of (u, v). Every other lift is uniquely $z_g h_g$ with $z \in H^1(G_K, \mathbb{F}_3) = \mathcal{A}$. Conjugating by an element with abelian coordinates (r, s) adds the central character $rv - su$. Central conjugations do nothing. The exact subgroup of differences generated by all

geometric conjugations is therefore $\langle \kappa_3(A), \kappa_3(b) \rangle$. This proves the entire torsor description (E35.2), including each of the dependent cases just treated. For completeness, the coordinate computation (E35.2a) uses the Euclidean norm $N(a+b\zeta) = a^2 - ab + b^2$ on $\mathbb{Z}[\zeta]$. Rounding the two rational coefficients of any quotient to nearest integers leaves errors r, s of absolute value at most $1/2$, and $r^2 - rs + s^2 \leq 3/4 < 1$. The division algorithm proves that this ring is a principal ideal domain and a unique factorization domain. Its norm-one units are $\{\pm 1, \pm\zeta, \pm\zeta^2\}$, so their quotient by cubes is generated freely over $\mathbb{F}_3$ by ζ. Factoring the numerator and denominator of an element of $K^\times$ now proves both the injectivity and surjectivity of (E35.2a). Row reduction gives an actual isomorphism from its quotient by $[A], [b]$ to the stated direct sum: normalize the pivot entries to one and all other pivot entries to zero, then send a vector to its unique representative with every pivot entry zero. This also describes the inverse by inclusion of those nonpivot coordinates.

At $(a, b) = (1, 7)$, equation (E35.1) would assert that 7 is a norm from $K(\zeta_9)$, contradicted by the local calculation at $(3 + \zeta)$ in E.31. For $A = 2, b = 3$, the element $1 + \alpha$ has norm $1 + \alpha^3 = 3$, proving the positive example. The program `e_intrinsic_finite.py` checks the norm polynomial, this solution and the relations in (E35.8)–(E35.9), and enumerates every conjugation orbit for all two-generator assignments in H_3. The finite intrinsic quotient map of E.31 remains the comparison to the full cubic problem; no inverse to that quotient map is asserted. □

Every fiber of the cuspidally central cubic quotient

E.36. A class-two quotient retaining all ten abelian coordinates, with every fiber evaluated [Proved]

Let $E = X_3^+$, let D be its nine coordinate cusps, and put $O = [1 : -1 : 0]$. Use the tangential base point at O with parameter $s = z/x$ and tangent one, both on $U_3 = E \setminus D$ and on $E \setminus \{O\}$. Write

$$N = \pi_1(U_{3,\overline{\mathbb{Q}}})^{(3)}, \qquad V = N^{\mathrm{ab}} = T_3 J_{E,D}, \qquad T = T_3 E, \qquad I = \ker(N \longrightarrow T).$$

Closures are understood in commutator subgroups. Set

$$N^{\mathrm{cc}} = N/[N, I], \qquad C = \pi_1((E \setminus \{O\})_{\overline{\mathbb{Q}}})^{(3)}/\gamma_3.$$

The maps induced by the open immersion $U_3 \hookrightarrow E \setminus \{O\}$ and by abelianization give an actual, based, $G_{\mathbb{Q}}$-equivariant isomorphism

$$N^{\mathrm{cc}} \xrightarrow{\sim} C \times_T V, \qquad n \longmapsto (q_O(n), n^{\mathrm{ab}}). \tag{E36.1}$$

In particular the quotient retains all ten coordinates of V and its nonzero elliptic commutator. The exact quotient from the full intrinsic class-two group is

$$1 \longrightarrow \ker\Big(\bigwedge^2 V \longrightarrow \bigwedge^2 T\Big) \longrightarrow N/\gamma_3 N \longrightarrow N^{\mathrm{cc}} \longrightarrow 1. \tag{E36.2}$$

The displayed kernel is a free $\mathbb{Z}_3$-module of rank 44, with the Galois action induced on the exterior square; the remaining central commutator module is $\bigwedge^2 T \simeq \mathbb{Z}_3(1)$.

Every geometric-abelian section class $\alpha \in H^1(\mathbb{Q}, V)$ lifts to a section of $N^{\mathrm{cc}} \rtimes G_{\mathbb{Q}}$. For every such α, the complete fiber of geometric conjugacy classes is a torsor under

$$\mathcal{R}_{\mathbb{P}}(\mathbb{Z}_3) = \left\{ (a_p)_{p \text{ mrational prime}} \in \prod_p \mathbb{Z}_3 : \begin{smallmatrix} \text{for every } m,\ a_p \in 3^m\mathbb{Z}_3 \\ \text{for all but finitely many } p \end{smallmatrix} \right\}. \tag{E36.3}$$

The parameter $a=(a_p)$ acts by the central Kummer cocycle of the compatible classes $\prod_p p^{a_p}$ modulo $\mathbb{Q}^{\times 3^m}$. These finite products modulo each 3^m define the indicated element of the completion for multiplication by powers of three. Thus this is a computation of every fiber over the entire rank-ten abelian section group evaluated in E.24, including its ramified classes. The classes obstructed in E.31 also lift to (E36.1), while their further lifting through (E36.2) is obstructed by the explicit quotient of E.31.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379). The quotient maps (E36.1)–(E36.2) specify exactly the intrinsic cubic quotient evaluated here.

Proof. First compute the geometric maps integrally. An oriented presentation of the punctured complex elliptic curve is

$$N=\langle a,b,c_O,c_1,\dots,c_8 \mid [a,b]c_Oc_1\cdots c_8=1\rangle_{\text{pro-3}}. \tag{E36.4}$$

Here a,b map to a basis of T; the c_d are the peripheral loops. Eliminating c_O gives the free pro-three group of rank ten. The kernel I is normally generated by these peripheral loops. Passing to $N/[N,I]$ makes all c_d central and leaves only the indicated relation. Consequently the geometric cuspidal kernel is

$$I/[N,I]\simeq \mathbb{Z}_3(1)[D], \qquad [a,b]=-\sum_{d\in D} c_d, \tag{E36.5}$$

with its actual Galois action: the cusps are permuted and each inertia generator has the cyclotomic twist. This action can also be obtained from the torus $\mathrm{Res}_{D/\mathbb{Q}}\,\mathbb{G}_m$. The central quotient is the one defined in Borne–Emsalem–Stix, §3.2; their Proposition 3.5 identifies it with the fundamental group of their torus torsor by the canonical map from the open curve [BES18].

For an entirely explicit check of (E36.5), take the two-step Lie module with basis $a,b,c_1,\dots,c_8,k$ and only bracket $[a,b]=k$. The law $u*v=u+v+[u,v]/2$ is integral over $\mathbb{Z}_3$. Sending c_O to $-k-\sum_{i=1}^{8}c_i$ satisfies (E36.4). Conversely its generators and brackets give precisely the relations of $N/[N,I]$. The two maps are inverse, so there are no further relations or torsion in the nine-dimensional central lattice of (E36.5).

The map q_O fills the other eight punctures. On (E36.5) it is projection to the O-coordinate. Abelianization quotients (E36.5) by the diagonal element $\delta=\sum_{d\in D}c_d$. The maps on the two geometric cuspidal kernels are therefore

$$\mathbb{Z}_3(1)[D]\longrightarrow \mathbb{Z}_3(1)\oplus\big(\mathbb{Z}_3(1)[D]/\mathbb{Z}_3(1)\delta\big), \quad t\longmapsto (t_O,\bar{t}). \tag{E36.6}$$

This is an isomorphism: in coordinates it sends $(t_O,t_1,\dots,t_8)$ to $(t_O,t_1-t_O,\dots,t_8-t_O)$, with inverse $(u,v_1,\dots,v_8)\mapsto(u,v_1+u,\dots,v_8+u)$. It is $G_{\mathbb{Q}}$-equivariant because O is rational and the diagonal element is invariant apart from its cyclotomic twist. Both sides of (E36.1) map onto T, and (E36.6) identifies their kernels. This proves (E36.1). The isomorphism is induced by the stated morphisms of curves and abelianization, with the same tangent throughout. It therefore identifies the actual arithmetic extensions, not just their abstract geometric groups. There is no change of section origin in (E36.1).

The commutator module of the free class-two pro-three group on V is $\bigwedge^2 V$. The bracket surviving in (E36.5) is the exterior-square map induced by $V\to T$. In the basis from (E36.4) it sends $a\wedge b$ to one generator and sends the other 44 basic wedges to zero. Its kernel is therefore exactly $[N,I]/\gamma_3 N$, free of rank $\binom{10}{2}-1=44$; its image has index one. The orientation identifies the image with $\bigwedge^2 T=\mathbb{Z}_3(1)$, compatibly with the Weil pairing. Functoriality proves the asserted arithmetic action and (E36.2).

We next establish the existence of all the sections in the statement. Use the rational point O as origin on the proper elliptic curve. Its full geometric fundamental group is $T_\bullet E = \prod_\ell T_\ell E$, and its arithmetic fundamental group is the corresponding semidirect product with $G_\mathbb{Q}$. For a given α, let $\beta \in H^1(\mathbb{Q}, T_3 E)$ be its image, and choose compatible cocycle representatives. There is a specified full proper-curve section

$$g \longmapsto \big((\beta(g), 0, 0, \ldots), g\big), \tag{E36.7}$$

where every component at a prime other than three is zero. The inclusion of the three-primary factor is continuous and $G_\mathbb{Q}$-equivariant, so (E36.7) is an actual section of the full arithmetic fundamental group of E.

Borne–Emsalem–Stix, Theorem 5.10, states that for a smooth projective positive-genus curve over $\mathbb{Q}$ and a boundary which is a union of torsion sub-packets, every section lifts to its cuspidally central quotient [BES18]. Apply the theorem to E and the singleton boundary $\{O\}$. This is a torsion sub-packet by their Definition 2.3: the only difference of two points in its support is zero in the Picard group. All hypotheses are now satisfied. The cuspidal kernel for a once-punctured elliptic curve is the geometric commutator subgroup; making it geometrically central is precisely quotienting by γ_3. Projecting the resulting full lift of (E36.7) to its geometric pro-three quotient gives a section c_β of $C \rtimes G_\mathbb{Q}$ above β.

If this section projects to a representative conjugate to the chosen β, conjugate it by a geometric lift of the relevant element of T. Its projection then equals the chosen representative. Pairing it with the given abelian section α gives a section of the fiber product (E36.1). This proves existence for every α, without imposing any ramification condition.

Finally the kernel of $N^{\mathrm{cc}} \to V$ is the central copy of $\mathbb{Z}_3(1)$ generated by the elliptic commutator. After choosing one lift of a fixed representative of α, the other lifts differ by continuous one-cocycles with values in this module. Central geometric conjugation changes the cocycle by an ordinary coboundary. There is no additional identification from geometric elements with nonzero image in V: such an element could fix the prescribed abelian cocycle only if its image lay in $H^0(\mathbb{Q}, V)$, and this group is zero. Here is a direct verification. In the exact sequence $0 \to T_D \to J_{E,D} \to E \to 0$, the rational torsion of $T_D \simeq \mathbb{G}_m^2 \times (\mathrm{Res}_{K/\mathbb{Q}}\, \mathbb{G}_m)^3$ is finite, being a product of groups of roots of unity. The group $E(\mathbb{Q})$ has order three by E.24. Thus $J_{E,D}(\mathbb{Q})[3^\infty]$ is finite. An invariant in $T_3 J_{E,D}$ is a compatible sequence in this finite three-primary group under multiplication by three. Every such sequence is zero. Hence $H^0(\mathbb{Q}, V) = 0$ as asserted.

The exact fiber is consequently a torsor under $H^1(\mathbb{Q}, \mathbb{Z}_3(1))$. Kummer theory identifies this group with $\varprojlim_m \mathbb{Q}^\times / \mathbb{Q}^{\times 3^m}$. Unique factorization in $\mathbb{Q}^\times$ gives the isomorphism with (E36.3): the sign is a cube, each prime valuation is reduced modulo 3^m, and every such residue vector has finite support. The inverse constructs the compatible Kummer classes from precisely these finite-support residue vectors. This proves the complete coordinate description of all fibers.

The matrix in (E36.6) has determinant one, and the exterior-square map in (E36.2) has Smith diagonal one and kernel rank 44. The program `e_cuspidally_central.py` verifies these integral lattice computations. The existence theorem and the arithmetic fiber calculation above establish the infinite section assertions. □

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